

A DESIGN THEORY FOR CONTEXT-AWARE INFORMATION SYSTEMS

Improving IS Design Science Outcomes Through Theory-Driven Design

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Statement of Original Authorship

The work contained in this thesis has not been previously submitted to meet requirements for an award at this or any other higher education institution. To the best of my knowledge and belief, the thesis contains no material previously published or written by another person except where due reference is made.

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Abstract

Design theory and design theorising are an area of increasing interest in design science research in the information systems discipline. Based on the exemplary design problem of context and context-driven adaptation of systems, I demonstrate how prescriptive guidelines can be developed from justificatory knowledge that promise result in more effective information systems. I proceed in five steps. First, I review the current body of the literature in IS and IS reference fields to identify and compare divergent views of context and information systems. Second, I then explore and contrast two examples of context-driven adaptation based on a multiple-case study of two organisations in different industries. Third, using theoretical and empirical insights, I measure the completeness and clarity of an exemplary IS design method with respect to context and formulate design propositions regarding its extension. Fourth, based on these propositions I develop a set of prescriptions and, fifth, evaluate their applicability and ability to generate actionable recommendations. Finally, I argue that the specific approach taken in the refinement from justificatory knowledge, design propositions, prescriptions, and their evaluation represents the first example of theory-driven design in IS.

Keywords

Design theory, design science research, case study research, information systems, context, context-aware information systems

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List of Abbreviations

Abbreviation	Explanation
CAIS	Context-Aware Information System
CAQDAS	Computer-Aided Qualitative Data Analysis Software
CS	Computer Science
DREPT	Design-Relevant Explanatory Theory
DSRIS	Design Science Research in the IS Discipline
HCI	Human Computer Interaction
IS	Information System
ISDT	Information System Design Theory
MD	Meta-Design (a component of ISDT)
MR	Meta-Requirement (a component of ISDT)
MS	Management Science
OS	Organisation Science
SSC	System of Systems Concept
STS	Socio-Technical Systems
TP	Testable Design Proposition (a component of ISDT)
WSM	Work System Method

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No man is an island, entire of itself.

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Karsten Ploesser

Chapter 1: Introduction

1.1 MOTIVATION

Over the past two decades, the notion of "context" and the consideration of *context in system design* has received much attention in the reference fields of the Information Systems (IS) discipline. The consideration of context has come from the observation that the design of systems will benefit from an explicit study of the context in which its users work (Nardi, 1995). However, while there exists a general agreement regarding the importance of an explicit understanding and consideration of context in system design, much less clarity exists on the concrete measures that must be undertaken to arrive at systems that are context-aware. Academic research in the Computer Science (CS) and Human Computer Interaction (HCI) disciplines has led to two divergent views of context (Dourish, 2004). Different assumptions and conceptualisations of context and systems, in turn, have led to different prescriptions regarding the construction of context-aware systems. These range from explicit context representation (Bettini et al., 2010) to research on tailorable technology and user context (Mørch & Mehandjiev, 2000).

IS research, if understood as the study of phenomena that arise at the *intersection of organisation, people, and technology* (March & Storey, 2008), can benefit from this discussion of context in system design. The study of information systems is ostensibly about work systems (Alter, 2001). Much of the concepts in IS research such as organisation, people, information, and technology point to IS being a special case of work system (Alter, 2008). Furthermore, there have been repeated calls for considering IS as socio-technical systems embedded in their social and organisational context (McKay, Marshall, & Hirschheim, 2012). According to this view, technology must be viewed as situated in a context (Suchman, 1987) and design as the practice of giving "equal weight to social and technical issues when new work systems are being designed" (Mumford, 2000, p. 125). An explicit understanding and study of system context thus is instrumental in developing more effective systems and, in particular, systems that adapt to context changes. It is therefore surprising how little research on context as a construct in IS design has been published.

Despite the seeming relevance of context as a construct for IS research and design, *little conceptual and empirical work* has been conducted on the topic and the phenomenon remains poorly understood (Rosemann, Recker, & Flender, 2008). Published literature reviews of context and context in system design remain isolated and mostly confined to research streams in

artificial intelligence (Akman & Surav, 1996; Brézillon, 1999), human computer interaction (Dourish, 2004; Nardi, 1995), and mobile or pervasive systems (Abowd, Mynatt, & Rodden, 2002; Bettini et al., 2010; Dey, Abowd, & Salber, 2001). The topic of context has received some consideration in research streams that are generally associated with IS such as process modelling (Bessai, Claudepierre, & Saidani, 2008; Rosemann & Recker, 2006; Saidani & Nurcan, 2007), contextualisation and improvement of business processes (Ramos & Santoro, 2010; Rosemann et al., 2008), and context in process-aware information systems (Hallerbach, Bauer, & Reichert, 2009; 2010). While each contribution makes statements about specific design problems such as workflow management, they do not generalise to prescriptive guidelines for the design of context-aware IS.

The motivation for the present research study is therefore the development of a *design theory* for the design of more effective context-aware information systems (CAIS). In recent years, design theorising (Gregor & Jones, 2007; Kuechler & Vaishnavi, 2012; Walls, Widmeyer, & Sawy, 1992) has been recognised as an important research activity that is promoting relevance in IS research (Kuechler & Vaishnavi, 2011). Design theorising complements the behavioural cycle of research in IS (Hevner, 2007) with prescriptive statements regarding the design of systems on the basis of academic kernel theory (Gregor, 2006). Prescriptions arise from the fact that each design statement is derived from justificatory knowledge in the social or natural sciences (Gregor & Jones, 2007) and related to actions that, if followed, will lead the design to achieve an expected outcome. For example, justificatory knowledge in the social sciences about the behaviour of people can be used to create prescriptive statements about the design of recommender systems (Arazy, Kumar, & Shapira, 2010). I argue that a similar approach can be applied to create prescriptive statements about context-aware IS.

1.2 RESEARCH PROBLEM AND OBJECTIVE

The problem addressed by this study is the development of *prescriptive statements* about a new class of information systems, i.e., the class of context-aware information systems (CAIS). Design theories refer to an integrated prescription consisting of a class of problems or "meta-requirements", a class of solutions that are hypothesised to resolve these problems, and testable propositions that can be evaluated in system practice (Walls et al., 1992). As such, design theories provide guidance to system practice in managing a set of design problems and a basis for system research to make predictions about the design of effective information systems (Markus, Majchrzak, & Gasser, 2002). In IS, there is a long tradition for the labelling of recurring design problems as system classes as evidenced by the use of concepts such as Transaction

Processing Systems (TPS), Decision Support Systems (DSS), or Executive Information Systems (EIS). In the spirit of Walls *et al.*, Markus *et al.*, and the practice of labelling system classes, the study introduces and defines a new class of Context-Aware Information Systems (CAIS).

However, developing prescriptive guidelines for system classes must not only meet the needs of system practice but must also *adhere to the strict standards* of scientific research. One of the fundamental problems of design theorising in the IS discipline is tracing the link between justificatory knowledge, to prescriptive statements about a class of systems, to utility provided by a concrete instantiation of guidelines in an artefact (Kuechler & Vaishnavi, 2012). Kuechler and Vaishnavi identify three levels of abstraction that affect the nature and applicability of research outputs. At the most abstract level, design-relevant explanatory theories translate academic kernel theories (Gregor & Jones, 2007) into system explanations and utility claims. For example, Kasper (1996) maps behavioural theory into system explanations. Second, IS design theories formalise design knowledge into prescriptions about a class of problems and system solutions. This is evidenced by the work of Walls *et al.* (1992). Finally, design artefacts define the most immediate and manifest occurrence of IS design (Hevner, March, Park, & Ram, 2004).

The research objective of this study is therefore the *theory-driven design* of CAIS as a set of prescriptive statements as opposed to the more traditional view of design research as the creation of artefacts (Hevner *et al.*, 2004). Following an increasing trend in IS design research (Arazy *et al.*, 2010; Baskerville & Pries-Heje, 2010; Gregor & Jones, 2007; Kuechler & Vaishnavi, 2012; Müller-Wienbergen, Müller, Seidel, & Becker, 2011), I view design theory as the primary output of design research studies. Unlike most of these works, on the other hand, I do not focus on one level of abstraction alone. Similar to Yin's (2009) notion of creating a "chain of evidence" in a qualitative research study, I provide complete traceability of design knowledge by mapping empirical knowledge and concepts in kernel theories, to system explanations and utility claims in explanatory design theory, to meta-requirements and meta-design in IS design theory. The stepwise refinement of concepts to prescriptions follows related work on representation theory and theory evaluation (cf. Weber, 2012). I refer to this approach as theory-driven design and provide a concrete example in the following in the form of a design theory for context-aware IS.

1.3 RESEARCH QUESTIONS

In order to meet the research objective of defining prescriptive guidelines for the design of more effective, context-aware information systems, I identify three relevant research questions. Research questions focus the attention of the researcher and guide the research study. I

relate each research questions to views and contributions on context and system design in the literature and provide a motivation for why answers to each will contribute to our knowledge.

First, divergent views of context and assumptions about context in system design require me to define what is *relevant information system context*. Research in context has so far not led to a conclusive definition of context and its use in system design (Dourish, 2004). Two prevalent views and the assumptions they make about the nature of context can be identified. First, context can be viewed as information about the situation of entities at a given moment in time (Dey, 2001). This view defines context as information that can be captured, encoded, and represented much like other forms of information. Examples for the types of entities processed by information systems include people, processes, technology, customers, and products or services (Alter, 2008). Second, context can be viewed as an occasioned property that cannot be represented (Dourish, 2004). This view defines context as a property shared by people which must be considered in system design (Nardi, 1996). Examples for this understanding of context in IS can be found in discussions of tailorability (Germonprez, Hovorka, & Collopy, 2007).

I summarise this discussion in the following research question:

- **Research Question 1.** What is relevant information system context?

Second, divergent views of the nature of information systems and the variables classes of which they are composed require me to consider *where does context change affect information systems*. Since its inception, research in the IS discipline has struggled to define its core subject matter (Alter, 2008; Benbasat & Zmud, 2003; Orlikowski & Iacono, 2001). Two prevalent views of information systems and assumptions about their design can be identified. First, the artefact view describes IS as the application of information technology to support tasks embedded in structures which are embedded in a social and organisational context (Benbasat & Zmud, 2003). Design research that subscribes to this view assumes that IT artefacts are innovative combinations of hardware and software that are surrounded by the organisational context in which they are used (Hevner et al., 2004). Second, the systems view describes IS as work systems in which people perform work by using information and technology to deliver outputs to customers (Alter, 2008). IT defines one component of such systems and their design (McKay et al., 2012).

I summarise the discussion of information system variables and the role of context in the following research question:

- **Research Question 2.** Where does context change affect information systems?

Finally, the discussion and divergent views of context and information systems lead to the question of how information systems *should be changed in order to adapt to relevant context change*. System change and failure have long been viewed as critical to the effective implementation of information systems (Bostrom & Heinen, 1977; Markus, 1983). Again, two views can be identified. First, rationalist views of technology (Kling, 1980) view IT as a driver of change and the creation of internal efficiencies (Hammer, 1990). Similarly, technology-driven views of context (Weiser, 1993) view information technology as a driver for more adaptive systems. However, the effectiveness of this view and its ability to lead to more effective system design has been critiqued (Suchman, 1987). Second, views of information systems as work system (Bostrom & Heinen, 1977) emphasise the need of joint optimisation of the technical and social system (Mumford, 2000). Following this view, information technology alone does not lead to more adaptive systems. Rather, adaptiveness must align both people and technology to context.

I summarise the discussion of how information systems should be adapted to context change in the following research question:

- **Research Question 3.** How should information systems be changed in order to adapt to relevant context change?

1.4 DEFINITIONS

The objective of this chapter is to provide a single location for defining core concepts and to provide examples that justify the idea of Context-Aware Information System (CAIS). The idea of context and context-aware computer systems is not new (cf. Lieberman & Selker, 2000; Dey et al., 2001). However, what is new is to consider context in the design and implementation of Information Systems (IS). Before proceeding to define prescriptions concerning how such systems should be designed, it is therefore necessary to bring attention to four concepts that are fundamental to understanding the development of the argument in my thesis. These are Information Systems, Context, Adaptation, and Context-Aware Information Systems. Foreshadowing the more detailed work in subsequent chapters, I define each concept in the following and give illustrative examples.

1.4.1 KEY CONCEPTS: IS, CONTEXT, ADAPTATION, AND CAIS

The term Context-Aware Information System (CAIS) is a combination of several concepts that have each long been discussed in the academic literature. First, information systems have been studied since the early 1970's when computerised systems were introduced into organisations

(Ackoff, 1968). Several interpretations of IS emerged of which my work embraces the notion of information-based work systems (Alter, 2008). Second, advances in information processing and storage capabilities led to an increasing interest in context (Lieberman & Selker, 2000) as means to adapt systems to the needs of their users.

Information systems are systems the purpose of which is to deliver information-based products or services to internal or external customers. Two views of information systems are prevalent in the IS literature. First, views of IS as artefact suggest IS are tools that support tasks that are embedded in structures that are part of a larger social context (Benbasat & Zmud, 2003). This view has received much attention in artefact-centric approaches to IS design (Hevner et al., 2004) but fails to consider the complex, socio-technical nature of IS (McKay et al., 2012). Second, views of IS as systems have been prevalent in the early literature on IS (Bostrom & Heinen, 1977) and have received increasing attention in recent years (Alter, 2013). This view emphasises IS as complex, socio-technical systems.

In the context of my research enquiry, I adopt a systems view of IS and more specifically the view of IS as information-based work systems. A systems view of IS includes the problem of environment as one of the central tenets of systems theory. The environment of a system is the set of elements that are not part of the system but a change in which leads to a change in the state of the system (Ackoff, 1971). Following systems theory, I turn to Alter's work on positioning IS as information-based work systems to define the nature of IS and their interaction with elements in the external environment:

An information system is "a system in which human participants and/or machines perform work (processes and activities) using information, technology, and other resources to produce informational products and/or services for internal or external customers", (Alter, 2008, p. 5)

Second, context is a specific type of information that is not explicitly provided to the system as user input. Two views of context have emerged in recent years (Dourish, 2004). First, research in Human Computer Interaction suggests context is an occasioned property that emerges from particular situations at particular times involving particular people (Nardi, 1996). According to this view, context cannot be encoded as it does not generalise across the situation in which it occurs. Second, research in Artificial Intelligence defines context as "information that can be used to characterise the situation of an entity" (Dey, 2001, p. 5). According to this view, context can be captured, stored, and processed much like other forms of information.

In the remainder, I adopt the definition of context as information, and thus regard context as an aspect of information systems that can be designed. Following the earlier definition of information systems as complex socio-technical systems, viewing context as information relates to Ack-

off's comment that "a system's environment consists of all variables which can affect its state" (1971, p. 663). The problem defining context thus relates to the problem of identifying relevant variables in a system's environment. I therefore turn to and extend Dey's definition of context to the domain of information systems:

"Context is any information that can be used to characterise the situation of an entity", (Dey, 2001, p. 5), where that entity is of relevance to the production of informational products and/or services.

Third, adaptation is the modification of system variables to changes in the context of the information system. Little conceptual work on the adaptation of information systems to context has been conducted so far. Rosemann, Recker, and Flender (2008) discuss the context-driven adaptation of business processes to context. This has prompted research on the context-driven configuration of process models (Hallerbach et al., 2010) and data mining techniques to explore contextual variables in process-aware information systems (Ghattas et al., 2010; Ramos et al., 2010). In order to provide a unifying view of the problem of adaptation, I turn to the seminal work of Ackoff (1971) on systems theory. Ackoff describes the ability of (general) systems to adapt to context changes as follows:

"A system is adaptive if, when there is a change in its environmental and/or internal state which reduces its efficiency in pursuing one or more of its goals which define its function(s), it reacts or responds by changing its own state and/or that of its environment so as to increase its efficiency", (Ackoff, 1971, p. 668)

Finally, a context-aware information system is a type of information system the capability of which is to facilitate the adaptation of work in an information system to changes in its context. First, I defined information systems as information-based work systems. This includes the human participants, machines, processes, information, technology, and other resources required in the conversion of informational inputs into outputs (such as informational products or services). Second, I defined context as information that is of relevance to this conversion. For example, information about weather, a contextual variable, may be relevant to the conversion of customer orders into bookings for a sporting event. This information is not explicitly provided by system users but must be collected by the booking system.

Using the above definitions of information system, context, and adaptation, I thus define context-aware information systems as a class of information systems with the following properties:

A context-aware information system is an adaptive system which responds to changes in its context that reduce its ability to produce informational products and/or services by modifying the way human participants and/or machines perform work using information, technology, and other resources.

Table 1.1 summarises each concept defined above and presents diverging views in the literature as well as the view embraced in my work. The inclusion of alternative views allows the comparison of the definition followed in the remainder of the manuscript with other views in the literature.

Concept	View embraced	Alternative view(s)
Information System	Information system as system in which people perform work by using technology and information	Information system as tool that supports tasks embedded in a structure part of a social context
Context	Context as information that can be encoded much like other forms of relevant information	Context as specific to combination of situations, people, and time that cannot be encoded
Adaptation	Adaptation as ability of systems to recover from loss in efficiency	n/a
Context-Aware Information System	Adaptive systems which respond to context change by modifying people, process, or technology	n/a

Table 1.1. Definition of key concepts

1.4.2 ILLUSTRATIVE EXAMPLES

This section presents five examples extracted from secondary evidence of published case studies to justify the concept of Context-Aware Information System. First, I define the criteria that guided my selection of cases and discuss sources from which I sampled secondary evidence. Each case is introduced and key elements of the narrative are summarised in a vignette. I then justify why each case a) represents an instance of the concept of information system and b) demonstrates the importance of considering relevant context in system design. In particular, I view and interpret case data through the lens of the four concepts defined earlier. In a final step, I compare both instances of information systems and define changes to their form and function required to respond to context.

I selected the two cases presented in the following based on a) their suitability as instances of information systems and b) their difference in form and function. Following Eisenhardt (1989), site selection allows the researcher to fill theoretical categories and provide examples of polar types. Cases are chosen to replicate specific situations or emergent theoretical concepts. At the same time, cases need to be substantially different to allow comparison. I therefore selected cases that are similar insofar as they are instances of information systems as defined by

Alter (2013). The cases represent examples of context-driven adaptation. Furthermore, they are different insofar as the nature of work in each example differs and the cases respond to different contexts.

Finally, I collected secondary evidence about each case from archival records such as newspaper clippings. When selecting archival evidence for data collection in case studies, the researcher must be careful to evaluate the usefulness and accuracy of records (Yin, 2009). Archival records are often produced for specific purposes and specific audiences. I thus took care to a) collect records from more than one source to avoid bias and b) where possible corroborated findings with peer-reviewed content about the case. My objective at this stage was exploratory in nature and did not encompass the causal explanation of phenomena (Yin, 2009). In other terms, the aim of my research was to answer “what” are examples of context-aware systems rather than “how” can we explain them.

Case Alpha: Operations Forecasting & Planning

The first case is an example of an information system used for flood mitigation and operations of a hydro-electric generating facility. The case represents an example of the information system concept as it combines the work of operations engineers, field stations that measure fill levels of the facility, and software for the real-time monitoring and forecasting of flood events. The product produced by the system is information-based, i.e., it delivers recommended levels of water releases on a daily or hourly basis to an internal customer, the facility operator. The case also represents an example of context as recommended releases must be adapted in anticipation or response to weather events such as heavy rainfall. However, the case cannot be considered context-aware as shown below.

Table 1.2 illustrates a recent event involving the flood mitigation system (Alpha) and demonstrates a failure to respond to relevant context. In this example, a lack of context-awareness (weather) leads to a failure to adapt water release plans produced by the system, resulting in a severe flood event.

Alpha, Water Resources, Australia

On January 13, 2011, the Brisbane river broke its banks, flooding properties and businesses in Australia's third largest city and creating a total of AUD 2.55 billion in flood damage. The event is widely recognised as a dam release flood resulting from large amounts of water released from the city's primary dam operated by Alpha in the wake of significant rainfall throughout the area.

Dam releases are largely automated by the operator's dam control system and rely on the actual water level in the dam as well as models of future water levels.

However, meteorological data and rainfall sensors in the catchment area of the dam were not sufficiently considered in the decision-making process by dam engineers in the days leading up to January 13. By the time water levels in the dam reached critical levels as a result of massive inflow made up of surface run-off in its catchment area, dam engineering had to release large quantities of water contained in the dam.

As a result, dam releases led to widespread flooding and damage in urban communities, affecting the major Australian urban centres of Brisbane and Ipswich.

Table 1.2. Vignette of case Alpha

The failure to monitor external context as defined by current weather events impacts on the ability of the system to adapt release plans and suggests it cannot be considered an example of CAIS. First, while the system monitors fill levels in the facility to forecast capacity, it does not consider rainfall and thus cannot respond to short-term variations following significant rainfall events. The inclusion of rainfall data into the system and adjustment of forecasting models has the potential to alert operations engineers to potential capacity issues. Second, the operating manual of the facility clearly defines strategies and thresholds for water releases. Combining context information with the release strategies of the operator will allow the system to better respond to changes in the weather context.

Case Beta: Foreign Exchange Swap Processing

The second case is an example of an information system used for foreign exchange swaps in financial institutions. The case represents an instance of the information system concept as it comprises the work of clerks in the treasury department of the bank, transaction processing software, and infrastructure for electronic data interchange. The system produces an information-based output, a foreign exchange swap, which trades identical amounts in one currency for another with a preferred trading partner to hedge against currency risk. Furthermore, the case is an example of context as the solvency of the trading partner is a relevant factor if swap orders are executed by the system. However, for the reasons listed below, the case represents a failure to respond to relevant context.

Table 1.3 illustrates an extreme event involving the foreign exchange swap system (Beta) and shows a failure to respond to relevant context. In this event, a lack of context-awareness (sol-

venency) leads to the execution of swap orders with a trading partner that is bankrupt, leading to a significant loss.

Beta, Financial Services, Germany

Beta, a German financial institution, lost EUR 350 million when its systems triggered a transaction with its trading partner on September 15, 2008, after the latter declared bankruptcy. Scheduled transactions such as swap agreements are coordinated by the trading and transactional systems of financial organisations and form large part of a bank's business.

On July 10, Beta had entered into an agreement with its trading partner to swap an agreed amount of US dollars into Euro. The agreement was subsequently entered into the bank's information systems and set to expire on September 15. From September 8 to 11, discussions about a liquidity crisis at the trading partner increased in frequency, leading to a board meeting at Beta on September 12 to discuss the bank's current dealings and future relationship with its trading partner. However, no changes were made with respect to current agreements in the bank's information systems.

On September 15, US news agencies reported Beta's trading partner was ready to file bankruptcy. The scheduled transaction at Beta still went ahead as planned.

Table 1.3. Vignette of case Beta

The failure to monitor external context (trading partner solvency) and the apparent inability to adapt foreign exchange swap processes makes this case clearly not context-aware. First, monitoring trading partner solvency could alert back office clerks about impending issues with swap orders. This process could use information from public or commercial sources about the partner and integrate it with information about orders that are scheduled in the system. Second, based on this information, the system could compute the risk of conducting a swap transaction and depending on thresholds execute an order or defer the order to a treasury clerk. In case of a concrete solvency risk, a treasury clerk would therefore be able to adapt swap transaction processing to changes in trading context.

Case Gamma: Production Order Processing

Gamma is an example of an information system for production order processing in manufacturing organisations. It is an information system as it combines the work of production engineers, software, hardware, and information about material, plants, capacity, and workload to schedule and complete orders. The relevant context of the system includes parameters such as external events (supply chain disruption, natural disasters) and internal events (such as the workload and current capacity of the plants in the virtual network). Furthermore, Gamma can be considered an example of a CAIS as the tracking of context allows production engineers to adapt the production of orders in the network.

Table 1.4 illustrates the ability of the production order processing system (Gamma) to monitor context and demonstrates its capability respond to relevant context. In this example, changes in process or resource parameters or external event lead to an adaption of production orders.

Gamma, High Tech, United States Gamma, a high tech manufacturer of computer chips, uses an advanced production system that enables the firm to resume production in neighbouring plants if a plant should go offline. Each plant is identical in the processes, technology, and infrastructure it utilised to produce wafers. Wafers that are partially completed in one plant can be finished in another plant as if the two plants were a single virtual production facility. The manufacturer introduced its “copy exactly” factory strategy in the mid-1980’s and completed adoption in its global network of manufacturing plants by the late 1990’s. This allows the firm to copy the process flows, equipment sets, suppliers, and training methodologies used during chip development to its global network of high-volume production plants. The monitoring of process and output parameters in each plant allows the manufacturer to move production orders between plants with the benefit of rapid productivity and financial gains. By continuously sharing information across plants, the “virtual” factory can respond to events in its context by adjusting process and resourcing configurations. Examples include man-made or natural disasters that affect its global supply chain.

Table 1.4. Vignette of case Gamma

The monitoring of variables in the internal (capacity, workload) and external system context (supply chain, disasters) and adaptation of processes (production orders) and resources (plants, equipment) achieved by the system demonstrate its ability to serve as an example of CAIS. First, the capability to exchange information in the global network of plants allows production engineering to monitor the relevant context in which production orders are scheduled and processed. Second, the identical sets of processes, hardware, and manufacturing equipment in each plant enable the simultaneous adaptation of processes and resources to events in the context. The system is therefore able to minimise time lost when events occur. This capability maps to the definition of CAIS provided earlier.

Case Delta: Work Schedule Planning

Delta is an example of an information system used for work scheduling and rostering in the HR department of local councils. It is an example of an information system as it combines the work of clerks in the department, hardware, software programs, and infrastructure required to determine work schedules, define rosters, and distribute timesheets to staff. The system produces information-based outputs, i.e., capacity planning sheets and staff rosters to meet the anticipated demand in the council. The relevant context of the system includes internal and external factors that impact staff availability. In the example illustrated below, rising fuel prices

impact on the ability of council staff to travel to work. This impacts on capacity planning and work schedules as described in the following.

Table 1.5 illustrates the adaptation of work schedule planning to rising fuel prices as an example of context-driven adaptation of work schedules. In this example, rising fuel prices impact on the cost of commuting of staff. Council officials respond by introducing a four-day work week.

Delta, Public Services, United States Delta, a local council in the United States, is switching to a four-day work week following rising fuel prices that make uneconomic for staff to travel to work. The council responds to concerns from its workforce that recent rises in fuel cost of about \$1-1.5 per gallon represent a significant increase in the cost of commuting. The council sought approval from the county’s Board of Commissioners to install a four-day, 40 hour work week as opposed to current arrangements for a five-day work week. According to this plan, total work hours for a single work day would increase to 10 hours as opposed to 8 hours currently defined in the five-day plan. This would save its 800 workers travelling to work an estimated \$300,000 over the course of a single year by working longer hours per work day. The plan is intended to remain in place for the foreseeable future.

Table 1.5. Vignette of case Delta

The adaptation of work processes (work scheduling and rostering) to relevant context demonstrates the ability of the case to monitor and respond to context. First, the context change faced by the system is one-off and permanent for the “foreseeable future”. It is therefore not an example of a more dynamic type of context that required continuous adaptation. The benefit from more effective context monitoring using, e.g., information technology is expected to be low. Second, the adaptation made to work schedules is likely to remain in force for a considerable time given the effort of setting up and communicating changed working hours to staff. Again, the benefit of designing mechanisms for more effective adaptation such as rule-based systems is low given the low frequency of change.

Case Epsilon: Payroll Processing and Control

Epsilon is an example of an information system used for payroll processing in higher education, i.e., an Australian university. It involves the work of clerks in the HR back office of the university, payroll software, hardware, and infrastructure for electronic interchange with banks, i.e., components of an information-based work system. It produces payroll runs that compute staff remuneration using the payroll rules defined for a country and posts payments to accounting. The relevant context of payroll includes the rules and legislation for remuneration in different countries and external factors such as the availability of downstream systems to which it is

connected (accounting, systems for payment settlement of banks). Changes in this context impact on the system's ability to calculate payroll runs.

Table 1.6 illustrates a recent example in which changes in the downstream context of the system leads to manual intervention and adaptation of payroll runs. In this example, errors in payment processing of banks lead to missed payments and a need to financially support staff.

Epsilon, Higher Education, Australia

Epsilon, a higher education institution in Australia, left a significant number of employees without pay after the downstream payment processing systems of banks that settle payroll runs failed in an overnight run. The payroll runs are computed by the university's payroll system and transferred into accounting systems before being paid out into staff bank accounts with the help of the payment processing systems of banks. An error in the processing of payments that occurred overnight had staff of the banks and university wake up to a high number of missed payments that affected not only university staff but staff of employers Australia-wide.

The missed payments prompted management at the university to liaise with technical departments at the banks and to notify its staff of the missed payments. The director of finance of the university sent an e-mail apologising to staff and offering financial support to staff experiencing hardship.

The incident was part of a wider failure caused by a software glitch at a bank processing payroll payments that left thousands of Australian workers without pay. The bank promised to process payments by the end of the business day and to cover any fees incurred as a result of the delayed payment processing.

Table 1.6. Vignette of case Epsilon

The system demonstrates the ability of manual response (notification of staff, follow up banks, offer of financial support) to changes in downstream context (processing errors at bank). First, the payroll and accounting departments of the university are notified by the bank of the processing error in the overnight payroll run. Automating the acquisition of relevant context information (status updates for payments in the bank's systems) presents an opportunity to accelerate a context response. Second, the university manually coordinates the alerting of management in payroll and accounting, the notification of staff, and setting up of financial support for staff experiencing hardship. Defining rules and strategies for adaptation upfront would help implementation and improve context-awareness.

1.4.4 COMPARATIVE ANALYSIS OF SECONDARY CASE EVIDENCE

In this section, I arrange each case discussed in the previous section into a case-ordered display to compare the dimensions of context-driven system adaptation. Case-ordered displays array data by case to understand differences between cases (Miles & Huberman, 1994). In other terms, they show differences between high, medium, and low cases as defined by the phenomenon of interest. In my enquiry, the phenomenon of interest is adaptation of systems to changes in relevant context and the boundaries of context-aware systems.

Accordingly, I used the four concepts introduced above as the measuring device against which to compare cases. This includes the application of the work systems view of IS, relevant context, adaptation to context, and whether the system is an instance of CAIS.

Table 1.7 arrays each case discussed above into a case-ordered display to compare the dimensions of context-driven adaptation observed in each example. First, Alpha and Beta show strong evidence for context and the need for context-driven adaptation but both systems fail to respond to relevant context change in time. Making changes to form and function such as monitoring data feeds or defining rule-based adaptation strategies would move these cases to “high”. Second, Delta and Gamma show weaker evidence of context compared to other cases and response strategies tend to be manual and in the case of Delta one-off. This makes these cases a “medium” candidate for CAIS. Finally, Gamma shows the strongest support for both context and adaptiveness, making the case a “high” candidate.

1.4.5 DISCUSSION

In this chapter, I presented a summary of the key concepts discussed in the remainder of my work and presented two illustrative examples to justify my ideas. First, I presented definitions of four key concepts (Information System, Context, Adaptation, Context-Aware Information System) that guide my work. In particular, I compared diverging views of each concept held in the literature and justified the interpretation adopted in my work. Taken together, these definitions provide the foundation for understanding the research execution presented in subsequent chapters. Second, I presented two examples based on secondary evidence of published case studies to substantiate these ideas. I used the five concepts as lens to view and interpret data and to compare my findings.

The conceptual discussion in this chapter points to diverging views of key concepts in the literature and supports the call for a more detailed conceptual analysis made in later chapters. First, diverging views of context in the literature emphasise the need to more properly define context against the background of IS research. Context can be defined from the viewpoint of information or interaction between humans and devices. This raises the question how we define context in IS and how context impacts our understanding of information-based work systems. Second, little conceptual work on adaptation in IS exists and more prescriptive work on how IS should adapt to context is needed. This relies on a sound theoretical foundation for reasoning about adaptation and making prescriptions.

The discussion of five illustrative examples highlights the need to develop a more detailed empirical understanding of context-awareness. The mapping of the concepts identified and dis-

cussed above to secondary evidence of published case studies revealed shortcomings in how systems monitor and adapt to relevant context. The examples of context identified included information about weather events and trading partner solvency. These shortcomings lead to a failure of the studied systems to deliver desired outputs (accurate water release plans and foreign exchange swap orders). Based on this initial analysis, more detailed work is required to understand the processes of monitoring and adapting to context. This includes the selection of successful examples of context-driven adaptation.

1.5 STRUCTURE OF MANUSCRIPT

Table 1.8 illustrates how each chapter in the manuscript builds on the findings of the previous chapter and contributes to finding answers to the research questions of the study. First, introduction defines the research problem and derives relevant research questions that guide the study. Second, program of research positions the study in the wider context of research in IS and design science research in particular. Third, background reviews the current body of knowledge in IS and its reference fields to define context, information systems, and to select appropriate kernel theories for theorising about context-driven adaptation. Fourth, exploration presents a multiple-case study conducted with two organisations that have the requirement of rapid adaptation of work systems to context. Fifth, theory development maps insight derived from kernel theory and case data to contemporary conceptualisations of IS. Finally, theory-driven design and theory evaluation develop and test prescriptive guidelines while conclusions summarises the study's contributions, limitations, and future research.

Chapter	Description	Mapping to research question
Ch. 01. Introduction	Motivation of research, problem statement, and definition of research questions of the study	Definition of relevant research questions guiding the study
Ch. 02. Program of Research	Identification of research components of design research in IS and definition of a research plan	Definition of research plan to generate findings and provide evidence for each question
Ch. 03. Background	Review of the peer-reviewed literature regarding key concepts and kernel theories for context	- Definition and explanation of what is relevant context - Definition and explanation of system and the adaptation of systems to context change

Chapter	Description	Mapping to research question
Ch. 04. Exploratory Case Study	Collection and analysis of empirical evidence about context and context-driven system adaptation	• Empirical evidence about relevant system context • Empirical evidence about affected system variables • Empirical evidence about system adaptation strategies
Ch. 05. Theory Development	Identification of deficiencies in contemporary design relevant explanatory theories regarding context and system adaptation	• Identification of deficiencies in understanding of context • Identification of deficiencies in understanding of context-driven system adaptation
Ch. 06. Theory-Driven Design	Theory-driven design of prescriptions regarding the effective design of context-aware information systems	• Prescriptive statements about modelling relevant context • Prescriptive statements about modelling system adaptation • Prescriptive statements about activating system change • Prescriptive statements about learning from context-driven system adaptation
Ch. 07. Theory Evaluation	Evaluation of these guidelines and the utility claims they make	• Evaluation of findings for each question in the form of utility
Ch. 08. Conclusion	Summary of study findings, contributions, limitations, and the suggestion of future research	• Review of the findings made for each research question • Identification of limitations and outlook on future work

Table 1.8. Mapping of manuscript structure to research questions

Chapter 2: Program of Research

2.1 INTRODUCTION

In this chapter, I map the research objective of prescribing effective guidelines for the *design of context-aware information systems* to a research plan and define how this plan is operationalised in the present research enquiry. Design as an activity and research subject has a long tradition in the IS discipline (Iivari, 2007). A general distinction between studying design as an activity (i.e., studying the methods, cognition, and education of designers) and using design as a research method to generate new design knowledge has been established in the literature (Vaishnavi & Kuechler, 2004). My research falls firmly into the latter class and thereby qualifies as design science research (DSR) as opposed to design research (DR). The objective of prescribing effective guidelines for the design of context-aware systems must thus be viewed as an instance of DSR in the IS discipline (DSRIS) and thereby is subject to established guidelines in the IS field for conducting such forms of research. In the following, I review the philosophical foundations of DSR and position my program within the conversation on DSRIS.

My first step in identifying suitable frameworks for the definition of a research plan is to map my research questions to the *components of design theorising* in DSRIS. Three components of design theorising have emerged in the DSRIS literature in recent years (Gregor & Jones, 2007; Kuechler & Vaishnavi, 2012). First, using design as a research method must draw from the body of justificatory knowledge in the natural and social sciences to justify the assumptions and principles underpinning design artefacts. This includes "kernel theories" (Walls, Widmeyer, & Sawy, 1992) by which a phenomenon of interest can be explained but also informal, experience-based insight into a design problem (Hevner, March, Park, & Ram, 2004). Second, design-relevant explanatory theories (Baskerville & Pries-Heje, 2010; Kuechler & Vaishnavi, 2012) map the high-level nature of kernel theories (e.g., theories of cognitive psychology) to a concrete domain of application (e.g., decision support systems) and clarify the relationship between design and its desired effect. Third, the codification of design knowledge in IS design theories (Walls et al., 1992) prescribes guidelines on how the effect can be achieved.

Furthermore, I define the operationalisation of the research plan as a "research system" and map the findings made by each research activity to corresponding chapters in the manuscript. The development of a well-defined flow of argument is a highly desirable property of good research work and enhances the clarity and readability of a manuscript (Mullins & Kiley, 2002).

The hierarchical decomposition of research questions to components of the research plan and of components in the research plan to chapters of the manuscript allows the reader to follow the flow of argument and the specification of findings in corresponding chapters.

2.2 PHILOSOPHICAL GROUNDING OF DESIGN SCIENCE RESEARCH (DSR)

Research in multi-paradigmatic fields of research such as the IS discipline (Benbasat & Weber, 1996; Vaishnavi & Kuechler, 2004) must reflect on the philosophical foundations on which the research enquiry rests. This is particularly true of design science research in the IS discipline, the metaphysical assumptions of which are unique and require explanation (Vaishnavi & Kuechler, 2004). Most generally, research is defined as an activity the purpose of which is the creation of new knowledge concerning some phenomenon of interest (Kuhn, 1996). Knowledge is a precursor to prediction and the presence of both is generally considered as understanding. Three basic assumptions about reality, knowledge, and value underpin the activity of research. Ontology is the study of what constitutes the nature of reality and the basic concepts and relationships from which reality is composed. Epistemology is the study of how we obtain knowledge and limitations in our understanding of the world. Finally, axiology is the study of what we collectively value as a discipline and what we thus accept as contribution to our knowledge.

Much has been said about the philosophical foundations of design science research in the IS discipline and its unique position in the continuum of world views and scientifically accepted paradigms (Iivari, 2007; Vaishnavi & Kuechler, 2004). First, design science researchers are most commonly viewed as assuming an objective and stable reality into which design artefacts are introduced. This differs from positivist and interpretive views of reality in that reality is viewed as consisting of alternative states (similar to multiple realities) which can be obtained by introducing design artefacts that follow basic natural and social laws (similar to positivist views of reality). Second, the approach taken by design science researchers to the creation of knowledge is referred to as "iterative circumscription" (Vaishnavi & Kuechler, 2004), i.e., the creation of design knowledge by iterating between the design and explanation of artefacts. Finally, design science researchers as a community value design as a fundamental approach to create knowledge and bring about improved states in the world by designing and testing artefacts (March & Smith, 1995).

Table 2.1 maps the assumptions underpinning my program to the four dimensions of research according to the philosophy of science. The program of research assumes that phenomena in the world can be viewed as systems and that their behaviour is governed by general law that hold irrespective of their specific kind. This view is consistent with the design science view of

reality (Vaishnavi & Kuechler, 2004) and addresses ontological questions of what constitutes reality. Second, the program of research assumes that alternative states in this reality exist and that they can be brought about by the careful design and application of artefacts. This view maps to epistemological considerations in design science and thus differs from other ways of knowing in the IS discipline. Third, the creation of a new artefact follows a cycle from developing an empirical understanding of a design problem, to theorising about the problem, to constructing normative guidelines. Finally, the program of research intends to bring about a change in how we construct and use context-aware information systems.

Philosophical dimension	Design research assumption	Mapping to research program
Ontology	Single, stable (physical) reality underlying multiple possible world states	Assumption of general laws governing adaptation in natural and artificial systems described in system of systems ontology
Epistemology	Knowing through making theory-ingrained artefacts	Knowing through the successive mapping of system laws to normative guidelines for context-aware information system design
Methodology	Cycles of artefact design, construction, and evaluation	Cycle of empirical observation, theory development, design of normative guidelines, and evaluation of their applicability
Axiology	Improvement in our understanding of systems and thus improvement of their utility	Improvement in our understanding of context-aware IS and their applicability in practice

Table 2.1. Philosophical positioning of the research program

2.3 POSITIONING THE PROGRAM OF RESEARCH

IS research is a diverse field that studies multiple phenomena of interest and draws from many theoretical foundations and reference fields (Benbasat & Weber, 1996). It is therefore prudent to position the program of research and the research enquiry itself in the wider efforts of the IS community. Principally, IS research can be partitioned into multiple concerns that deal with the development of information systems, their positioning and contribution to organisational strategy, the issues of group work and decision support, and the theoretical foundations of IS (Taylor, Dillon, & Van Wingen, 2010). As such, it draws from multiple reference disciplines such as computer science for engineering and design, the management and cognitive sciences for foundations of decision-making processes, organisation science for research on social proc-

esses, and economics for studies of the economic efficiency of information systems (Benbasat & Weber, 1996). Furthermore, the IS discipline is characterised by diverging methods of knowledge acquisition such as empirical methods (Benbasat & Zmud, 1999) and methods that focus on designing artefacts (Hevner et al., 2004).

The positioning of the present program of research in the wider context of the IS discipline is characterised by two issues. First, the program identifies a gap in our understanding of how context-aware IS are designed. Context and context-aware design have been the subject of much discussion in the reference fields of IS such as artificial intelligence (Brézillon, 1999), human computer interaction (Nardi, 1996), and pervasive computing (Schilit & Theimer, 1994). However, little progress has been achieved in defining a common foundation for the design and construction of context-aware systems in these fields let alone in IS research (Rosemann, Recker, & Flender, 2008). Second, the program of research intends to close the breach in our understanding of IS development by developing normative guidelines that are theoretically complete and consistent. In other terms, it aims to make explicit the "justificatory knowledge" (Gregor & Jones, 2007) based on which effective systems can be constructed. This differs from approaches in which such knowledge is tacit (Hevner et al., 2004) and not articulated in an explicit way.

2.4 RESEARCH PLAN

One point of convergence in the IS literature in recent years has been the *desirability of developing design theories*, i.e., a set of prescriptive statements about how information systems ought to be constructed. Design theorising marks a departure from the preoccupation with behavioural and managerial aspects in IS research (Orlikowski & Iacono, 2001) and a shift toward research that provides utility through constructing innovative systems and artefacts (Baskerville & Pries-Heje, 2010).

While the desirability of design theory is generally accepted, less clarity exists on *how design theorising is achieved* and what its mandatory components are. Gregor (2006) and Gregor and Jones (2007) note that design theorising must ground in "justificatory knowledge" of IS and its reference fields. This includes relevant "grand theories" (Mills & Gitlin, 2000) and an empirical understanding of the relevance of design problems. Baskerville and Pries-Heje (2010) and Kuechler and Vaishnavi (2012) argue that design theorising must explain why artefacts work (i.e., why they generate a desired effect) based on knowledge about their domain of application (as codified in mid-range theories; cf. Mills & Gitlin, 2000). Furthermore, Walls, Widmeyer, and El Sawy (1992; 2004) argue that the design knowledge generated by theorising must be codi-

fied in "prescriptive statements" based on which effective information systems can be designed. Finally, design knowledge must be instantiated and evaluated to test its feasibility and utility in practice (Hevner et al., 2004).

Table 2.2 summarises the scholarly discussion on design theorising in four components, the questions asked by each components, and key contributions on each in the IS literature. A considerable overlap exists between each contribution and authors generally focus on describing more than one component. However, each author makes a unique contribution to design science research in the IS discipline by emphasising a particular aspect of relevance.

Theorising component	Type of questions asked	Key reference
Justification	What is the theoretical (and empirical) foundation of a design problem?	Gregor, 2006; Gregor & Jones, 2007
Explanation	Why do artefacts lead to effects and how can this be explained through theory?	Baskerville & Pries-Heje, 2010; Kuechler & Vaishnavi, 2012
Prescription	How can more effective artefacts (and systems) be designed following theory?	Walls, Widmeyer, & El Sawy, 1992, 2004
Evaluation	How can the ability of the artefact to generate the effect be evaluated in practice?	Hevner, March, Park, Ram, 2004

Table 2.2. Components of design theorising

Following the discussion of design theorising components, I *map the research objective* of providing normative guidance on designing context-aware IS to each component and, ultimately, to a research plan. First, *justification* must provide the theoretical foundation for context and context-aware IS and must be grounded in empirical knowledge about the problem of designing such systems. Second, *explanation* must map the theoretical foundation of context-aware IS to the domain of work systems. This includes (1) developing an understanding of work systems based on the IS literature and (2) articulating whether or not contemporary conceptualisations are able to explain the phenomenon of context and context-awareness. Third, *prescription* must map the understanding of our current knowledge about work systems to normative guidelines for the construction of context-aware IS that are (1) grounded in theory and (2) provide innovative solutions to relevant problems. Finally, the applicability of these guidelines must be evaluated to ensure their feasibility and relevance.

Figure 2.1 illustrates the successive mapping of design theorising components to a research plan based on and extending an example by Kuechler and Vaishnavi (2012). Each design theo-

rising component is mapped onto a component of the research plan and criteria for operationalising the necessary research activities for each plan component are derived from the supporting literature. Four plan components can be identified. First, justificatory knowledge consists of two research activities: the identification and selection of relevant kernel theories and the generation of empirical knowledge by means of field work. Second, explanatory design theory maps the theoretical foundation to the domain of work systems and, based on empirical evidence, identifies and closes gaps in our current understanding of work systems. Third, the extended understanding of context-aware work systems is mapped to a set of prescriptive statements about how such systems must be constructed. Fourth, these statements are evaluated for their relevance and utility to information system practice.

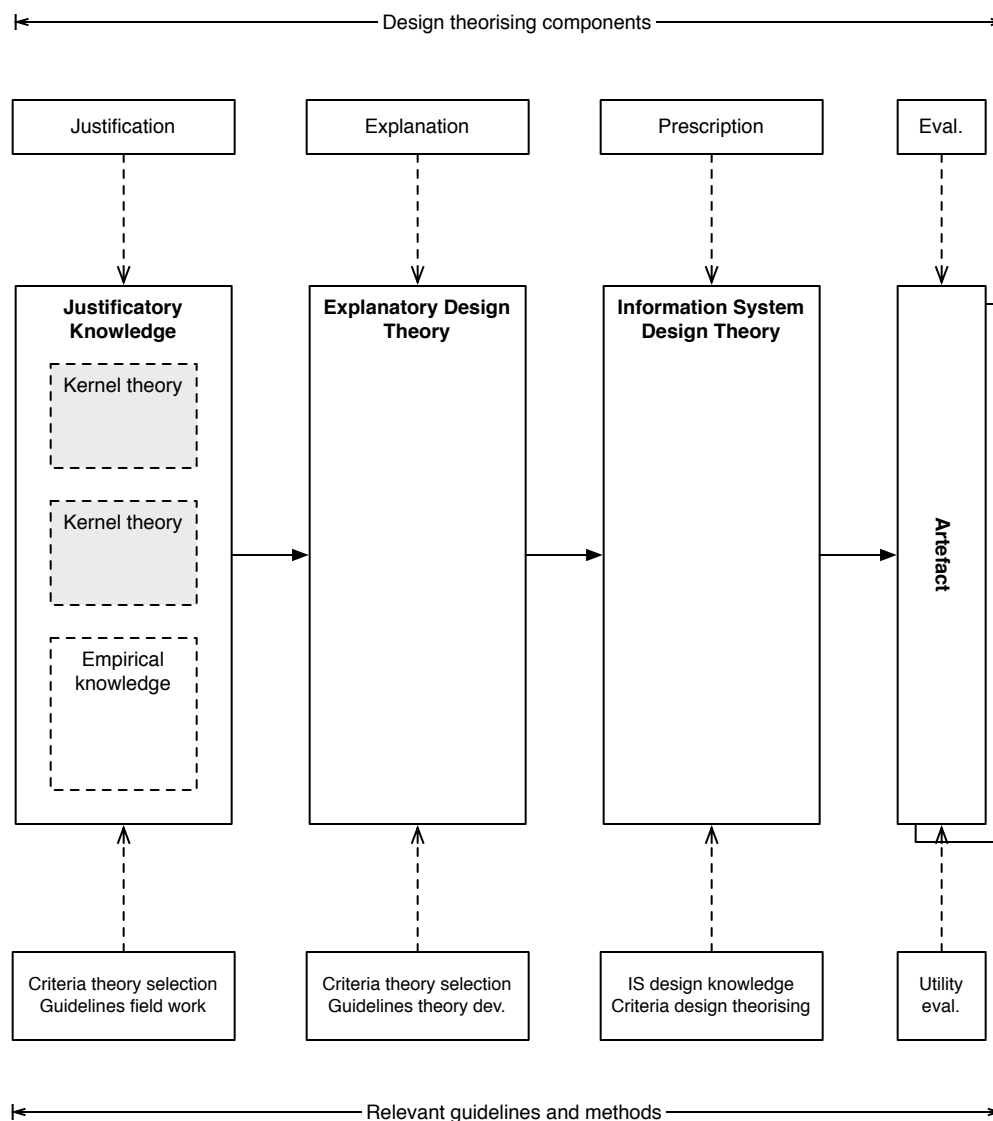


Figure 2.1. Research plan (based on Kuechler & Vaishnavi, 2012)

In the following, concerns surrounding the operationalisation of each plan component in research activities and the criteria for adhering to standards of scientific research are discussed. Each issue is briefly introduced and decisions are motivated without specifying the concrete measures undertaken to implement them in the respective research activities. The mapping of decisions to concrete measures of implementation is discussed in the respective chapters.

2.4.1 LITERATURE REVIEW: KNOWLEDGE GAPS AND KERNEL THEORY SELECTION

Literature reviews provides a *firm foundation for theory development* and close areas of research where sufficient research already exists (Webster & Watson, 2002). I turn to literature review as a method for identifying knowledge gaps in constructing context-aware systems and for selecting suitable kernel theories based on which the problem can be justified. Four independent but jointly correlative issues can be identified. First, an understanding of the "work system" construct and its relationship with information systems is required to define the core phenomenon of the program of research and to position it within IS research (Orlikowski & Iacono, 2001). Second, an understanding of how information-based work systems are constructed today is required to clarify the current state of the art. This leads to a review of the literature in IS and its reference fields concerning (1) the definition of context and (2) the identification of knowledge gaps in how context-aware systems are constructed. Finally, a review of the foundational literature on IS must reveal appropriate kernel theories based on which this breach can be closed.

2.4.2 CASE STUDY RESEARCH: EMPIRICAL KNOWLEDGE ABOUT DESIGN PROBLEM

Case study research is a qualitative method of enquiry that allows the research to study a *contemporary phenomenon* in its natural setting (Yin, 2009). I turn to case study research (1) to explore the problem of designing context-aware systems and (2) to generate empirical knowledge in a domain in which conceptual progress has been limited and no commonly accepted definition and theoretical foundation of context has emerged (Brézillon, 1999). The advantage of the case method over other forms of empirical enquiry is its ability to study a relevant problem in its natural setting (Benbasat, Goldstein, & Mead, 1987) and to provide the foundation for theory development from cases (Eisenhardt, 1989) where little theoretical progress has so far been achieved. Three issues in designing case studies can be identified (Yin, 2009). First, the criteria underpinning the selection (and exclusion) of case sites and the rationale of the case study design must be specified. Second, protocols for field work must be specified that detail the type of data collected. And third, strategies for coding and analysing data must be defined.

2.4.3 THEORY DEVELOPMENT: MAPPING KERNEL THEORIES TO DOMAIN OF APPLICATION

Theory development is the process of creating new knowledge and is considered as the central goal of scientific research in general and the IS field in particular. I approach theory development through the lens of representation theory and recent frameworks for assessing the "goodness" of theory with respect to some phenomenon of interest (Weber, 2012). The phenomenon of interest in my research enquiry is the construction of context-aware information systems and in this context, two relevant types of theory can be identified (Kuechler & Vaishnavi, 2012). Meta-theory or grand theory provides a complete and consistent account of a phenomenon and generalises across a wide range of settings. For example, general system theory is a "meta-theory" that describes general laws governing natural and artificial systems (Kast & Rosenzweig, 1972). Mid-range theory, on the other hand, applies the premises of meta-theory to a concrete domain of application. For example, socio-technical theory (Bostrom & Heinen, 1977) and work system theory (Alter, 2008) apply system thinking to the domain of work system and IS design.

2.4.4 THEORY-DRIVEN DESIGN: DERIVING NORMATIVE GUIDELINES FOR DESIGN

Finally, theory-driven design is the process of deriving prescriptive guidelines for the construction of effective information systems from academic kernel theory and mid-range theory in the IS discipline. System design is an issue that is at the core of research in the IS discipline (Taylor et al., 2010). In recent years, a renewed interest in system design has led to the definition of guidelines for conducting design science research in the IS discipline. These guidelines generally recognise the importance of design justification and rigour in the execution of design science research but differ in the extent to which the "justificatory knowledge" that underpins a specific design is made explicit (Gregor & Jones, 2007; Kuechler & Vaishnavi, 2012). In artefact-driven approaches to design science, justificatory knowledge often remains tacit and is not explicitly represented (Kuechler & Vaishnavi, 2012). Theory-driven approaches, on the other hand, aim for the explicit representation of justificatory knowledge in "design theories" (Walls et al., 1992) that generalise across a class of artefacts and which describe normative guidelines for constructing artefacts.

2.5 MAPPING RESEARCH PLAN TO MANUSCRIPT STRUCTURE

Following the discussion of how the research objective of creating normative guidelines for context-aware system design can be obtained, I now discuss when and in what order each of these steps is executed in the context of my research enquiry. The development of a well-

structured argument is a highly desirable property of research work in general and a thesis manuscript in particular (Mullins & Kiley, 2002). In order to provide clarity and a well-structured flow of argument, I map each of the research requirements identified in the research plan to a flowchart defining (1) the inputs flowing into a specific research stage identified by the plan, (2) the specific method used to operationalise a research stage and the skills and responsibilities required of the research team during that stage, (3) the outcomes expected from a phase of research and how such outcomes influence subsequent phases of research in the overall research enquiry. Furthermore, each stage is mapped back to the original research question that provided the starting point of the enquiry and to which it must ultimately provide an answer.

Figure 2.2 illustrate the flow of research activities, their operationalisation through research methods and guidelines, and the inputs and outputs flowing from one activity to another. The research enquiry is marked as a system of activities defined by a boundary or "scope" and inputs and outputs. The inputs into the research system are the experience-based hunches, insights, and research questions that give rise to the research enquiry. The outputs of the research system are a set of prescriptive statements concerning context-aware IS (CAIS) and the specification of implications for system practice, theory, and research methodology (in terms of operationalising theory-driven design in DSRIS). Each inputs is converted into one output by a sequence of five research activities that map to the requirements of the research plan. Each research activity consumes inputs such as guidelines for conducting field research and produces partial outputs such as empirical knowledge about the design problem.

The structure of thesis manuscript follows the flow of research activities as illustrated in Figure 6.1 and the context of each chapter is briefly introduced in the following. Each chapter introduces the problem addressed by a research activity and defines guidelines for its resolution.

Chapter 1. Chapter 1 introduces the design problem on the basis of informal insights into the design problem and hunches and articulates the research questions guiding the enquiry. This chapter sets the stage for the research enquiry and provides an initial discussion of the research problem and relevant research questions.

Chapter 2. Chapter 2 defines the program of research and draws the boundary of my research enquiry within the overall program of research. Issues such as philosophical foundations of the research, its positioning in the stream of IS research, and the definition of a research plan and its operationalisation in stages are discussed.

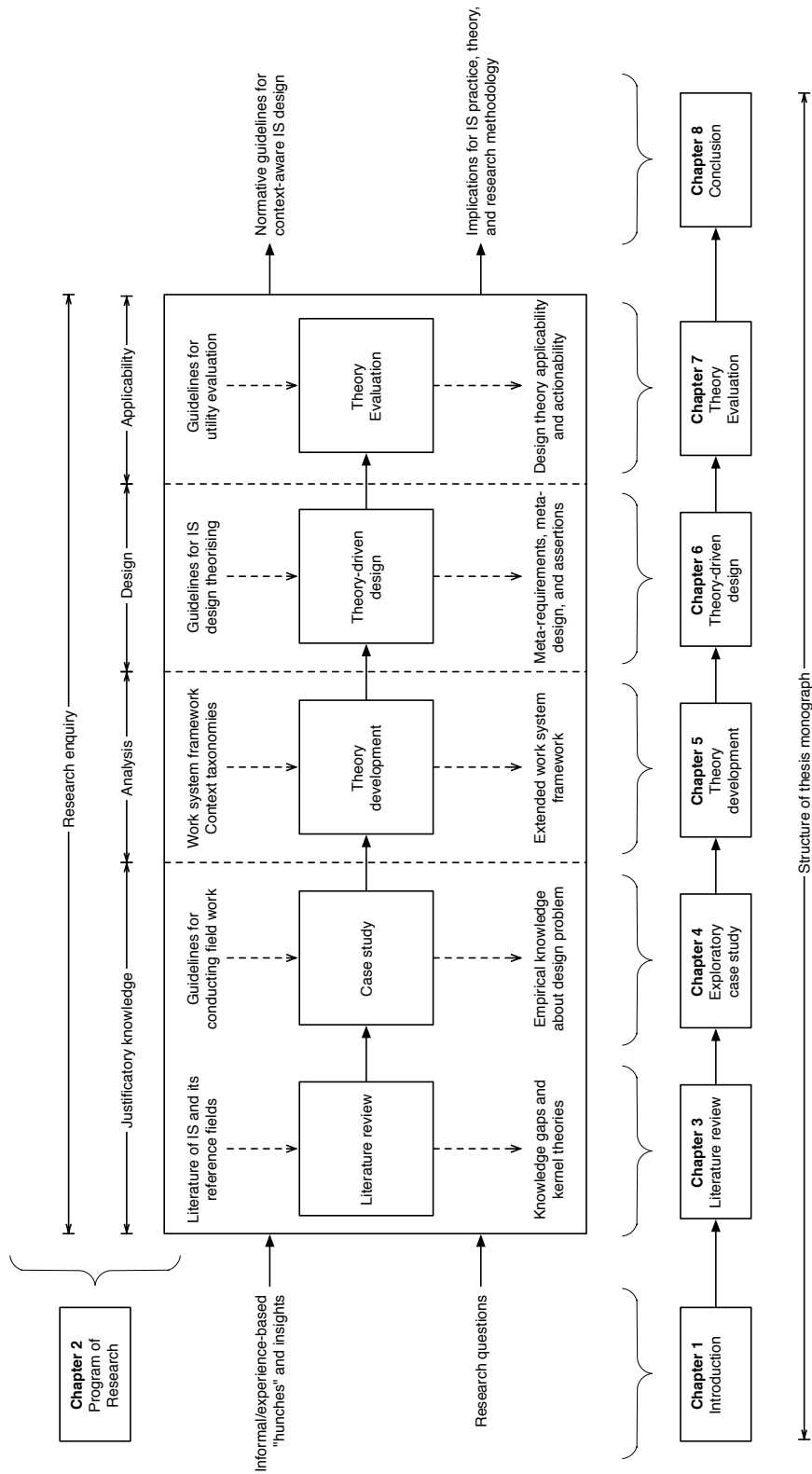


Figure 2.2. Illustration of research organisation and manuscript structure

Chapter 4. Chapter 4 develops an empirical understanding of context-aware system design by means of an exploratory case study and cross-site comparison of findings in two work systems. Exploratory case studies are justified in fields of enquiry in which there does not yet exist an established theoretical basis for research (Yin, 2009). Context-awareness falls under this category and research in context-aware system design has drawn from a variety of theories such as situation theory (Akman & Surav, 1996), situated action frameworks (Suchman, 1987), and system theory (Rosemann et al., 2008). I first define guidelines for conducting field work by reviewing guidelines for case study research (Eisenhardt, 1989; 1991; Yin, 2009) and qualitative data analysis (Miles & Huberman, 1994; Saldana, 2009). I then proceed to discuss the operationalisation of guidelines and the findings made in each case site. This covers the impact of context on the work systems of two organisations in separate industries (insurance carrier, airport operator). Finally, I analyse similarities and differences between case sites.

Chapter 5. Chapter 5 combines the findings from Chapter 3 and Chapter 4 by introducing the construct of context in system design and extending contemporary models of work systems. In order to enhance the theoretical completeness and clarity of the extended work system model, Ackoff's system of systems ontology (1971) is identified as a common foundation for describing adaptive systems and each construct in the ontology is carefully mapped to a contemporary model for reasoning about IS and IS design, i.e., the work system model (Alter, 2008). An approach of mapping system of systems (SSC) to the work system model (WSM) is presented that is loosely based on representation theory and more recent additions in analysing and measuring the representational quality of theories (Weber, 2012). For example, five hypothetical situations are identified that occur in mapping constructs from meta-theories such as SSC to "mid-range" theory such as WSM (Kuechler & Vaishnavi, 2012). Examples for each situation are provided and a final, theory-conform model of context-aware work systems is introduced.

Chapter 6. Chapter 6 then derives a set of prescriptive guidelines from the extended work system model that guide the construction of context-aware artefacts and which are firmly grounded in system thinking and socio-technical theory. Developing such guidelines takes the form of an Information System Design Theory (ISDT), a special kind of theory that explains artefacts as instances of a generalisable class of design problems and their resolution by a generalisable class of solutions (Walls et al., 1992). Each member of the class of problems relates to justificatory knowledge in the natural and/or social sciences. For example, designing context-aware systems relates to the problem of designing adaptive systems and the nature of work systems as "contrived" systems that consist of multiple parts that are interrelated (Bostrom & Heinen, 1977). Each problem, in turn, maps to a set of features or "meta-design" which all sys-

tem instances must possess in order to be viewed a solution to a given problem. For example, system modelling and context modelling provide solutions to context-driven adaptation.

Chapter 7. Chapter 7 applies the set of normative guidelines for context-aware information systems (CAIS) to the case data in order to verify the applicability of the theory and evaluate improvement areas in current system practice. In particular, a context-aware insurance claim system is constructed on the basis of case data of the first of two case sites by applying the CAIS guidelines. A comparison of the work system as observed in the case study and an improved system constructed by means of the CAIS guidelines reveals weaknesses in practice such as an incomplete representation of the context of claims handling. Similarly, a context-aware passenger facilitation system is constructed by following the CAIS guidelines and findings from the case study. This system is then compared with the system observed in the case study and weaknesses such as the lack of a formal context learning mechanism are defined. Finally, I discuss the discrepancy between the work systems observed in the multiple-case study and the improved systems defined on the basis of CAIS as a roadmap for future research.

Chapter 8. Finally, I summarise the findings of the research enquiry in Chapter 8 and define implications for system practice, theory, and research methodology for theory-driven design in the IS discipline. Limitations of the study are identified and a roadmap for future research is suggested for questions that still elude answers.

2.6 DISCUSSION

Two open questions remain in the discussion of design research and theory development in DSRIS. These are summarised in the following:

First, the starting point of much design science research are "informal or experience-based insights" (Kuechler & Vaishnavi, 2012) and knowledge about "design problem relevance" (Hevner et al., 2004). However, knowledge about the specific design problem at hand often remains tacit in the minds of the design researcher and is not made explicit in the structure of DSRIS projects. The absence of a more formal way of articulating and codifying a design problem leads to problems. Such problems include a lack of empirical evidence for a design problem collected by means of qualitative or quantitative approaches to empirical research in the IS field (Benbasat & Zmud, 1999) and codified using established methods of data analysis (Miles & Huberman, 1994). In order to improve the relevance of design science research in the IS discipline and its applicability to the community of practice (Rosemann & Vessey, 2007), it is therefore necessary to embrace empirical methods for data collection about a design problem before se-

lecting appropriate kernel theories (Eisenhardt, 1989) and commencing with the iterative circumscription of new artefacts.

Second, one point of convergence in DSRIS community is the need for theory development (Kuechler & Vaishnavi, 2012) and codification of design knowledge (Gregor & Jones, 2007) based on academic kernel theories. However, little to no guidelines exist to date on how the mapping of kernel theories to design-relevant theories and the mapping of design-relevant theories to normative guidelines for artefact design can be operationalised. The absence of such guidelines lead to problems. In particular, the mapping of kernel-theoretical constructs to design-relevant constructs, to prescriptive design statements, and to applicable design artefacts as is suggested by some frameworks (Kuechler & Vaishnavi, 2012) requires clarification. For example, knowledge about how such mappings are performed and measures for the theoretical quality (Weber, 2012) of the resulting design theories are missing. Furthermore, the high-level and general nature of kernel theories makes a direct mapping of high-level to design-relevant constructs highly unlikely. More attention must thus be paid to mapping situations and their desirable resolution.

I address each of the two points in the context of the following chapters. In particular, I address issues pertaining to the empirical study of design problems in Chapter 4. Issues pertaining to the quality of design-relevant explanatory theory and for mapping them to normative guidelines are discussed in Chapter 5 and 6.

2.7 CONCLUSION

In this chapter, I discussed the positioning of the study in the wider context of academic IS research and the notion of design science research in the IS field in particular. Four components of design theorising were identified (Gregor, 2006; Gregor & Jones, 2007; Kuechler & Vaishnavi, 2012). These are the justification or theoretical foundation of a design problem, why artefacts lead to certain effects and how this can be explained through theory, the prescription of guidelines that generate more effective artefacts if followed in practice, and the evaluation of utility claims made by design theory. I mapped each component of design theorising to concrete research activities and demonstrated how research outputs of one activity inform and guide the subsequent activities of the research study. Finally, I identified open questions in design theorising and mapped each research activity to a chapter in the manuscript where these questions are addressed.

Chapter 3: Background

3.1 INTRODUCTION

In this chapter, I turn to a review of the literature as the basis for identifying *knowledge gaps* and the foundation for *design theory development* in later chapters. As stated before, my objective is the development of prescriptive guidelines for the design of effective context-aware information systems. In keeping with a cumulative tradition in IS research (Webster & Watson, 2002), I therefore commence by reviewing the literature of IS and its reference fields. IS is a diverse (Benbasat & Weber, 1996; Taylor, Dillon, & Van Wingen, 2010) and inherently transdisciplinary (Galliers, 2003) field of study. IS researchers draw from a variety of reference fields including but not limited to computer science, management science, organisation science, and economics (Benbasat & Weber, 1996). Furthermore, contributions in the IS literature can be associated with different subfields including the design, development, and implementation of systems; the alignment of IS with organisational strategy; and topics in group work and decision support (Taylor et al., 2010). Following my objective, my enquiry falls broadly within the first field and into design science research.

In particular, I review conceptualisations of *information systems*, *context*, and theoretical frames for explaining *context-aware information systems* as topics of immediate relevance to my enquiry. A lack of clarity concerning its core subject matter has troubled research in the IS discipline since its inception (Benbasat & Zmud, 2003; Orlikowski & Iacono, 2001). Various interpretations of IS as information technology (IT) artefacts (Benbasat & Zmud, 2003) and IT-reliant work systems (Alter, 2003) have emerged and each suggests a different focus for DS researchers and IS researchers in general (McKay, Marshall, & Hirschheim, 2012). Before proceeding to a review of context as a relatively new phenomenon in the IS literature (Rosemann, Recker, & Flender, 2008), I therefore review and compare contemporary definitions of information systems in the literature. In addition, context has received little attention in the IS literature and has so far been contained to subfields such as process modelling (Rosemann & Recker, 2006). I therefore review and compare conceptualisations of context in IS reference fields before turning to the selection of kernel theories for theorising.

Finally, I compare each of the conceptualisations of information systems and context and identify *three questions for further enquiry*. First, much of the work on context has focussed on individual and group settings (Nardi, 1995) but not on a work system level. If design science re-

search in the IS discipline is to take on a work system view of IS (McKay et al., 2012), then empirical evidence for the context construct in work system design is missing. Second, work system theorising has evolved from informal use of the construct in socio-technical design (Bostrom & Heinen, 1977a; 1977b) to a conceptual framework for analysis and design of IT-enabled work systems (Alter, 2006a; 2009). Questions remain to what extent contemporary work on conceptualising work systems has taken into account the constructs context and context-driven adaptation. Finally, given a firm, empirical understanding of context and an understanding of the theoretical completeness of work system theory with respect to context and adaptation, how can we construct and evaluate context-aware information systems?

3.2 RESEARCH APPROACH

Literature reviews provide an overview of areas of prior research and identify relevant gaps in our understanding of the phenomena of interest in a field of enquiry. They close areas where sufficient research has been conducted and allow the researcher to focus on gaps in our understanding of concepts that provide the opportunity for theory development and to advance the body of knowledge of a discipline (Webster & Watson, 2002). In the following, I present the primary concerns of literature reviews (source selection, review structure, and presentation of findings) and define their instantiation in the context of my research study by following the guidelines defined by Webster and Watson. In particular, I define the strategy for selecting sources that is underpinning my review of the literature and reiterate the key concepts of the research enquiry. I also present the criteria for including and excluding sources and articles in the review process based on the objective of providing prescriptive design guidelines. Finally, I discuss matters relating to tone and the presentation and discussion of findings of the review.

3.2.1 SOURCE SELECTION

Webster and Watson (2002) argue that high quality, complete reviews of the literature focus on *concepts* and integrate contributions from *different* sets of methodologies, journals, and geographies. Three primary concepts underpin my research enquiry: information system, context, and work system. Since my objective is the theory-driven design of prescriptive guidelines for the construction of context-aware information systems, I focus on the understanding of information system and context in the design science (DS) relevant literature. This includes discussions of these concepts in the Design Science Research in the IS field (DSRIS) literature and its reference fields in Computer Science (CS) and Human Computer Interaction (HCI). It also includes developing an understanding of the conceptual foundation of work systems in Organisational Science (OS) and Management Science (MS). Following Webster and Watson, source

selection must not only focus on peer-reviewed journals but should include other peer-reviewed sources such as the proceedings of premier conferences.

3.2.2 STRUCTURING THE REVIEW

Following Webster and Watson (2002), I use a *concept-centric approach* to reviewing the peer-reviewed literatures of IS research and its reference fields in CS, HCI, OS, and MS. The IS field is an inherently diverse field that is characterised by the lack of a central and generally accepted paradigm (Taylor et al., 2010; Vaishnavi & Kuechler, 2004). Clear criteria for the inclusion and exclusion of articles in the review process are therefore required. First, I included contributions which define the information system concept and discuss the impact of our understanding of information systems on their design. For example, this includes diverging views of information systems held by various scholars in design science research in IS (Alter, 2003; Hevner, March, Park, & Ram, 2004; McKay et al., 2012). Second, the concept of context has received little attention in the IS literature as a design construct but has exhaustively been discussed in the reference fields of CS and HCI. In my review of the literature, I therefore included contributions that treat context as a design construct. This includes conceptualisations of context as representation or interaction problem (Dourish, 2004). Finally, I review the foundational literature of IS for conceptualisations of work systems and theories of adaptation.

3.2.3 PRESENTATION OF FINDINGS

The literature review presented in the following provides a neutral viewpoint on the different streams of research and conceptualisations in the literature that are relevant to the concepts information system, context, and work system. Following Webster and Watson (2002), my objective is to inform readers about the diverging views held by scholars in the peer-reviewed literature concerning these concepts. The objective is not to criticise a single view but to define the foundation on which prescriptive guidelines for the design of context-aware information systems can be developed and, thereby, to make a contribution to the cumulative tradition of research. The review is therefore presented in the order of the three constructs that guide the enquiry. At each stage, findings from the literature review are summarised and presented to the reader. This includes, in particular, the identification of the major streams of research and the views scholars hold concerning a concept. Finally, I compare and discuss each view in the discussion section and define questions for further enquiry.

3.3 WHAT ARE INFORMATION SYSTEMS?

In the following section, I review *definitions of information systems* in the IS literature and their *implications* for DS in the IS discipline. The lack of a commonly accepted definition of information system has troubled the IS field since its inception. On the one hand, IS have been conceptualised as Information Technology (IT) artefacts (Benbasat & Zmud, 2003; Orlikowski & Iacono, 2001). This view has given rise to the dominant interpretation of DS as the construction of IT artefacts (Kuechler & Vaishnavi, 2008) as evidenced in the framework of Hevner *et al.* (2004). According to Hevner *et al.*, DS provides solutions to topical problems through the innovative combination of hardware and software. On the other hand, information systems have been conceptualised as a special case of work systems (Alter, 2003; 2006b; 2008) that combine people, information, and technology in the context of organisational work. The view of information systems as work systems is evidenced in more recent conceptualisations of DS as human-centred activity (McKay *et al.*, 2012) and has its roots in the tradition of socio-technical design (Clegg, 2000; Mumford, 1971). Socio-technical design places the joint optimisation of technical and social variables at the centre of design.

3.3.1 INFORMATION SYSTEMS AS ARTEFACTS

One influential view of IS is that they are *information technology (IT) artefacts* (Benbasat & Zmud, 2003; Orlikowski & Iacono, 2001). According to this view, IS are ensembles of hardware and software that encapsulate the tasks, structures, values, and norms evident in their organisational context. For example, a budget planning tool is an encapsulation of the process of specifying and analysing divisional or corporate budgets and their embedding in the wider context of the organisation within which such a tool is used. Benbasat and Zmud identify four features of IT artefacts. First, IT artefacts are combinations of hardware and software. They combine the hardware, networking, and software resources of an organisation in an IT-enabled system. Second, IT artefacts extend the capabilities of users by enabling the tasks performed by them. Third, these tasks are embedded in task structures which are, fourth, embedded in an organisational context. The interactions between artefact and task context and between the elements of the context determine its usage and affect the impact of artefacts on context.

Benbasat and Zmud's (2003) definition of IS as IT artefacts is one example of the artefact-centred view and is evidenced in the identification of four properties of IT artefacts. Their definition emphasises the technical nature of IT artefacts and their "embedding" in the social and organisational context of use:

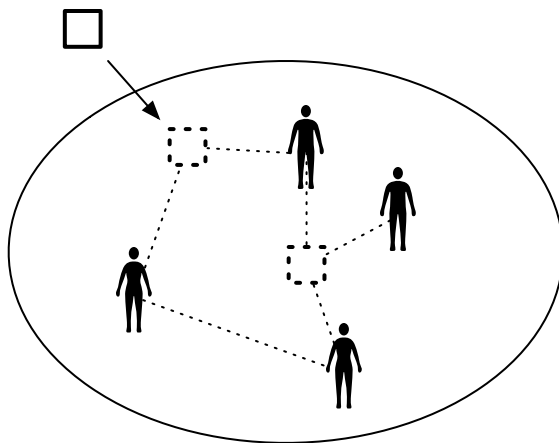
We conceptualize the IT artifact as the application of IT to enable or support some task(s) embedded within a structure(s) that itself is embedded within a context(s).

The view of IS as IT artefacts has been embraced by much of DS and is most evident in its most widely accepted framework (Hevner et al., 2004). Hevner *et al.*'s work on design and design science has received widespread attention and to date is the most widely accepted framework (Kuechler & Vaishnavi, 2008) for conducting DS in the IS discipline. The framework introduced by the authors gives central significance to the design and construction of artefacts as innovative combinations of hardware and software to solve topical problems. The construction of such artefacts provides novel and innovative solutions to relevant problems as perceived by practitioners and design researchers. These solutions differ from routine design by generating new knowledge through their application of the laws of natural and social science. Their application and implementation in the organisational context of use affects the behaviour of its users and therefore feeds back into the process of behavioural research. Both design science and behavioural research in the IS discipline are thus viewed as mutually reinforcing cycles of activity.

According to Hevner *et al.*, IT artefacts are innovative combinations of hardware and software that *remain separable* from the wider social and organisational context in which they are deployed and evaluated. The authors define IT artefacts as constructs, models, methods, and their instantiation in artefacts. Constructs define the vocabulary and concepts of the domain of application. For example, the entity-relationship model (P. P.-S. Chen, 1976) is a set of constructs for reasoning about data and as such has had a fundamental impact on system analysis and design (K. E. Kendall & Kendall, 2010). Models are representations and abstractions of these constructs in software systems. For example, tables are mechanisms for representing data defined using the constructs of entity-relationship models. Methods define specific algorithms and practices by which models can be created from constructs. In the case of entity-relationship modelling, this includes normalisation and languages such as SQL (Codd, 1970) for data retrieval. Finally, instantiations of constructs, models, and methods in artefacts make them available for use.

Figure 3.1 illustrates the view of IS as IT artefacts and the implication of this view for system design. The design researchers observes a topical problem in a social setting (requirement) which is translated into an innovative combination of hardware and software (solution) following the researcher's understanding of the problem and the state-of-the-art in design knowledge in a given domain. The resulting artefact is then "injected" into the social setting for testing and iteration. Through multiple iterations, the researcher refines the artefact until incremental learn-

ing is achieved. The application and use of the artefact henceforth is subject to behavioural research. For example, this may include research concerning the "intention to use, perceived usefulness, and impact [of IT artefacts] on individuals and organizations" (Hevner et al., 2004, p. 77). Notably, the design and behavioural science cycles of research are viewed as two separate but complementing cycles. Design science is primarily concerned with the design of artefacts whereas behavioural research studies their application in a particular context.



IT artefact as "surrounded" by organisational context of use

Figure 3.1. Illustration of DS as design of IT artefacts (based on McKay et al., 2012)

3.3.2 INFORMATION SYSTEMS AS SYSTEMS

A second view of information systems is that they are *IT-reliant work systems* (Alter, 2003; 2006b; 2008). According to this view, IT is but one part of a wider system in which people perform work using information, technology, and other resources. For example, a customer relationship management (CRM) package is but one component in a work system maintaining customer relationships. Understanding CRM as a work system must include understanding the properties of customers, the products and services they procure, the technology and infrastructure available for processing orders, and the properties and relationships among staff and customers. Alter defines nine relevant features of work systems. These are the customers of a system, the kinds of products or services it produces; the coordination of people, information, and technology in processes; and the infrastructure, strategy, and environment on which it relies. The analysis and design of effective work systems must thus not only consider the parts in isolation but their interaction and interdependence in the system.

One example for viewing IS as work systems is Alter's (2008) treatise on the socio-technical nature of IS and their conceptualisation as a special case of work systems. Alter gives special

significance to the role of people in systems as recipients of system outcomes and actors using information and technology:

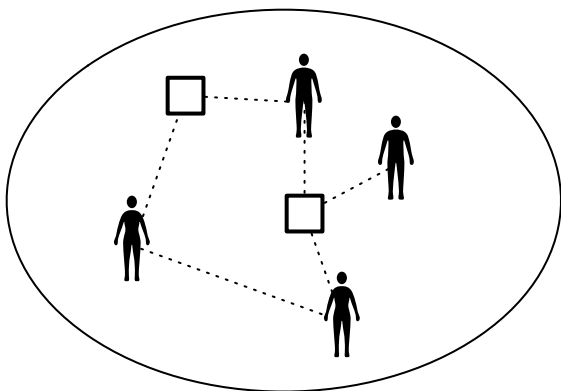
A work system is a system in which human participants and/or machines perform work (processes and activities) using information, technology, and other resources to produce specific products and/or services for specific internal or external customers.

The view of IS as work systems has received less attention in the DS literature but is evident in repeated call for socio-technical (Alter, 2009; 2010) or "human-centred" (McKay *et al.*, 2012) design in IS. McKay *et al.* review the IS literature for conceptualisations of design and find that much of DS is premised on the IT artefact. The central objective of DS in the IS discipline is viewed as the construction of IT artefacts and their evaluation in the organisational context of use. McKay *et al.* refer to this view of DS as the "construction-centred" view: following advocates of this view of DS, the construction of IT artefacts can be separated from its organisational context. The authors turn to conceptualisations of design in other fields and suggest that the view of design as construction is limiting. Instead, design should be "human-centred" and include the conceptualisation of IS as complex systems of people, information, and technology. This view is shared by Alter (2009; 2010) who seeks to bridge the divide between the views of IS as artefacts and as work systems in IS design.

According to Alter and the work of McKay *et al.*, viewing IS as work systems implies viewing IT artefacts as an *essential but non-exclusive* part of a system of people, information, and technology. According to Alter (2008), treating IS as work systems recognises their socio-technical nature. Unlike views of IS as IT artefacts, the work system view "includes people both as system participants and as internal or external customers of the system" (2008, p. 453). By considering people as internal and external participants (as opposed to technology users) of work systems, their design and implementation tends to include topics such as the skills, interests, incentives, and social relations that exist among people in the system. Similarly, McKay *et al.* (2012) argue that DS must balance the economic interests of organisations with sociocultural issues of technology use. According to the authors, designers should aim for a "joint optimization between technology and the individuals who must use that technology" (2012, p. 134) in the tradition of socio-technical design.

Figure 3.2 illustrates the view of IS as work systems and the implication of this view for DS in the IS discipline based on McKay *et al.*'s work. In this illustration, IS are systems combining people, information, and technology in the performance of work. The objective of work in the system is to produce and deliver products or services to other people, i.e., the system's internal or external customers. The special role Alter assigns to people as participants in the system

as opposed to users of technology shifts attention to the properties of people and their interaction with technology. The IT artefact is therefore not an isolated bundle of hardware and software. Instead, it must be viewed as an essential but non-exclusive part in the analysis and design of IT-enabled work systems. In this sense, technical artefacts cannot be separated from their context of use. As McKay *et al.* suggest, "it is within this context that the function of the IT artifact is socially constructed" (McKay et al., 2012, p. 134). The task of the researchers is thus defined as shaping the context by changing the way IT artefacts are used.



Socio-technical view of IS as technical and social system

Figure 3.2. Illustration of DS as design of IT-reliant work systems (based on McKay et al., 2012)

3.3.3 COMPARISON OF ARTEFACT-CENTRED AND SYSTEM-CENTRED VIEWS

Table 3.2 compares contributions in the design and design-related literature of the IS field and their conceptualisation of the information system concept. The contributions were extracted from premier peer-reviewed journals of the IS discipline including MIS Quarterly (MSQ), the Journal of the Association of Information of Information Systems (JAIS), Communications of the Association of Information Systems (CAIS), and other publications that have published design or design-related research in IS in the past. Contributions are sorted by year of publication to allow the reader to develop an understanding of the evolution and focus of the scholarly discussion over time. Furthermore, a brief summary of key propositions made by the author(s) of an article is provided in order to justify the association of the article with one of the two overarching conceptualisations of information systems as Information Technology (IT) artefacts (Benbasat & Zmud, 2003) or IT-reliant work systems (Alter, 2003). Overall, the comparison allows an understanding of key contributions to defining the IS concept.

Author(s)		Conceptualisation of information system in DS	
		<i>as artefact</i>	<i>as system</i>
	<i>Year</i>		

Author(s)		Conceptualisation of information system in DS	
Mumford	1971		System design must place equal weight on technical and social optimisation
Bostrom and Heinen	1977		IS design to embrace joint optimisation of technical and social factors to avoid issues
Walls, Widmeyer, and El Sawy	1992	Design theory is a set of hypotheses to be proven by construction of an artefact	
March and Smith	1995	IT artefacts are technology for processing information embodied in IT systems	
Alter	2003		IS should study and design IT-enabled work systems rather than IT artefacts
Hevner, March, Park, and Ram	2004	IT artefacts involve design of constructs, models, methods, and instantiations	
Walls, Widmeyer, and El Sawy	2004	Although IS involve more than technology, factors are designed and thus artefacts	
Gregor and Jones	2007	IS studies the design and use of artefacts in human-machine systems	
Iivari	2007	IS as a design science must be based on sound ontology, i.e., ontology of IT artefacts	
Alter	2010		Work system theory bridges gap between technical and socio-technical IS design
Aier and Fischer	2011		IS consist not only of a technical and social system but include their interaction
Kuechler and Vaishnavi	2012	IS are ensembles of hardware, software, users, and processes of artefact use	

Author(s)		Conceptualisation of information system in DS	
McKay, Marshal, and Hirschheim	2012		IS are work systems and design science studies how artefacts enable work tasks

Table 3.2. Comparison of artefact-centred and system-centred views

3.4 WHAT IS CONTEXT?

Furthermore, I review *definitions of context* in the literature of the Computer Science (CS) and Human Computer Interaction (HCI) fields and their *implications for system design*. Context has received little attention in the IS literature but has been exhaustively discussed in the literature of its reference fields. Two views of context and their corresponding assumptions dominate the literature of these fields (Dourish, 2004). First, the representational view of context defines context as information that can be encoded and represented much like other forms of information. This view is evidenced in the work of Schilit and Theimer (1994) and Dey (2001) who view context as information about the situation of entities that can be "used" to provide relevant information or services to users of computer applications. Second, the interactional view of context defines context as an occasioned property arising from the interaction of people, artefacts, and groups. This view of context is evident in Suchman's (1987) work on situated action and Nardi's (1995) reconceptualisation of HCI.

3.4.1 CONTEXT AS REPRESENTATIONAL PROBLEM

One view of context is that it is *information about the situation of entities* such as objects or people. Interest in context as information has been high in the fields of pervasive computing (Dey, 2001; Schilit & Theimer, 1994) and artificial intelligence (Brézillon, 1999). In their article on context, Lieberman and Seltzer (2000) lament that despite the pervasiveness of IT in everyday life, computers are still notoriously weak in understanding the context in which people use them. They trace the insufficiency of computers to recognise context to the traditional input/output model underlying their architecture and design. According to the authors, the solution lies in enabling computers to not only process explicit input (provided by the user via direct manipulation interfaces) but implicit inputs or context (provided, e.g., via sensors). This view is shared by others such as Schilit and Theimer or Dey who treat context as information about the situation of entities. Similarly, Brézillon views "modeling, representation and use of context" as one of the key challenges in artificial intelligence (Brézillon, 1999, p. 47).

The view of context as information is reflected in Dey's (2001) definition of context as information about the situation of entities. Following Dey, context information is used to adapt the behaviour of software applications such that they provide information and services of relevance to users in a specific context:

Context is any information that can be used to characterise the situation of an entity. An entity is a person, place, or object that is considered relevant to the interaction between a user and an application, including the user and applications themselves.

This view has been embraced by much of the pervasive computing community and, lately, the IS community (Adomavicius, Sankaranarayanan, Sen, & Tuzhilin, 2005; Rosemann et al., 2008). Early examples of context include the usage of location to adapt the behaviour of software applications (Schilit & Theimer, 1994) and to provide more relevant information and services to their users (Dey, 2001). This led to the development of platforms for rapid development of context-aware computing applications (Dey, Abowd, & Salber, 2001) and approaches to context modelling and representation (Henricksen & Indulska, 2006). More recent examples of context include the work of Rosemann *et al.* on context in process modelling (2008; 2006) and Adomavicius *et al.* (2005; 2010) in the domain of context-aware recommender systems. The authors focus on the representation and classification of context to adapt the behaviour of information systems. The problem of context has also been recognised in artificial intelligence, involving its formalisation (Akman & Surav, 1996) and application to problem solving (Brézillon, 1999).

The basic assumptions of this view regarding system design include the representation of context information and parameterisation of systems such that they can be adapted. Bettini *et al.* (2010) survey the literature and identify four classes of context representation techniques that are commonly used in the design and implementation of pre-defined adaptation. Such techniques include key-value pairs (Strang & Linnhoff-Popien, 2004), object-role based models of context information (Henricksen & Indulska, 2006), spatial models of context as location (Schilit, Adams, & Want, 1994), ontology-based techniques for representing and reasoning about context knowledge (H. Chen, Perich, Finin, & Joshi, 2004; Gu, Pung, & Da Qing Zhang, 2005), and the application of temporal logic and situation abstractions to reason about context in mobile applications (Henricksen & Indulska, 2006). Once encoded and represented in software systems, contextual information can be used to adapt the information and services they provide to users to context. Examples for architectures supporting context-aware system adaptation include application toolkits (Dey et al., 2001) and context-aware middlewares (Gu et al., 2005).

3.4.2 CONTEXT AS INTERACTIONAL PROBLEM

Another view of context is that it is an *interactional problem* that arises from the interaction of people, artefacts, and groups. In a critique of the prevalent view of context as information, Dourish (2004) argues that the focus on encoding and representing context in much of the research in pervasive systems stems from a misreading of the definition of context and understanding of its nature in the social sciences. The limitations of traditional approaches to the design of interactive systems and the problem of context has been recognised by Suchman (1987) and Nardi (1996) in the study of human computer interaction. The observation that many interactive systems fail to respond to the context of their use was subsequently picked up by researchers in fields such as computer science to design systems that hold and adapt to explicit representations of context (Lieberman & Selker, 2000). However, Dourish suggests that readings of social science theory and its application to CS (Abowd, Mynatt, & Rodden, 2002; Weiser, 1993) fail to realise that what is meant by context is not what can be represented.

The view of context as an interactional problem is reflected in Nardi's (1996) commentary on the study of context in system design. She brings focus and attention to the difficulty of generalising context and abstracting it from the particular situation and setting in which technology-mediated action occurs:

Taking context seriously means finding oneself in the thick of complexities of particular situations at particular times with particular individuals. Finding commonalities across situations is difficult because studies may go off into so many directions, making it problematic to provide [a] comparative understanding across domains.

The view of context as an interactional problem has been particularly strong in the human computer interaction community and has had a lasting impact on our understanding of the design and implementation of interactive systems. Suchman (1987) critiques the preoccupation with plans and purposeful action in her analysis of interactive systems and suggests human activity is situated. Humans construct and reconstruct their reality through constant interaction with the social and material world. Similarly, Nardi (1995) brings attention to the role played by artefacts in everyday life and explains their integration into social practice on the basis of activity theory. These foundational works lead to a breadth of research on context in system design. For example, the situated action concept has found application in research on technology-mediated work in organisational settings (Orlikowski, 2007), work-oriented participatory design (Blomberg, Suchman, & Trigg, 1996; Suchman, Blomberg, Orr, & Trigg, 1999), and tailorability in the field of computer-supported collaborative work (Mackay, 1990; MacLean, Carter, Lövstrand, & Moran, 1990; Mørch & Mehndjiev, 2000).

The basic assumptions of this view regarding system design include the *inability to create representations of context* and, as a result, the requirement to create tailorable systems that users can adapt. According to Dourish (2004), context is the shared knowledge of users concerning the settings of tasks. This knowledge cannot be separated from the concrete task and context does not generalise across settings. The main purpose of system design thus becomes the ability to tailor systems to the needs of users. Mackay (1990) describes scenarios of sharing system customisation settings and advocates the design of tailorable systems. MacLean *et al.* (1990) bring attention to a two-pronged approach to system tailorability including a "tailoring culture" that encourages users to adapt systems to their work practices and system features that support tailoring. Mørch and Mehandjiev (2000) describe tailoring as a collaboration between users and developers and Germonprez *et al.* (2007) develop a design theory for tailorable technology.

3.4.3 COMPARISON OF REPRESENTATIONAL AND INTERACTIONAL VIEWS

Table 3.3 compares contributions in the IS and IS related literature, their conceptualisation of context, and design-relevant assumptions of integrating context into system design. As the context concept has received limited attention in the premier outlets of the IS field, peer-reviewed journals and conference proceedings from the related fields of CS and HCI have been selected. Discussion of the concept have been strongest in the outlets Personal and Ubiquitous Computing (PUQ), Pervasive and Mobile Computing (PMC), Computer-Supported Cooperative Work (CSCW), and publications of the ACM and IEEE. Contributions are sorted by year of publication to convey an understanding of the evolution of the scholarly discussion of context over time. Furthermore, a brief summary of each contribution is provided concerning the conceptualisation of context proposed by the author(s) in order to justify its association with one of the two overarching views of context as representational and interactional problem (Dourish, 2004). This allows identifying key definitions of context across fields.

Author(s)	Conceptualisation of context in IS, CS, & HCI			
	Year	Field	as interaction problem	as representation
Suchman	1984	HCI	Conventional system design ignores the settings of system use	
MacLean, Carter, Löfstrand, and Moran	1990	HCI	Users should tailor systems to their work practices or context of use	

Author(s)		Conceptualisation of context in IS, CS, & HCI		
Weiser	1994	CS		Context is information relevant to the interaction of user with system
Schilit and Theimer	1994	CS		Context is information about the location of users and objects
Schilit, Theimer, Want	1994	CS		Context is information about the location of use and nearby people
Nardi	1995	HCI	A particular situation at particular times with particular individuals	
Brézillon	1999	CS		Context constrains problem solving without explicitly intervening in it
Lieberman and Seltzer	2000	CS		Context is implicit information gathered next to direct user input
Mørch and Mehandjiev	2000	HCI	Organisational work is situated, collaborative, and constantly changing	
Dey	2001	CS		Context is information used to characterise the situation of entities
Dey, Abowd, Salber	2001	CS		Information about people, applications, and physical environment
Abowd, Mynatt, and Rodden	2002	CS		Context is information acquired by devices next to explicit user input
Germonprez, Hovorka, and Collopy	2007	IS	Users modify and redesign technology to tailor it to its context of use	
Dourish	2004	HCI	Context is occasioned property and cannot be predetermined	

Author(s)		Conceptualisation of context in IS, CS, & HCI		
Bettini, Brdiczka, Henricksen, Indulska, Nicklas, Ranganathan, and Riboni	2010	CS		Context is abstraction of situation and can be represented as information

Table 3.3. Comparison of representational and interactional views

3.5 THEORISING ABOUT WORK SYSTEMS AND ADAPTATION

Finally, I review the foundational literature of IS in the OS and MS disciplines (Benbasat & Weber, 1996; Taylor et al., 2010) for conceptualisations of the problems of work, work systems, and adaptation. The IS discipline is ostensibly about "systems" (Alter, 2001). Systems theory and systems thinking have a long tradition in the reference fields of IS (Ackoff, 1971; 1973) and despite criticism (Kast & Rosenzweig, 1972) remain a viable paradigm for research (Ashmos & Huber, 1987) in many fields. Most generally, systems are sets of interrelated elements that are themselves part of a larger whole. Different classifications of systems, their properties, and behaviour have been proposed in the literature (Ackoff, 1971; Boulding, 1956; J. G. Miller, 1995). Purposefulness, adaptation, and learning are properties that are unique to higher level systems such as organisations and organisational work systems. Furthermore, systems can be distinguished according to the nature of the components they contain. This requires a foundation for understanding work systems as combinations of people, information, and technology.

3.5.1 FEATURES OF ADAPTIVE SYSTEMS

The System of Systems (SSC) concept studies the origins of purposeful behaviour in systems and the ability of some systems to adapt to relevant changes in the environment (Ackoff, 1971). The classification of systems based on their common properties have a long tradition in systems thinking (Ashmos & Huber, 1987). Some examples include the works of Boulding (1956), Miller (1995), and Ackoff (1971). Ackoff applies systems thinking to define the purposeful behaviour of systems such as organisms and organisations. According to Ackoff, organisations (and as such organisational work systems) are comprised of purposeful parts and a system-control function by which achieved outcomes are compared with desired outcome. Organisations correct their course of action by measuring the discrepancies between achieved and desired outcomes in an environment and correct their course of action accordingly. The principles of environmental feedback, system control, and learning allows organisations to improve their efficiency in obtaining desired outcomes over time.

Ackoff distinguishes among *five key features of adaptive systems* by which system adaptation at a specified moment in time can be described. Although the SSC concept comprises a total of 32 concepts, only five of these concepts are introduced below. The purpose of reducing the number of concepts presented below is to focus the discussion on concepts relating to adaptive, organisational systems as opposed to other classes of systems discussed in Ackoff's text:

Environment. The concept of environment has received much attention ever since the open systems view of organisation began to take hold in the literature (Katz & Kahn, 1978). The environment of systems is generally defined as those "elements and their relevant properties, which [...] are not part of the system but a change in any of which can produce a change in the state of the system" (Ackoff, 1971, p. 663). For example, elements in the environment of organisational systems generally include the organisation's suppliers, customers, competitors, and capital providers, or forces such as the labour market, government, and the characteristics of the natural and social environment (Rüegg-Stürm, 2005). The history of systems and the environment is coupled, i.e., the state of a system and the environment evolves over time and one cannot be explained without the other (J. G. March & Simon, 1993). Interdependency with the environment and coupling is a property of open systems that requires us to view them as part of a larger whole (Ackoff, 1973). Failing to do so leads to a failure to understand their behaviour.

State. The state of a system (and the environment) is the set of relevant properties the system (or the environment) has at a moment in time. Ackoff points to the necessity to reduce complexity by focussing on the essential properties of systems. Applying systems concepts to problems requires us to make models which almost inevitably leads to making simplifications (Ashby, 1956). The choice of elements and their relevant properties therefore is subjective to some extent and must be guided by the purpose of the enquiry. Three general laws concerning systems and their properties can be identified (Ackoff, 1973). First, the properties or behaviour of one part has an effect on the properties or behaviour of all other parts in the system. Second, the properties or behaviour of a system's part and their effect on the system as a whole depend on the state of all other parts. No part has an independent effect on the whole. Finally, these two laws hold for all divisions of a system into subsystems. Each division has an effect and none has an independent effect. They are thus interdependent.

Event. An event is a change in the relevant properties of a system (and the environment) over a specified duration of time. The behaviour of open systems is generally defined as having a cyclical nature. System behaviour commences with the importation of energy and information from the environment and leads to the production of some output that is discharged back into the environment (Katz & Kahn, 1978). The cyclical nature of system behaviour is reflected in

Ackoff's concept of system and environmental events. Environmental events are changes in the state of the environment (as defined by the state of its elements) at a moment in time that leads to a change in the state of the system. For example, a change in the state of an organisation's customers leads to a change in the state of the organisation in the form of changes to its order backlog. System events are events for which an environmental or another system event are either necessary or sufficient. The system thereby evolves through a series of iterations that are defined by the importation of inputs and the discharge of outputs.

Adaptation. Adaptation is a system event by which a system adapts its behaviour as a reaction or response to a change in its environment. Decision-making in purposeful systems is characterised by the concept of efficiency. According to Ackoff, efficiency is defined as the probability with which a system obtains its goals and objectives in an environment. Changes in the state of the environment or system can lead to variances and thereby decrease the efficiency of the system. For example, changes in the labour or capital markets can affect an organisation and decrease its efficiency in producing and delivering products or services to its customers. Adaptation thus is the ability of a system to adapt its behaviour such that the loss in efficiency is at least partially recovered. Ackoff distinguishes four types of adaptation in systems. The first two types of adaptation define reactions or response to changes in the environment by modifying the internal or external system state. The third and fourth type deal with modifications of the system or environment following changes in the state of the system.

Learning. Finally, learning is the ability of purposeful systems to improve their efficiency in obtaining outcomes under constant conditions. The ability of memory and learning have been characterised as important properties of organisational systems (Argyris & Schon, 1978; Nonaka, 1994). In systems thinking, learning is defined as a property of higher order systems such as purposeful systems that exhibit memory and will. Memory enables a system to memorise the specific condition of actions and the outcomes its action achieved given these circumstances (Ackoff, 1971). For example, an organisation may memorise the specific condition of its interaction with customers and the outcome of these interactions. Second, "will" describes the ability of the system to correct its own actions following the discrepancy between actual and desired outcomes. According to Ackoff, the system improve its ability to obtain desirable outcomes over time under constant conditions. For example, an organisation learns over time to improve the quality and outcome of interactions with its customers (Treacy & Wiersema, 1997).

3.5.2 FEATURES OF SOCIO-TECHNICAL SYSTEMS

Socio-Technical Systems (STS) views organisational work systems as two independent but correlative interacting systems (Bostrom & Heinen, 1977a): the *technical* and *social*. The observation of behavioural issues following the introduction of technology into work systems (Trist & Bamforth, 1951) and a critique of rationalist assumptions about organisation and people (Bostrom & Heinen, 1977a) characterise the socio-technical view. Socio-technical theory and design places the human at the centre of the enquiry and argues that economic efficiency can only be achieved by the joint optimisation of the technical *and* social systems. The technical systems comprises the tasks, technology, tools, and information required in the process of transforming inputs (material, information, services, labour, capital) into outputs (products or services) that are discharged into the environment. The social system comprises the properties of people, their relationships, reward systems, and the authority structures governing their behaviour. These ideas have found application in a variety of fields including the study of IS (cf. Bostrom & Heinen, 1977b; Mumford, 2003).

STS distinguishes among *four distinct features of socio-technical systems* which, taken together, form the foundation of socio-technical theory, analysis, and the design of work systems. These are the tasks, technology, people, and structures of which work systems are composed. The interactions among these features and between work system and the environment determines the economic efficiency and behavioural outcomes that is obtained by a given work system design:

Tasks. The task feature of work systems denotes how work is done and usually includes the description of the inputs, outputs, and processes by which inputs are transformed (Bostrom & Heinen, 1977b). Tasks and the coordination of tasks in work systems has received much attention in the past in the reengineering (Davenport, 1993; Hammer, 1990) and process management (Harmon, 2007) literature. In these literatures, structured tasks and processes are generally understood as "a structured, measured set of activities designed to produce a specific output for a particular customer or market" (Davenport, 1993, p. 5). However, tasks in work systems must not necessarily be structured (Markus, Majchrzak, & Gasser, 2002) and limitations of an overly technical and structured approach to task and process design have been documented in the literature (Davenport & Stoddard, 1994). Interdependencies of tasks with other system features include the requirement to align technology with work practices (Hammer, 1990), the importance of removing structural silos (Harmon, 2007), and providing participants with authority (Hackman & Oldham, 1975).

Technology. The technology feature of work systems denotes the problem-solving artefacts used to facilitate transformation and include Information Technology (IT) and other forms of technology. The importance of technology in creating internal efficiencies and enabling the transformation of work has been recognised on multiple occasions (Davenport, 1993). However, technology is not neutral and the creation of efficiencies and transformative effects depends on the social context in which technology is deployed and used (Markus, 1983). While the socio-technical literature generally remains cautious with respect to technology, it does recognise its economic necessity (Mumford, 2000). Guidelines for socio-technical design (Cherns, 1976; 1987; Clegg, 2000) thus emphasise issues such as information flow and support congruence. If information technology is used to rationalise information processing activities in work systems, then information should be presented in a way that is consistent with team structures and a system should integrate with surrounding work practices and systems.

Structure. The structure feature of work systems denotes the formal systems of authority, communication, and workflow as well as actual, informal patterns of behaviour that occur in work systems. For example, this may include the mapping of the organisational or departmental hierarchy and the preparation of a role network illustrating the actual flows of communication between work system participants (Bostrom & Heinen, 1977b). Organisational hierarchies describe the formal organisation of work and the flow of information and work between various units in the organisation. Three primary concerns in decomposing organisations into smaller units are known (E. J. Miller, 1959): technology, territory, and time. Furthermore, hierarchical structures guarantee the formal flow of information and work between various organisational units (Huber, 1982). Communication role networks, on the other hand, map the task-based interactions of people within and between departments (Bostrom & Heinen, 1977b). They provide an overview of interactions between people, information needs, and potential gaps in communication.

People. People, their properties, and relations among people are the at heart of socio-technical analysis and design. Much of the socio-technical literature is premised on the discrepancy between assumptions of system designers about organisation and people and behavioural outcomes resulting from the design decisions they make (Bostrom & Heinen, 1977a). Two basic assumptions can be identified (McGregor, 1960). Theory X assumes people are recipients of orders and should operate within tightly defined boundaries with limited authority. This view is also referred to as the machine view of organisation. Theory Y, on the other hand, posits that people are responsible, self-achieving individuals and that control will stifle their innate creativity and flexibility. This view is also referred to as the human resource view. These assumptions

lead to a different understanding of the causes of variance and the need for control. Following the socio-technical view, better outcomes are achieved if designers focus on the individual needs, characteristics, and abilities of people in the work system (Bostrom & Heinen, 1977b).

3.6 DISCUSSION

Four clusters of context research can be defined by combining prevalent views of context and information systems in the literatures of the IS, CS, and HCI fields. A review of the literature of the IS field and discussions of context in the reference fields of CS and HCI have revealed two diverging views about the properties of information systems and context. First, information systems can be viewed as IT artefacts designed, deployed, and used in the context of tasks, structures, and the social and organisational setting of artefact use (Benbasat & Zmud, 2003). On the other hand, information systems can be viewed as complex systems that combine people, information, and technology to produce and deliver products to customers (Alter, 2006b). While context has received limited attention in the IS field (Rosemann et al., 2008), it has garnered much attention in CS and HCI as a variable in system design. In these fields, the problem is described as either a problem of representation or one of interaction (Dourish, 2004). In this section, I view and discuss contributions against the background of these views.

The first cluster comprises contributions that focus on *tailorable artefacts* that users can adapt to the changing context of use. For example, Mackay (1990) describes tailorable artefacts in the context of computer-supported cooperative work. Tailorisation is achieved by means of customisation files for the software applications used in such work environments. Mackay identifies several implications for using customisation files that are of relevance for a system's design. Managers must recognise the role of translators, i.e., subject matter experts, in making adjustments to software packages; they must understand that not all sharing is beneficial; and they must provide opportunities for the sharing of customisation files among members of the organisation. Similarly, designers must provide tools to evaluate the effectiveness of a user's customisation settings; they must offer well-tested examples for customisation; and they must provide tools to support sharing and the tasks of translators. Similar results have been published in the domain of process and system design (La Rosa, van der Aalst, Dumas, & Hofstede, 2008).

The second cluster comprises contributions that focus on *pre-defined context* and the *adaptation of artefacts* such as software applications to context. This cluster is characterised primarily by research on the context of mobile device usage and the parameterisation of software applications on devices to the current context of their use. For example, Schilit and Theimer (1994)

describe a system that subscribes to and monitors information about context in order to adapt and filter information that it provides to its users. This includes notifying users about interesting objects nearby, detecting location-based information such as messages left by other users at a particular location, and using information about nearby objects and people to show reminders or actions set to be triggered in their presence. On a similar note, Dey, Abowd, and Salber (2001) present a toolkit that enhances the behaviour of software applications by informing them about the context of their use. Applications of the toolkit enable the display of context and the presentation or retrieval of contextually relevant information.

The third cluster comprises contributions that focus on providing *tailorability* not only on the artefact level but encourage a culture of tailoring in which work systems are continuously adapted to participants' needs. For example, MacLean *et al.* (1990) describe tailorability as a two-pronged problem and describe the requirements of systems which let users adapt the system to their specific work practices and encourage a culture in which users feel in control of the system and in which tailoring is the norm. The authors distinguish among three groups of people with different skill sets and differences in their ability to apply tailoring: workers, tinkerers, and programmers. The authors then proceed to discuss specific techniques by which tailorability can be made accessible to these user groups. Similarly, Germonprez, Hovorka, and Collopy (2007) define four characteristics of tailorable technology: dual design, user engagement, environments, and component architecture. The authors present design principles and empirical evidence for building such systems in practice.

Finally, the fourth cluster combines *pre-defined context* with *adaptation on the work system level* but has so far received little attention in the IS field, CS, and HCI literatures. Work systems are complex systems combining people, information, and technology to produce and deliver products or services to customers (Alter, 2008). Focussing on the representation of user context or focussing on tailorable systems and creating a culture of tailoring (MacLean *et al.*, 1990) thus only provide partial solutions to the problem of context-driven adaptation. More work is required on understanding and capturing the relevant context of work systems. For example, Rosemann, Recker, and Flender (2008) identify four layers of context and describe their impact on the technical system of work (as defined by processes, tasks, software applications, and resources used by the system). While this has created much interest in representing context (Ghattas, Soffer, & Peleg, 2010; Ramos & Santoro, 2010) and configuring workflows (Hallerbach, Bauer, & Reichert, 2010; Saidani & Nurcan, 2007), empirical evidence for work system context is missing.

Figure 3.3 illustrates the four clusters as combinations of viewing IS as artefacts or systems and of viewing context as a representational or interactional problem. Cluster I (*Tailorable IT Artefacts*) summarises contributions that emphasise tailorability as a technical property of IT artefacts. This is represented by, e.g., discussions of exchanging customisation files as a means of artefact adaptation to user context (Mackay, 1990). Cluster II (*Pre-Defined Artefact Adaptation*) summarises work that conceptualises context as representational problem but specifies adaptation on an artefact level (e.g., displaying user context, context-based information filtering and retrieval). This cluster is represented by much of the work in the pervasive computing community (Dey et al., 2001; Schilit & Theimer, 1994). Furthermore, cluster III (*Tailorable Work Systems*) extends the idea of tailorable artefacts to include social aspects such as the creation of a culture of tailoring (MacLean et al., 1990). Finally, cluster IV (*Pre-Defined System Adaptation*) combines context with adaptation at the work system level.

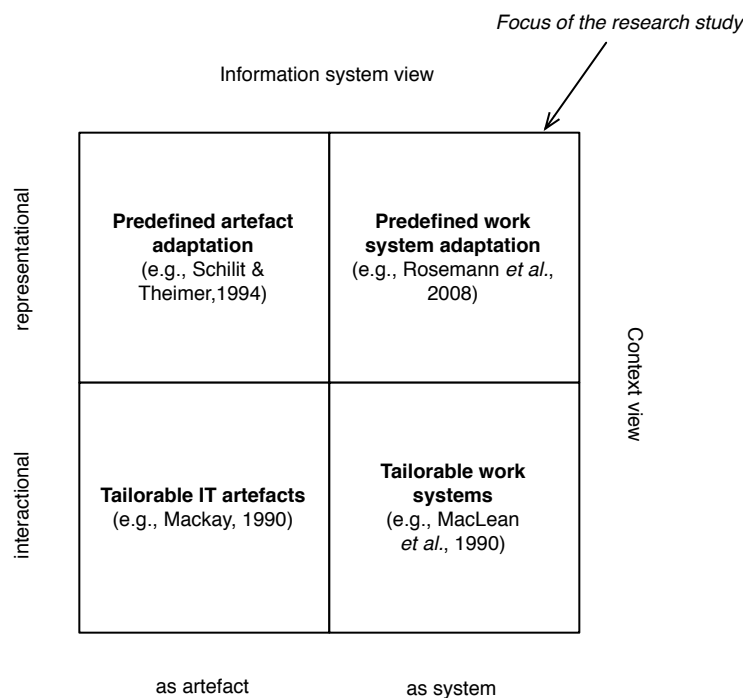


Figure 3.3. Categorisation of research on context systems with examples in the literature

3.7 CONCLUSION

In the following, I focus on cluster IV (*Pre-Defined System Adaptation*) as presented in Figure 2.3. The conceptualisation of IS as work systems (Alter, 2008) and the need for human-centred design (McKay et al., 2012) brings attention to their socio-technical nature. Studying context and the design of adaptive work systems therefore cannot rely on adaptation of the technical system alone. By the same token, the joint optimisation of the technical and social aspects of

work must balance flexibility, control, and ultimately profit generation. As Mumford points out, "managing complexity requires flexibility and diversity while profit generation requires efficiency and control" (2000, p. 125). In practice, pre-defined approaches to system adaptation therefore seem more likely if rapid adaptation of systems to changing contexts is the objective of system design. However, more evidence is required on the relevant context of work systems and the practicability of context representation. This includes an analysis of contemporary design methods and guidelines for effective system design.

Chapter 4: Exploratory Case Study

4.1 INTRODUCTION

In this chapter, I present and discuss empirical findings from a multiple-case study covering the adaptation of work systems to context in two organisations as a *precursor* to design theory development. A review of the academic literature on the topics of work systems, context, and the theoretical explanation of context-aware information systems presented in Chapter 03 revealed predefined context and the joint optimisation of technical and social systems as crucial problems that deserve further enquiry. Work systems consists of two jointly independent but correlative interacting components: the social and technical systems of work (Bostrom & Heinen, 1977a). The optimisation of work systems must give equal attention to both systems. Mumford argues that "managing complexity requires flexibility and diversity while profit generation requires efficiency and control" (2000, p. 125). I therefore suggested that predefined approaches to context are more likely than situated approaches to satisfy the needs of work system designers. I review these notions in the following and draw insights from case data.

In particular, I view and compare similarities and differences in interpretations of the emergent concepts *strategy*, *context*, the adaptation of *tasks*, *technology*, *people*, *structure*, and *learning*. Both organisations I study are commercial entities and customer satisfaction, employee engagement, and profit generation are three commonly accepted but often conflicting objectives. This is evidenced by interpretations of strategy and interrelationships between strategy and context. Balancing these objectives in a changing context represents a formidable challenge to both organisations and prompts the rapid adaptation of the work systems through which they produce and deliver products or services to customers. This is evidenced by interpretations of the concept context and its interrelationships with the work system variable classes task, technology, people, and structure. I find instances of these classes in the case data and discuss the challenges of both organisations in adapting them to context. Finally, I review the concept of learning from context in multiple iterations by looking at two learning strategies.

I conclude by observing a need for further *theory development* and the *extension* of work system theory (Alter, 2001; 2008) toward context. Views of information systems as IT-reliant work systems (Alter, 2001) and the call for socio-technical or "human-centered" design (McKay, Marshall, & Hirschheim, 2012) have long argued in favour of an increased consideration of technical and social issues in the design of IS. This has led to much work on conceptualising work sys-

tems (Alter, 2001; 2003; 2008), methods for analysing and improving IS (Alter, 2006a), and tools that support such analyses (Alter, 2009; 2010). However, context has played no role other than as a passing reference (Alter, 2001). The relevance of the concept in work system design (as evidenced by examples drawn from case data and presented throughout this chapter) point to a significant deficiency in work system theory that limits its applicability to system practitioners. Taken together, these arguments suggest much more work is required in enhancing our understanding of context in work system design.

4.2 RESEARCH APPROACH

The case method is a research strategy which focuses on developing an empirical understanding of a phenomenon by studying it in the social context in which it occurs. Case studies have been used for different purposes such as the empirical validation of existing theory (Yin, 2009), exploration, and the development of new theory from empirical data (Eisenhardt, 1989). Case studies make use of data collection methods such as interviews, archival records, documentation, or participant observation to collect evidence and typically 1) use multiple sources of evidence, 2) organise evidence into an assembly referred to as case database, and 3) maintain a strict "chain of evidence" that link research questions to data sources, to conclusions drawn (Yin, 2009). Furthermore, case studies make use of qualitative (and in some case quantitative) analysis strategies for examining, categorising, and testing evidence. While data collection strategies have received much attention, data analysis still is considered the least codified and to some extent subjective aspect of applying the case method in enquiries (Miles & Huberman, 1994).

I turn to the case method as a specific strategy to develop an empirical understanding of context in work systems and to provide the foundation on which design theorising can occur. In Chapter 02, I identified four essential components of design theorising and described their relationship. I emphasised the importance of justificatory knowledge for design theorising (Gregor & Jones, 2007) and identified different kinds of such knowledge including academic kernel theories and empirical knowledge about a design problem (Kuechler & Vaishnavi, 2012). However, the design science literature provides little guidance on obtaining empirical knowledge about a design problem. Most often, knowledge is viewed as informal, experience-based insights into a particular technology or setting (Kuechler & Vaishnavi, 2012). This leads to a situation in which the initial understanding and assumptions about a design problem remain implicit. By drawing from approaches to theory development from data (Eisenhardt, 1989; Glaser &

Strauss, 1967), this limitation can be overcome. In the following, I describe their operationalisation in the context of my enquiry.

4.2.1 SITE SELECTION

I selected the two organisations based on their similarities and differences. Following guidelines for case study research and theory development from case data (Eisenhardt, 1989; Glaser & Strauss, 1967), sites must be selected in accordance with their theoretical relevance and the research purpose. Studying claims handling and passenger facilitation systems in the selected organisations gave me the opportunity to observe context-driven adaptation in an organisational setting. Both organisations responded to my call for participation as context is a considerable concern in how they design, implement, and operate work systems. They must coordinate technical (call handling, claims management, check-in, and immigration systems) and social systems (staff or contractors such as call centre agents, claims handlers, ground handlers, security contractors, immigration officers) to produce and deliver outcomes to external customers (policyholders, passengers). They view context as information about situations to which they must adapt. This includes external and internal variables such as weather, traffic, or equipment.

By the same token, I selected sites that are different despite their obvious similarities. The purpose of my research enquiry is to develop prescriptive guidelines for the construction of Context-Aware Information systems (CAIS). These guidelines must generalise across application domains to qualify as a generalised class of systems that map to a generalised class of design problems (Walls, Widmeyer, & Sawy, 1992). Following the imperative to select sites that would reveal insights into context in different settings, I manipulated the dimensions industry, organisational size, organisational culture, type of work, and degree of automation. Both organisations vary in industry (financial services and transportation), size (18,000 and 300 staff), culture (knowledge work and operations), and the nature of the work they do (handling information and people). The organisations created AUD 10.1 billion (SGM) and AUD 457 million (BTA) in revenue from insurance, aviation charges, and other sources in 2011. The differences in organisational conditions allowed me to challenge emerging concepts and draw insights from different work settings.

Finally, I used process maturity (de Bruin, 2007) as a proxy for the organisational and technological maturity of both organisations with respect to context-driven process adaptation. De Bruin argues that the maturity of organisations with respect to their work processes can be classified according to the structure of their process management approach. The four structures de Bruin identifies are: enterprise-wide program, centre of excellence, project-based, and

ad hoc process management. I excluded case sites that had only an ad hoc or project-based approach to process management as the lack of structured approaches to process management would have a negative impact on their ability to rapidly adapt processes to context change. By the same token, I included case sites that had either a centre of excellence or enterprise-wide approach. As the unit of analysis are work systems, I did not consider enterprise-wide programs as such but focussed on the structure of process management in the work systems I studied. This allowed me to filter cases with a low degree of structure in their approach to process management.

Table 4.1 summarises the structure of process management at both case sites based on the discussion of process maturity above. The first case, an insurance company, exhibits a high degree of structure in its approach to process management. This is evidenced by the central coordination of process design activities and the coordination of programs across the enterprise. This allowed me to collect empirical evidence in a setting in which process knowledge is captured, managed, and improved in a structured fashion. The second case, an airport operator, exhibits a high degree of structure in some areas such as passenger facilitation whereas other processes exhibit a lower degree of structure. This is evidenced by the coordination of passenger facilitation design and improvement by the airport in collaboration with its aviation tenants. In the multiple-case study, I therefore focussed on passenger facilitation as an area of high process maturity. This enabled me to study context-driven process adaptation in one area without the need to consider maturity at the organisation level.

Case site	Process maturity	Description
Sigma Group	Enterprise-wide	Processes are centrally coordinated not only for the claims management work system studied in the case study but across other functions as well. This allows the study of context-driven process adaptation in a mature organisation with respect to process management.
Beta Corporation	Centre of excellence	Processes are centrally coordinated only for the work system studied (i.e., passenger facilitation) in the case study and efforts are not integrated across the organisation. This allows the study of context-driven process adaptation in an organisation that has not yet reached the highest level of process maturity but which has a structured approach to managing some processes.

Table 4.1. Process maturity for Sigma Group and Beta Corporation

4.2.2 DATA SOURCES

I collected data in both case sites using unstructured and semi-structured interviews, documentation review, and observations made during my field visits to the organisational headquarters. Many authors (Eisenhardt, 1989; Yin, 2009) emphasise the need for data triangulation to provide multiple perspectives on issues and to strengthen research findings. I focussed on unstructured and semi-structured interviews held with respondents in the case sites, documents that were provided to me by these respondents, and observations made during my visits to organisational headquarters. Initial contacts were established through a trusted individual, a member of the IS department of the insurer and university staff working with the airport operator on a joint project respectively. Through the help and advice of these individuals, I had access to work system participants in the claims handling and passenger facilitation systems including management, system analysts, and staff. Field research followed ethical guidelines set out by the university and interviews were recorded and transcribed in preparation for data analysis.

Data collection focussed on the design objectives, history, and challenges experienced by system participants in aligning work system behaviour with context. I sought information on relevant variables in the internal and external context of the work system, the importance of rapid modification of work system variables to context, and the specific strategies applied by respondents to coordinate adaptation and learning. Interviews lasted between 60-90 minutes and ranged from unstructured (mostly general overviews of respondents' roles, responsibilities, and work environment) to semi-structured (following a protocol and covering a set of specific topics such as information retrieval strategies, work system design, and learning strategies). A total of 12 interviews with 13 respondents were conducted over the course of 6 months. I complemented this data with documents provided to me by respondents and field notes made during my site visit. Data collection proceeded iteratively from an initial understanding of a case site to developing a concrete picture of goals and problems associated with context.

In particular, interview questions focussed on the following topics. I asked respondents for the permission to record interviews and made notes about people a respondent would recommend or documents handed to me by respondents. This allowed me to schedule follow-up meetings or to obtain documentation. The questions were as follows:

- *The role and responsibilities of respondents:* Asking questions concerning roles and responsibilities allowed me to gain insight into the work performed by respondents and their task-based interaction with other participants of the work system.

- *The work objectives and targets of respondents:* Asking questions concerning work objectives and targets allowed me to understand the motivation and incentives guiding the behaviour of respondents and to elicit the rationale underlying system adaptation.
- *Relevant context of system adaptation:* Asking questions pertaining to relevant variables in the internal and external environment of work systems gave me insights into the initial conditions and contextual circumstances for the occurrence of the phenomenon of interest, i.e., system adaptation.
- *Strategies for adapting system variables:* Probing respondents about specific strategies they applied provided insight into the degree of planning and the specific areas of adaptation such as workflow, information, systems, and staffing models.
- *Strategies for learning from system adaptation:* Finally, asking respondents on their approach to learning from system adaptation in a given context gave me insight into different strategies for organisational learning and capturing organisational knowledge applied by the organisations.

Table 4.2 summarises the type and amount of interviews conducted at the first case site, Sigma Group. The study was conducted in the context of my research enquiry and involved members of the insurer's IS department (through which the study was initiated), members of the claims design team of the insurer, and line managers and staff responsible for call centre and back office operations in the area of claims. The study unfolded over the months of June and July in 2009 and was conducted at the insurer's headquarter in Brisbane, Queensland. After initial meetings with the IS department to set the scope of the study, I was introduced to members of other units with whom I scheduled interview appointments. In particular, this included members of the insurer's claims design team responsible for process design and customer experience, management staff for the call centre and claims back office functions and staff in the insurer's call centre. This provided me with a comprehensive view of the technical and social components of the claims handling system and its context.

Respondent	Position	Gender	Date, duration
A	Disaster programme coordinator, Home Claims, Sigma Insurance, Sigma Group	Male	10/06/2009, 90 min
B	Executive manager, Home Claims, Sigma Insurance, Sigma Group	Female	11/06/2009, 90 min
C	Operations manager, Claims Experience, Sigma Insurance, Sigma Group	Male	16/06/2009, 90 min

Respondent	Position	Gender	Date, duration
D	Operations manager, Claims Experience, Sigma Insurance, Sigma Group	Female	16/06/2009, 90 min
E	Operations manager, Home Claims, Sigma Insurance, Sigma Group	Male	17/06/2009, 90 min
F	Business analyst, Claims Experience, Sigma Insurance, Sigma Group	Female	18/06/2009, 90 min
G	Business analyst, Claims Experience, Sigma Insurance, Sigma Group	Female	18/06/2009, 90 min
H	Executive manager, Claims Experience, Sigma Insurance, Sigma Group	Female	07/07/2009, 90 min

Table 4.2. Type and amount of interviews at Sigma Group

Table 4.3 summarises the type and amount of interviews conducted at the second case site, Beta Corporation. The data was collected as part of a larger research initiative involving multiple faculties at the university and drew from prior findings made concerning the operator's passenger facilitation process. The study was conducted in the months of December 2009 to January 2011 with some follow-up interviews and took place at two sites on the airport premises. After initial meetings with members of the joint research project team, I was introduced to staff in different functions including passenger facilitation, airport security, terminal operations, IS management, and security contract management with whom I scheduled interview appointments. This included mainly the responsible manager for passenger facilitation at the international terminal, the managers for terminal operations and security strategy at the airport, and staff in the security screening and IS units. This provided me with a comprehensive view of the context, technical, and social components of passenger facilitation.

Respondent	Position	Gender	Date, duration
A	Terminal facilitation manager, Terminal operations, Beta Corporation	Male	19/11/2009, 60 min
B	Security services manager, Security services, Beta Corporation	Male	27/11/2010, 45 min 04/03/2010, 50 min
C	Operations systems manager, Terminal communications and systems, Beta Corporation	Male	07/01/2010, 85 min
D	Security screening manager, Security service, Beta Corporation	Male	06/06/2011, 90 min

Respondent	Position	Gender	Date, duration
E	Terminals Co-ordination Manager, Terminal operations, Beta Corporation	Male	09/06/2011, 90 min

Table 4.3. Type and amount of interviews at Beta Corporation

After recording and transcribing each interview, I inserted case data together with supporting documents and field notes into a case database. Yin (2009) points to the need for maintaining a complete chain of evidence in case study research. Providing a complete case database ensures traceability of findings from data points in the case data to the final conclusions drawn in the course of qualitative data analysis. In order to support me in maintaining and organising a record of my work, I used the Computer-Aided Qualitative Data Analysis (CAQDAS) software HyperRESEARCH. Each case interview was recorded and subsequently transcribed before inserting it into the case database. Furthermore, I appended supporting documents such as organisational charts, descriptions of operational procedures, data sets extracted from information systems, and secondary evidence such as annual reports and financial statements of the organisations into the database. Copies of these documents were stored on a personal computer to which only I had access in order to guarantee information privacy and security.

4.2.3 DATA ANALYSIS

I analysed data first within each site and then across both sites to identify *similarities* and *differences* between emerging concepts. My strategy for data analysis followed guidelines for building theory from data (Glaser & Strauss, 1967) and searching for cross-case patterns (Eisenhardt, 1989). Glaser and Strauss (1967) define the process of building theory from data as three sequential stages of coding and analysis. First, open coding examines the data collected during field work for salient categories of information or "codes". Second, axial coding explores relationships between emerging categories in the data and develops a model describing the connection of initial conditions, phenomena, intervening conditions, context, action strategies, and action consequences as described by respondents. Finally, selective coding selects a central category and interprets its relationships with other categories in the data through theoretical lenses. Data analysis commences with the within-case analysis of categories before analysing similarities and differences in categories across sites (Eisenhardt, 1989).

Open coding is the process of examining data collected during field work for *salient categories* of information. I performed open coding for each case separately beginning with data collected at Sigma Group and proceeding to Beta Corporation. I used in vivo coding to maintain a close link between codes and data. In vivo coding uses respondents' own vocabulary instead of con-

ceptual terms. For example, I applied terms such as daily weather, event module, or terminal duty manager used by respondents to code text fragments line by line. I continued coding each line in this manner until all textual data had been coded. This led to 250 codes in the Sigma Group case and 100 codes in the Beta Corporation case. Using this code base, I then started searching for leading categories in the data. For example, I grouped all occurrences of issues pertaining to the tracking of weather and traffic variables into the emerging concepts weather and traffic respectively. I continued until I was unable to further aggregate concepts, which led to 18 codes for Sigma Group case and 18 codes for Beta Corporation.

Axial coding is the process of *exploring relationships* between emerging concepts in coded data and mapping them into a model. Following Glaser and Strauss (1967), I focussed on exploring the relationship between the initial conditions, phenomena, intervening conditions, context, strategies, and outcomes surrounding system adaptation for each case separately and the across both sites. For example, I analysed variance in work system outcomes (initial condition), decisions to modify system variables such as workflow and staffing (phenomenon), the context in which adaptation occurred (context), constraints underlying decisions such as work objectives (intervening condition), strategies for modifying system variables (action strategies), and the impact of outcomes on system learning (consequence). I started with analysing such relationships within Sigma Group on the basis of system adaptation in response to a surge in claims volume and then proceeded to Beta Corporation on the basis of system adaption to variations in passenger throughput to compare similarities and differences.

Finally, selective coding is the process of identifying a *single category* as the main phenomenon and systematically relating the core category to other categories in the data. At this stage, I enfolded theoretical concepts drawn from the definition of adaptive (Ackoff, 1971) and socio-technical (Bostrom & Heinen, 1977a) systems in the literature to strengthen emerging concepts. I selected system adaptation as the central category of interest in accordance with the objective of my research. I then applied concepts from Ackoff's system of systems concept and Bostrom and Heinen's conceptualisation of IS as socio-technical systems to reason about 1) changes in the external and internal work system environment, 2) the influence of work objectives on system adaptation, 3) strategies for the modification of technical and social system variables, and 4) the process of learning under constant conditions. This led me to identify similarities in the interpretation of context, the role of customers and shareholders, planned and ad hoc adaptation, single and double-loop learning as well as differences in the implementation strategies.

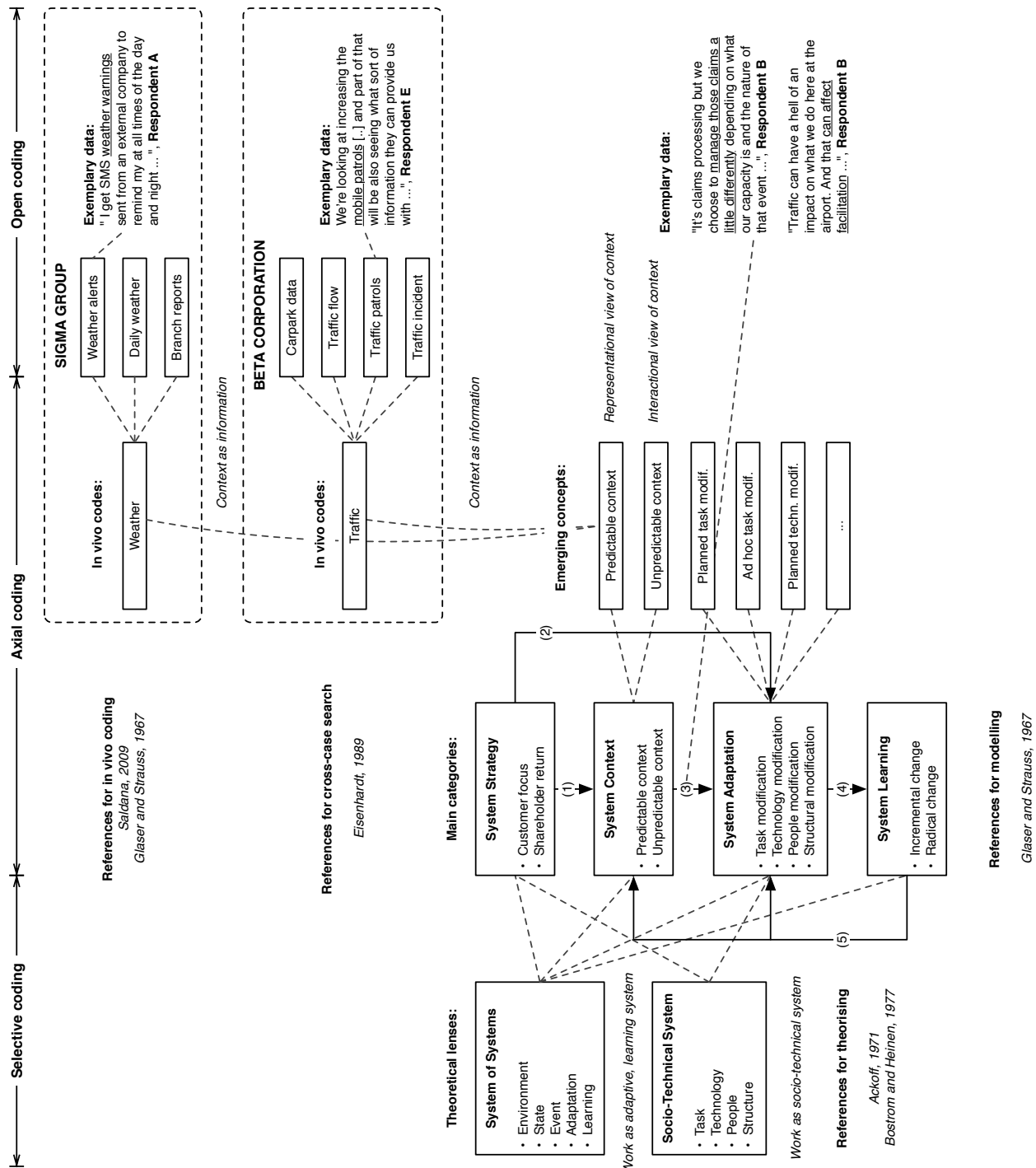


Figure 4.1. Illustration of coding process

Figure 4.1 illustrates the process of open, axial, and selective coding starting with the within-case analysis of data and extending to cross-case pattern search. I began with coding fragments in the text relating to issues such as weather alerts, daily weather, and branch reports (SGM) or carpark data, traffic flow, traffic patrols, and traffic incidents (BTA) as an example for

features of the internal and external work system environment. I then summarised those codes that related to a similar issue, e.g., extracting information about weather or extracting information about traffic. I proceeded line by line until all data were coded and codes could no longer be refined. I then began comparing diverging interpretations of context as information about an issue or object and as situation, which led me to define predictable context (measurable, informational) and unpredictable context (situational, occasioned) as emerging concepts in the data. On the basis of emerging concepts, I began comparing their relationships by enfolding literature and data and finally mapped each concept to theoretical categories.

4.2.4 CASE VALIDITY

Obtaining valid, reliable findings from data is regarded as the key concern in successfully implementing qualitative research in general and the case method in particular. The literature encourages researchers to 1) maintain a chain of evidence tracing the link between research questions, data, and the findings made by analysing data and 2) using multiple sources of evidence and data collection methods to strengthen findings (Yin, 2009). In addition researchers are asked to incorporate multiple points of views or explanations (Gibbert & Ruigrok, 2010) and to explain the rationale underlying the generalisation of findings beyond the concrete setting in which they were made (Walsham, 1995). In the following, I describe the measures I undertook to ensure the reliability and validity of my findings.

First, I created a *chain of evidence* (Yin, 2009) by maintaining a close link between data, in vivo coding of interview transcripts, and the detailed description of the coding process which led to findings. Yin brings attention to the issue of reliability in case study research. Readers of a case study must be in a position to trace the process from research questions, to the data collected, to the findings made in order to assess the reliability of the study's findings. In the context of my enquiry, I enhanced reliability by defining a field protocol specifying data collection techniques, the types of data sources sought, and the type of questions to be asked during case interviews. I then used in vivo coding to maintain a close link between data and codes. Furthermore, I followed recommendations for coding textual data (Saldana, 2009), for building theory from data through coding (Glaser & Strauss, 1967), and for cross-case pattern search (Eisenhardt, 1989) as illustrated in Figure 4.1. Finally, I maintained all data, codes, and coding memos in a CAQDAS software package (Hesse-Biber, Dupuis, & Kinder, 1991).

Second, I corroborated findings by using *multiple data collection techniques* (interviews, case documentation, site visits) and *multiple data sources* (respondents from different units cooperating in the context of a work system). Yin (2009) points to the need to make use of multiple

sources of evidence and to use multiple strategies for data collection. In the context of my enquiry, I used qualitative interviews (H. J. Rubin & Rubin, 2011) and involved multiple respondents from different units of the participating organisations to collect data. I corroborated my findings with documents provided to me such as process documentation, organisational charts, and operational guidelines to compare respondents' interpretations of roles, actions, and responsibilities with documented evidence. Furthermore, I used secondary evidence such as newspaper clippings, industry reports, and annual reports extracted from sources available to me through the university. Combining these sources of evidence allowed me to gain a better understanding of the industry and organisational context, and organisational performance in the market.

Furthermore, I recognised *multiple views* and *alternative explanations* of findings (Gibbert & Ruigrok, 2010) by enfolding rationalist and socio-technical interpretations of technology (Kling, 1980; Markus, 1983). Multiple points of view concerning technology and the use of technology in organisations exist (Kling, 1980). In my analysis of data I applied both rationalist and socio-technical views of technology and the use of technology in a work setting. For example, respondents in the Beta Corporation case seemed to follow a more rationalist interpretation of technology and its impact on work. Technology was viewed by many as a means to create efficiencies and to drive down labour cost. Interpretations of technology in the Sigma Group case, on the other hand, tended to be less enthusiastic, focussing on rigidity as a result of the adoption of information technology. This view correlates with the socio-technical view of technology (Bostrom & Heinen, 1977a). However, both organisations face competitive pressure and rely on the joint optimisation of people *and* technology to achieve work objectives and goals.

Finally, I used a multiple-case design to increase the generalisability of findings across a single case setting. Generalisability of findings from case study research remains a concern in the design of case studies and presentation of results (Yin, 2009). I followed Eisenhardt's (1989) recommendations for developing theory from data and the generalisation of findings across case sites. Site selection focussed on a single phenomenon (the adaptation of systems to context) in two organisations selected from a different organisational background (size, industry). This allowed me to study similarities of system adaptation to context in two different backgrounds while understanding differences in how the organisations responded to context change. Following Eisenhardt's advice, I compared emerging concepts across both sites, placing emphasis on similarities and subtle differences in how respondents viewed a concept.

4.3 SIGMA GROUP LTD (SGM)

In this section, I present the within-case findings of my study of the *claims handling system* of Sigma Group, a large Australian financial services and general insurance provider. The insurer supplies insurance services to private households and organisational customers under several insurance brands and offers different levels of cover including home, content, and motor policies, amongst others. The case study took place in the claims department of the organisation that is responsible for receiving customer calls pertaining to losses, that computes and updates estimates pertaining to loss value, and that handles correspondence with customers regarding additional information such as proof of loss or ownership documentation. In turn, the department maintains the claims backlog of the insurer, performs reporting on claims estimate if requested, and acknowledges the receipt of additional documentation or follows up pending requests. System performance is closely monitored by executive management as it determines the profitability of the insurer and impacts on customer satisfaction and advocacy.

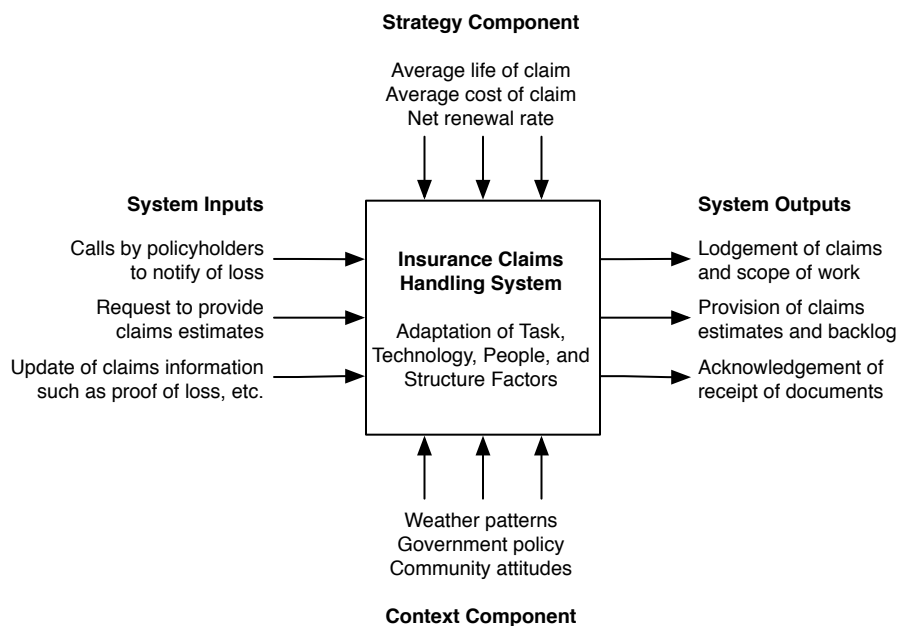


Figure 4.2. Illustration of claims handling system (based on Bostrom & Heinen, 1977b)

Figure 4.2 illustrates the components of the system based on and extending an example by Bostrom and Heinen (1977b). The organisation receives and processes calls by its customers at two different locations along the eastern Australian seaboard. Calls are handled according to a context-specific script and call centre agents record details of the customers and the loss incurred using a claims information system. This includes, for example, information about the identity of the customer, the type of loss that occurred, and the type and validity of the policy

against which a claim is being made. Once basic information is recorded, a claim is lodged in the system and is then forwarded to one of several claims handling teams (following predefined rules) at the organisation's headquarter for further processing. Several context triggers such as weather events, changes in government policy, or community attitudes can prompt the claims department to perform rapid changes to its claims process, claims information system, and staffing levels to recover system performance.

4.3.1 STRATEGY COMPONENT

Customer satisfaction. The construct customer satisfaction is described by respondents as a key outcome of work systems and covers the total experience made by the customer from the sales process, to claims and policy renewal. Customer satisfaction is an important determinant of a) whether customers recommend an insurance brand or product to their social network and b) whether a customer will renew her policy after making a claim. Respondent B explains: "The test comes when you put in a claim and we look to get between 1 and 2% differential between people who renew at a set rate, because it's different after a claim". In the interviews, respondents stress that achieving customer satisfaction requires deep understanding of customer requirements, clear communication, and managing expectations. Respondent H describes the nature of customer satisfaction as managing relationships and expectations across a number of activities in her example:

But the expectations setting is not just at the time of claim, it's about the whole sales process, so sales through to if you need to do a policy change. If you're sending out renewals to the contract expired, so where do you get the opportunity to touch the customer's expectations. Understand what they want, understand what motivates them so when you come to the time of claim, the claims experience matches the expectations of the customer, because we can't do everything, at the end of the day.

The aspect of measuring customer satisfaction is attributed by respondents to two external and a range of internal metrics. External metrics include customer surveys and the monitoring of policy renewal ratios. Respondent B points out that "our customer survey process is a constant scanning of customers with a mindset around how they perceive services that we offer them". Internal metrics relate the performance of front-office staff in handling customer requests. According to respondent C, "following that initial recruitment, training and induction stage it's then the consultants working with their leaders to improve their own performance and understand the expectation aspects". For example, front-office performance can relate to "how quickly we answer the phone, or create a service as it's called". However, respondent G highlights con-

flicts arising from how staff are incentivised in front-office and back-office functions in her comment:

It's like, if you're getting pressured to say I've got to get my number of, my open claims book down as quickly as possible, then what do you have to do to do that? Do it, irrelevant of the KRA, but it's very hard because you get, these guys are measured on it so, and there's never any, well we're going to change the KRAs for this week, is it?

Shareholder value. Respondents emphasise the role of shareholder value in how the insurance delivers products and services. Respondents H contends that the insurance finds itself "in a very different position than a lot of other big corporates in that we have mums and dads shareholders who live in Queensland, who are insured with us and who have all their business with the bank". Work system performance affect shareholder value in three different ways. Profitability encompasses the efficient handling of claims and the reduction of leakage (overpaying). Respondent B suggests "every time we touch a claim and go through that fulfilment process there's an option there for leakage and ... then there's an opportunity to increase your cost base". Next, the timely updating of claims estimates by work system participants "has a financial impact for not only monthly reporting but mid year and end year reporting to the analysts", explains respondent E. Finally, the insurer must create adequate reserves for losses. Respondent B summarises these issues in her comment on "business acumen":

I think you need a general business understanding of how the business works, financially, and how this business works. You need to understand it doesn't work if you're not profitable, so what are the key elements that make it profitable? Like charging more than you have to pay out. I think you need people who've got a good sense of efficient process design, okay. Making sure that you're running a not excessively lean business, but a smart business.

Community recovery. A third outcome of the claims handling system is community recovery, i.e., the fast and efficient delivery of financial relief by the private sector to communities affected by natural disasters. Respondents note that state and federal governments rely on the insurance system to share the burden of financial relief payed out to constituents in the aftermath of a natural disasters. In the interviews, respondents describe differences in the approaches taken by state governments to manage natural disasters. For example, respondent A notes difficult working relationships with one state government he describes as "trigger happy, or they'll jump straight off the bat there and they want to have one-stop locations for 50 people". Other state government, in the eyes of respondents, take more efficient approaches such as setting up insurance task forces to create a common response to disaster. Respondent B gives an example in her observation:

The government of Queensland has also worked out that it's prudent to be kind to the insurance companies because if they piss you off, they've worked out that the customer goes, hang on a minute this insurance thing is crap. I don't really need to be insured, the government will look after me, the government will give me a nice big handout, so the government has worked out it costs them a lot more money to have an adversarial relationship with us, so we work closely with them.

4.3.2 CONTEXT COMPONENT

Weather patterns. The construct weather patterns relates to the annual cycle of severe weather events that affects wide parts of rural and metropolitan Australia, in particular the state of Queensland. Respondent H observes: "well obviously, so Queensland can be called the peril state, you get a lot of rain, regardless of where it's going to be, ... you get storms, the peril, that's natural peril, perils, storms, cyclones". These events require the insurer to swiftly adapt work systems to provide efficient relief to the affected communities and quickly start the re-building effort. Each event is different and losses incurred as a result of the event require a different response, as outlined by respondent B: "we choose to manage those claims a little differently depending on what our capacity is and the nature of that event". Respondent E suggests that events can generally be distinguished between those that require an immediate response and those that build up more slowly:

If a storm has hit, people generally ring us straight away after a storm's finished, which would be 24 hours later, but when it's flood sometimes they're evacuated and they can't get to their phone line for a few days, so that's when you start seeing the calls coming through and if there's a power outage and most phones these days are all electric, people can't ring in until the power is restored ... Bushfire, house burns down, you've got no phone so you know, we got a lot of claims being lodged at these one stop shops ... Hailstorm, generally, sooner, they can be over within 30 seconds, hail damage, people on the phone straight away, so generally they don't get any warning of that stuff, particularly on the weekend.

Respondents point out that among weather patterns, the annual storm season from October to March has the single most import impact on productivity and performance of the insurer. Each year, thunderstorms and flooding create widespread damage in rural and metropolitan communities and require the insurer to handle large number of claims and payouts. Respondents stress that while the storm season used to be a distinct pattern, severe weather events are increasingly a year-round phenomenon. This increasingly puts the capacity of the insurer to analyse events and improve work system performance under pressure. Respondent E highlights the increasing frequency and impact of events in this comment:

To be fair to it, it's hard to be proactive, in that sense because you're so busy with events. I mean we've had, I mean look at the dollars paid out over the last 12 months to 2 years. Almost half a million dollars in payments on events from Blacktown hail, to Mackay floods, to bushfires, the Brisbane storm, Ingham floods, Coffs Harbour floods, the second Brisbane storm you know, you just get pulled into [it] ... It's only between the months of maybe May to say September or October that are supposed to be our quiet period that we can get to look at the business and cost.

Government policy. The construct government policy is a factor that is closely coupled with weather patterns and that relates to changes in expectations and requirements imposed by government authorities on insurers. I observed two interpretations of the construct in the case data: ex ante and ex post. Government pressure "ex ante" relates to the pressure exerted by local, state, or federal government on insurers to respond to relevant events and ensure that constituent receive appropriate aid. This includes the creation of "task forces" and "one stop shops" in the affected communities to facilitate claims lodgement. Government pressure "ex ante", on the other hand, relates to changes in building codes, environmental regulations, and pressure to standardise coverage terms in the aftermath of events. Respondent H relates to such changes in her comment on disaster relief and legislative changes in its aftermath:

The government, as an example, would be as part of the government legislation around climate change and the green response, right, so any repairs where the damage had been over a certain amount, we now have to put in a solar hot water, as an example, right. Or, this is the new building code that comes out from government, so after the bushfires as an example, new building codes were being written.

Community attitudes. I summarised social aspects such as lifestyle choices, consumer behaviour and attitudes I observed in the case data under the construct community attitudes. Community attitudes relate to a variety of factors prevalent in the wider community that influence how the insurer delivers work system outcomes to its customers. The interpretations provided by respondents broadly fall into two categories. Customer expectation relates to differences in the interpretation of what is acceptable service by different customer segments. Respondent H relates to these differences in her comment: "people in the country, yeah no worries mate, I know it takes a bit longer, and they'll go yeah. City, what do you mean it's going to take you more than 3 months?". Customer behaviour, on the other hand, relates to social norms and accepted behaviour in communities. Respondent H points to the problem of opportunistic fraud in her comment: "it's okay to pad your claim, the bastard insurance companies, so it's all right isn't it, ... it was a 9ct gold ring, it's now an 18ct gold ring".

Economy. Finally, the construct economy summarises interpretations provided by respondents concerning changes in the wider economy that affect work system outcomes. Respondents

provide a variety of examples concerning economic impact on work systems. For example, a decline in household wealth often entails an increase in the total number of smaller value claims. As a result, the insurer must cater for the additional claims volume as indicated by respondent B: "because of the global financial crisis we've actually needed more resources than we've really budgeted ... because customers' behaviours have changed". The impact of macroeconomic effects such as inflation can also be measured in the rising costs to fulfil claims. Respondent H relates to the impact of inflation in her comment: "so there's an inflationary impact in terms of cost that your groceries go up, petrol goes up, the same thing will happen to the types of things in insurance". Finally, an economic downturn may affect staff recruitment or replacement as a result of a "recruitment freeze".

4.3.3 TASK COMPONENT

Cycle time. The construct cycle time measures the time taken by work system participants to respond to and fulfil a customer request and impacts on customer satisfaction. Respondents share the view that context changes such as weather events or financial pressure on households increase the average time required to settle a claim (i.e., the "life of claim"). For example, respondent B, who is the responsible manager for the claims backend, notes: "you're handling 2 or 3 times the volume that you normally handle and so you get slower, so what happens is, instead of an average life of claim looking like that, say 60 days ... your average looks like 90 to 120 days". Respondents also stress the impact of context changes on the quality of work system output (i.e., the "grade of service"). Context events generally have a negative impact on quality as staff come under pressure from an increased backlog of claims and process changes. Respondent G provides an example of how pressure and communication between management and staff increase error rates with respect to a) service levels or b) policy coverage in this comment:

One minute you're getting told, okay, if you're doing this you've got to cover flood, and you've got to process a payment, if somebody calls around, so there was so much stuff being heaped at the front end that you've got to wonder how much of it gets used and how much of it, it's a moving target. A moving target of information, so much of it, they couldn't take any more.

Reputation. The potentially adverse effect of context changes on reputation is a concern shared by many respondents. Respondents are aware of the often conflicting stakeholder expectations and perceptions of the insurer's actions following an event. Interpretations of stakeholder expectations fall into two categories. Respondents stress that community expectations are focussed on quick recovery, an expectation that is often propagated by media and government. For example, respondent E describes the issue of negative press: "these guys are getting calls

every week to tell them how their claim was going, and that's our biggest risk from a brand point of view, because these guys would ring up ABC and Channel 7 and say, look my roof is still not fixed". Shareholders, on the other hand, expect financial performance. Respondent B notes: "in short, people lodge a claim, we fix it for them ... we try and do that as quickly as possible at the lowest possible cost for our shareholders". Respondent A describes how the insurer continuously scans for reputational risk:

One thing that we found for in this readiness stage, which was highlighted with the Victorian Bushfires, was a GIO marketing campaign that showed a couple standing in front of a burnt out house that was going in the Saturday Courier Mail. You get the Weekend Magazine, that had 350,000 copies printed in the week prior to the Victorian Bushfires and it was our ad in there of the couple in front of the burnt out house that was going into circulation in Victoria and New South Wales, so that posed an adverse risk to our brand because we'd been shown as insensitive, having the bushfire. The readiness stage is understanding what's going on in the business and what's a possible risk to us and what we can work around to reduce that risk.

Leakage. The construct leakage (overpayment) emerged as a strong pattern in the interpretations provided by respondents. Leakage is the degree to which an insurer overpays a claim and can have various reasons. Respondent B explains: "you could be being gouged or padded like we discussed before, from repairers, suppliers or from customers". The degree of leakage in work processes negatively affects overall financial performance of work systems, an issue respondent B reflects on in this comment: "every time we touch a claim and go through that fulfilment process there's an option there for leakage and ... then there's an opportunity to increase your cost base. I suppose, you're riding the margin". Leakage is generally higher in events: "when you get really busy and you're under pressure and under-resourced the leakage goes right up considerably". The way an insurer can control leakage is by applying tighter process controls to its work systems, which effectively increases time. Respondent H describes the relationship between both as a trade-off:

So I think trying to say that you're going to have a circumstance where there's not leakage when you're dealing with this type of jump is you know, nice if you think you can do it but the reality of the situation is you will get some. It's how you control it to as little as possible.

4.3.4 TECHNOLOGY COMPONENT

Claims information system. The construct of claims information system relates to interpretations of impact on the information and communication technology infrastructure underlying work systems of the insurer. Two issues emerge in the case data. On the one hand, respondents point to the impact of rigid processes on staff productivity and performance. Respondent

A, who supervises the event leadership team's activities, explains: "[claims processing] was quite in depth, scope of work, what happened, policy obligations, it was quite lengthy". The experience with the insurer's information system led respondent A to instigate the development of an "event module", which he explains in this comment "what the system actually does is it asks a series of questions we predetermined at the event leadership team ... and as the context becomes available to that event, we can then change those questions and dictate the event [response]". Another impact is the ability to deliver claims processing capabilities outside the insurer's headquarters, an issue highlighted by respondent G:

Deploying an onsite team to the actual location of the event and getting set up and okay, if we delay that by a day or two days what, if we don't get the team there within the first three days what's the possible impact of that cause. There's all that aspect of it as well, which from a coordinator's perspective we should be able to help enable decisions that impact process.

Event module. The construct of event module relates to an extension to the claims information system designed, developed, and implemented by the insurance carrier to more effectively respond to context changes. Respondent A describes the motivation behind constructing an extension: "we focussed on fire, storm including hail, wind including cyclone and flood, and we thought to ourselves, one thing that came out of fire was that the government and the insurance council wanted us to categorise that properties into A, B and C". Categorisation is achieved by asking context-specific questions, i.e., different sets of questions callers are asked when their call relates to a specific event such as a recent thunderstorm, bushfire, hailstorm, or flooding. The module also enables call centre agents to provide context-specific information to customers: "you would think that our category A customers would give them a lot more information than our category C customers, plus it gave us the ability to have upfront information to change our process quite quickly". Respondent E concurs, but points to some issues:

What that meant was that everyone can take more claims in a short period of time but the problem with that is when again the claim is lodged you take a limited piece of information [...] so great customer service at the front end, but then it suffers at the back end.

4.3.5 PEOPLE COMPONENT

Workforce capacity. The construct of workforce capacity relates to the impact of context changes on workforce planning and availability of staff in lodgement, adjustment, and fulfilment functions. Events such as thunderstorms, hailstorms, or flooding can overnight create surges in claims volume which require the insurer to scale its operations and reduce the claims backlog quickly. In the interviews, respondents highlight the principal challenge of the organisation to balance financial factors and scalability. Respondent A describes the process of workforce

planning in his comment: "you really want to create a plan that would, say if you get hit with a 2.5 thousand claim event, these are the numbers you're going to need and this is where you want to get them from". Respondents note the importance of temporary staff to fill capacity bottlenecks. Respondent E, who is receiving temporary staff in his team, explains: "you're going to resource up at the busy time and use that, use the temps to do that and then let them off when claims go down, and that's a financial decision because it's less expensive that way than carrying staff throughout the year". However, respondent A notes that this approach compromises quality:

But what we also need to change is our mentality on our resourcing because you find that the temporary staff that get employed to come in and process claims and even within our, your business, your events role is such a learning curve, it takes some time to get skilled and again that compromises experience because you've got somebody on the other end at the time that isn't up with it, and the customer perceives that they don't know what they're talking about, that has a flow on effect.

Workforce skills. The people factor of work systems is also evident in respondents' interpretation of the impact of events on workforce skills. Events can create shortages of skilled staff for different activities in the work system. Respondents draw a general distinction between skill sets required for front-office and back-office activities. Respondent C provides an example for call assignment based on the skill sets of front-office consultants: "we can attach skill sets to consultants so they can, skill set for motor, a skill set for home and so on and this event number, we have a skill set for that as well ... depending on the type of event". Back-office activities, on the other hand, are described as more complex and requiring a different skill set. Respondent H, for example, describes the difficulty of scaling the complex activity of handling claims in the back-office during an event: "if we can get that skill in terms of the back end, and then find a way to scale it up and scale it back down then I think we would have much better control over the claims cost and leakage". Another concern voiced by the respondents is the skill set required to deal with situations of distress and grief as described by respondent H:

So if you're dealing with somebody who has lost their house, has lost family members, you know, new building codes and regulations coming in and has a significant asbestos risk on the property, we need to have a team to actually deal with those claims on a one on one basis now, so all bushfire into that team and it's having that flexibility to process to identify the type of event that we're going to skill it, with the most appropriate skill set and staff that you need.

4.3.6 STRUCTURE COMPONENT

Event leadership team. The construct event leadership team relates to respondents' interpretation of temporary changes made to structure during an event. The event leadership team coordinates the insurer's response to a weather event and determines required changes to claim handling procedures and staffing levels as a result of the event. Respondent A describes the process of activating the team: "typically the first ELT would be called within an hour of the event actually striking or passing through a region, so the information would be very sketchy, you won't have all the information to make correct determinations or decisions on, they'll be stock standard what we do every time". The team then meets regularly through the event to provide updates and progress of changes to process, people, and technology: "so we'll throw, out of that last meeting about ten minutes, of giving all the stakeholders a rundown of what's actually happening and then we'll go around the room and we'll talk about what each business unit is actually achieving". Respondent D points to issues in how the team operates:

I think we should clarify the roles of everybody that's in the ELT team. Yeah I, yeah clarify the roles up front and then if there's any change that happens as a consequence of that, that workshop, then we need to understand that my role is this and your role is this and so on and so on.

Government task force. The construct government task force relates to respondents' interpretation of working groups created by local government to coordinate disaster response and recovery. Local government exerts pressure on insurance companies to respond quickly to natural disasters such as storms or flooding. The insurer takes a very active approach to managing relations with government concerning its claims process and disaster response strategy. Respondent B describes a recent event in her comment: "[we] did a round of all the local politicians in the area, to go through each of them what our claims response is, what our policy decisions were and what we could do with them. So the local government members as well as the federal government members understood what our response would be and how we were actually managing that event". Natural disasters is viewed by respondents as a politically sensitive topic. Respondent H describes the approach taken by local government representatives to ensure the needs of their constituency are met in the aftermath of such an event:

So you get pressure from local MPs to say, my constituents are affected, what are you doing to help them? And then you get state pressure from the state government, pressure from councils and you get, sometimes you get pressure from the federal government depending on circumstances, so you get it from everyone.

4.3.7 LEARNING COMPONENT

Post-incident review. Post-incident reviews are meetings held by respondents in the wake of events that led to system adaptation in order to collect feedback and continuously improve event preparedness. The meetings are called by the event coordinator and assemble key representative from each unit of the work system. Their purpose is to review system changes and the decisions that led to these changes being implemented. Respondent G describes the reason for conducting such reviews: "there were so many things we did wrong and you know, so many moving of resources that people were confused totally, you know, the front end and stuff like that and, I think customers were too". The team reviews process changes, changes to staffing levels, and other decisions made in preparation for subsequent system adaptations. However, not all team members are satisfied with the meeting outcomes. Respondent E laments the lack of effectiveness: "it's not an effective debriefing session, in my view and in any event there's very little change that takes place". Respondent G concurs:

I highlighted it at the end of this event season, I actually highlighted and said, I would love to go back over each of the ELTs and see the decisions that were made.

Event simulation. Event simulations are workshops held by respondents in anticipation of events to simulate the impact of events and the level of preparedness of the work system. Each workshops commences with a scenario and assembles representatives from all organisational units including participants of the work system itself and stakeholders from marketing and government relations. Respondent B describes the point in time and participants involved in event simulation workshops in her comment: "towards the end of winter or preparation for the next storm season, we get everyone in a room for a day, all the key stakeholders it has to be, and we say this is what's happening, okay, it's either do your job inside the room, learn how to respond". The workshops enable respondents to enact scenarios and learn about potential weaknesses in the preparedness of the work system: "we do these preparations like a war room information so that we can flesh out any gaps and realise that too many people are trying to do the same thing". Respondent A suggests the workshops enable people to learn:

It was good to see the team functioning together because normally we're at many sites, we're not all in the one place together, we're all very different, apart from the simulation where we all came together and it was fantastic. I think people don't get the feel for an event or our event leadership team in the room.

4.3.8 FINDINGS

Table 4.4 summarises code frequency and emerging concepts in the data collected at Sigma Group. Three issues emerge. First, work objectives and value creation plays an important role in the interpretation of respondents. The claims department is viewed by many as the first touch point a customer has with the insurer after signing a policy. Interpretations of customer satisfaction and the need to balance satisfaction with profitability thus rank high in the interviews. Second, changes in context such as weather events or government policy require the department to act quickly to prevent variance in claims volume and claims complexity from impacting on its ability to satisfy work objectives. This is evidenced by interpretations of weather events, surges in call volume, and the necessity to cooperate with local government and communities in handling natural disasters. Third, modifications to system variables are not confined to technical variables but extend to social variables. This is evidenced by the emphasis on structural aspects (i.e., event leadership team) and workforce properties.

Construct	Respondent							
<i>Supporting codes</i>	A	B	C	D	E	F	G	H
STRATEGY								
<i>Customer satisfaction</i>	•	•	•		•		•	•
<i>Shareholder value</i>	•	•	•		•		•	•
<i>Community recovery</i>	•	•		•	•	•	•	•
CONTEXT								
<i>Weather patterns</i>	•	•	•	•	•	•	•	•
<i>Government policy</i>		•	•			•	•	•
<i>Community attitudes</i>		•					•	•
<i>Economic downturn</i>	•	•			•			•
TASK								
<i>Cycle time</i>		•	•	•	•	•	•	•
<i>Reputation</i>	•	•			•	•	•	•
<i>Leakage</i>	•	•			•	•	•	•
TECHNOLOGY								

Construct	Respondent							
<i>Claims information system</i>	•		•	•		•	•	
<i>Event module</i>	•	•	•	•	•	•	•	•
PEOPLE								
<i>Workforce capacity</i>	•	•	•	•	•	•	•	•
<i>Workforce skills</i>		•	•	•	•	•	•	•
STRUCTURE								
<i>Event leadership team</i>	•		•			•	•	•
<i>Government task force</i>		•						•
LEARNING								
<i>Post-incident review</i>	•		•		•	•	•	
<i>Event simulation</i>	•	•	•			•	•	

Table 4.4. Code frequency: context in claims handling

4.4 BETA CORPORATION (BTA)

In this section, I present the within-case findings from my study of the *passenger facilitation system* of Beta Corporation, the main airport serving Brisbane, the state capital of the Australian federal state of Queensland, and its surrounding region. The airport operator owns and operates two terminals serving international and domestic destinations. The case study took place in the passenger facilitation and security units responsible for coordinating passenger facilitation in close collaboration with the airlines. The airport processes several thousand passengers per hour departing or arriving from multiple destinations and the facilitation of passengers requires the coordination of airport staff, airline ground handlers, federal authorities, and security contractors. Passengers either check in individually or as a group and must be routed through the security, immigration, and departure stages before they board an aircraft. Furthermore, the units can be requested to provide updates on current load at the terminal. The performance of the system in terms of throughput and security impacts on airport satisfaction and rating.

Figure 4.3 illustrates the components of the system based on and extending an example by Bostrom and Heinen (1977b). Passengers departing on an outbound flight arrive at the terminal either by private drop-off, motorcab, or the public transport system. They are then received by

terminal ambassadors and routed to the check-in counter of the airline they are departing with. At this stage, the credentials of the passengers are verified and boarding passes are issued in case passengers possess valid travel documents. They are then given information pertaining to on-board security and proceed to one of several security screening points for security screening and inspection of personal belongings. Once cleared by security contractors, passengers proceed to immigration and finally departure where they can make use of "dwell time" to purchase food or duty free items. Context triggers such as weather or traffic events outside the terminal prompt the units to make rapid changes to passenger process, resources, and staffing levels to recover from a loss in throughput.

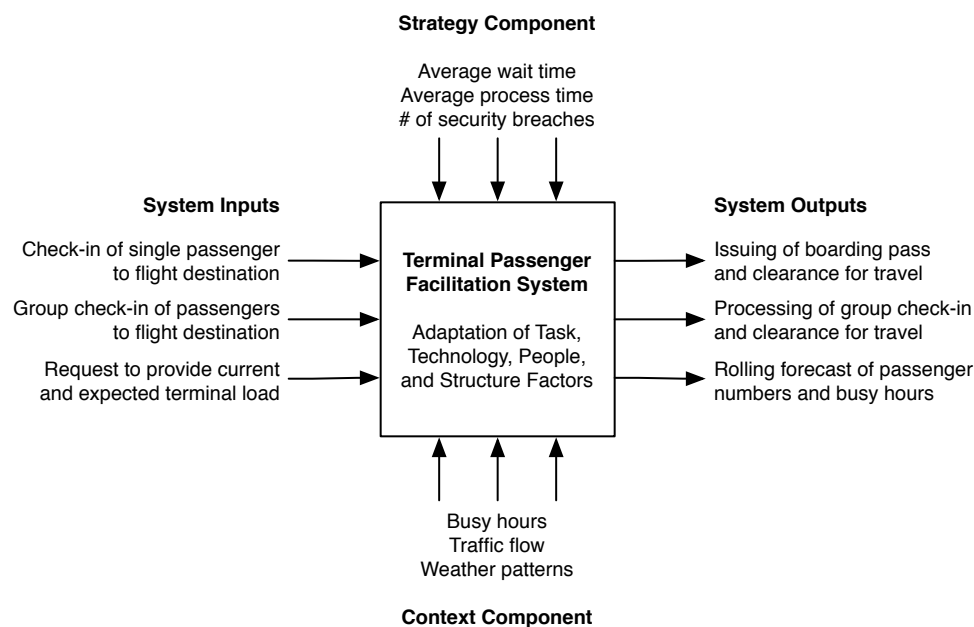


Figure 4.3. Illustration of passenger facilitation system (based on Bostrom & Heinen, 1977b)

4.4.1 STRATEGY COMPONENT

Operational efficiency. The construct operational efficiency is described by respondents as an important measure of performance and a primary outcome of the airport's passenger facilitation system. Operational efficiency refers to the efficient and effective utilisation of organisational resources such as the terminal infrastructure, equipment, information systems, and workforce used to process departing and arriving passengers. The attractiveness of the airport depends to a large extent on its ability to support airlines in turning around flights in a quick fashion. Respondent B explains: "airlines in particular look at their [turnaround], it's certainly one of their KPIs, you know, turnaround". This is why the airport has established performance agreements with airlines that define targets for work system measures such as process throughput, infra-

structure availability, and level of passenger comfort. Measures are defined in a common document ("aviation services and charges") and outline the obligations of all system stakeholders. Respondent B explains the agreements in his comment:

The simple fact is that we have performance agreements. So the agreements we have with the airlines in order to help them turn around their flights in a quick fashion. Sometimes they create problems by arriving late or arriving early. Sometimes the problems might be with us. There might be equipment breakdowns or there might be a terminal evacuation, or things like that, that will impact on that.

Shareholder return. A second outcome of the passenger facilitation system of the airport is the issue of profitability and shareholder return. The way the airport utilises its infrastructure to process passengers affects its profitability and impacts on future growth. The airport is thus constantly looking for opportunities to optimise the way it can improve processes, technology, and workforce. This might mean, for example, utilising information technology to reduce the number of "people decisions". Respondent B summarises the continuous effort of the airport to increase shareholder return in his comment: "you know, have we got the right technology, are we making the most of our technology, do we have the right people, do we make the most of our people". A second reason for optimising work systems is the cost of infrastructure expansion and limits to which the airport can invest in new infrastructure as defined by respondent B:

The other issue ... because we're providing infrastructure here, we don't want to ... one of the reasons for making everything more efficient, not only passenger awareness and that, but we don't want to build more terminal because it's so costly. We want to make a profit. So and continue to make a profit and shareholder return.

Security outcomes. A final outcome of passenger facilitation at the airport is the enforcement of security outcomes defined by the Australian aviation policy and regulation. The key objective of security outcomes is to protect airport infrastructure and the livelihood of its workforce and passengers against security threats such as terrorism. This involves security standards and procedures that affect the accreditation and background screening of workforce and the screening and inspection of passengers. Respondent B summarises the issue of security outcomes in this comment: "it's people as being part of the security, whether they are being screened or whether they are actually part of the security equation". The application and enforcement of security standards is being enforced by the Australian federal government, creating pressure on airports to achieve compliance as explained by respondent B in this comment:

What this is actually talking about is more about national consistency on the application of the security outcomes. So, for example, the security regulations are written on an outcomes-based basis. So

you do get differences between ports on how they interpret the regulations. The legislator tries to ... or the regulator tries to make it as even as possible. But really it's about ... but also that performance measure is not only the deliverable of the security outcome but it's also about the people and it's the testing standard of the people, the testing of procedures for equipment, testing procedures for people, and testing procedures for process.

4.4.2 CONTEXT COMPONENT

Busy hours. The construct busy hours relates to an operational pattern that is consistently described by all respondents. Busy hours are daily peaks during which the bulk of passengers depart or arrive at the terminal and have a significant impact on the performance and effectiveness of the work system. The daily peaks are determined by flight schedules, i.e., timetables that define the arrival and departure times of aircraft. Flight schedules are published half-yearly by federal authorities and depend on a variety of upstream factors such as demand, the geopolitical situation, and seasonality. For example, geographical factors play a role in how airlines bid for landing slots and operate aircraft at Australian airports. Respondent A explains the relationship between geographical location and busy hours at the international terminal:

Because where Australia is positioned in the world, most of your airlines want to arrive into Australia first thing in the morning. So from 6 am in the morning till ... here's an example. This is a bay plan basically for the day. From 6 am in the morning through till 10 ... So that's your peak in the morning, basically in the morning, and that's your evening peak, which is not so big.

The airport tracks and measures busy hours by extracting data from a number of information systems used by different stakeholders to process passengers or to record flight movements. The complexity of these information systems, however, prevents the airport from having access to complete and consistent data. For example, respondent C notes that the airport has to rely on an indirect measure to track when a passenger arrives at the terminal. This data is not available from the airline check-in systems but must be "simulated" by extracting baggage data from the baggage sortation system based on which an approximate number can be calculated: "there's about a 1.1 ratio of bags to passengers. So, that's a fairly good indication of the number of passengers going through check-in".

Weather patterns. A second factor that impacts on the effectiveness of the work system to handle passengers in a fast and efficient manner is the construct of weather patterns. Weather patterns relate to seasonal events such as storms in summer and fog in winter, an issue mentioned consistently by all respondents. While weather patterns do not have a direct impact on the terminal, they affect passenger flow as aircraft are delayed by whatever time the event is in effect. This is because occupational health and safety regulations require labour working out-

side the terminal in jobs such as aircraft maintenance and inspection must be withdrawn during a thunderstorm. The duration of such events can be from 30 minutes to several hours. During this time, passengers that have cleared the security screening and immigration will be confined to the terminal's departure lounge. Respondent A provides a recent example of a thunderstorm:

Everything is running along as per plan up until the weather hits. The airlines will then withdraw their labour off the tarmac. The aircraft will be delayed for whatever length of time that weather is happening. Now if it's a quick storm and it's over in 30 minutes, then the aircraft won't even be delayed for 30 minutes. If it's a storm that goes on for four hours like it did the other night, then everything will possibly be delayed for four hours. And it's all to do with Occupational Health & Safety.

Traffic flow. The construct of traffic flow refers to issues respondents ascribe to traffic arriving at the terminal via public transport, taxicab, or private drop-off. Passengers departing from the terminal are required to arrive within a certain time window to allow for them to be processed, cleared for departure, and for potential issues to be sorted out. Traffic patterns such as congestions in the road network can cause passengers to arrive late or miss flights. This affects both passenger handling as delayed passengers can hold up flights and cause security issues as they vent their frustration at ground staff. The airport is investing in retrieving more accurate traffic information. Respondent E explains: "there's always room for improvement and we look at how we can improve it. CCTV for one thing, it could certainly be improved and we're looking at that. At the moment we've got a number of systems that have been fully integrated so you know we're looking at seeing how we can improve that". Participant B describes the impact of traffic events on terminal throughput and security:

Traffic is a really good example. Traffic can have a hell of an impact on what we do here at the airport. And that can affect facilitation, people running late, missing flights, causing delays, those delays have a knock-on effects within the airline network. Even from a security perspective, those delays from a security perspective for example, people caught in traffic, they're delayed, they get to the airport, they're running late. They tend to take out their frustration on the people they meet.

Security threat. The construct security threats refers to any risk to infrastructure and livelihoods of workforce and passengers at the terminal. This area is looked after by respondent B: "it's really protecting the infrastructure, the airport infrastructure. But then also it's, you know, it's people as being part of the security". While direct risk of terrorist acts on Australian territory is perceived as relatively low by respondents, there can be flow-on effects of attacks in overseas airports on airport operations. Respondent A explains: "disasters can have affect as well. Like 9/11, when that happened, that had an effect on the business". Federal and international authori-

ties provide constant updates on security threats and define security standards and procedures that must be met by the airport. In his comment, respondent B describes the tasks and responsibilities he associates with his work in the security area as follows:

In my area, it's mainly security and some emergency. But security from a day to day basis is the key one. Now that's really the one we see on a day to day basis. But there's lots of other compliance requirements as well, because that's regulatory compliance. There's also compliance with our own service level agreements that we have with our customers and that we have with our contractors and I think that's really important to note.

Commercial events. The construct commercial events summarises events to which participants ascribe long-term, commercial impacts on the airport. Such events do not provoke immediate operational responses at a work system level but impact on passenger growth and security outcomes in the long term. Respondents show a general awareness of these factors but do not relate commercial events to immediate effects on work system outcomes. Respondent A explains that "other than weather or a critical infrastructure issue, everything is slow change". Respondents thus dedicate relatively little time and effort in the tracking of events that fall under this category. However, respondent C notes that other organisational units responsible for long-term planning and development of the airport access and analyse data concerning commercial events: "I don't personally do that but our commercial people do because they have the time to analyse the root, they do a lot of root analysis".

4.4.3 TASK COMPONENT

Throughput. The tasks performed by work system participants may need to adapted to increase or decrease the throughput of passengers through the terminal. Each task in the passenger facilitation work system is designed to handle a certain number of passengers per unit of time, i.e., to achieve a certain throughput. Operational events such as group check-ins, passengers delayed in traffic, or the breakdown of critical systems such as check-in or baggage sortation can affect that throughput and thus require the modification of tasks. The airport has little control over check-in as this is handled by the airlines. Respondent A explains: "It's controlled by your check-in process. We don't control it. It's just if the airlines don't put enough check-in agents up there". However, the airport can control security up to a certain extent. Respondent D notes: "I turn the switch up on security to high then everybody is going to be slow going through the facilitation. And high means everything is checked, every single thing will be checked to the highest level, every bag will be searched, every person will be frisk searched".

Temporary accommodation. Furthermore, the airport operator may be required to make ad hoc changes to passenger facilitation such as providing temporary accommodation to passengers in a weather event. Respondents explain that the passenger facilitation system is designed for throughput. However, in some cases passenger flow stops as the result of an external event such as a thunderstorm. In such cases, passengers that have already been cleared for travel must be temporarily accommodated in the departure lounge. Respondent A explains: "that's all to do with employees but it impacts on the passengers. And once it impacts on the passengers, it impacts on the terminal or on the facilities". However, there are limits to what the airport is required to do as respondent A notes:

You got to start providing maybe pillows and blankets, whatever you can find, and so ... But that's not the airport who has to find that, the pillows and blankets side of it, it's the airlines' responsibility. The comfort for the airport I suppose is just providing enough seating, and the food and beverage outlets, and your toilets. So the rest of that, anything else outside of that, then becomes basically the airlines.

4.4.4 TECHNOLOGY COMPONENT

Screening settings. Security threats to airport infrastructure and the livelihood of the traveling public can require swift adjustments to technology settings and security standards. Security threats generally relate to the risk of a terrorist attack on the airport infrastructure or members of the travelling public. While no major terrorist event has occurred in Australia, the threat of terrorism to international aviation remains high. This has flow-on effects on operations at the airport and security standards, procedures, and technology settings in particular. For example, the federal government may issue restrictions on the quantity or nature of goods that can be taken on board an aircraft. Respondent B provides a relevant example in his comment: "you might recall some years ago with the LAGS requirements, Liquids, Aerosols, and Gels, in Europe that was done over night. That was implemented over night, that whole process". The government may also require the airport to change the technology settings for security screening or increase the level of intervention. Respondent B describes such changes:

But in general terms if there's a risk that's been identified that is, you know, that's a risk that's going to be in Australia, we can escalate. So there's whole range of things we can do. We can increase the standard of screening, because there's different standards of screening that's set particularly from a technology perspective. So we can change the technology setting. We can increase the level of intervention with passengers.

Information displays. Information displays are used by the airport to inform passengers about departing and arriving flights but respondents suggest the airport is increasingly looking at them

as a means to communicate other information as well. This can become necessary when process changes are planned in a specific context, the airport is closed for some reason, or delays are expected in the lead-up to major public holidays. Respondent E describes the efforts undertaken by the airport to repurpose the displays in different contexts: "the other thing we're looking at is this airport operating database and it's using our flight information displays. At the moment it's very difficult to put any free text [...] on the [displays] so this new system will allow us to do that so we can get information out to, not just the public but the staff". Information about process changes and delays is important. Respondent B explains: "we want to get the person [...] into their seat by the appropriate time. We also want them here in sufficient time so that if there are problems, they can be sorted out at check-in or whatever it may be".

4.4.5 PEOPLE COMPONENT

Staffing levels. Variance in passenger numbers means the airport operator must plan and adjust staffing levels in different contexts often at the last minute. The planning of staffing levels relies on a rolling forecast of passenger numbers and busy hours extracted from several terminal information systems. Respondent C describes how passenger information is used by the various stakeholders at the airport: "certainly helps the organisation to better plan and to better understand and really to help our people but also the airlines and customs to better understand those flows [...] they do load [reports] into their resource planning tool and [...] probably along with some other data and then they, you know, it produces a roster of some sort for them". Appropriate planning and preparedness for context changes is important for avoiding bottlenecks in security screening. Respondent B explains: "we'll always one machine that's open 24/7 but at different times the people [...] we will only bring people for, say, 8 hours or four hours depending on the timeframe that they're needed for it to cope with that scalability issue".

Occupational safety. Complying with and operating within occupational safety standards in the aviation environment is a key concern to the airport. Occupational safety relates to delivering a safe work environment for staff in the landside and airside areas of the terminal and the passenger facilitation system in particular. This includes, among others, the safety of terminal facilities, aerobridges, aircraft parking bays, and taxiways. For example, the safety of workforce in the airside environment is impacted by weather events and the danger of lightning strikes. Occupational safety regulations thus require the airport and airlines to withdraw all workforce from these areas should a thunderstorm affect the airport. This effectively brings operations at the terminal to a halt and impacts on the passenger facilitation system. The airport has introduced an incident management system to track incidents reported by staff, to improve safety, and to

ensure compliance with federal regulations. Respondent A describes the impact of regulation on passenger facilitation and changes he observed over several years at the airport:

And it's all to do with Occupational Health & Safety, it's not do with anything else. Years ago, they used to just work through the weather and it didn't make any impact on the process or anything else. It's only in the last probably 10 years that due to OH&S issues, the work stops out there. If it's only raining, work keeps going on. It's only the lightning that stops everything.

4.4.6 STRUCTURE COMPONENT

Incident response team. The construct terminal incident response team relates to interpretations of ad hoc teams assembled by the terminal duty manager in order to respond to incidents at the airport in a timely and effective manner. This is the responsibility of the terminal duty manager role. Respondent E, who is the international terminal manager supervising the duty managers, explains this role: "I also provide leadership to the duty terminal managers who are here, with a seven day, twenty-four hour day to day role here at the airport". Part of the responsibility of the duty manager is to assemble incident response teams which consists of representatives from multiple entities operating at the airport such as airline ground staff, security provides, and law enforcement. In the words of respondent B: "when things go wrong, that person has to be notified, that person will pull together this management team or assessment team or probably change it to ... and you get the key people". The incidents to which these teams must respond are various, as explained by respondent E in his comment:

I mean we, we deal with a lot of incidents. Sometimes maybe once in a lifetime, others on a fairly regular basis. We've got a train that operates to the airport so it does bring in all sorts of people and because we're open twenty-four seven, some people use the terminal as a refuse so we get some mental health issues at the airport. I had a lady yesterday, an elderly lady that was, she wasn't travelling but she had no idea why she was here and you know, but she was heading out through customs and security.

Airline working group. The construct airline working group relates to interpretations of ad hoc working groups created by the airport to cooperate with airlines on bounded problems. Multiple airlines are operating from the airport and the airport is seeking to make most of its infrastructure "common use", i.e., usable by more than one airline. This includes issues such as check-in counters, check-in systems, and the software used by airlines to process passenger data and to issue boarding tickets to departing passengers. Issues that arise in the operation of such infrastructure can cause concerns and the airport may decide to create a working groups to resolve such issues. Respondent B explains the purpose of such working groups in the context of common use facilities at the terminal:

You might put a working group working with the airlines to address it ... to address that particular problem but what we're trying to do is to make everything common, what we call common user or we want to try and make all the processes for everybody as close to the same as possible. So for example, now check-in is always going to be a bit different between the airlines but, you know, that's up to the airlines what they have on their software.

4.4.7 LEARNING COMPONENT

Continuous improvement. Continuous improvement relates to efforts by the airport operator to continually improve process, technology, and people factors and to make more efficient use of its resources. Respondent B summarises the improvement program of the airport as follows: "people, technology, process is what we always refer to [...] the key issues on fixing problems". However, the organisation takes a pragmatic approach to improvement and consistent documentation of changes remains low: "Documentation isn't very good. We're very light on the ground. We do document more these days than we have in the past". Change is introduced to create internal efficiencies through, e.g., the automation of decisions. However, some areas of the passenger facilitation system are less prone to changes. In particular security screening underlies regulation by aviation security authorities. Respondent D explains: "we don't very often change things unless we absolutely have to and we have good reason to and that would come from the department of transport". Respondent A gives his view of change as follows:

One thing, remember. There's never two days the same at the airport. I've been here 37 years or 38 year that I've been here and I never had two days the same. That's what makes it so good. And you might meet new changes and you might meet new things, but there's never two days the same.

Incident simulation. Incident simulation relates to the simulation of security or safety incidents that affect operations at the terminal. For example, respondent A describes the kind of incidents for which simulations are performed and gives an account of a recent event: "well if you have a fire in that part of the terminal that is not a big incident. We did an exercise yesterday for a bomb going off on level 4, which impacted on ... it closed down level 2, 3, and 4 in the terminal. So that did have a big impact on the process in the terminal". However, the responsibility of the airport operator to respond to and coordinate security incidents is limited. Respondent D explains the consequences of an impact such as an explosion in the terminal on airport operations: "and in the event we did have a major terrorist incident, if it happened here then everything would be handed over to the police and then we'd be seeking guidance and [...] it'll go through to Canberra and then it goes under the national counter-terrorism rule anyhow and then we've lost our control". So the airport operator focusses mainly on safety issues.

4.4.8 FINDINGS

Table 4.4 summarises code frequency and emerging concepts in the data collected at Beta Corporation. Similarly to the insurer case, three issues emerge. First, competition between airports in particular for international travel requires the airport to balance profitability and passenger satisfaction. This is evidenced by interpretations of passenger flow measures such as queue time and process time on the one hand and efforts to make passenger facilitation as efficient as possible on the other hand. Second, variance in passenger volume and throughput following external events such as weather and traffic require the operator to coordinate rapid changes to passenger facilitation and to prevent flow-on effects into the wider aviation network. This is evidenced by interpretations of occupational health requirements and the impact on traffic flow on terminal operations. Third, as in the Sigma Group case, changes are not isolated to the technical system but include social variables. This is evidenced by occupational safety regulations and incident management procedures at the airport terminals.

Construct	Respondent				
<i>Supporting codes</i>	A	B	C	D	E
STRATEGY					
<i>Operational efficiency</i>	•	•	•	•	•
<i>Shareholder return</i>	•	•	•		
<i>Security outcomes</i>		•		•	
CONTEXT					
<i>Busy hours</i>	•	•	•		
<i>Weather patterns</i>	•	•	•	•	•
<i>Traffic flow</i>	•	•	•		•
<i>Security threats</i>		•		•	
<i>Commercial events</i>	•	•		•	
TASK					
<i>Throughput</i>	•	•	•	•	•
<i>Temporary accommodation</i>	•	•			
TECHNOLOGY					

Construct	Respondent				
<i>Screening settings</i>		•		•	
<i>Information displays</i>		•			•
PEOPLE					
<i>Staffing levels</i>	•	•	•		
<i>Occupational safety</i>	•	•			•
STRUCTURE COMPONENT					
<i>Incident response team</i>	•	•	•	•	•
<i>Airline working group</i>	•	•			
LEARNING					
<i>Continuous improvement</i>		•		•	•
<i>Incident simulation</i>	•	•			•

Table 4.5. Code frequency: context in passenger facilitation

4.5 DISCUSSION

In this chapter, I viewed and explained the *context-driven adaptation* of work systems through the lens of socio-technical theory (Bostrom & Heinen, 1977a). The theory of socio-technical systems stresses the importance of adaptation of work systems to organisational objectives such as value creation (as evidenced by views of the concept "strategy" in the data) and the requirement of joint optimisation of the technical and social system of an organisation (evidenced by interpretations of modifications to task, technology, structure, and people variables in the data). I furthermore argued that variables in the context of the work systems impact on the ability of work system participants to obtain work objectives and that the modification of work system variables has the objective of recovering some of the loss in performance experienced as a result of context change. For example, this is evidenced by interpretations of variance in inbound call handling and the fulfilment of insurance claims in a weather context. Similarly, passenger handling performance at the airport varies as a result of external factors such as traffic.

However, socio-technical theory is *but one view of the use of information technology* in an organisational context (Kling, 1980) and the literature encourages the researcher to actively seek alternative explanations (Gibbert & Ruigrok, 2010). One such explanation is the rationalist view of technology as an instrument that is used by its users to rationalise information processing

tasks and thereby to create internal efficiencies (Markus, 1983). Interpretations of IT as an instrument have been discussed in Chapter 03 (interpretations of IS as tools) and evidence for this view can be found in the responses provided by some respondents in the case data. In the Sigma Group case, respondent H points to the importance of technology in supporting claims handlers in identifying opportunistic fraud and thereby in reducing the potential for leakage. Similarly, respondent B in the Beta Corporation case points to IT as an opportunity for rationalising information processing and to make airport processes more efficient. These interpretations focus on the role of IT as an instrument deployed to rationalise otherwise manual, human tasks.

The role of IT as an instrument deployed to rationalise information processing tasks in the work system is evidenced by the following comment. Respondent H is the executive manager for claims experience and in this role supervises all efforts made by the insurer in the areas claims process and customer satisfaction:

"Now, in the fraud area we have got quite good technology in there to be able voice record, tests for timbre in their voices and that type of thing so, you, there are a whole number of ways that you can start to harden and try to cut", **Respondent H, Sigma Group**

Similar, the following comment of respondent B, the responsible manager for security strategy at the airport, points to views of IT as an instrument to rationalise human information processing or "people decisions". Using technology is viewed as contributing toward making processes more speedy and efficient:

"We might be able to use technology to take some of the people decisions out of the equation. We might be able to increase the level of customer service with some of our people. We might be able to improve processes between technology [...] to hasten and speed it up so it's more efficient", **Respondent B, Beta Corporation**

While rationalist views of technology account for the ability of technology to contribute to profitability, they do not account for issues that arise at the interface of the technical and social systems of an organisation (Bostrom & Heinen, 1977b). Such issues can include resistance to change if technology is conceptualised as the sole driver for work system optimisation (Markus, 1983). Similarly, the context-driven adaptation of work systems extends beyond the technical system perspective and must include changes to the social system such as task authority, flexibility, and training. This is evidenced by interpretations of the role of people and the need for flexibility in a dynamic context. In the Sigma Group case, respondent H suggests "staff have amazing power". The ability of the insurer to handle variance in call volume and claims complexity does not only depend on technology but also on the motivation of people. Similarly, respon-

dents in the Beta Corporation case speak fondly of the aviation environment. For example, respondent B suggests people "step up to the plate" and take on tasks as the need arises.

The following comment by respondent B, the executive manager for the claims department, reflects the need to balance organisational objectives and people requirements. The insurer relies on its staff as the point of contact with customers and knows employee engagement impacts on customer satisfaction:

"Making sure that you're running a not excessively lean business, but a smart business. I think you must have good people management capabilities because this is a people driven business and if you were Attila the Hun you'd lose the hearts and minds of people", **Respondent B, Sigma Group**

Flexibility and the importance of people is also recognised by respondent B, the manager for security strategy at the airport, in the following comment. Aviation is a highly dynamic environment and respondents stress on multiple occasions that flexibility is paramount in dealing with uncertainty in day to day terminal operations:

"It's a very tight resourced, tightly resourced organisation. So people do step up to the plate and do any other supporting activities and so forth. But it is very tight. The public service will have a lot more latitude [...] but in the aviation environment, you don't have any of that", **Respondent B, Beta Corporation**

Socio-technical theory can thus be considered as a *more appropriate frame for theorising* about actions of both case sites as it considers both the importance of value creation and the need for joint optimisation of their technical and social systems (Mumford, 2000), an element that is missing from rationalist views. The joint optimisation of technical and social systems in context-driven adaptation of work systems emerges as a strong pattern in the case data collected at the two sites. Modifications to the technical system such as the adaptation of tasks or changes made to technology in different work contexts enable both organisations to rapidly respond to variance (in call and claims volume, in passenger volume and throughput) and to recover from a loss in efficiency. Modifications to structure and people variables ensure they are informed and have the necessary authority and training to implement and enact these changes. Going forward, more work is required to understand whether views of IS as work systems (Alter, 2001) and methods for their analysis and design (Alter, 2006a) reflect these findings.

4.6 CONCLUSION

In concluding, I observe a need for further *theory development* and *extension of work system theory* toward context. In Chapter 3, I identified a increasing call for socio-technical or "human-centred" (McKay et al., 2012) design in IS. Human-centred design views IS as work systems in

which people perform work by using information, technology, and other resources to produce and deliver products or services to customers (Alter, 2006b). However, discussions of work systems (Bostrom & Heinen, 1977a) and work system theory (Alter, 2001; 2008) in the literature fail to conceptualise context. In light of the relevance of context for the analysis and design of work systems (as evidenced by the examples drawn from case data collected in the multiple-case study), this failure leads to significant deficiencies in the utility the theory provides to work system designers (including managers, system analysts, and staff). The emergent concepts developed in the case study promise to provide a foundation on which the deficiencies can be closed by applying guidelines for theory development in IS (Weber, 2012).

Chapter 5: Theory Development

5.1 INTRODUCTION

In this section, I develop *design propositions* from the analysis of empirical evidence in Chapter 04 and the selection of kernel theories in Chapter 03. In Chapter 04, I studied two work systems in a multiple-case study of organisations in different industries. The commonality of these cases is their requirement of context-driven adaptation of work system variables to changes in weather, traffic flow, or equipment failure. The analysis of empirical evidence led me to conclude that each case can be conceptualised as a socio-technical system. Furthermore, I observed that rapid adaptation of the technical and social component parts requires system members to continually monitor relevant context variables and acquire context-related information. In Chapter 03, I reviewed the academic literature and identified conceptualisations of adaptive systems (Ackoff, 1971) and socio-technical theory (Bostrom & Heinen, 1977a) as conceptual foundations of the context-driven adaptation of IS. I then develop design propositions from theory and data that form the basis of the prescriptions presented in Chapter 06.

Furthermore, I map each proposition derived from kernel theories of adaptive systems (Ackoff, 1973) to an *exemplary method of socio-technical analysis* in the IS discipline (Alter, 2006; 2008; 2009). Socio-technical analysis and design has a long tradition in the IS discipline (Bostrom & Heinen, 1977b; 1977a). Furthermore, socio-technical theory has been applied as theoretical lens for analysing information systems (for a recent example cf. Seidel, Recker, & Brocke, 2013). I turn to a contemporary application of socio-technical analysis to the IS domains, the Work System Method (WSM), as reference of our current understanding of work systems in IS. My objective in selecting WSM is to understand the degree to which current IS design-relevant theory (Kuechler & Vaishnavi, 2012) explains context and context-driven adaptation of information systems. To this end, I develop a set of mapping guidelines by drawing from existing work on representation theory and analysis (Weber, 2012). This allows me to define criteria for measuring completeness and clarity of IS design-relevant theory with respect to kernel theory.

In a final step, I relate design propositions to empirical evidence to measure their relevance in the data I collected across the two case sites studied in Chapter 04. In Chapter 03, I selected Ackoff's systems classification and theory of adaptive systems (1971) as relevant justificatory knowledge for the design of context-aware information systems. However, as with many theories that classify as kernel theory, Ackoff's work remains on a level of abstraction that must be

considered insufficient for the development of applicable system prescriptions (Kuechler & Vaishnavi, 2012). It is therefore necessary to "ground" design propositions derived from kernel theory in empirical evidence in order to assess their merit in the design and construction of information systems. I therefore map each of the design propositions developed in the previous step to occurrences in the interview data collected in Chapter 04. The objective is not to re-evaluate findings generated in Chapter 04, but to find empirical evidence for a design construct and its relevance in the interpretations provided by respondents in the case study.

5.2 RESEARCH APPROACH

In this section, I describe the *criteria* for IS design theory development used in the remainder and demonstrate their *application* to the problem of explaining the nature of context and adaptation. First, IS design theorising must ground in sound academic kernel theory and present justificatory knowledge (Gregor & Jones, 2007). I turn to related work on theory evaluation in general (Weber, 2012) to define concrete criteria for IS design theory evaluation in particular. Theory evaluation has the objective of ascertaining the quality of behavioural theory. I apply and refine these criteria to reason about the quality of contemporary IS design theory with respect to context and adaptation. Second, I demonstrate how I applied these criteria in the evaluation of work system theory (Alter, 2009), a method for the analysis, design, and implementation of IT-enabled work systems. I turn to system classification and the system of systems concept (Ackoff, 1971) as kernel theory for the explanation of open adaptive systems. Finally, I show how prescriptions about normative system outcomes can be generated in three basic steps.

5.2.1 MAPPING CRITERIA

Weber (2012) argues that high-quality IS theory is a) complete with respect to its phenomenon of interest and b) parsimonious, i.e., uses a very small number of constructs to reason about its focal phenomenon. In the following, I argue that these properties equally apply to design theory in IS (Kuechler & Vaishnavi, 2012) and define a set of criteria according to which the quality of design theory with respect to its underlying kernel theory can be measured and explained.

Before introducing and defining each mapping criterion and its relevance for design theorising, I define the concepts of theoretical completeness and clarity of design theory based on Weber's work. This definition is necessary to apply the generic notion in Weber's discussion (the level of IS theory in general) to the specific domain of application in IS design theorising, i.e., design-relevant explanatory theory as motivated by Baskerville and Pries-Heje (2010) and defined by

Kuechler and Vaishnavi (2012). Following this introduction, the definition of theoretical completeness and clarity is as follows:

- *Theoretical completeness* is a desirable feature of design-relevant explanatory theory developed as part of a DSRIS study and measures the degree to which each construct of a kernel theory on which the study relies is reflected in a design proposition of the design-relevant explanatory theory.
- *Theoretical clarity* is a desirable feature of design-relevant explanatory theory developed as part of a DSRIS study and measures the degree to which the design propositions of the study unambiguously interpret and apply kernel theory constructs to the domain of application.

Theoretical completeness and clarity form ideals of design-relevant explanatory theory in DSRIS that must be evaluated in practice to assess the quality of such theory. It is therefore necessary to identify situations in which both theoretical completeness and clarity are violated. Based on such an understanding, scholars can evaluate the quality of decisions made during design theorising and identify potential deficiencies in design-relevant explanatory theory with respect to the underlying kernel theory. In the following, I introduce and define five such situations and provide recommendations how they are to be mitigated during design. I relate each mapping criterion to Weber's general discussion of evaluating the quality of IS theory and, where necessary, provide examples for design theorising.

First, design theory deficiencies that diminish theoretical completeness include construct deficit. Weber defines construct deficit as a situation in which not all of the aspects of a phenomenon are covered by matching theoretical constructs. Real-world problems can be characterised by a set of entities that Weber refers to as ontology (2012). Theoretical constructs are representations of such problems in a theoretical system (including design-relevant explanatory theory in IS). Following this definition, construct deficit is a situation in which an information system design problem (phenomenon) perceived by a system designer has no correspondence in the propositions offered by a design theory. This represents a problem as the application of design propositions in system design and construction is necessary incomplete. The IS design theory can therefore not be regarded as a complete representation of its underlying kernel theory. For example, design theories for socio-technical design have a construct deficit if they ignore one of the four variables of socio-technical systems (Bostrom & Heinen, 1977a).

Second, design theory deficits that diminish theoretical clarity include *construct overload*, *construct redundancy*, and *construct excess*. Weber suggests that a scope for ambiguity and mis-

representation in theory can arise if a) two separate phenomena are described by the same construct (i.e., construct overload), b) the same phenomenon is described by more than one construct (i.e., construct redundancy), or c) there exists a construct for which there is no phenomenon in the real-world (i.e., construct excess). Following this definition, construct overload is a situation in which a particular design proposition maps to two or more constructs of the kernel theory on which the design theory relies. Second, construct redundancy is a situation in which two or more design propositions map to the same kernel theory construct. Finally, construct excess is a situation in which there is no theoretical precedence for a design proposition. All three situations therefore lead to a lack in theoretical clarity with respect to instantiating the design propositions within the boundary described by the underlying kernel theory.

Having defined theoretical completeness and clarity of design propositions and deficits that diminish these properties, I now proceed to identify and define five patterns of theory-driven design which improve the completeness and clarity of new and/or existing design theories as formulated by a DSRIS study,

Construct mapping. Construct mapping maps a single kernel-theory construct (phenomenon) to exactly one design-theory construct (design solution). Kernel-theory constructs are the relevant constructs defined by the kernel theory underlying a information system design theory (Gregor & Jones, 2007). Each kernel-theory construct describes a pertinent information system design problem that must be resolved by the design theory. For example, the theory of socio-technical systems (Bostrom & Heinen, 1977a) defines four variables that must be considered to improve the performance and adaptiveness of systems that combine social and technical aspects of work. Mapping each of the four variables — people, structure, tasks, and technology — to an information system design theory (Alter, 2009) ensures instantiations of the design theory in artefacts lead to a complete representation of all phenomena defined by socio-technical theory. The resulting design theory can be considered to have a high degree of theoretical completeness as defined above.

Construct extension. Construct extension introduces a new design-theory construct based on a kernel-theory construct if it is missing from the design theory. A missing representation of a kernel-theory construct in a design theory implies artefacts constructed from the design theory are incomplete and the resulting situation is an instance of construct deficit as defined earlier. Extending the design theory by a new design-theory construct that is an effective and economical solution to the phenomenon defined by the missing kernel-theory construct thus removes the situation of construct deficit. For example, an information system design theory (Alter, 2009) that does not define the attributes of people, the relationships among them, reward

mechanisms, and authority structures (Bostrom & Heinen, 1977a) must be regarded as an incomplete representation of socio-technical theory. Adding design-theory constructs for each missing kernel-theory construct improves completeness and increases the probability of successful artefact instantiations.

Construct specialisation. Construct specialisation splits one design-theory construct into two or more specialised construct if it maps to two or more of the construct of the underlying kernel theory. Situations in which one design-theory construct is overloaded with two or more kernel-theory constructs decreases the clarity of the design-theory construct and leads to construct overload. For example, the construct "automated agents" introduced by Alter (2009) in his information system design theory is defined as a participant in a work system that is automated, i.e., a non-human actor. This leads of an overloading of the participant construct in the design theory with two kernel-theory constructs: people and technology. Removing the automated agent construct while retaining "technology entity" reduces unclarity concerning the participants of information-based work systems.

Construct reduction. Construct reduction removes one or more redundant design-theory constructs from a design theory if two or more design-theory constructs map to a single kernel-theory construct. Two or more design-theory constructs mapping to a single kernel-theory construct is a sign of construct redundancy and diminishes the clarity of an information system design theory. Removing redundant constructs from the design theory thus increase clarity and improves the probability of successful artefact instantiation. For example, the work of Alter (2009) on the socio-technical design of information systems introduces two constructs — customer participant and non-customer participant — for a single construct (people) in socio-technical theory. This leads to unclarity pertaining to the attributes of work system participants and customer participants in work systems and their impact on overall system performance. Removing one construct (customer participant) reduces unclarity and improves applicability of the theory.

Construct elimination. Construct elimination removes a design-theory construct if it maps to none of the constructs of the underlying kernel theory. A design-theory construct that maps to none of the constructs of a kernel theory is a sign of construct excess. In other terms, the design-theory construct is not necessary to explain a phenomenon of interest described by its underlying kernel theory. For example, in his work on information-based work systems, Alter introduces the construct "infrastructure" to describe the human, information, and technical resources provided by an organisation that are shared among its work systems. However, from the point of view of socio-technical theory, infrastructure is not a construct that is necessary to

view and explain (superior performance and adaptiveness of) socio-technical systems. Eliminating the infrastructure from the design theory as defined by Alter thereby would reduce the unclarity pertaining to the use of infrastructure to explain and improve the behaviour of socio-technical systems.

Table 5.1 illustrates the five patterns of theory-driven design and contrasts them with the notions defined by Weber for evaluating the quality of theory in IS research. A graphical illustration is provided that indicates the direction of analysis (representation vs. interpretation of a theory) and shows the steps taken to improve theoretical completeness and clarity through theory-driven design. A circle represents a construct in either the kernel theory or design theory. A solid line represents an existing mapping between constructs in the kernel and design theory. A dashed line represents an altered mapping between kernel and design theory as per the rules defined above. Finally, a filled circle represents a new construct whereas one that is crossed is deleted.

Ontological analysis	Theory-driven design	Illustration
Construct completeness. <i>One theoretical construct maps to exactly one artefact construct.</i>	Construct mapping. <i>Maps the theoretical construct to exactly one artefact construct.</i>	
Construct deficit. <i>A theoretical construct exists for which there is no matching artefact construct.</i>	Construct extension. <i>Introduces a new artefact construct based on the theoretical construct.</i>	
Construct overload. <i>More than one theoretical construct maps to one artefact construct.</i>	Construct specialisation. <i>Splits the artefact construct into two or more specialised constructs.</i>	
Construct redundancy. <i>One theoretical construct maps to two or more artefact construct.</i>	Construct reduction. <i>Retains one artefact construct and removes redundant constructs.</i>	

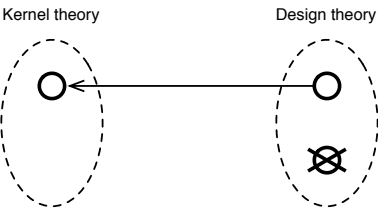
Ontological analysis	Theory-driven design	Illustration
Construct excess. <i>An artefact construct exists for which there is no theoretical construct.</i>	Construct elimination. <i>Removes the excess artefact construct.</i>	

Table 5.1. Five patterns of theory-driven design

5.2.2 THEORY SELECTION

Based on the outcome of Chapter 04, I now turn to the selection of appropriate *conceptual lenses* to reason about and explain the context-driven adaptation of work systems. In Chapter 04, I identified the emerging categories of strategy, context, adaptation (of task, technology, people, and structure variables), and learning from the empirical analysis of data. Furthermore, I stated that each emerging category can be subdivided into the concepts customer focus, cost focus, predictable context, unpredictable context, planned adaptation, ad hoc adaptation, incremental, and disruptive context learning. In this section, I now turn to the selection of kernel theories based on which the categories and interactions among categories can be viewed and explained. I identified and reviewed four such kernel theories in a review of the foundational IS literature (Benbasat & Weber, 1996). IS research is a diverse discipline grounding in multiple reference theories and research fields. These include organisational, managerial, and system-theoretical frames (Taylor, Dillon, & Van Wingen, 2010) from which the four kernel theories were sampled.

Each kernel theory is briefly introduced below and matched against the four emerging categories identified in Chapter 04. First, I summarise the key premises of the theory. Second, I discuss whether the theory has been used as a conceptual frame in the analysis of information systems. I then evaluate expressiveness of the theory with respect to the four categories. Finally, I summarise my findings and make recommendations regarding theory application.

System of Systems Concept (SSC). The System of Systems Concept (SSC) is a classification that generalises the behaviour of abstract and concrete systems into four distinct classes. It is based on the work of the system thinker Ackoff (1971) and distinguishes among state-maintaining, goal seeking, multi-goal or purposive systems, and purposeful systems. Systems thinking is one of the foundational theories of the IS discipline (Taylor et al., 2010) and its concepts have been applied to the study of information systems (Ackoff, 1968). Goal-orientation is explained by SSC on the basis of purposefulness in system behaviour, one of the core attrib-

utes of artificial (as opposed to natural) systems. Context is explained by SSC as the state of the environment or system at a point in time and the consequence of state transitions in the environment on the state of the system (i.e., events). Finally, SSC explains adaptation and learning as the ability of purposeful systems to modify their behaviour in response to environmental events and to improve the efficiency of system behaviour under constant conditions.

Socio-Technical Systems (STS). Socio-Technical Systems (STS) describes organisational work systems as characterised by the interdependence of two systems: the technical and social. It grounds in the tradition of socio-technical analysis (Mumford, 2000) and design (Clegg, 2000). STS defines two systems composed of four variables including tasks, technology, people, and structure. The use of STS in IS research has a long tradition (Bostrom & Heinen, 1977a) that continues to impact on system analysis and design today (Alter, 2010). Goal-orientation is an implicit attribute in STS and is characterised by the struggle to balance economic and social outcomes of systems (Mumford, 2000). Although STS makes implicit references to the system environment, it does not discuss context and the impact of context on system performance. Furthermore, adaptation in STS is defined as the joint optimisation of both the technical and social systems to overcome the problem of resistance (Markus, 1983). Finally, STS defines learning as the ability of system members to improve system outcomes by creating shared plans.

Complex Adaptive Systems (CAS). Complex-Adaptive Systems (CAS) describe the phenomena of emergence and self-organisation in complex systems on the basis of the local adaptive behaviour of agents. CAS has garnered much attention in recent years to reason about information systems (Merali, 2006) and how information systems are designed (Nan, 2011). CAS posits that complex systems are characterised by networks of agents that exchange information and learn by modifying the cognitive schemata underlying their actions (Anderson, 1999). CAS does not describe goal-orientation. The outcomes at the level of systems are entirely emergent and arise from the local adaptive behaviour of agents (which, however, direct their actions according to a utility function). Context in CAS is the local information gleaned by agents from their connections with other agents in the network. This information is processed by agents according to mental maps based on which they take or delay actions. Agents learn by constantly evolving the maps in response to local information and the outcome of their actions.

Dynamic Capabilities (DC). Finally, the Dynamic Capabilities (DC) framework describes the processes by which firms build internal and external competencies to sustain their advantage in a dynamically shifting market. DC grounds in the Resource-Based View of the firm (Barney, 1991) and has found some applications in the context of IS research (Wade & Hulland, 2004). Both DC and RBV posit that the competitive advantage of firms is defined by whether they

possess and sustain a set of unique resources and capabilities. DC then describes the processes by which firms renew these resources in a dynamic market. Goal-orientation in DC is defined as creating and sustaining a competitive advantage. However, DC does not describe firm context other than referring to what Eisenhardt and Martin call a "dynamically shifting market" (2000). Adaptation in DC emerges from the constant pressure on firms in such markets to integrate, reconfigure, and release resources. Learning in DC is described by equifinality and path dependence, i.e., the observation that similar conditions lead to different outcomes.

In concluding, Table 5.2 summarises and compares the expressive power of each of the kernel theories in terms of the conceptual categories defined in Chapter 04. SSC has the most expressive power when compared to the four kernel theories selected. It supports all four categories and is therefore most suited to reason about context-driven process adaptation. STS is more suited to reason about information systems as it describes four discrete variables of which all technology-enabled social systems are composed. However, it lacks sufficient expressive power to reason about system context and the reasons for system adaptation. Furthermore, CAS describes the adaptive behaviour of (artificial) systems and includes notions such as context and learning. The theory does, however, not define goal-orientation but views system behaviour as emergent from the local adaptive behaviour of agents. System adaptation only partially meets the evaluation criteria as it is not planned but emergent. Finally, DC lacks expressive power to reason about context and operates on firm level.

Theory	RQ1		RQ2 & RQ3	
	Strategy	Context	Adaptation	Learning
System of Systems (SSC)	•	•	•	•
Socio-Technical Systems (STS)	•	–	•	•
Complex Adaptive Systems (CAS)	–	•	(•)	•
Dynamic Capabilities (DC)	•	–	•	•
Legend	Explanation			
•	Evaluation criteria met			
(•)	Evaluation criteria partially met			
–	Evaluation criteria not met			

Table 5.2. Evaluation of kernel theories

For the sake of completeness, Table 5.3 summarises the concepts of SSC and organises them into four distinct clusters based on Ackoff's (Ackoff, 1971) original text. First, the system cluster describe the types and structure of systems found natural and artificial systems. Concepts such as system and environment are defined and their interaction is defined in the basis of state, duration, and events. Second, system change describes the action taken by systems in response to changes in the environment and Ackoff defines two conditions (necessary, sufficient) based on which system behaviour can be classified. This leads into a discussion of different types of systems in the behavioural classification cluster in which concepts such as process, outcome, goals, objectives, and ideals are introduced and defined. Finally, SSC discusses the processes underlying system adaptation and learning on the basis of function, the efficiency of systems in obtaining outcomes, and the ability to learn by acting purposefully, i.e., by improving their adaptive behaviour under constant conditions.

SSC concept cluster	Definition of concept (based on Ackoff, 1971)
SSC concept	
SYSTEM	
System	Set of elements and relationships connecting each member in the set
Abstract system	A system all of whose elements are concepts
Concrete system	A system, some of the elements of which are objects
State of system	Relevant properties of a system (and its elements) at a moment in time
Environment	Set of elements which are not part of the system but to which it relates
State of environment	Relevant properties of environmental elements at a moment in time
Closed system	A system which does not have an environment (and is self-contained)
Open system	A system which interacts with elements in its environment
Event	A change in the properties of a system/environment of known duration
Static system	A system which does not change and on which no events occur
Dynamic system	A system whose state does change over time and on which events occur
Homeostatic system	A static systems whose elements and environment are dynamic
SYSTEM CHANGE	
Reaction	A system event for which another (internal/external) event is sufficient

SSC concept cluster	Definition of concept (based on Ackoff, 1971)
Response	A system event for which another event is necessary but not sufficient
Act	A system event for which no other event is necessary or sufficient
Behaviour	A system event which is necessary and sufficient for another event
BEHAVIOURAL CLASS	
State-maintaining system	A system which reacts to different events in a deterministic way
Goal-seeking system	A system which has memory and can respond differently to an event
Process	A sequence of behaviour which has a goal-producing function
Multi-goal-seeking system	A goal-seeking system which seeks different goals in different states
Purposive system	A multi-goal-seeking system the goals of which have a common purpose
Purposeful system	A system that can change goals under constant conditions (has choice)
Relative value of outcome	Probability a system will produce a specific outcome in a specific state
Goal	Preferred outcome of a system that can be obtained in the near term
Objective	Preferred outcome of a system that can be obtained in the long term
Ideal	Preferred outcome of a system that cannot be obtained in practice
Ideal-seeking system	A purposeful system which upon obtaining its goals seeks new ones
ADAPTATION/LEARNING	
Function	Productions of the outcomes that define system goals and objectives
Efficiency	Probability with which a system selects actions which produce a goal
Adaptive system	A system which modifies itself or the environment if either has changed
Other-other adaptation	Adaptation of the environment to an environmental change
Other-self adaptation	Adaptation of the system to an environmental change
Self-other adaptation	Adaptation of the environment to a system-internal change
Self-self adaptation	Adaptation of the system to a system-internal change
Learning	Ability of a system to increase its efficiency under constant conditions

Table 5.3. System of systems concept (based on Ackoff, 1971)

5.2.3 MAPPING METHOD

I now turn to the work of Wand and Weber (1995) and Weber (2012) on representation theory and analysis as a framework for operationalising theory development in my study. Representation theory seeks to evaluate the representational quality of IS theory (Weber, 2012) and IS grammars (Weber, 1997) in terms of a common ontology of information systems. Two analytical steps can be identified. First, representation mapping maps ontological constructs to theory or IS grammar to evaluate the completeness with which they are represented. I use representation mapping in the following to evaluate the completeness with which SSC constructs (ontology of open adaptive systems) are represented in WSM (a specific example of IS grammar). Second, interpretation mapping views and interprets theory or IS grammar in light of ontological constructs to evaluate the clarity with which they are represented. I use interpretation mapping in the following to identify deficiencies in WSM from the perspective of SSC. Finally, I use empirical evidence to develop a set of improved propositions for IS design.

I select the Work System Method (WSM) and metamodel (Alter, 2009) as an IS grammar that helps organisations in understanding, analysing, and designing socio-technical systems and therefore is applicable to context-driven process adaptation. In Chapter 04, I defined that context-driven process adaptation affects the technical and social variables of information systems. Rapid adaptation of processes to context change requires the modification of technical variables such as tasks and technology. Based on the interpretations provided to me by respondents, I concluded that such modifications affect the state of the social system (overwork, communication issues, fatigue) and that context change requires the joint modification of variables in both systems (i.e., technical and social). Work system theory provides a conceptual frame through which such interdependencies can be viewed and explained (Alter, 2008). By the same token, it provides support for IS design and analysis based on a common model of IS and work system constructs (Alter, 2009). This serves as the foundation for my own work.

In the first step of my analysis, I map the 32 constructs defined in Ackoff's (1971) original text to constructs in work system theory (Alter, 2009). The mapping process unfolds in three steps. First, I use focussed reading of Ackoff's original text to familiarise myself with the definition and meaning of each of the constructs in the system of systems ontology. For example, I familiarise myself with the definition of system and deepen my understanding by reading additional sources (Ackoff, 1973). Second, I extend my literature review to identify and compare definitions of system constructs in the work of Alter. This draws from two different sources. On the one hand, the conceptual work by Alter (2008) on positioning work system theory serves as my foundation for viewing information systems as information-based work systems. On the other

hand, I use the conceptual modelling work by Alter (2009; 2010) to clarify and refine the definition of constructs in work system theory. For example, I map the construct of 'environment' to three distinct interpretations of environment in WSM (work system/organisational/enterprise environment). Finally, I synthesise my findings into a set of proposition following Section 5.2.

In the second step of my analysis, I interpret the 32 constructs in WSM in light of the propositions made by Ackoff (1971). The purpose of the step is to clarify their meaning and identify deficiencies in the definition of work system constructs in Alter's work that affect theoretical clarity. This unfolds in three stages. First, I familiarise myself with the definition and meaning of key constructs in Alter's (2008; 2009; 2010) work. For example, I deepen my understanding of work systems from both a conceptual level of analysis as well as from the detailed level of work system decomposition and cardinalities. Second, I interpret the definitions in light of the constructs discussed by Ackoff to identify whether they are consistent and clear. For example, I identify that the definition of three types of environment in WSM (work system/organisational/enterprise environment) leads to ambiguities and that they must therefore be regarded as redundant. Third, I synthesise my findings in a set of 8 deficiencies of work system theory with respect to systems thinking. The issues pertain to the four types of deficiencies to completeness and clarity discussed in Section 5.2.1. I conclude by discussing their impact on CAIS.

The final step of my analysis is concerned with measuring the relevance of system constructs and limitations in work system theory in case data collected in an exploratory case study (cf. Chapter 4). The purpose of this step is to identify differences and similarities in how these constructs are interpreted across the case sites. The analysis unfolds in three steps. First, I familiarise myself with the coded data and findings produced as an outcome of the exploratory case data. For example, I familiarise myself with interpretations of environmental elements and work system activities provided to me by respondents in the case interviews. Second, I map interpretations in the case data to the 10 essential constructs of context-aware information systems identified in Section 5.4. For example, I map interpretations of environmental elements to the construct "environment" and discuss similarities and differences that can be observed in different case sites. Finally, I extract revelatory quotes from the case data substantiating these similarities and differences by using the coded data I produced as a result of the qualitative analysis presented in Chapter 4. This allows me to trace system constructs, their limitations in work system theory, and measurement in case data.

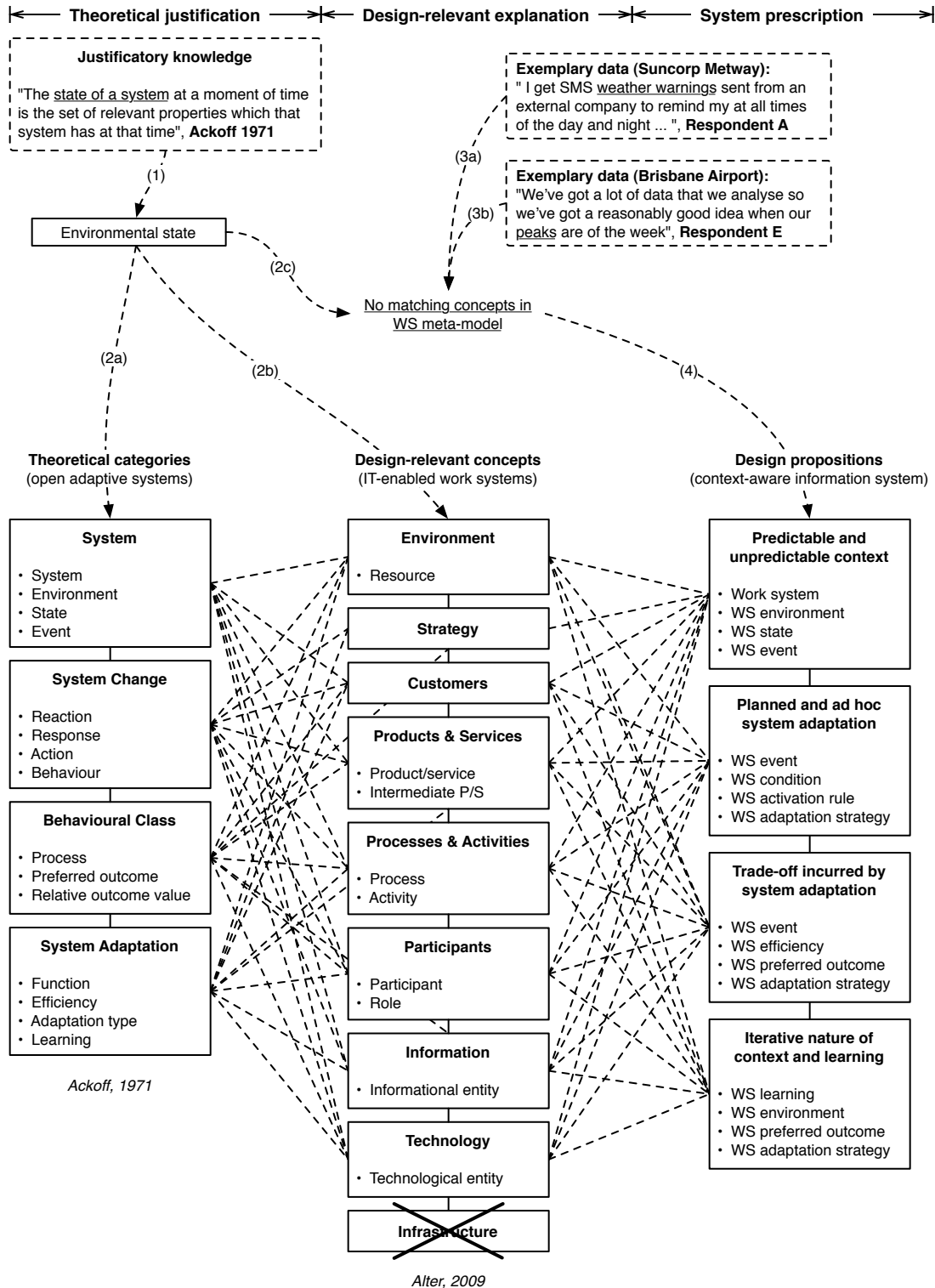


Figure 5.1. Mapping process

Figure 5.1 illustrates the different stages and concerns of mapping justificatory knowledge, to design-relevant explanations, to the definition of design propositions. First, I identify and define

relevant theoretical categories in Ackoff's original text. I refine the textual description into concepts that can be mapped to the domain of application (*arrow 1*). Second, I view and interpret these concepts (*arrow 2a*) in the IS domain by comparing them to conceptualisations of information systems (*arrow 2b*). I then highlight each instance of a mapping criterion as defined earlier. For example, I highlight the state of the environment as an instance of construct deficit in WSM (*arrow 2c*) that must be rectified. Third, I draw from empirical data to understand the implications of theoretical categories in the domain of application and map, extend, specialise, reduce, or eliminate concepts as defined by the mapping criteria. For example, environmental state is an instance of concept extension (*arrow 4*). Finally, I discuss specific issues to be addressed by context-driven adaptation of processes.

5.3 MAPPING ADAPTIVE SYSTEM CONCEPTS

This section maps each construct cluster and relevant construct in system theory to work system theory to define the theoretical foundation of context-aware information systems (CAIS). I begin by clustering the 32 constructs defined by Ackoff (1971) to describe the properties and behaviour of systems. Four such clusters can be identified based on Ackoff's original text: system, system change, behavioural classification, adaptation and learning. I then assign each construct to its cluster (following the structure in Ackoff's text) and search constructs in work system theory that correspond to its meaning. This search process is guided by a focussed reading of Ackoff's text and Alter's work on a conceptual model of work systems (2009). Where necessary, I include other sources (Ackoff, 1973; Alter, 2006; 2008) to clarify the meaning of constructs. Three situations can be identified. First, a system-theory construct maps to a work system construct. Second, a system-theory construct does not map to any of the work system constructs. Third, a system-theory construct maps to more than one work system construct (and vice versa). Finally, I summarise the findings of the mapping in 21 proposition.

The result of this mapping is a subset of 10 system-theory constructs and their representation in work system theory that inform CAIS design. These constructs are as follows. Environment and environmental elements define relevant elements such as customers, suppliers, and the larger whole of which an information system is a part. The structural state is the set of relevant properties of elements in the system environment and of the system itself at a particular moment. Internal and external events are transitions in the state of the system and/or the environment which affect the ability of the system to obtain goals. The relative value of system outcomes, goals, and objectives are measures that define preferred outcomes of systems and guide system behaviour toward their attainment. Finally, efficiency, adaptation, and learning are

capabilities of adaptive systems that distinguish them from other types of systems. An adaptive system has the ability "to modify itself or its environment when either has changed to the system's disadvantage" (Ackoff, 1971, p. 668). The system modifies its properties or those of the environment to regain its ability to perform the function specified by its goals. Taken together, these 10 constructs form the foundation of CAIS.

5.3.1 SYSTEM

System. Ackoff defines a systems as "a set of interrelated elements [...] an entity which is composed of at least two elements and a relation that holds between each of its elements and at least one other element in the set" (1971, p. 662). Each element in the set is directly or indirectly connected to every other element. For example, the elements of the human body are connected by way of the vegetative system. Furthermore, systems can be decomposed into one or more subsystems such that "no subset of elements [in a system] is unrelated to any other subset [in the same system]" (1971, p. 662). The different subsystems of the body such as vision, motion, etc. do not exist in isolation but are interconnected. The discussion of systems as sets of interrelated elements and their decomposition into subsystems corresponds with the definition of work systems as systems "in which human participants and/or machines perform work [...] using information, technology, and other resources" (2009, p. 3). Work systems "can be decomposed into at least several layers of smaller work systems" until "there is no benefit from further decomposition" (2010, p. 11) and may themselves be part of a larger system called "organisation".

Proposition P-01. Work systems are systems composed of processes, activities, participants, roles, information, and technology.

However, the distinction made by Alter between the work system being studied and *other work systems* and the definition of *enterprises* or *organisations* as a work system raises questions. On the one hand, introducing the construct "other work system" to specify work systems on which the work system of interest depends for certain outcomes leads to confusion. The discussion of the construct by Alter does not suggest other work systems are substantially different from "regular" work systems. On the other hand, the definition of a firm or organisation as a work system raises the question of practicability. Alter recognises this issue in his comment that "attempting to analyze or design an entire organization or enterprise at a detailed level is simply too overwhelming" (2009, p. 16). It thus seems reasonable to suggest that the inclusion of "other work systems", "organisation", and "enterprise" in the work system model is redundant.

Proposition P-02. The inclusion of other work systems, organisations, and enterprises in the work system model is redundant.

Finally, the distinction between *abstract* and *concrete systems* established by Ackoff leads me to suggest that work systems are concrete systems. Ackoff defines a concrete system as "one at least two of whose elements are objects" (1971, p. 662). Work systems are composed of "human participants and/or machines [performing] work" (2009, p. 3). It is thus unreasonable to suggest that they are an analytical construct comprising only "concepts".

Proposition P-03. Work systems are concrete systems.

Environment. Next to distinguishing abstract and concrete system, Ackoff also draws a distinction between systems that are *open* (i.e., interact with elements in their environment) and that are *closed* (i.e., that are entirely self-contained). In fact, much of the discussion in the organisation science suggests that social and sociotechnical systems are open systems (Anderson, 1999). The fact that organisations and their work systems are open systems is also recognised by Alter. He suggests that work systems interact with internal and external "customers", i.e., "recipients of a work system's products and services", and receive resources such as "relevant human, information, and technical resources that are used by the work system" from their enclosing organisation (2009, p. 9). Furthermore, work systems "should support the organization's strategy" (2009, p. 35) and work system performance is affected by "a series of factors including external stakeholders, culture, policies and procedures, history, internal politics, competition, standards, regulation, and demographics, and technology trends" (2009, p. 34). It is thus reasonable to suggest that work systems are open system that interact with elements in the environment.

Proposition P-04. Work systems have an environment.

However, Alter does not provide a complete and exhaustive discussion of the *elements* of which the work system environment is composed. Ackoff defines the environment of systems as "set of elements and their relevant properties, which elements are not part of the system but a change in any of which can produce a change in the state of the system" (1971, p. 662). In organisation science, discussions of the environment of organisational systems include (1) the stakeholders of an organisation such as customers, suppliers, competitors, and shareholders or (2) the general discourse in the society regarding shared issues such as society, nature, technology, and economy (Rüegg-Stürm, 2005). Alter, on the other hand, defines the work system environment as a "series of factors including external stakeholders, culture, policies and procedures, history, internal politics, competition, standards, regulation, and demographics, and

technology trends" (2009, p. 34). This does not include the constructs of "customer", "infrastructure", and "strategy", albeit it appears from Alter's discussion that they form an essential part of the environment as well. It is thus reasonable to suggest a deficit in the definition of constituent elements of the environment.

Proposition P-05. The constituent elements of the work system environment are not (unambiguously) defined.

Structural State. Furthermore, Ackoff suggest that the *structural state* of systems and the environment at a particular moment in time can be defined as the "set of [a system's/the environment's] relevant properties at that time" (Ackoff, 1971, p. 663). This definition is recursive and may equally be applied to the entities and/or elements of which a given system/the environment is composed. For example, the structural state of an order processing system at a moment in time can be expressed as the value of its order backlog and throughput of its activities at that time. Similarly, the environmental state of the order processing system may be defined as the inventory levels and order propensity of its customers at that moment in time. Unfortunately, the discussion of "relevant properties" of a work system, its environment, and its constituent components such as participants, technology, and information cannot be considered complete and exhaustive in the work system model (Alter, 2009). Alter discusses the relevant properties of work systems to some extent in earlier work (the author uses the term 'performance indicator') but classifies them as "useful ideas" (Alter, 2006). This is not sufficient to provide guidance to system analysts. It thus follows that:

Proposition P-06. The structural state of a work system and/or its environment at a moment in time is not defined.

Internal and External Events. Ackoff defines *internal* and *external events* as transitions occurring on the structural state of a system and the environment over a specified duration of time. For example an external event can be an "out-of-stock" event occurring to a customer of an order processing system. The event is defined by a transition from the structural state "in-stock" of the customer to "out-of-stock" over a period of time (i.e., until the inventory is replenished). In the case of the order processing system, external events lead to a series of internal or "system" events (a change in the order backlog, the delivery schedule, and accounts payable state) following the occurrence of the "out-of-stock" event. Following proposition P-06, the structural state of a work system and environment is not defined. It is therefore reasonable to suggest that transitions in the structural state, i.e., "events", cannot be expressed until this deficiency is rectified.

Proposition P-07. Internal and external events occurring to a work system are not defined.

Finally, the distinction between *static* and *dynamic systems* established by Ackoff leads me to suggest that work systems are dynamic systems. Ackoff defines dynamic systems as systems "to which events occur, whose state changes over time" (1971, p. 663). This corresponds with Alter's comment that "work systems change over time in a series of iterations" (2009, p. 3). It is therefore not reasonable to suggest that work systems are immutable.

Proposition P-08. Work systems are dynamic systems.

5.3.2 SYSTEM CHANGE

Reactions, Responses, and Acts. Ackoff describes the *reaction*, *response*, and *action* of a system to an event as "a system event for which another event that occurs to the same system or its environment is sufficient [and/or necessary]" (1971, p. 664). A reaction by a system is a system event for which another event in the system/environment is sufficient. For example, the sending of an order acknowledgement is the reaction of an order processing system to a customer event (order requisition). A response, on the other hand, is a system event for which another event in the system/environment is necessary but not sufficient. For example, a customer event (order requisition) is necessary but not sufficient for an order processing system to send a delivery notification. Finally, actions of a system are system events for which no other external event is required. For example, an availability-to-promise check is a system event for which no customer event is required. However, the specification of reactions, responses, and actions of a system requires the system analyst to identify and describe events occurring to the system of interest. As the construct of "event" is absent from the work system model, such specifications cannot be created.

Proposition P-09. The reaction, response, and action of a work system to an internal/external event is not defined.

Behaviour. According to Ackoff, the *behaviour* of a system is "a system event(s) which is either necessary or sufficient for another event in that system or its environment" (1971, p. 664). For example, the creation of a sales order is a necessary event for the production of goods/provisioning of services and the billing and invoicing of a customer. The construct of behaviour as defined by Ackoff maps to the construct of "activity" in the work system model. An activity is a unit of work "performed by a participant using technology or by a technology acting as an automated agent" (Alter, 2009, p. 19). Typically, activities produce one or more outcomes including "physical things, information, and actions that are received or used by the work system's customers" (Alter, 2009, p. 9). The creation of a sales order can therefore be described as

an "activity" performed by an order processing system that has as its outcome a new sales order record. The creation of this record may in turn trigger additional activities. Alter suggests the occurrence of activities within a work system is "initiated by a known condition, such as the completion of a previous activity" and that "a series of related, clearly specified steps" is a "process" (2009, pp. 22-23).

Proposition P-10. The behaviour of work systems is described by one or more activities.

5.3.3 BEHAVIOURAL CLASSIFICATION

Process and Outcomes. Ackoff suggests that the processes occurring in a system can be defined as "a sequence of behavior that constitutes a system and has a goal-producing function" (1971, p. 666). Each unit of behaviour in the process brings the system closer to a goal it seeks to obtain. For example, each activity in an order system brings the system closer to completing a customer order, removing it from the order backlog, and collecting payments relating to the order. The construct of "process" as defined by Ackoff maps to the notion of work process (Alter uses the more generic term "process") in the work system model. Alter defines processes as "a set of two or more related activities that occur within the same work system" and argues that the analysis of the behaviour of a work system "emphasizes activities rather than processes because any reasonably deep analysis of a work system needs to determine which activities are performed by which participants using what technologies and information" (Alter, 2009, p. 23).

Furthermore, Alter recognises that "many important situations rely heavily on human judgment and improvisation and therefore are not totally structured" (2009, p. 22). The structure of processes can thus vary with highly structured processes at one end of the spectrum and semi-structured and unstructured "decision-making" at the other end of the spectrum. It thereby follows that:

Proposition P-11. Work systems produce outcomes through one or more structured, semi-structured, or unstructured processes.

Furthermore, the classification of system behaviour into *state-maintaining*, *goal-seeking*, *multi-goal-seeking*, *purposive*, and *purposeful* systems leads me to suggest that work systems must be regarded as purposeful systems. Ackoff defines purposeful systems as system "which can produce the same outcome in different ways in the same (internal or external) state and can produce different outcomes in the same and different states" (1971, p. 666). This type of behaviour is exclusive to purposeful systems and corresponds to Alter's comment that work systems

improve "over time in a series of iterations" (2009, p. 3) and that "many important situations [in work systems] rely heavily on human judgment and improvisation" (2009, p. 22).

Proposition P-12. Work systems are purposeful systems.

Relative Value. Systems (i.e., purposeful systems) which can "choose between different outcomes can place different values on different outcomes" (Ackoff, 1971, p. 666). In other terms, a system showing purposeful behaviour can choose between different outcomes in a particular situation and decide among these outcomes by virtue of their *relative value*. For example, the relative value of a customer transaction at a particular moment in time may depend on (1) the transactional cost incurred by the work system and/or (2) the lifetime value received by the work system over the duration of the customer relationship. The work system may therefore (1) prioritise outcomes that minimise the transactional cost and/or (2) prioritise outcomes that deliver service experience and thus improve long-term customer satisfaction (Treacy & Wiersema, 1997). According to Alter, if and how work systems prioritise outcomes based on their relative value cannot be explained by analysing the system alone: "strategy at the work system level should support the organization's strategy" (2009, p. 35). It follows that:

Proposition P-13. The choice of a work system between multiple possible outcome is determined by its (organisational) strategy.

Goals, Objectives, and Ideals. Ackoff defines the goals, objectives, and ideals of a "purposeful system in a particular situation is a preferred outcome that can be obtained within a specified [or unspecified] time period" (1971, p. 666). Goals relate to preferred outcomes of a system at a moment in time that can be obtained in a "specified", i.e., finite, duration of time. For example, the goal of a complaints processing system is to provide minimal response and wait times to calling customers. The "objective" of the same system, e.g., to provide superior customer experience, cannot be obtained over a specified duration of time (a day, a week, a month). It is the result of a consistent delivery of minimal response and wait times over a longer period of time. Finally, "ideals" are outcomes that cannot be obtained in practice. While Alter acknowledges the importance of goals and objectives for system analysis and design, he argues that "showing goals and metrics attached to many of the elements would turn [the work system model] into an incomprehensible jumble" (2010, p. 9). It follows that:

Proposition P-14. The goals, objectives, and ideals of a work system with respect to its outcomes are not defined.

Finally, Ackoff's definition of *ideals* and *ideal-seeking* systems as systems "which, on attainment of any of its goals or objectives, then seeks another goal and objective which more

closely approximates its ideal" (1971, p. 667) leads me to suggest that work system do not necessarily satisfy these criteria. Alter suggest a work system's "goal is to provide value for its customers, not just to operate consistent with its own original specifications" (2009, p. 5). However, it is unreasonable to suggest that all work systems follow some higher ideal such as the creation of "perfect" value. It follows that:

Proposition P-15. Not all work systems are ideal-seeking systems.

5.3.4 ADAPTATION AND LEARNING

Function. Ackoff defines the *function* of a system as the "production of the outcomes that define its goal(s) and objective(s)" (1971, p. 668). Unlike the construct of "process" (a sequence of behaviour having a goal-producing function), the function of a system is invariant. Given two structurally different types of behaviour in the same or different environment, the function of a system is defined by its ability to produce the same outcomes. In other words, whereas processes describe "how" an outcome is achieved, the function of a system defines "what" the system produces. This corresponds with Alter's definition that "a work system's purpose is to produce products and/or services for customers" (2009, p. 25). Alter suggests that work systems produce "specific products and/or services for specific internal or external customers" (2009, p. 3) and that the production of outcomes by the work system is achieved through its activities. The emphasis of analysis is on "activity" and not "process" because "the work that is performed may not be structured or sequential enough to be called a process" (2009, p. 23). It is thus reasonable to suggest that the function of a work system is its primary activity.

Proposition P-17. The function of a work system is its primary activity, which can be subdivided into a set of activities or processes.

Efficiency. The *efficiency* of a system is the probability with which it obtains preferred outcomes in the same or different environment. Ackoff defines efficiency as the sum of the products of (1) the "probabilities that the system will select [particular] courses of action" and (2) the "probability that course of action [...] will produce a particular outcome" (1971, p. 668). For example, the efficiency of a customer complaints handling system is the sum of the products of (1) the probability with which customer service representatives will select a particular action and (2) the probability that this action will satisfy customer needs and expectations in a particular context. This requires the ability to measure the probability of a certain course of action taken by a work system as well as the extent to which the course of action produces preferred outcomes as defined in propositions P-12 to P-14. As some of these constructs are not (sufficiently) defined and the work system model lacks a discussion of the relevant properties and

measurement of work system elements, the efficiency of a work system cannot be formally captured by a system analyst. It is thus reasonable to suggest the construct of work system efficiency as a measure of performance is not defined.

Proposition P-18. The efficiency of a work system with respect to its function is not defined.

Types of Adaptation. Further from the construct of efficiency, Ackoff defines four *types of adaptation* that specify the "ability of a system to modify itself or its environment when either has changed to the system's disadvantage so as to regain at least some of its lost efficiency" (1971, p. 668). Systems react or respond to external and/or internal events by modifying their structural state and/or the state of the environment. For example, a call centre may respond to a surge in call volume by (1) adding additional staff (changing its structural state "call handling capacity") and/or (2) asking customers to call a dedicated hotline (changing the behaviour of customers, i.e., the state of the environment). However, while it seems reasonable to suggest that most work systems exhibit the four types of adaptation suggested by Ackoff (other-other adaptation, other-self adaptation, self-other adaptation, self-self adaptation), the work system model does not define essential constructs such as "event" (the cause of adaptation) and "state" (a modification of which is the effect of adaptation). System analysts are therefore not able to capture the adaptation of work systems and classify it according to Ackoff's four types. It is therefore reasonable to suggest that:

Proposition P-19. Work system adaptation is not defined.

The definition of *adaptive system* by Ackoff and the (informal) discussion of work system adaptation by Alter (2006) leads me to suggest that work systems can be viewed as adaptive systems. Ackoff suggests a system is adaptive if, "when there is a change in its environmental and/or internal state which reduces its efficiency [...] it reacts or responds by changing its own state and/or that of its environment so as to increase its efficiency" (1971, p. 668). This corresponds to Alter's comment that work systems evolve through a series of "planned" and "unplanned" or ad hoc adaptations (Alter, 2006). It follows:

Proposition P-20. Work systems are adaptive systems.

Learning. Finally, Ackoff specifies *learning* as a system's ability "to increase one's efficiency in the pursuit of a goal under unchanging conditions" (1971, p. 669). For example, a call centre system learns by (1) refining and standardising call handling procedures for certain situations and (2) learning how to adapt these procedures if the environment changes. Despite the recognition of learning as an essential capability of organisations and work systems (Argyris & Schon, 1978), the construct of learning is absent from the work system model. Alter suggests that

work systems "change over time in a series of iterations" and that "increasingly, those iterations have led toward largely automated work systems". However, the actual process of learning is not defined. It therefore follows that:

Proposition P-21. Work system learning is not defined.

5.3.5 SYNOPSIS

In this section, I conducted a representation mapping of the 32 constructs of Ackoff's system of systems concept to the work system model (WSM) as defined by Alter (2009). The purpose of this mapping was to establish the degree of completeness (Weber, 1997) of the work system model with respect to system theory. I clustered the 32 constructs of the system of systems ontology into four cluster constructs (system, system change, behavioural classification, relationship between system and elements, and adaptation and learning) following the original text by Ackoff (1971) to facilitate the mapping. By mapping each of the 32 system of systems constructs to any of the 32 work system model constructs, I made 21 propositions that pertain to the completeness of Alter's model. For example, the definition of system on the level of SSC (a set of interrelated elements) corresponds to the construct of work system on the level of work system analysis (a system in which people perform work using information and technology). Notably, not all constructs in Ackoff's ontology are relevant for mapping work systems. For example, work system comprise concrete elements (people, technology) and can therefore not be defined as abstract.

Table 5.4 summarises the results of the representation mapping, omitting system of systems constructs that do not apply to work systems. Each construct in the system of systems ontology is clustered into one of the four construct clusters based Ackoff's (1971) text. The kernel theory construct is then mapped to zero, one, or many work system model constructs. For each mapping, a rationale is provided to substantiate the mapping decisions made during the enquiry. Finally, a list of all identified excess constructs is defined that do not correspond to theoretical constructs.

Adaptive system construct	Work system construct	Description
SYSTEM		
System	Work system, participant, process, activity, information, techn.	People performing work using information and technology
Environment	Work system environment, customer, infrastructure, strategy	Customers, infrastructure, and other factors in the environment

Adaptive system construct	Work system construct	Description
Structural state	—	Not defined
Event	—	Not defined
SYSTEM CHANGE		
Behaviour	Activity	Unit of work performed by a participant producing and outcome
BEHAVIOURAL CLASSIFICATION		
Process	Process	Sequence of unstructured, semi-structured, or structured activities
Relative value	Work system strategy	Alignment of work system behaviour with organisational strategy
Goal	—	Not defined
Objective	—	Not defined
ADAPTATION AND LEARNING		
Function	Activity, product/service	Unit of work the purpose of which is to produce a product or service
Efficiency	—	Not defined
Adaptation	—	Not defined
Learning	—	Not defined

Table 5.4. Synopsis of representation mapping

However, not all constructs in the system of systems ontology and work system model are relevant to define *context-aware information systems*. As defined earlier, context-aware information systems are systems which recover from a loss in efficiency (pertaining to the ability to produce/deliver informational products or services) in a given context (internal and external state) by modifying the way information is captured, transmitted, stored, retrieved, and so forth. The focus of information system analysis and design is therefore on transitions in the system context (internal and external state at a particular moment in time), how this change affects the ability of the system to produce and deliver outcomes, and how the internal and/or external state of the system must be modified to recover from a loss in efficiency. Following

the definition presented in the background section of this chapter, system analysts need to specify relevant events, system behaviour, and outcomes. Based on this, we can prioritise the list of propositions to exclude those that pertain to issues that are generalisable across all system types and to include those that are specific to context-awareness. The limitations and extensions defined later are thus specific to the phenomenon of interest.

Table 5.5 clusters the 21 propositions identified in this section based on whether they are relevant for the explanation of adaptive systems or not. The clustering is based on the definition of the type of adaptive systems and its corresponding constructs in Ackoff's (1971) text.

Relevance	Proposition	Explanation
Not relevant	P01-P04	System, system elements, and system boundary are common to all classes of information systems
	P09, P10	System behaviour and response to external events is common to all classes of information systems
	P11, P12	All classes of information systems are purposeful insofar as they can select among multiple process paths
	P15	Ideal-seeking is a property that is not relevant for the specification of a context-aware information system
	P16	The notion of function is common to all classes of information systems and comprises products/ services
Relevant	P05, P06	Understanding relevant elements in the environment and the internal and external state of systems at a particular moment is specific to instances of CAIS
	P07, P08	Specification of internal and external events is fundamental to the definition of context-driven adaptation
	P13, P14	Specification of relative value in a context and the constructs goals, objectives is fundamental to CAIS
	P17-P20	Efficiency, adaptation, and learning are essential constructs that are specific to the class of CAIS

Table 5.5. Propositions for further enquiry

5.4 DEFICIENCIES IN WORK SYSTEM PROPOSITIONS

The objective of this section is to identify deficiencies in work system theory that lead to unclarity and ambiguity with respect to the subset of 10 system-theory constructs defined in section 5.4. The subset of 10 system-theory constructs forms the foundation of the class of systems I refer to as CAIS. Interpretations of these constructs that are unclear or ambiguous lead to design artefacts that do not faithfully represent the meaning of the underlying kernel theory. This feature is undesirable and the following section is dedicated to identifying such deficiencies in contemporary conceptualisations of information-based work systems. My enquiry proceeds in two steps. First, I reread the definition of a work system construct and compare it with the definition of its corresponding system-theory construct. Three situations can be identified. First, the work system construct faithfully represents the meaning of the system construct. Second, a single work system construct represents the meaning of two or more distinct system constructs. Third, two or more work system constructs represent the meaning of a single system construct. I repeat this process for the subset of 10 system-theory constructs and define deficiencies by using the terminology of Section 5.2.1.

The result of this evaluation is a set of 8 deficiencies in work system theory that introduce unnecessary ambiguity with respect to constructs that explain the properties and behaviour of CAIS. This ambiguity limits the ability of system designers to create faithful representations of context-aware information systems and evaluate them against the propositions of system theory. The 8 deficiencies are as follows. Unclarity concerning elements of the work system environment, the omission of state, and lack of events in work system theory lead to a situation in which transitions in the internal and external state of work systems cannot be clearly defined. Furthermore, the omission of relative value, goals, and objectives in work system theory constrains the ability of system designer to specify desirable outcomes of work processes and how their relative value changes in different work contexts. Finally, efficiency, adaptation, and learning are not clearly defined by work system theory. This limits our ability to reason about why work systems should adapt to relevant changes in their context and what can be learnt from past adaptations. Overall, these deficiencies reduce the theoretical clarity of work system theory and limit its application to context-aware system design.

5.4.1 SYSTEM

Environment. More than one construct in the work system model map to the construct of environment as defined by Ackoff. Ackoff defines the environment of a system as a "set of elements and their relevant properties, which elements are not part of the system but a change in

any of which can produce a change in the state of the system" (1971, p. 663). According to Alter, elements in the environment of work systems can be partitioned into the work system environment, the organisational environment, and the enterprise environment. This leads to construct redundancy (which type of environment is necessary for the analysis of work systems) and confusion for system analysts with respect to the scope of analysis. For example, it remains unclear whether factors in the organisational environment affect a work system directly or primarily affect the organisation and thus have an indirect impact on the work system. Following the definition of environment proposed by Ackoff, the environment of a system comprises elements (including other systems) to which the system is connected. A complete specification of the work systems should thus list the typical elements found in their environment such as customers, suppliers, and shareholders.

Deficiency D-01. The definition of three kinds of environments in the work system model leads to construct redundancy.

Structural State. There are no corresponding constructs in the work system model for the kernel theory constructs internal and external state. Ackoff defines the internal and external state of a system at a particular moment in time as the "set of relevant properties which that system [and the system environment] has at that time" (1971, p. 662). For example, the state of an order processing system at a particular moment in time can be described by its order backlog, processing capacity, customer demand, and customer inventory levels. The definition of the external and internal state of a work system at a particular moment in time is important to (1) define the context in which work is conducted and (2) isolate factors that could impact on the ability of the work system to produce and deliver outcomes to its internal or external customers. However, a discussion of the structural state of work systems and/or their elements at a particular moment in time is missing from the work system model. Alter specifies only top-level constructs (work system, participant, informational entity, technological entity) but does not discuss their relevant properties. This leads to a construct deficit that limits the ability of system analysts to specify work system context.

Deficiency D-02. The omission of the structural state of entities in the work system model leads to a construct deficit.

Internal and External Events. It follows from the omission of the structural state of entities in the work system model that events occurring on a work system or entities in the environment cannot be expressed. Ackoff defines events as a "change in one or more structural properties of the system (or its environment) over a period of time of specified duration" (1971, p. 663). For example, a change in the demand and/or the inventory level of one or more customers to

which an order processing system is connected leads to a change in the system itself (an immediate or anticipated increase in the order backlog). Similarly, a change in the equipment (outage) and/or labour (recruitment freeze) available to the system affect its ability to process orders in the backlog. Both events are relevant context changes that require the system to modify its behaviour and/or that of elements in its environment to avoid a loss in customer experience and/or future revenue. However, a discussion of internal and external events affecting work systems and their elements is missing from the work system model. This leads to a construct deficit that limits the ability of system analysts to define relevant changes in the context of systems and their impact on system state.

Deficiency D-03. The omission of internal and external events affecting entities in the work system model leads to a construct deficit.

5.4.2 SYSTEM CHANGE

In the previous section, I presented two proposition pertaining to change occurring on work systems and the behaviour of work systems in reaction/response to change. These propositions argue that the reaction and response of work systems to internal and external events is not defined and that the behaviour of work systems is specified by their activities. However, as argued in the synopsis of the section, the propositions relate to issues that affect all classes of information systems and are not specific to the class of context-aware information systems. In other terms, all information systems react or respond to changes in their internal and external state by capturing, transmitting, storing, and retrieving information (and so forth). For example, an order processing system reacts and/or responds to changes in the inventory levels of customers, order documents, or delivery schedules by scheduling one or more actions (such as order handling, delivery notification, payment collection, and so forth). The ability of information systems to react and/or respond to events, on the other hand, is not sufficient to make them context-aware. A detailed discussion of the limitations of the work system model with respect to the propositions is therefore not in my scope.

5.4.3 BEHAVIOURAL CLASSIFICATION

Relative Value. There are no corresponding constructs in the work system model to express the relative value of an outcome in a context as defined by Ackoff. Ackoff defines the relative value of an outcome that is "a member of an exclusive and exhaustive set of outcomes" of the system as "the probability that the system will produce that outcome when each of the set of outcomes can be obtained with certainty" (Ackoff, 1971, p. 666). For example, the relative value of a service provided by a call centre system is defined by the probability with which the sys-

tem delivers a consistent customer experience. Transactions involving customers that deliver a low customer experience can be said to have a low relative value (approaching 0). Transactions involving customers that result in a high customer experience over time, on the other hand, can be said to have a high relative value (approaching 1.0). However, a discussion of the relative value of outcomes (products and services) delivered by information-based work systems is missing in the work system model. Alter suggests "a work system's goal is to provide value for its customers" (2009, p. 5) but does not specify how customer value is to be determined. This leads to a construct deficit.

Deficiency D-04. The omission of the relative value of work system outcomes leads to a construct deficit.

Goals, Objectives, and Ideals. Furthermore, the omission of the relative value of outcomes produced by a work system leads to a construct deficit with respect to goals, objectives, and ideals. Ackoff defines goals, objectives, and ideals as preferred outcomes "that can be obtained within a specified time period", that "cannot be obtained within a specified period but which can be obtained over a longer time period", and that "cannot be obtained in any time period but which can be approached without limit" (Ackoff, 1971, p. 667). For example, the preferred outcomes of a call centre system may include (1) to respond to customer enquiries within minimal time (goal), (2) to consistently respond to customer enquiries in minimal time across all encounters with the customer (objective), and (3) to deliver the best customer experience in the industry (ideal). Neither of these constructs is discussed and represented in the work system model. Alter suggests that "including goals and metrics attached to many of the elements would turn [the work system model] into an incomprehensible jumble" (Alter, 2010 p.9). Omitting them, on the other hand, leads to a construct deficit and limits system analysts to specify the goal-oriented behaviour of work systems.

Deficiency D-05. The omission of goals, objectives, and ideals that guide work system behaviour leads to a construct deficit.

5.4.4 ADAPTATION AND LEARNING

Efficiency. The omission of the relative value of work system outcomes and the goals, objectives, and ideals guiding work system behaviour lead to a construct deficit with respect to work system efficiency. Ackoff (1971) defines the efficiency of a system as the probability with which the system will select appropriate actions and that these actions will produce desirable outcomes in a given context. For example, the efficiency of a call centre system in delivering high customer experience is its ability to consistently respond to customer enquiries in minimal

time across all encounters with the customer. Its efficiency is low if it fails to respond in minimal time or only responds to customer enquiries in minimal time in certain contexts (e.g., during normal workload). Its efficiency is high if it is able to respond in minimal time consistently and regardless of its current context. However, there exist no constructs in the work system model to define and measure work system efficiency. Alter suggests the "real goal of analysis and design is improving business performance" (Alter, 2009, p. 16). Improving the performance of work systems requires system analysts to identify shortfalls in efficiency. The absence of efficiency as a construct thus leads to a deficit.

Deficiency D-06. The omission of the efficiency of work system behaviour leads to a construct deficit.

Types of Adaptation. None of the constructs of the work system model map to any of the four types of adaptation defined by Ackoff. Ackoff defines adaptation as the ability of a system "to modify itself or its environment when either has changed to the system's disadvantage so as to regain at least some of its lost efficiency" (1971, p. 667). For example, the ability of a call centre system to modify itself (call handling routines) or the environment (event hotline) in response to an external event (weather event) is an instance of adaptation. Alter acknowledges the importance of adaptation and suggests work systems evolve "over time through multiple iterations" and that "unplanned changes [are] part of a work system's natural evolution" (Alter, 2008, p. 467). However, the work system model omits three important constructs to define and analyse work system adaptation. First, the construct of system and environmental state is undefined. Internal and external events occurring on a system can therefore not be specified. Second, the omission of work system efficiency limits the ability to measure a loss in efficiency. Finally, actions taken by the system in response to a context change cannot be distinguished from other kinds of behaviour.

Deficiency D-07. The omission of adaptation as a special kind of work system behaviour leads to a construct deficit.

Learning. Finally, the construct of system learning as defined by Ackoff does not map to a corresponding construct in the work system model. Ackoff defines system learning as the ability to "increase one's efficiency in the pursuit of a goal under unchanging conditions" (1971, p. 669). Furthermore, Ackoff suggests that learning "requires an ability to modify one's behavior (i.e., to display choice) and memory" (1971, p. 669) to improve the efficiency of adaptation over time. For example, a call centre must keep records of its performance in different kinds of events in order to improve the efficiency of its call handling routines in different kinds of contexts. Without the ability to keep these records and without the ability to modify call handling

routines, the call centre system would be unable to learn and improve. Although the construct of learning is implicit in Alter's comment that work systems evolve "over time through multiple iterations" (Alter, 2008, p. 467), it is not made explicit in the work system model. Both a formal specification of work system efficiency and adaptation is missing from the work system model. It follows from this omission that the necessary constructs to define and specify work system learning are missing.

Deficiency D-08. The omission of work system learning leads to a construct deficit.

5.4.5 SYNOPSIS

In this section, I presented 8 limitations that reduce the theoretical completeness and clarity of work system theory with respect to reasoning about context-aware information systems. Each of the limitations presents an instance in which the definition of a work system construct does not adequately reflect the meaning ascribed to it by systems thinking. For example, the definition of environment in work system theory is ambiguous and prevents a meaningful analysis of elements in the relevant context of work systems. Using the terminology developed in Section 5.2.1, I labelled each deficiency and provided reasons for why it presents a shortcoming for the design of context-aware information systems. Furthermore, I suggested that these deficiencies must be addressed in order for work system theory to be extended to the domain of dynamic work context and context-driven system adaptation. Details of this discussion are out of scope for this chapter but are discussed in Chapter 6. I now proceed to the measurement of system constructs and work system deficiencies in evidence collected in the exploratory case study (cf. Chapter 4). The objective of this step is to identify differences and similarities in the interpretation of key constructs and deficiencies.

5.5 FINDING EVIDENCE FOR IMPROVED PROPOSITIONS

The final step in my analysis is concerned with measuring the occurrence of the set of 10 CAIS design constructs in data collected during the exploratory case study of two work systems (cf. Chapter 4). In a multiple-case study I studied the work systems of two separate organisations in different industries and found commonalities in how these systems responded to changes in their context. In the following, I map each of the 10 CAIS design constructs to the data I collected and coded as part of the case study. This proceeds in two steps. First, I reread the definition and meaning of a construct in Ackoff's (1971) original text. I then map the description of the phenomenon capture by the construct to interpretations provided in the case interviews. Three situations can be identified. First, interpretations of a construct in the case data match

consistently across case sites. Second, interpretations of a construct are different in the case sites. Third, interpretations of a construct in the case data match consistently but the case data suggests additional properties that are not part of the original description of the construct. I repeat this process for each of the constructs and define how issues in the case data are to be interpreted and resolved.

This leads me to suggest 3 refinements of constructs based on interpretations provided in the case data. First, interpretations of the work environment and its relevant elements suggest respondents track issues that affect the customer experience and cost of outcomes produced by work systems. For example, the current weather affects the needs and expectations of customers of a claims handling system. Similarly, traffic flow on the airport road network impacts on passengers and their expectations of passenger facilitation. Second, some state of a system (or the environment) are predictable whereas others are not. This is reflected in interpretations of weather patterns, traffic flow, and equipment breakdown on the one hand and natural disasters and other contingencies on the other hand. Finally, some changes in the relevant properties of the system (or the environment) happen rapidly while others unfold over a protracted period of time. For example, storms and lightning require rapid adaptation of work systems to either (1) handle additional call volume or (2) withdraw labour from exposed areas for workplace safety reasons. Other types of changes such as changes in government policy and industry legislation, on the other hand, unfold over time.

5.5.1 SYSTEM

Environment. Interpretations in the case data suggest respondents *selectively monitor elements* in the internal and external work system environment based on whether they affect customer satisfaction and cost. Ackoff defines the environment as "a set of elements and their relevant properties, which elements are not part of the system but a change in any of which can produce a change in the state of the system" (1971, p. 663). The case data suggests the relevant elements of the internal and external work system environment include customers, suppliers, and the availability and utilisation of resources. This is reflected in comments on the behaviour and attitudes of customers, supplier capacity, and availability of qualified staff in the Sigma Group case. In the Beta Corporation case, relevant constructs in the data include the behaviour and needs of passengers, airline expectations, regulatory requirements, and critical infrastructure of the airport (terminal buildings, baggage sortation and screening systems). A change in the relevant properties of these entities at a particular moment in time can lead to a change in the structural state of the respective work systems. This will be discussed in more detail in relation to structural system state and events.

Evidence for relevant elements in the environment of Sigma Group's claims handling system include the constructs *weather patterns*, *community attitudes*, *government policy*, and *recruitment freeze*. These elements affect the ability of the work system to deliver outcomes that are consistent with customer service and cost expectations set forth by the organisation. This is reflected in the comment by H, the executive manager for claims experience of the insurer:

In terms of you know, external risks that can impact in terms of our claims, so in respect in the context of claims I guess from a Queensland perspective, we have a lot of storms, that tends to drive costs into the claims business. So it really is awareness of what's happening within the state, looking at state development, looking at state guidelines, looking at building codes, so it's quite a high external gambit in terms of what our requirements would be to respond to customers in respect of something happening.

Evidence for relevant elements in the environment of Beta Corporation's passenger facilitation system include the constructs *busy hours*, *traffic events*, *weather patterns*, *special events*, and *incidents*. Changes in these elements tend to affect the airport's ability to process passengers on time for their departure and with reasonable cost per passengers as expected by the airport operator. This is reflected in the comments by B, the security manager responsible for the terminals:

But going back to the operational side, for example, this is day to day activity, I'm not talking about anything extraordinary, but day to day activity. Traffic is a really good example. Traffic can have a hell of an impact on what we do here at the airport. And that can affect facilitation, people running late, missing flights, causing delays, those delays have a knock-on effects within the airline network. Even from a security perspective, those delays from a security perspective for example, people caught in traffic, they're delayed, they get to the airport, they're running late. They tend to take out their frustration on the people they meet.

Structural State. Interpretations suggest not all relevant changes in the structural state of a work system and/or the work system environment *can be predicted*. Ackoff defines the structural state of a system or the environment at a particular moment of time as "the set of relevant properties which that system [or the environment] has at that time" (1971, p. 662). This corresponds with interpretations of small to medium variations in workload of a work system on the one hand and with major disruptions affecting one or all work systems of an organisation on the other hand. Small to medium variations in the workload of systems are largely predictable. This is reflected in comments of respondents in the Sigma Group case on seasonal events such as weather patterns and their impact on claims handling workload and throughput. In the Beta Corporation case, examples for small to medium variations in workload provided in the case interviews include traffic events, weather patterns, and equipment breakdown. Major disruption

tions, on the other hand, are contingencies and as such are largely unpredictable. For example, interpretations of natural disasters, pandemics, and terrorist threats in the case data suggest such events are difficult to predict.

Evidence for predictable and unpredictable states of Sigma Group's claims handling system include the constructs *business as usual*, *event*, and *state of emergency*. Normal, i.e., expected, operating states of the system are *business as usual* and *event* (response to a weather event) whereas *state of emergency* is a contingency (e.g., response to a major natural disaster). This is reflected in the comments by A, Sigma Group's disaster response coordinator:

Every day I log onto the Bureau and I look at what's happening around the company, that's one, if I get an alert message for South East Queensland I'll be on there every hour. The alert stage really is just a souped up readiness stage, I'll approach various business units and I will tell them there is some type of weather system on the radar, not the actual radar, but on the natural radar, and I will ask the state of their business, again.

Evidence for predictable and unpredictable states of Beta Corporation's passenger facilitation system include *peak hours*, *equipment breakdown*, *security breach*, and *security threat*. Respondents describe the state of passenger facilitation to be in a "continual flux" but some issues do stand out. These include predictable conditions such as peak hours and major disruptions such as an escalation in threat levels. This is reflected in the comments of E, the terminals manager:

Traditionally, we do a lot of survey so we've got a lot of data that we analyse so we've got a reasonably good idea when our peaks are of the week so we do tend to focus on those and you know I do get a daily report that tells me where the congestion issues were, if there's any traffic incident or failure at lights and so on [...] We're looking at trying to improve that. At the moment it's normally, you know they use different landmarks as in the traffic is back to the intersection at Moreton Drive. But we're also looking at identifying like you say, actual points that they can use to say look it may be back to I don't know it could be point A for example and we'll identify that on the roads so that the traffic officer will be able to say traffic back to point A you know point B and so on.

Internal and External Events. Finally, interpretations by respondents suggest changes in the internal and external context of work systems can happen *rapidly* or *gradually*. Ackoff defines internal and external events occurring on a system as "a change in one or more structural properties of the system (or its environment) over a period of time of specified duration" (Ackoff, 1971, p. 663). In other terms, events describe transitions in the structural state of an entity. Two kinds of state transitions can be distinguished in the case data. First, gradual change in a work system or the work environment occurs over a protracted period of time. Constructs in

the case data that map to gradual context change include policy change (aviation policy), regulatory action (security standards, workplace safety, prudential regulation), and court decisions (affecting the insurance industry). Second, rapid context changes in a work system or the environment often occur daily but last for shorter durations of time. Constructs in the data that map to rapid context change include seasonal patterns (storms, fog), weekly patterns (traffic congestion and road network events), and daily patterns (peak hours at the airport). The two kinds of change are dealt with using different strategies.

Evidence for rapid and gradual events affecting Sigma Group's claims handling system include the constructs *storm*, *flooding*, *cyclone*, *policy change*, and *court decision*. While all of the mentioned events impact on the ability of the claims handling system to meet customer service and cost objectives, they unfold on different time scales and require different adaptation strategies. This is reflected in the comment of E, Sigma's operation manager for events:

If a storm has hit, people generally ring us straight away after a storm's finished, which would be 24 hours later. But when it's flood sometimes they're evacuated and they can't get to their phone line for a few days. So, that's when you start seeing the calls coming through [...] Bushfire, house burns down, you've got no phone so you know, we got a lot of claims being lodged at these one stop shops, so they call them in under the government community centres where everybody gets there, so it depends on the event.

Evidence for rapid and gradual events affecting Beta Corporation's passenger facilitation system include the constructs *traffic event*, *weather event*, *equipment breakdown*, *incident*, *security breach*, and *policy change* among others. According to respondents, the terminal staff has to deal with a large number of events every day. However, some events unfold on a longer time scale such as policy change. A, the passenger facilitation manager, describes this as follows:

Other than weather or a critical infrastructure issue, everything is slow change. Let's do weather as a starter. I tell you you've got your scheduled services data. Let's use the other night, Tuesday night, as an example. Everything is running along as per plan up until the weather hits. The airlines will then withdraw their labour off the tarmac. The aircraft will be delayed for whatever length of time that weather is happening.

5.5.2 SYSTEM CHANGE

Following the comment that system change is a property that is common to all classes of information systems (cf. Section 5.3.2), I do not measure constructs that pertain to system change in the following. However, the measurement of constructs that pertain to adapting work systems to relevant changes in their work context is presented in the following sections.

These constructs are specific to the class of context-aware information systems and will thus be included.

5.5.3 BEHAVIOURAL CLASSIFICATION

Relative Value. Interpretations in the case data suggest that the relative value of work system outcomes such as customer service and cost changes in different work contexts. Ackoff defines the relative value of an outcome that is "a member of an exclusive and exhaustive set of outcomes" as "the probability that the system will produce that outcome when each of the set of outcomes can be obtained with certainty" (1971, p. 666). In other terms, a system will select the outcome that is most valuable at a particular moment in time if this outcome is within its reach. This corresponds with process decisions made by respondents in response to changes in their work context. For example, Sigma Group seeks to reduce claims handling cost by applying process controls in a business as usual context but removes these controls for event-related claims. Customer service takes precedence over cost in an event context. Similarly, the cost of labour is an important concern for the management team at Beta Corporation. The team seeks to automate manual decisions and reduce labour but will allocate additional resources in a public holiday or event to avoid delays and long-term impacts on the airport's reputation.

Evidence for measures of relative value concerning the outcomes of Sigma Group's claims handling system include the constructs *customer experience*, *shareholder value*, and *community relations*. Customer experience and community relations on the one hand and shareholder value on the other hand are not mutually inclusive. The management team must walk a fine line between service and cost. The executive manager for claims, respondent B, explains:

We have two budgets to work to, a working loss budget, which includes our normal losses as well as our major losses and then we run what we call a natural hazards budget and that's any kind of, like a little bit of rain comes it's a storm claim, or, there's been a major flood for example [...] So, there's some forgiveness to an extent of overdoing your natural hazard budget, a certain amount a month and you're spending way more than that because you've had some calamity that's beyond what was budgeted, there's a level of forgiveness with that. There's less forgiveness in a working loss budget, which is static, very much static throughout the year.

Evidence for measures of relative value concerning the outcomes of Beta Corporation's passenger facilitation system include the constructs *security outcomes*, *passenger satisfaction*, and *shareholder return*. The terminal management team enforces security outcomes in accordance with federal guidelines regardless of work context but must balance passenger service and cost per passenger in different contexts. The airport terminal manager, respondent E, explains:

We want it to be a good experience for the public and for the passengers so it's reputation which we take very seriously. We take our reputation very seriously, that's right. Very, very seriously. Because we don't want it to get out in the media that we're causing delays and congestion at the domestic terminal [...] So we obviously throw some additional resources at it and I've got Queensland police also managing the intersection because that's where we found was the major issue, the management of that intersection just before the domestic terminal. And just really an education you know, just making people aware that it is going to be busy on those days.

Goals, Objectives, and Ideals. Finally, interpretations in the case data suggest both case sites have implemented a performance measurement system to track and improve the outcomes delivered in different work contexts. Ackoff distinguishes among three kinds of preferred outcomes delivered by systems: goals, objectives, and ideals. Two issues can be identified in the case data that map to these constructs. First, the insurer measures service metrics such as wait time, response time, and financial metrics such as handling cost per claim. Similarly, the airport measures metrics such as average queue length, wait time, and cost per passenger. These measures provide a quantitative insight into actual work system performance and allow comparison against expected performance per customer or passenger. Second, these measures are aggregated over time and compared against strategic objectives of the organisations such as customer advocacy, airport pricing and competitiveness. The analysis of aggregated measures allows the organisations to identify improvement potential. It remains questionable, however, if work systems contribute to higher ideals over and above the objectives of their organisation and commitment to shareholders.

Evidence for work objectives and goals of Sigma Group's claims handling system include the constructs *oldest wait time*, *response time*, *cost of claim*, and *life of claim*. These measures are reflected in the comments of B, the executive manager of the home claims portfolio:

Our objectives for business? [...] The first one is cost of claims. Nothing's more important than cost of claims, right. The second thing, which is customer advocacy, are customers actively recommending our product. So we're measuring customer's interest in us in three ways: whether they consider our product, whether they choose that product and whether they recommend our product. I can't influence the consideration, that's done by our marketing group. I can't influence whether they choose because that's done in sales, I can only influence relations. I've had a claim with you and customers will claim every 6.6 years on average, so that's your moment of truth.

Evidence for work objectives and goals of Beta Corporation's passenger facilitation system include the constructs *waiting time*, *number of security breaches*, *length of security screening*, and *price per passenger*. This is reflected in B's comments, the terminals security manager:

There's also compliance with our own service level agreements that we have with our customers and that we have with our contractors and I think that's really important to note. The simple fact is that we have performance agreements. So the agreements we have with the airlines in order to help them turn around their flights in a quick fashion. Sometimes they create problems by arriving late or arriving early. Sometimes the problems might be with us. There might be equipment breakdowns or there might be a terminal evacuation, or things like, that that will impact on that [...]

5.5.4 ADAPTATION AND LEARNING

Efficiency. Interpretations in the case data suggest rapid changes in work context lead to a measurable impact on the *customer service* and *cost* delivered by work systems. Ackoff defines the efficiency of a system as the probability with which the system selects actions that lead to preferred outcomes in a particular situation. This corresponds with concerns raised by respondents in the case interviews that rapid work context changes affect the customer service and cost of processes in work systems. Two issues can be identified. First, events in the environment of work systems may affect customer needs and expectations and thereby impact on the ability of a work system to deliver superior customer service. For example, severe weather events require the insurer to handle claims from emotionally distressed customers while traffic events force the airport to deal with disgruntled passengers. Second, events in the environment of work systems may affect the cost of processing as a result of increased workload and/or shortages in qualified staff. For example, increases in workload following a weather event increase the potential of leakage for the insurer while delays in passenger processing impact on aircraft parking bays and aircraft turnaround time.

Evidence for the impact of context change on outcomes of Sigma Group's claims handling system include the constructs *return calls*, *leakage*, and *policy renewals*. Changes in work context generally increase the volume of return calls from customers if service levels are not met, increase the potential for missing overstated claims, and may have a long-term impact on policy renewals. This is reflected in the comment of B, the executive manager for home claims:

So the more people are claiming, there's a lot of handling costs and there's great opportunity for leakage [...] Every time we touch a claim and go through that fulfilment process there's an option there for leakage and if you're leaking \$10, \$50, you know \$5, \$100 [...] then there's an opportunity to increase your cost base. I suppose, you're riding the margin, so lower frequency is better business.

Evidence for the impact of context change on outcomes of Beta Corporations's passenger facilitation system include the constructs *terminal throughput*, *queue times*, *delays*, and *reputation*. Context changes increase the potential for blockages at points such as screening, decrease the throughput of passengers per unit of time, and thus lead to delays and long-term

effects on airport reputation. This is reflected in the comments of A, the passenger facilitation manager:

We currently we have, say, 14 parking bays. If it happens in a peak period where we got the 14 bays full and then all of a sudden the airport opens, which still got 14 aircraft sitting on the ground here. We might have another 14 aircraft behind, waiting to come in and land and park in those bays. So if the aircrafts are delayed on the bays, it impacts on next ones coming in and that's the infrastructure which we look after and manage. So that's where the impact is on us. But getting those aircraft out is out of hands as well, because we can't force the airlines to move their airplanes.

Adaptation. Furthermore, responses suggest that some actions taken to adapt work systems to context changes are *ad hoc* while others are *planned*. Ackoff defines adaptation as "the ability of a system to modify itself or its environment when either has changed to the system's disadvantage" (1971, p. 668). This corresponds with actions taken by respondents to adapt work systems to rapid context changes and mitigate their impact on customer service and cost. Two kinds of actions can be identified. First, ad hoc actions are actions that are specific to a single instance of context change and are not replicated across other instances. For example, product decisions and coverage levels are decided by Sigma Group on a per-event-basis and decisions to bring in additional staff are made by Beta Corporation depending on the actual workload and conditions on a particular day. Second, planned actions are actions that are specific to a certain type of context and are applied to all instances of context change. For example, severe weather events prompt radio reads and newspaper advertising by Sigma Group whereas Beta Corporation works together with airlines to educate passengers about expected workloads and arrival times on public holidays.

Evidence for ad hoc and planned actions taken by Sigma Group to adapt its claims handling system to context changes include the constructs *event module*, *product decisions*, *do-and-charge limits*, and *radio reads*. The insurer automatically triggers outbound communication, changes to call handling scripts, product decisions, and entitlements in response to a work context change. This is reflected in the comment by A, the disaster response co-ordinator of the insurer:

You don't fully understand the event that you're responding to until at least 24 to 48 hours have passed. Typically the first [event leadership team meeting] would be called within an hour of the event actually striking or passing through a region, so the information would be very sketchy. You won't have all the information to make correct determinations or decisions on, they'll be stock standard what we do every time for about our marketing, so about our government, talk about the front end and resourcing. After the first 24 hours or 48 hours has passed and the event unfolds and the information is a bit more free flowing then we can actually talk about process.

Evidence for ad hoc and planned actions taken by Beta Corporation to adapt its passenger facilitation system to context changes include the constructs *standard operating procedures*, *screening point settings*, and *passenger education*. Terminal staff may decide to bring on additional staff, change the settings of screening points, and educate passengers about required arrival times ahead of public holidays. This is reflected in the comments of B, the terminal security manager:

Operationally the airport is on its front foot. Every day and every minute of the day, there are decisions being made because things are out of the ordinary. And operationally, it's very, very hard to define what's the norm and what's normal. You got some data that will help guide you but it's only for your planning. You really don't know until the minute that it occurs, what those decision-making requirements are. And most of the staff are equipped, where at least they can make decisions or seek some guidance or information to make those decisions to meet those changes.

Learning. Finally, interpretations in the case data suggest learning from past events is crucial to improve the effectiveness of context-driven adaptation of work systems in the future. Ackoff defines learning as the ability "to increase one's efficiency in the pursuit of a goal under unchanging conditions" (1971, p. 669). This maps to the constructs "debrief session" and "documentation" found in the case data. Debrief sessions are management team meetings called by the disaster response coordinator of Sigma Group after an event to reflect on actions and outcomes of actions implemented during the event. However, some team members question the value of these sessions and perceive them as too short and not structured enough. Similarly, documentation is the process and outcome of documenting actions implemented by Beta Corporation in response to an event for learning and training purposes. However, some management team members are not satisfied with the level of documentation available when compared with industry standards and other organisations such as airlines. This suggests there is significant potential for improving the effectiveness of context-driven adaptation of work systems by learning from past context events.

Evidence for learning from past events in Sigma Group's claims handling system includes the construct *post-incident review*. Post-incident reviews are management team meetings called after an event to review the effectiveness of the insurer's response to a context event. Interpretations suggest not all team members view such meetings as effective. This is reflected in the comments by E, the operations manager for Sigma Group's event claims team:

There is a debrief session that they have but it's too short, because it's an hour session, too many people involved. They spend half an hour talking about the weather and football and so it's not effective. It's not an effective debriefing session, in my view and in any event there's very little change

that takes place with the decisions that are being made, so old ways, old behaviours, habits can kick in.

Evidence for learning from past events in Beta Corporation's passenger facilitation system includes the construct *documentation* and *incident reports*. The team documents changes in process and deviations from normal operating procedures and provides recommendations for the future. However, respondents suggest that the level of documentation can be improved. This is reflected in the comments by B, the terminal security manager responsible for both terminals:

Documentation isn't very good. We're very light on the ground. We do document more these days than we have in the past. So, for example, if there is a problem, that would generally be documented in some form of incident report and then from there, normally we would document something as an evaluation of how we might fix it [...] So, yeah, we're quite conscious of that. We may not document everything that goes by, but certainly document the final result.

5.5.5 SYNOPSIS

The analysis of differences and similarities in the interpretation of key constructs of context-aware information systems led me to suggest three extensions to system constructs and work system theory defined in the previous sections. First, work context can be defined as the state of relevant elements in the work system environment and work system. This raises the question of which elements to include and which elements to exclude from a work system analysis. Second, there exist differences in the predictability of state changes in relevant environmental or system elements. A basic distinction can be established between state changes that are predictable (weather events, peak hours, traffic flow) and unpredictable (state of emergency, equipment breakdown, security breach). Finally, the changes in the relevant and predictable work context of the observed work systems unfolds at a different pace. For example, changes in the weather or traffic context of the observed work systems changes at a rapid pace of several hours to a day. Gradual changes in work context, on the other hand, unfold over a protracted time period. Examples provided for such changes in the case interviews include policy change, regulatory action, and court decisions.

The three extensions point to the need for defining a revised model of work systems based on the concepts of systems thinking and context-awareness. Some of the phenomena in the case data such as relevant and non-relevant context, predictable and unpredictable state changes, and rapid and gradual context change cannot be explained on the basis of the work system constructs discussed by Alter (Alter, 2008; 2009; 2010). Additional work is required to define how these phenomena can be explained using systems thinking and how they should be rep-

resented in work system theory. A discussion of the details of this work is not in scope of this chapter. The theory-driven extension of WSM toward context-aware information systems will be discussed in Chapter 6.

5.6 DISCUSSION

In this chapter, I viewed and explained work systems through the lens of open adaptive systems as described by systems thinking (Ackoff, 1971) and defined deficiencies in work system analysis and design with respect to context. In order to provide sufficient clarity, I turned to the work of Weber (2012) on evaluating and developing theory in IS. In particular, I defined situations which lead to a lack of clarity and completeness in IS design theories with respect to the kernel theories on which their prescriptions rest (Gregor & Jones, 2007). I then turned to the Work System Method (WSM) as an established design-relevant theory to reason about work systems and their interaction with entities in their environment. The focus of my analysis was on understanding 1) whether the phenomena described by open system thinking can be represented in WSM and 2) whether WSM constructs are consistent with system theory. This led me to suggest that WSM has deficiencies that prevent it from being used to reason about context-driven adaptation. I used case data to further refine the propositions I had made.

The mapping of open system thinking constructs to work systems and the measurement of system constructs in case data leads me to suggest that more work is required in three discrete areas. First, prescriptions must be made regarding the selection of relevant context and the allocation of resources to the tracking of elements that lead to system change (as opposed to those that have no tangible impact on system performance). For example, this includes empirical methods (Ploesser, Recker, & Rosemann, 2011; Ramos & Santoro, 2010) and data-driven methods (Ghattas, Soffer, & Peleg, 2010; Ramos & Santoro, 2010) of context selection. Second, prescriptions must provide more guidance on dealing with events that are predictable (i.e., are certain) and those that are unpredictable (the occurrence of which is uncertain). For example, events in the multiple-case study in Chapter 04 that are predictable include weather and traffic whereas unpredictable events include natural disasters, pandemics, and terrorism. Finally, prescriptions must be made with respect to iterative context and the problem of learning. Each requirement is described in more detail below.

First, I defined context as the relevant state of system and environmental elements at a particular point in time and suggested (based on case data) that system participants *selectively* monitor context. This brings attention to the problem of context selection and raises questions regarding methods that support selection. Ackoff defines the state of the system or its environ-

ment at a given point in time as "the set of relevant properties which that system [or its environment] has at that time" (1971, p. 662). Furthermore, I suggested that, based on Ackoff's text, those elements are relevant to adaptive systems that "are not part of the system but a change in any of which can produce a change in the state of the system" (1971, p. 663). System participants must therefore draw a boundary between the elements that are included and those that are excluded in the design of a context-aware information system. In other terms, participants must have appropriate methods for context selection that allow them to make such decisions. Two methods can be identified that have the potential to do so.

Empirical approaches to context selection. One example for context selection is the use of empirical methods such as case studies and qualitative interviewing such as the one defined in Chapter 04. For example, Ramos *et al.* (2010) suggest qualitative interviewing as a method to collect data on the elements in the environment of work systems that impact on the efficiency with which it produces outcomes such as cost. The authors introduce and define the concept of key intelligence topics (KIT) and define a procedure through which external context variables can be identified. Context classification such as the one introduced by Rosemann *et al.* (2008) can provide the structure required to define interview protocols (H. J. Rubin & Rubin, 2011) and help to identify the specific layers in which such variables may be found. I used this approach in the multiple-case study presented in Chapter 04 in the context of semi-structured interviews with key decision-makers of two work systems. Furthermore, modelling techniques such as system dynamics can be used to illustrate context (Ploesser *et al.*, 2011).

Context selection and machine learning. Another example for context selection is the use of data mining and machine learning techniques such as the methods described by Ghattas *et al.* (2010) and Ramos *et al.* (2010). Ghattas *et al.* describe a conceptual model and approach to learn from context based on runtime data extracted from information systems. The conceptual model grounds in earlier work on goals in process modelling (Soffer & Wand, 2005), which it extends toward the explicit consideration of factors in the external process environment. Using data mining techniques, the authors then develop procedures by which contextual similarities between process executions in information systems can be computed. Ramos *et al.* combine qualitative and quantitative insight to mine process execution data for environmental variances. Qualitative approaches are used to determine key intelligence topics (cf. description above). This data is then used to apply mining techniques such as association rules to correlate process runtime data with information about factors in the system context.

Second, I used systems thinking to describe context change as events defined by a transition in the relevant state of elements at a time and suggested based on case data that some events

are *predictable* while others are *unpredictable*. This brings attention to the need for developing strategies in system analysis and design to respond to either type of contextual situation. First, I defined predictable context events as transitions in the state of relevant environmental elements that occur frequently. For example, weather events and traffic events are interpreted by respondents in the multiple-case study as predictable events that occur with frequency. Second, I defined unpredictable events as transitions in the state of relevant environmental elements that occur infrequently, i.e., that represent contingencies or risk (March & Shapira, 1987). For example, natural disasters, pandemics, and terrorism are interpreted as infrequent events that have a high uncertainty or risk. This has implications on the strategies applied to constructing information systems and more guidance must be provided for designing context-aware IS.

Context-as-usual or predictable events. Context-as-usual defines the set of context elements and predictable events that can be encoded in information systems through techniques such as context modelling and reasoning. Context modelling enables system analysts to abstract from the underlying context infrastructure through which contextual information is acquired and aggregated (Bettini et al., 2010). This includes, amongst others, sources of context information such as sensor networks and infrastructures for disseminating context information in distributed system (Gu, Pung, & Da Qing Zhang, 2005). Abstractions from low level sensor protocols and related technical challenges allow the analyst to focus on the specification of situational abstractions which can be integrated and reused across a wide range of context applications (Henricksen & Indulska, 2006). This is of particular relevance for information system design, where contextualisation approaches have been described for the adaptation of information artefacts (Analyti, Theodorakis, & Spyrtos, 2007) and processes (Rosemann et al., 2008; Rosemann & Recker, 2006) encoded in the information system structure.

Considering unpredictable events. Unpredictable context describes contingencies or risk cannot be encoded and represented in information systems as either no reliable model exists for their prediction or the information required to build models is not available. The application of techniques such as context modelling and representation is therefore of little use. In this case, techniques such as exception handling or risk modelling may support the system analyst in analysing and mitigating the impact of unpredictable events on information systems. Exception handling (Russell, van der Aalst, & Hofstede, 2006) defines exceptional situations in process execution and the runtime decisions that must be taken to mitigate the effect of such situations on system outcome. For example, the unexpected failure of information system hardware may require processing to be resumed in manual fashion. Risk modelling (Neiger, Churilov, &

Muehlen, 2006; Rosemann & Muehlen, 2005) defines risks and mitigation strategies that must be activated should the risk eventuate. Risk is generally defined as variance in the distribution of outcome (March & Shapira, 1987).

Finally, the definition of context must consider the *iterative nature of context*. Once elements in the system context are internalised, they become part of those elements that are tracked by a system and the focus of system design shifts to other factors in the system environment. The iterative nature of context can be explained on the basis of system theory and the construct of system boundary. In SSC, systems are defined as sets of elements and their relationships such that each of members in the set is connected to all other members. The environment of system, on the other hand, is defined as a set of elements that are not part of the system but a change in which produces a change in the state of the system. Once external elements are internalised in context-aware information systems, they can no longer be considered part of the system environment. Instead, the focus of system design shifts to identifying other elements that are not yet internalised in the system's structure. Context-aware information systems must therefore be constantly evolved to retain the property of context-awareness.

Figure 5.2 summarises the discussion above in a diagram that illustrates the problem of context, context selection, and learning in work systems. First, context is the set of elements $CE1-CE3$ and their states a change in which leads to a change in the state of a work system W . The set of context elements C can be further partitioned into the set of elements P that act as the source of predictable events and the set of elements U that act as a source of unpredictable events. For example, the set C may contain elements such as weather, location, or community attitudes for the claims handling system and traffic, weather, or flight schedules for the passenger facilitation system studied in Chapter 04. Context selection implies defining a subset of each set the elements of which will be tracked as part of tracking work system W 's performance. These elements and, more specifically, information about their state at a moment in time are thus "internalised". The system evolves over time by constantly "pushing" its boundary outward to select and internalise new context.

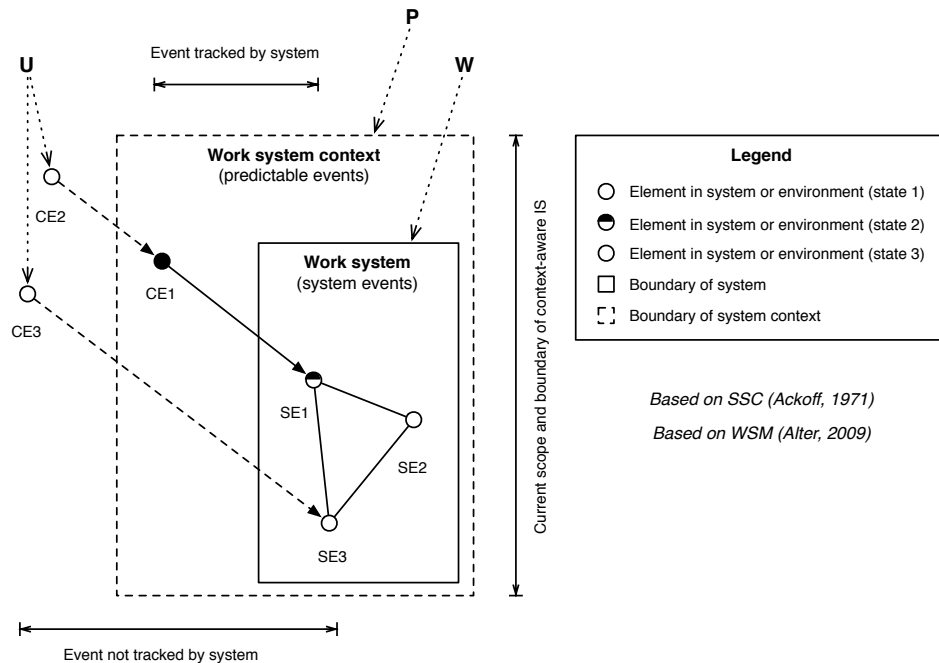


Figure 5.2. Model of context selection and learning in work systems

5.7 CONCLUSION

In this chapter, I presented a mapping of systems thinking (Ackoff, 1971) to work system theory (Alter, 2008) as a first step toward defining and explaining the class of context-aware information systems (CAIS). This mapping is based on the notion of theoretical completeness and clarity, an extension of the seminal work of Weber (Weber, 1997) to the domain of design theory. My analysis proceeded in three stages. First, I mapped the 32 constructs found in the system of systems ontology (Ackoff, 1971) to work system theory. This led me to suggest that 10 constructs are necessary and sufficient to explain the behaviour of context-aware information systems and distinguish them from other classes of information systems. Second, I identified 8 limitations in work system theory with respect to context-awareness that limit its application to dynamic work contexts and context-aware information systems. For example, I concluded that conceptualisations of work environment found in work system theory are redundant and lead to confusion. Finally, I measured the constructs and limitations to case evidence from the exploratory case study presented in Chapter 4 in order to identify differences and similarities in how they are interpreted in the data.

This leads me to conclude that additional work is required to extend work system theory and analysis to dynamic work contexts and the problem of designing context-aware information systems. For example, I identified that the issue of selecting relevant/non-relevant elements in

the work context, isolating predictable or unpredictable state changes in these elements, and defining the rapid/gradual pace with which these elements change is not defined in work system theory. Additional work is required to define how these phenomena can be explained using systems thinking and how they should be represented in work system theory. This was not in scope of this chapter. The theory-driven extension of WSM toward context-aware information systems will be discussed in Chapter 6.

Chapter 6: Theory-Driven Design

6.1 INTRODUCTION

In this chapter, I present prescriptive guidelines for the theory-driven design of *Context-Aware Information Systems* (CAIS) based on the outputs of Chapter 04 and Chapter 05. In Chapter 04, I presented empirical knowledge for the relevance of context in information system design. Based on an exploratory study of two work systems exposed to context changes, I concluded that system efficiency is improved by jointly adapting the technical and social systems of work in anticipation of changes in the system context. Furthermore, I argued that the process of context analysis and context-driven adaptation is supported by "maps" that provide guidance to system participants. In Chapter 05, I turned to adaptive systems (Ackoff, 1971) and work system theory (Alter, 2006; 2008; 2009) to measure the completeness and clarity of contemporary models of work systems with respect to context-driven adaptation. This led me to conclude that these models have deficiencies and I derived a set of design propositions based on a) academic kernel theories and b) empirical evidence.

Prescriptive design guidelines are a necessary output of Information System Design Theory (ISDT) and differentiate design theorising from routine design by the creation of new design knowledge. Several classes of design theorising can be distinguished (Kuechler & Vaishnavi, 2012) and range from guidelines for constructing and evaluating "artefacts" (Hevner, March, Park, & Ram, 2004), to describing how a set of design problems is solved by a class of solutions (Walls, Widmeyer, & Sawy, 1992), to explaining why an artefact behaves in a certain way (Baskerville & Pries-Heje, 2010). I briefly discuss each approach and then turn to the ISDT approach as described by Walls, Widmeyer, and El Sawy. ISDT breaks the process of reporting, communicating, and cumulatively developing design theory into three distinct problems. First, a set of problems is derived from kernel theory which describes the "meta-requirements" common to all instances of a system. Second, a set of features or "meta-design" that is hypothesised to solve these problems is defined. Third, testable hypothesis are formulated that test the ability of system instances to address each of the meta-requirements.

I go about creating these guidelines by mapping each of the design propositions presented in Chapter 05 to a set of theory-ingrained meta-requirements (Walls et al., 1992). Each meta-requirement describes a specific design goal that is to be satisfied by instances of context-aware information systems in practice. For example, I derive the meta-requirement of creating

representations for work system variables such as task, technology, structure, and people from related propositions presented in Chapter 05. Taken together, the set of meta-requirements provides a complete and exhaustive description of the design goals served by all instances of context-aware systems. Furthermore, I turn to system thinking, the principles of socio-technical design (Bostrom & Heinen, 1977), and contributions in the IS and CS literatures on a wide range of topics such as context modelling (Bettini et al., 2010) to derive a set of solutions hypothesised to meet the design goals of context-aware design. I conclude by defining a set of "testable product hypotheses" in the form of assertions that test the validity of the CAIS design theory.

6.2 RESEARCH APPROACH

Design theorising is of increasing importance in IS research and three overarching approaches to theorising about information system design are documented in the IS literature (Kuechler & Vaishnavi, 2012). Artefact-driven approaches to design produce new design knowledge by constructing and evaluating "artefacts" that provide a novel and innovative solution to a relevant problem (Hevner et al., 2004). This approach was inspired by earlier calls for combining the design and natural science types of research to explain the behaviour of information technology artefacts while addressing the real world problems faced by practitioners (March & Smith, 1995). Theory-driven approaches to design, on the other hand, seek to abstract from the construction of artefacts by providing prescriptive, theory-ingrained guidelines how more effective artefacts are to be constructed (Walls et al., 1992) or explanations of why an artefact "works" (or does not work) in a specific way (Baskerville & Pries-Heje, 2010). These approaches form a continuum (Kuechler & Vaishnavi, 2012) from (1) the creation of new knowledge by designing artefacts to (2) abstract design guidelines.

In the following, I turn to theory-driven approaches and in particular information system design theory (ISDT) as a guideline for the theory-driven design of CAIS. ISDT defines a framework whereby prescriptive design guidelines can be derived from justificatory knowledge that generalise to a class of systems rather than an artefact and provide prescriptive rather than explanatory knowledge about how systems must be constructed (Kuechler & Vaishnavi, 2012). ISDT predates much of the work on artefact-driven design and has been described by Walls, Widmeyer, and El Sawy in two seminal papers (1992; 2004). As such, ISDT provides a middle ground between approaches that emphasise artefact construction and evaluation as primary research activity (Hevner & Chatterjee, 2010) and those that define the explanation and prediction of artefact behaviour as primary purpose (Baskerville & Pries-Heje, 2010; Kuechler &

Vaishnavi, 2012). In other terms, ISDT offers enough guidance such that artefact instantiations can be created by following its instructions while addressing a class of problems instead of a single problem instance.

6.2.1 MAPPING CONSTRUCTS TO META-REQUIREMENTS

Constructs describe the basic units of interest (Gregor & Jones, 2007) defined by a mid-range design theory (Kuechler & Vaishnavi, 2012) and provide the underlying rationale for the design of an information system. Each construct represents one phenomenon of interest that must be addressed by artefacts constructed on the basis of an information system design theory (ISDT). For example, the task construct introduced as part of design theorising in Chapter 5 describes the tasks and relationships between tasks that characterise the work of system participants. ISDT seeks to bring about an improvement in the overall state of the world by defining design goals that are to be realised by instantiations of system artefacts (Walls et al., 1992). In the case of tasks, such as design goal can be to improve the transparency of work tasks such that a shared understanding of work is created among participants. Furthermore, ISDT do not address a single instance of a design problem but seek to generalise to a class of problems. This generalised class of problems is defined as a "meta-requirement" of system design.

Figure 6.1 illustrates the process of mapping mid-range design theory constructs to meta-requirements of the CAIS information system design theory. Each construct of the mid-range design theory is mapped to at least one meta-requirement in the ISDT such that the units of interest in the mid-range theory are fully represented in the design theory. In some cases, high-level constructs must be split into two or more meta-requirements in the ISDT.

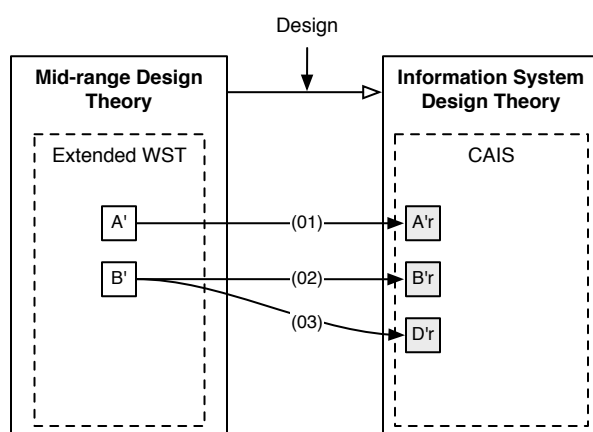


Figure 6.1. Mapping mid-range theory constructs to meta-requirements

6.2.2 MAPPING META-REQUIREMENTS TO META-DESIGN

Whereas meta-requirements describe the class of design problems addressed by ISDT, the ISDT meta-design defines a class of solutions that hypothesises an effect if a set of prescriptive guidelines is followed (Kuechler & Vaishnavi, 2012). For example, process modelling is generally regarded as a solution to the problem of system transparency that is hypothesised to bring about an increase in the transparency and shared understanding of organisational processes (Bandara, Gable, & Rosemann, 2005) by following a set of modelling guidelines (Mending, Reijers, & van der Aalst, 2010). Most importantly, the ISDT meta-design does not describe the properties of a concrete artefact but defines prescriptive guidelines by which more effective systems can be constructed (Gregor & Jones, 2007). In other terms, each feature defined by an ISDT meta-design does not describe an instantiation of a solution but the guidelines necessary to construct a generalised set of artefacts. It is therefore not restricted to a single problem instance but solves a generalisable class of problems (Walls et al., 1992).

Figure 6.2 illustrates the process of mapping ISDT meta-requirements to meta-design features of the CAIS information system design theory. Each meta-requirement is mapped to at least one meta-design feature in the ISDT such that each of the design goals of the ISDT is fully represented in the CAIS meta-design. In some cases, high-level meta-requirements must be refined into two or more meta-design features in the ISDT meta-design.

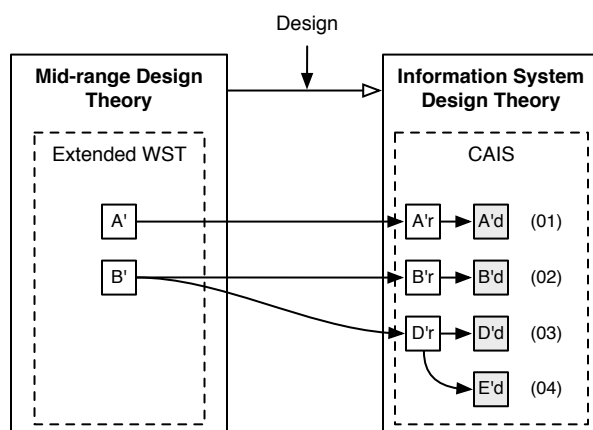


Figure 6.2. Mapping meta-requirements to meta-design features

In Section 6.5, I map each meta-requirement defined in Section 6.3 to one or more features that characterise the "meta-design" of ISDT (Walls et al., 1992). Each of the generalisable design problems of CAIS is mapped onto generalisable solutions on the basis of (1) theoretical knowledge about the construct underpinning a design goal, (2) prior design knowledge in the IS and CS literature on the construction of effective and efficient artefacts, and (3) a deep empirical

understanding of the design problem of contextualisation resulting from findings of the exploratory case study in Chapter 4. For example, mapping the meta-requirement of "system adaptation" to the domains of work systems draws from the theoretical understanding of adaptation in natural and artificial systems in system theory (Ackoff, 1971), the definition of four common viewpoints in information system architecture (Bernus, 2006), and the empirical understanding of adaptation trade-off resulting from the analysis of case interviews. This process is repeated until all meta-requirements are mapped to solutions in the CAIS meta-design.

6.2.3 MAPPING META-DESIGN TO TESTABLE PROPOSITIONS

Finally, testable design propositions articulate assertions or truth statements (Gregor & Jones, 2007) about an information system design theory that can be empirically validated. For example, process modelling asserts that an overall increase in the transparency and shared understanding of system behaviour can be achieved by depicting the tasks and relationships between tasks in work systems (Bandara et al., 2005). This increase is achieved by carefully selecting modelling grammars on the basis of their representational capabilities (Recker, Indulska, Rosemann, & Green, 2010) and following prescriptive guidelines for process modelling (Mendling et al., 2010). The combination of assertion and prescriptive guidelines in ISDT can be empirically validated in practice by a range of methods (Hevner & Chatterjee, 2010). These include experimentation by means of control groups (Arazy, Kumar, & Shapira, 2010), focus groups to validate the utility and relevance of design artefacts (Rosemann & Vessey, 2007) with practitioners, or other techniques such as action research (Sein, Henfridsson, Purao, & Rossi, 2011). Validation ensures the assertions made by ISDT are met when testing concrete instantiations of artefacts.

Figure 6.3 illustrates the process of mapping ISDT meta-design features to testable design propositions for empirical validation of the CAIS design theory. Each meta-design feature is mapped to exactly one testable design proposition in the ISDT such that each of the solutions proposed in the ISDT can be tested against the design goal specified by a meta-requirement. Each proposition is formulated as an assertion or truth statement about the ISDT.

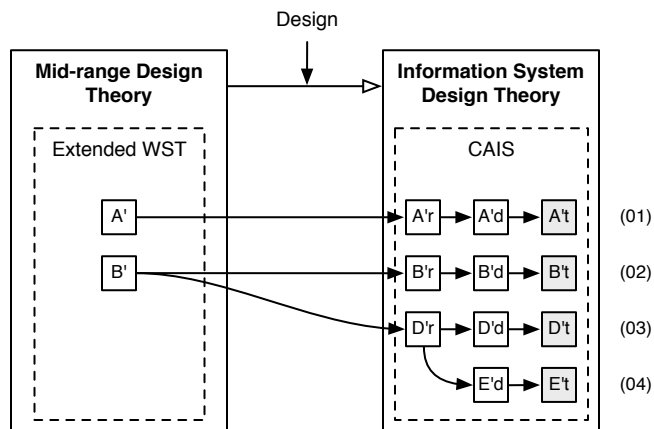


Figure 6.3. Mapping meta-design features to testable design propositions

In Section 6.6, I map each meta-design feature defined in Section 6.5 to exactly one testable design proposition that summarises the assumptions underlying CAIS design as a testable assertion. For example, the assumption that the explicit representation of the shared understanding of system context by system participants increases the general awareness of context by means of context modelling is defined as an assertion that can be tested in practice. The assertion holds that the design goal of increasing context awareness is achieved by following the prescriptive guidelines for context modelling introduced in Section 6.5. This process is repeated until all meta-design features in the CAIS design theory are mapped to exactly one testable design proposition. At each stage, design or theoretical knowledge in the IS literature is consulted to define an assertion. Most importantly, one assertion is defined for each meta-design feature in CAIS such that artefact instantiations created from the design theory can be tested in practice. Finally, the set of assertions is summarised and presented in a table.

6.3 META-REQUIREMENTS OF CONTEXT-AWARE IS (CAIS)

In the following, I derive the set of generalisable design problems or "meta-requirements" (Walls et al., 1992) for the class of context-aware IS (CAIS). The design problems at hand are derived from the propositions in systems theory and work system theory as summarised in Chapter 5. The set of five meta-requirements introduced below provides the foundation for the theory-driven specification of the CAIS meta-design that is discussed in detail in Section 6.5.

First, work systems comprise of multiple components and the interaction of these components determines system behaviour. This is reflected by propositions P-01, P-07, and P-11 as defined in Chapter 05. According to the propositions, work systems are characterised by the interaction between processes, functions, people, information, and technology (Alter, 2008). The potentially large number of components and component interactions in work systems requires ana-

lysts to deconstruct their complexity by creating abstractions (H. Smith & Fingar, 2006). Normative guidelines are therefore required to conclude which components must be considered and how to specify their interaction. This topic has received considerable attention in the IS literature and will be thus treated on a summary level:

MR-01. *Establish representation of work system.*

Second, work systems interact with a set of entities in the environment and changes in the state of the environment induce changes in the state of a system. This is reflected in propositions P-04, P-05, and P-06 in Chapter 05. Similar to propositions P-01, P-07, and P-11, this may involve a potentially large set of entities in the work system environment and requires mechanisms for identifying, classifying, and abstracting relevant context entities. Normative guidelines are thus required to conclude which entities must be considered and how they are to be represented in information systems. This maps to problems that are addressed by the representational view of context (Dourish, 2004) and requires the extension of existing context classification frameworks to the work system domain:

MR-02. *Establish representation of relevant work system context.*

Third, the adaptation of work systems to context change may involve both the modification of the system components and/or the modification of entities in the system environment. This is reflected in propositions P-12 to P-18. According to the propositions, systems respond to changes in the internal or external context by modifying their own state or the state of entities in the environment. This requires guidance on (1) selecting from a wide range of possible modifications and (2) identifying potential trade-offs that arise when activating a particular strategy (van der Aalst, Rosemann, & Dumas, 2007). Some guidelines for classifying strategies exist (van der Aalst et al., 2007; Weber, Reichert, & Rinderle-Ma, 2008) but their coverage of work system components is incomplete and generally lacks the ability to link strategy to context:

MR-03. *Specify strategies for behavioural adaptation of system.*

Fourth, work systems receive signals concerning the state of entities in the environment and must translate these signals into action by activating one or more adaptation strategies. This is reflected in proposition P-09 and P-10. According to this proposition, systems "sense" changes in the state of external entities by receiving signals, filtering them, and invoking actions based on cognitive schemata (Holland, 1992). This requires efficient mechanisms for capturing context, filtering context, and activating adaptation strategies. Three possible mechanisms can be identified (Krogstie, Sindre, & Jørgensen, 2006; van der Aalst et al., 2007). In manual activation,

sensing, filtering, and activation is performed by humans. In semi-automated or automated activation, the degree of human involvement is gradually reduced:

MR-04. *Match contextual situations with behavioural adaptation strategies.*

Finally, the structural state of work systems evolves through the process of learning from past adaptation to context change. This is reflected in proposition P-20 in Chapter 05. The literature on organisational learning distinguishes between two cycles of learning (Argyris & Schon, 1978; Nonaka, 1994). Single-loop learning is the process by which individuals, groups, or organisations learn to modify actions by comparing the discrepancy between expected and obtained outputs. Double-loop learning, on the other hand, is the process of questioning assumptions underlying actions. According to these propositions, context learning must modify the representation of context and system adaptation strategies when new information about their efficiency and relevance becomes available to the system.

MR-05. *Specify learning by modifying context representation and adaptation strategies.*

Table 6.1 summarises each of the meta-requirements discussed above and relates them to constructs and propositions made in Chapter 5. Each construct discussed in Chapter 05 relates to at least one ISDT meta-requirement in the CAIS design theory. Each meta-requirement, in turn, is a normative statement about the common requirements of context-aware information systems that generalises across all instances of such systems in practice.

Construct	Code	Meta-requirement (MR)	Related proposition (Ch. 05)
System	MR-01	Increase process awareness by modelling system variables and defining their interdependency	P-01, P-07, P-11
Context	MR-02	Increase context awareness by modelling context variables and the states assumed by variables	P-04, P-05, P-06
Adaptation	MR-03	Increase requisite variety by creating a repertoire of context-driven rapid adaptation options	P-12, P-13, P-14, P-16, P-17, P-18
Activation	MR-04	Decrease latency by specifying a set of rules and thresholds to trigger process adaptation in response to an event(s)	P-09, P-10
Learning	MR-05	Increase learning by improving response to context change and by considering new context	P-20

Table 6.1. CAIS meta-requirements

6.4 EVALUATING DESIGN-RELEVANT KNOWLEDGE IN IS KNOWLEDGE BASE

In the following, I briefly discuss the state-of-the-art in IS engineering and design with respect to the five meta-requirements and constructs defined in Section 6.3. This is followed by a detailed discussion of the application and extension of the state-of-the-art in the construction of context-aware information systems.

Context modelling and reasoning. The literature on context modelling and reasoning provides insight into the representation of context, situational abstractions, and the ability to draw inferences from context information. Context modelling and reasoning is particularly strong in the fields of pervasive computing (Bettini et al., 2010) and artificial intelligence (Akman & Surav, 1996; Brézillon, 1999). The core phenomenon studied by this research is context, with much attention given to how context can be captured, encoded, and processed in computerised systems. The problem of where context impacts on systems (e.g., process) or how systems should be adapted is often left unspecified (e.g. left to application developers as suggested by Dey, 2001). As a result, much of context research does not make explicit assumptions about the constructs process and adaptation. The discussion of adaptation mechanisms thus often remains on an abstract level (cf. Dey, Abowd, & Salber, 2001). The problem of learning has received some attention but focusses on improving context recognition (Brdiczka, Crowley, & Reignier, 2007) and not system adaptation.

Process modelling and configuration. The literature on process modelling and configuration is concerned with the representation of process knowledge and the configuration of enterprise systems to meet organisational requirements. Research in process modelling studies, among others, the analysis of representational capabilities of modelling grammars (Recker et al., 2010) and guidelines concerning the quality of process models (Mendling et al., 2010). Process configuration provides insight into how enterprise systems can be configured on the basis of process models (Dreiling, Rosemann, van der Aalst, Heuser, & Schulz, 2006; Rosemann & van der Aalst, 2007) to better meet the requirements of organisations. The core phenomenon of this research is the construct of process, with little attention given to context, adaptation, or learning. Context is only considered as a static parameter in the configuration of processes (La Rosa, van der Aalst, Dumas, & Hofstede, 2008). Similarly, adaptation of process models is considered only as a build time but not runtime decision (Rosemann & van der Aalst, 2007). Finally, learning does not appear as a construct in the literature on process modelling and configuration.

Process flexibility and change patterns. The literature on process flexibility and change patterns discusses how flexible process-aware information systems can be designed and constructed.

Process-aware information systems (Dumas, van der Aalst, & Hofstede, 2005) are a class of systems the design and implementation of which is governed by explicit process models. Research on process flexibility (van der Aalst, Pesic, & Schonenberg, 2009; Weber, Reichert, & Wild, 2009) and change patterns (cf. Weber et al., 2008) defines extensions to such systems that enable the flexible adaptation of process models by system users. Similar to process modelling and configuration, process models are the core phenomena studied by this research. Although process adaptation is an explicit construct in much of process flexibility research, little attention is given to the contextual situations in which adaptation must occur (Rosemann, Recker, & Flender, 2008). As a result, context remains an underspecified construct in the research stream. In recent years, learning has received increasing attention (Weber et al., 2009) and has focussed on model-driven support for process adaptation by system users.

Rules modelling and event processing. Furthermore, the literature on rules modelling and event processing discusses how flexible systems can be built on the basis of managing rules and events. For example, Kovacic (2004) points to the need for the explicit modelling of rules to address the requirements of organisations in dynamic and turbulent environments. Event management, on the other hand, combines a range of technologies to detect patterns and reduce latency between when operational data is acquired to when decisions regarding process changes can be made (Chaudhuri, Dayal, & Narasayya, 2011). The emphasis of both fields of research is on data and the actions taken by organisations in response to data. For example, rules act on changes in data (Muehlen & Indulska, 2010) to alert decision makers. Similarly, events capture changes in data (Chaudhuri et al., 2011) based on which action must be taken. Context is not considered as an explicit construct as is learning. The focus is instead on near real-time adaptation and activation of changes based on processing large volumes of transactional information.

Context mining and machine learning. Finally, the literature on context mining and machine learning studies how variance in process runtime data can be mined and correlated with events in the external context of systems. For example, Ghattas *et al.* (2010) define a formal model of context learning based on which procedures can be derived for the mining of context factors in process runtime data. Similarly, Ramos *et al.* (2010) define the construct of key intelligence topics and demonstrate how context factors can be mined from data about process runtime executions. The notion of context represents the core phenomenon studied by these contributions. Systems, on the other hand, only appear as data sources from which context variance can be mined. Accordingly, adaptation and activation plays a minor role in these contributions as the main focus is on contextual analysis and discovery. The construct of learning is viewed as es-

sential. This includes qualitative, knowledge-driven (Ramos & Santoro, 2010) and quantitative, data-driven approaches (Ghattas et al., 2010) to context learning.

In concluding, Table 6.2 summarises and compares the completeness of different streams of research in the IS knowledge base in terms of the constructs identified in Section 6.3. Contributions are evaluated on the basis of whether they support a relevant meta-requirement, whether they partially support the meta-requirement, or whether they cannot be considered to support the meta-requirement. For example, research in process flexibility has considered the problems of system, adaptation, and activation of adaptation options. Context has been considered in some contributions (Hallerbach, Bauer, & Reichert, 2010) but an overall approach to context selection and classification (Rosemann et al., 2008) is missing in the research stream. Accordingly, learning approaches (Weber et al., 2009) focus mostly on capturing knowledge that supports system users in selecting adaptation options but do not explicitly consider context. This is repeated for each of the research fields and technologies discussed above. Section 6.7. discusses future research required to more fully support CAIS requirements.

Design knowledge	RQ1 & RQ2		RQ3		
	Context	System	Adaptation	Activation	Learning
Context modelling and reasoning	•	–	–	•	(•)
Process modelling and configuration	–	•	(•)	–	–
Process flexibility and change patterns	(•)	•	•	•	(•)
Rules modelling and event processing	–	(•)	•	•	–
Context mining and machine learning	•	(•)	–	–	•
Legend	Explanation				
•	Evaluation criteria met				
(•)	Evaluation criteria partially met				
–	Evaluation criteria not met				

Table 6.2. Evaluation of prior IS design knowledge

6.5 META-DESIGN OF CONTEXT-AWARE IS (CAIS)

The meta-design component of ISDT describes the properties or "features" underlying a class of solutions to a given design problem (Walls et al., 1992) and their realisation in an information system artefact. For example, process modelling is a solution to the design problem of increasing transparency in work systems that disentangles tasks and relationships between tasks by depicting them in a formal process modelling grammar (Recker et al., 2010). Similarly, context modelling is a solution to the problem of increasing awareness of changes in the context of systems by capturing context variables and interdependencies that exist between variables in the system context in a formal modelling grammar (Bettini et al., 2010). By mapping meta-requirements to ISDT solutions, we define the solution space that is addressed by the design theory (Kuechler & Vaishnavi, 2012) and provide the search path by which the most effective and economic solution to a design problem is identified (Walls et al., 1992). A meta-design does not describe concrete artefacts but generalisable properties of a class of artefacts that solves a given problem.

The search path for solutions defined by ISDT consists of theoretical knowledge about a construct, prior design knowledge about efficient strategies for design and implementation, and empirical knowledge about the design problem (Hevner et al., 2004). Theoretical knowledge is knowledge about a unit of interest in the academic literature which guides the process of design (Gregor, 2006). In my case, theoretical knowledge is derived from conceptualisations of open adaptive systems in system thinking (Ackoff, 1971) and the application of system thinking to the domain of work systems. Existing design knowledge is knowledge about the guidelines and methods of constructing artefacts in the IS designs science and CS literatures (Hevner & Chatterjee, 2010). This includes the literature on the design, construction, and introduction of process-aware information systems (Dumas et al., 2005) and contributions in context modelling (Bettini et al., 2010) and the construction of context-aware applications (Henricksen & Indulska, 2006). Finally, I turn to the exploratory case study presented in Chapter 4 to extract empirical knowledge about designing context-aware systems.

Following the identification of five clusters of meta-requirements concerning the representation of work system variables, the representation of variables in the work system context, the creation of a repertoire of action, the automation of decision-making processes for context-driven adaptation, and the specification of context-based learning, I discuss the solutions provided by CAIS to each problem based on theoretical, design, and empirical knowledge in the following.

6.5.1 SYSTEM COMPONENT

Modelling system behaviour. Modelling system behaviour has received significant attention in the IS literature and issues pertaining to creating representations of it will be summarised for the sake of completeness. The literature defines work systems as characterised by two jointly independent but mutually correlative subsystems (Bostrom & Heinen, 1977), i.e., the technical and social system of work. Work systems respond to a set of inputs, a change in the state of which leads to a transformation of inputs into outputs. Input-output transformation (Kast & Rosenzweig, 1972) is the result of the interaction between two or more variables in work systems as described by socio-technical theory. For the technical subsystem, these variables are the tasks performed in the process of transformation and technology used to convert an input into an output. For the social subsystem, the relevant variables are the structure specifying the hierarchical organisation and reward mechanisms governing people and the properties of individuals (people) performing work in a system such as their needs, interests, skills, and capabilities.

Following Bernus et al. (2006), four distinct concerns or "views" of information system behaviour can be identified. The function and control view describes the transformation of inputs by an information system into intermediate or final outputs. The resource view defines the organisational resources involved in the input/output transformation (such as organisational units, their roles and responsibilities, and technological assets such as computing hardware, networking infrastructure, etc.) and their coordination in a hierarchy or network. The data view captures the data entities processed by the functions of an information system and messages a system exchanges with its environment. Finally, the output view specifies the material or immaterial outputs of processing in the system. Modifying system behaviour in response to context changes thus requires modifications to be made to all of the four viewpoints and requires the consideration of interdependencies between views and potential trade-offs arising from their adaptation (van der Aalst et al., 2007).

Table 6.3 defines how the meta-requirements of CAIS for representing task, technology, structure, and people variables are mapped to the four viewpoints of system behaviour defined in the IS literature. The requirement to capture the properties of tasks in work systems maps to the output, function, and control views in the IS literature. Tasks are derived through functional decomposition and their interdependencies are defined in the control view (Scheer, Thomas, & Adam, 2005). Each task contributes to one or more outputs defined by the output view. Furthermore, the resource view specifies the technological assets used in a work system such as computing hardware, and networking resources. The information produced by the assets in the course of input conversion is stored in the entities defined by the data view. Finally, structural

and people aspects of the work system are represented in the organisation hierarchy, role definitions, and job descriptions specified by the resource view. Each viewpoint thereby provides a solution to the meta-requirement of capturing work system variables.

Meta-requirement	Meta-design feature	Problem solution
Increase transparency by describing tasks performed by a system and their interdependencies	Output view	Capture outputs produced by the conversion of inputs in systems
	Function view	Capture functions performed by system in order to convert inputs
	Control view	Capture the coordination of inputs, functions, and outputs
Increase transparency by describing technology used to convert inputs by system participants	Resource view	Capture the technological assets used by system participants
	Data view	Capture the informational output by these technological assets
Increase transparency by describing organisation hierarchy, units, and roles relevant to a system	Resource view	Capture organisational hierarchy and ownership of resources
Increase transparency by describing the identity, availability, skill sets and relationships of people	Resource view	Capture the identity, roles, and skill set of system participants

Table 6.3. Mapping work system variables to modelling views

Figure 6.4 illustrates the mapping of constructs in the extended system model to entities in an ER model that can be used to specify information system behaviour. The model extends concepts found in socio-technical theory and refines similar efforts by Alter (2009). Concepts found in Alter's model are shaded in "grey" whereas extensions to Alter's model have no shading. The construct "work system" in the extended work system model maps to the entity "work system" and defines the boundary of the system of interest (01). As defined above, work systems consist of multiple variables each of which is represented by one or more entities. The construct "task" is refined into the entities "process" and "activity" whereby each process contains at least two activities (02). The construct "technology" is refined into the "technical artefact" and "information artefact" consistent with Alter's work. Furthermore, the construct "structure" is mapped

to "role" and the construct "people" is refined into "participant" and "capability" consistent with socio-technical theory.

Specific extensions to Alter's model include the entities "work variable", "work variable state", and "work system event". The extensions derive from Ackoff's (1971) discussion of the concepts "system element", "system state", and "system event" and their specialisation in the domain of work systems. Each work system thereby consists of a set of variables which define its state and changes occur on it through changes in the state of the relevant variables.

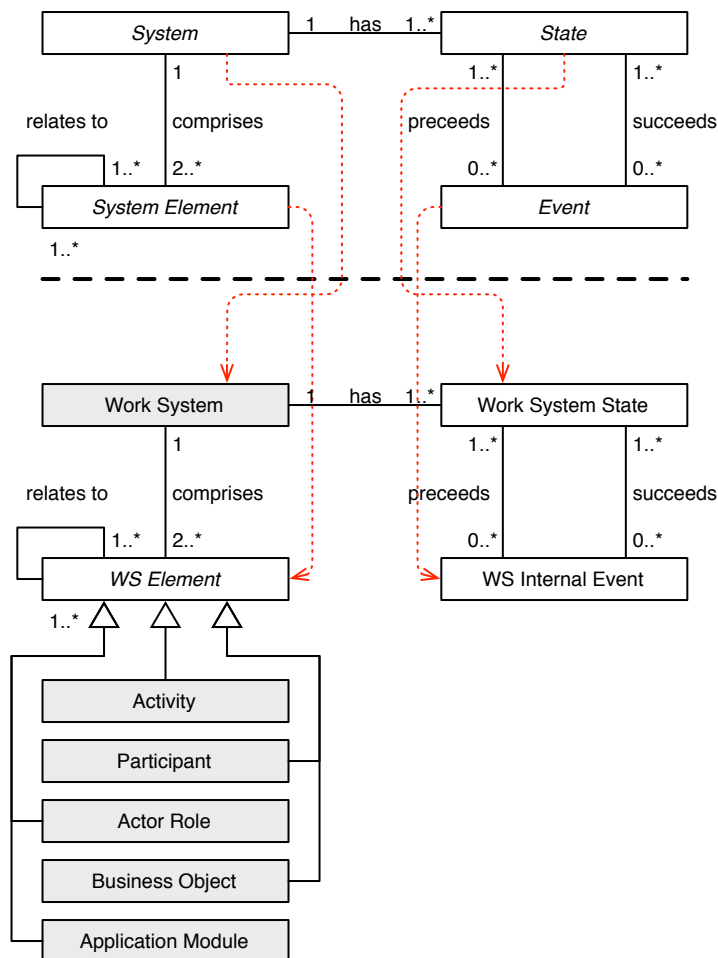


Figure 6.4. ER model of work system behaviour

Analysing system behaviour. Furthermore, analysing work system behaviour often relies on the creation of mental maps or representations for communication and learning purposes (Bostrom & Heinen, 1977). Many techniques have been proposed in the socio-technical design literature that aid analysis and joint optimisation of the two subsystems of work (Alter, 2006; London, 1976; Mumford, 1971). One approach that is increasingly receiving attention in the analysis and design of works systems is the use of conceptual modelling tools (Alter, 2009). Contemporary

examples for such tools include event-driven process chains (Scheer et al., 2005), activity diagrams, and swimlane diagrams (Object Management Group, 2011). Such tools generally cover multiple viewpoints of system behaviour which include the coordination of tasks in time and space, the assignment of organisational units to tasks, information that is processed by a work system, and the inputs and outputs transformed by it (Bernus, 2006). A wealth of literature exists on specific issues of modelling (Mendling et al., 2010; Recker et al., 2010).

6.5.2 SYSTEM CONTEXT COMPONENT

Modelling system context. In the following, I turn to work system theory and contingency theories of work systems to refine the meta-requirement of modelling work system context. Following work system theory (and reflecting the fundamental concept of system hierarchy and decomposition in system theory, cf. Kast & Rosenzweig, 1972), work systems are complex systems characterised by two interrelated subsystems (Bostrom & Heinen, 1977), the technical and social systems. By the same token, work systems do not exist in a vacuum but are interrelated with variables in the larger environment or "suprasystem" of which they form a part (Weill & Olson, 1989). Such variables may include the organisation structure, size, technology, and strategy which provides direction to tasks performed in a work system. It follows that considering context in work system design must not be restricted to intrinsic variables of work systems such as the properties of tasks, technology, structure, and people. Following system thinking, a holistic analysis of work system context must include the definition of extrinsic variables in the environment.

Figure 6.5 illustrates the hierarchical decomposition of work system context along the four dimensions defined in the extended work system model presented in Chapter 5. Each dimension of context (immediate, internal, external, and environmental) is further refined into context dimensions of finer detail following guidelines in the context modelling literature (Adomavicius, Sankaranarayanan, Sen, & Tuzhilin, 2005) until no further decomposition is possible or desirable. For example, the immediate context dimension of work systems can be decomposed into task, technology, structure, and people context. Once the desirable level of detail and refinement is achieved, the system analyst isolates the relevant properties of "context features" of a dimension. Context features represent concrete and measurable entities for which information can be extracted from one or more context information sources. By using conceptual modelling such as object-role models, system analysts can abstract from the underlying infrastructure used to acquire context information (Bettini et al., 2010; Henriksen & Indulska, 2006).

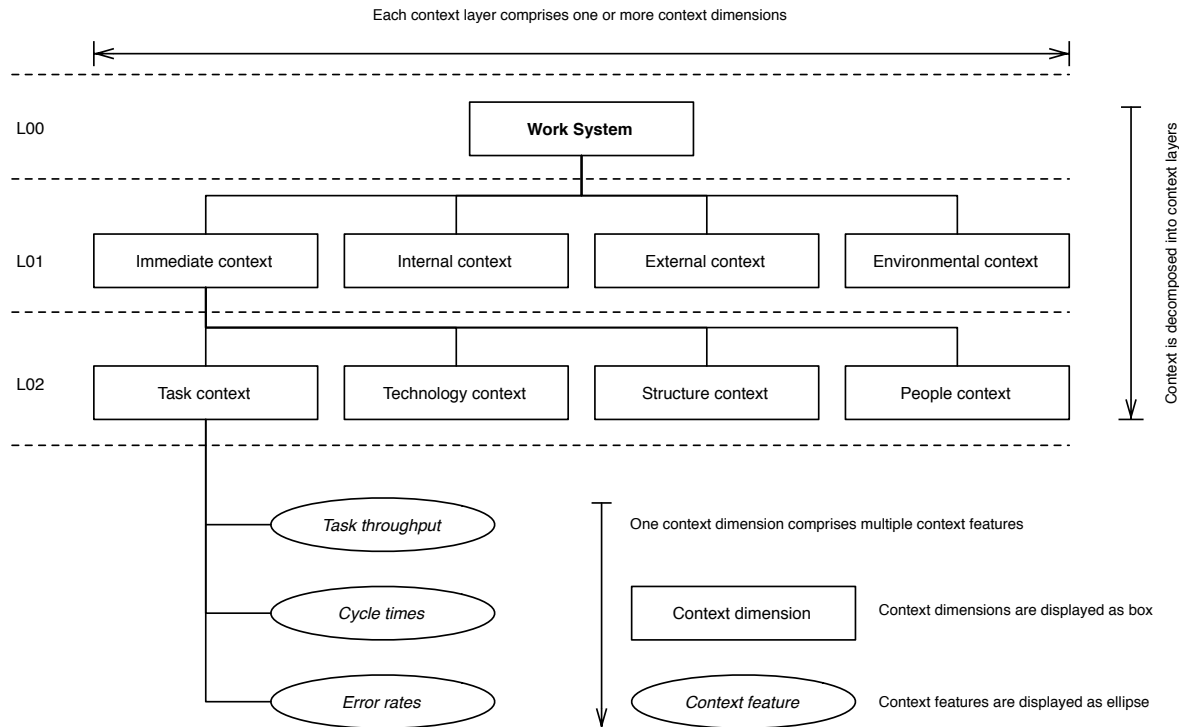


Figure 6.5. Hierarchical decomposition of work system context

Figure 6.6 clarifies propositions in Ackoff's (1971) original text as ER model and extends the work system model to capture work system context. Work systems consist of two subsystems and their respective variables classes (01) each of which define a dimension of the immediate work system context. Furthermore, work systems are connected to elements in the environment (02) and their behaviour is affected by contingency variables in the internal environment of an organisation. The two layers of immediate and internal context map to the hierarchical decomposition of work systems into their subsystems and their dependency on the organisational supra-system for resources and direction. Furthermore, work systems interact with stakeholders in the external environment of the market (03) including customers, suppliers, competitors, and regulators, which define the conditions of input/output transformation and benefit from the outputs created by work systems. Finally, environmental context such as weather, traffic, location, and time affects all other context layers.

Analysing context impact. Furthermore, the potentially unlimited number of dimensions and features of the work system context requires system analysts to isolate relevant context from context that does not affect work system behaviour. Ackoff (1971) argues that the analysis of general systems requires us to discern relevant properties of system elements and elements in the environment. This maps to the discussion of "relevant" context in work systems and procedures by which the impact of context on work system behaviour can be identified (Rosemann

The diagram illustrates the relationships between various entities in a system, organized into two main sections separated by a dashed line. The top section shows general system components, while the bottom section details work system components and their contexts.

Top Section (General System Components):

- Environment** (1) *has* **Extrinsic Variable** (1..*)
- System** (1) *has* **Extrinsic Variable** (1..*)
- Subsystem** (1) *has* **Intrinsic Variable** (2..*)
- Extrinsic Variable** (0..*) *has* **Environmental State** (0..*)
- Extrinsic Variable** (1) *has* **Intrinsic Variable** (1)
- Environmental State** (1) *affected by* **System State** (1)
- System State** (1) *has* **Element State** (0..*)
- Intrinsic Variable** (1) *has* **Element State** (1..*)

Bottom Section (Work System Components and Contexts):

- Work Sys. Environment** (1) *has* **Extrinsic Work Variable** (1..*)
- Work System** (1) *has* **Work Sys. Context** (1..*)
- Technical Subsystem** (1) *has* **Task** (1..*)
- Task** (0..*) *has* **Technology** (0..*)
- Technology** (1..*) *has* **Task** (1..*)
- Social Subsystem** (1) *has* **Structure** (1..*)
- Structure** (1..*) *has* **People** (1..*)
- People** (0..*) *has* **Structure** (0..*)
- Extrinsic Work Variable** (0..*) *has* **Internal Context** (0..1)
- Work Sys. Context** (0..*) *has* **Context Layer** (0..1)
- Context Layer** (0..1) *is a* **Immediate Context** (0..1)
- Environmental Context** (0..1) *has* **Extrinsic Work Variable** (0..*)
- External Context** (0..1) *has* **Extrinsic Work Variable** (0..*)
- Internal Context** (0..1) *has* **Extrinsic Work Variable** (0..*)

Relationships and Constraints:

- Red Dashed Arrows:** Indicate secondary or contextual relationships, such as from **Environment** to **Work Sys. Environment**, **System** to **Work System**, and from various context entities to **Extrinsic Work Variable**.
- Aggregation:** **Context Layer** is an aggregation of **Immediate Context**.
- Generalization:** **Context Layer** is a generalization of **Immediate Context**.

Figure 6.7 illustrates the process of *determining context impact on work system behaviour* by relating the four layers of work system context with the four variable classes found in work systems (Bostrom & Heinen, 1977). The horizontal axis enumerates the variable classes task,

technology, structure, and people that characterise work systems and a change in which leads to an overall impact on work system behaviour. The vertical axis defines the four layers of work system context starting with immediate context and ending at the outer layer of environmental context. By relating features of one context layer *A* with system variables *B* or other features on lower layers of the context hierarchy, system analysts can determine inter-layer context impact (01). Similarly, changes in the value of a feature *B* can lead to an intra-layer impact on other features *C* in the same context layer or on relevant system variables. For example, changes in the natural context (weather feature) can lead to increases in workload (task variable) and thereby affect the value of other system variables (people variable).

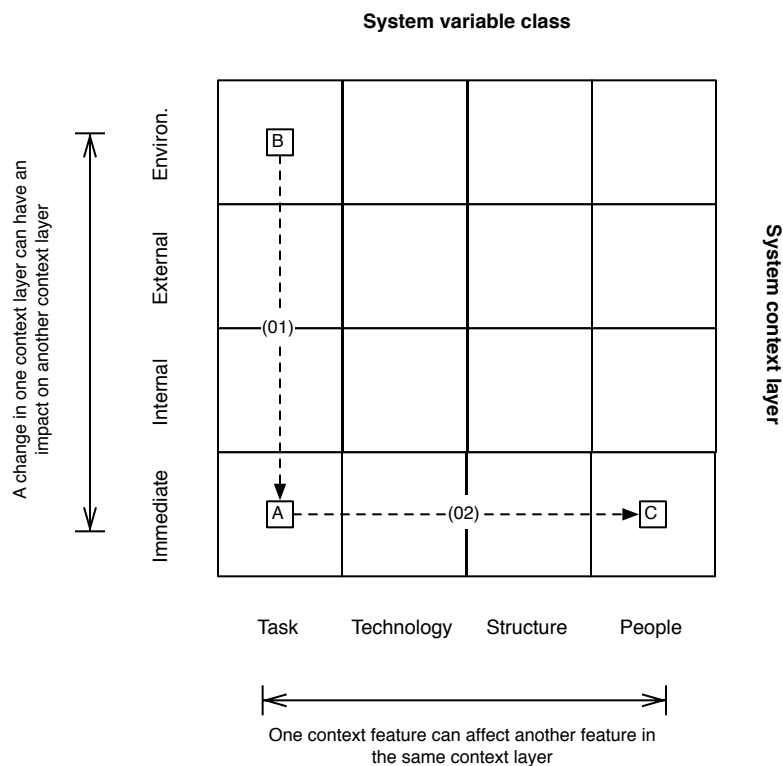


Figure 6.7. Context impact matrix

Finally, Table 6.4 summarises the mapping of meta-requirements concerning the representation of work system context to features of the CAIS meta-design presented above. By using the hierarchical decomposition of work system context presented in Figure 6.5, the four context dimensions defined by the extended work system model in Chapter 5 can be refined into a set of context features that are measured and tracked by accessing information from context sources. Context modelling allows system analysts to focus on the abstraction of relevant context features without the need to consider the technological infrastructure through which context information is obtained (Henricksen & Indulska, 2006). Furthermore, the context impact

matrix introduced in Figure 6.7 allows system analysts to specify interdependencies between context variables and determine their impact on any of the four variables of work systems. This allows for relevant variables to be distinguished from those that do not have measurable effect.

Meta-requirement	Meta-design	Problem solution
Increase awareness by describing variables in the immediate context and their impact on work	Immediate context view	Capture intrinsic variables within work systems and their impact on other intrinsic work variables
Increase awareness by describing variables in the internal context and their impact on work	Internal context view	Capture extrinsic variables within the organisation context and their impact on intrinsic work variables
Increase awareness by describing variables in the external context and their impact on work	External context view	Capture extrinsic variables within the external market context and their impact on the organisation
Increase awareness by describing variables in the environmental context and their impact on work	Environmental context view	Capture extrinsic variables in the natural and social environment and their impact on the market

Table 6.4. Mapping work context meta-requirements to meta-design features

6.5.3 ADAPTATION STRATEGY COMPONENT

Modelling adaptation strategies. Adaptation strategies define the measures to be taken by system participants in order to modify the information system state in a contextual situation. The process of adaptation in systems is described by Ackoff (1971) along two dimensions: the cause for adaptation and focus of state modifications. Two causes for adaptation can be identified. First, systems may respond to changes in their external environment or "context". In work systems, such changes may include changes in the state of the infrastructure, organisational policies, or needs and expectations of customers. Second, changes in the internal state of systems may require them to adapt their behaviour. For example, this can include changes to the skill sets and capabilities of participants or the properties of technology used within the work system. Furthermore, two foci of state modification can be identified. Modification of the external system state includes actions that modify the system environment. Modification of the internal state, on the other hand, includes actions that lead to changes in a system.

Figure 6.8 illustrates the selection of appropriate adaptation strategies based on (1) the reason for adaptation, (2) the desired focus of adaptation, and (3) the system viewpoint affected by the

modification of system state. Following Ackoff's taxonomy of system adaptation, system analysts view and identify the reason for adapting system behaviour (internal, external) and decide on the desired scope of adaptation (internal, external). For example, external changes in the environment may prompt the system to modify the state of external entities in order to neutralise the effect of a context change. By considering the impact of modifications on any of the four system viewpoints, system analysts can define the specific actions to be taken by system participants in response to a context change and potential interdependencies between these actions. For example, modifying the state of external entities in response to a change in the external context of an information system may be achieved by performing specific functions in that context such as triggering outbound communication.

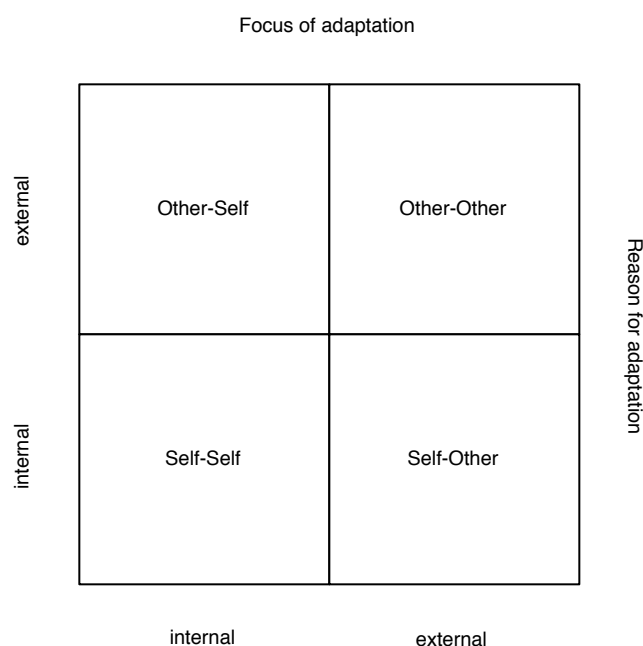


Figure 6.8. Selection of system adaptation strategies (based on and extending Ackoff, 1971)

Specifying adaptation strategies. Furthermore, the effectiveness of context-aware information systems depends on their ability to capture an exhaustive set of adaptation requirements. As defined earlier, information system behaviour must consider four concerns or "views" (Bernus, 2006). Context-aware information systems must therefore provide an adequate support for adaptation requirements on any of the four viewpoints of system behaviour. However, existing approaches to specify adaptation requirements often focus on the process or function perspective (Hallerbach, Bauer, & Reichert, 2009; Weber et al., 2008), neglecting other perspectives such as the organisation and data views (van der Aalst et al., 2007). A mechanism is therefore required by which the set of normative adaptation requirements that must be supported by all instances of CAIS can be identified for all four viewpoints and potential implementation strate-

gies can be suggested. Similar efforts have been made to identify patterns in process modelling and execution (van der Aalst & Hofstede, 2003) and the allocation of resources to functions in processes (Russell, van der Aalst, Hofstede, & Edmond, 2005).

I turn to prior work on specifying *behavioural flexibility in information systems* and *empirical evidence* to define the normative set of adaptation requirements in context-aware systems. Two concerns can be identified. First, a requirements specification captures the needs and boundary conditions for adaptation in any of the four system perspectives. For example, the need to deal with an increase in workload following a context change can prompt changes to the ordering of the functions performed by a system. Empirical evidence suggests that such changes lead to a trade-off that must be captured as part of the specification such that it can be used in the selection of appropriate adaptation strategies. This is evidenced by (1) increases in claims handling cost and "leakage" following adaptation of function flow (SGM) and (2) the utilisation of resources in passenger facilitation close to their maximum capacity (BTA). Second, guidance must be provided on the implementation of adaptation requirements in work systems covering a wide variety of technology such as workflow management and enterprise systems.

In the following, I define 11 adaptation strategies that represent the ideal set of capabilities that should be supported by instantiations of context-aware systems. Each strategy is (1) classified in terms of the reason and scope of adaptation as define in the taxonomy provided by Ackoff (1971), (2) associated with the information system viewpoint to which it applies, and (3) related to known patterns of adaptation defined in the IS literature for completeness. Furthermore, each strategy is measured in case data to identify the boundary conditions and trade-off of adapting system behaviour. Finally, an exemplary implementation is given to demonstrate the feasibility of implementing the strategy in context-aware systems using the Business Process Model and Notation (BPMN) grammar.

Adaptation strategies of the function and control view. The function and control view defines the types of functions performed by information systems and their ordering in time and space (Bernus, 2006). Adaptation strategies that apply to the function and process view thus modify three parameters: (1) the types of functions performed, (2) the dependencies between these functions, and (3) their ordering in time and space. By modifying these parameters, a system may achieve (1) an increased (or decreased) throughput by performing less functions or (2) mitigate the risk of performing functions that lead to adverse outcomes in a context such as regulatory compliance or customer satisfaction risks. Where applicable, I provide synonyms in the process adaptation and change literature.

Perform function fragment (CAF-01). Context changes can prompt systems to perform a function or process fragment to modify the state of an entity in the system environment. This strategy mitigates the risk of a particular state change in the entity occurring and is evidenced in the case data by the scheduling of functions to communicate (1) with policyholders to provide guidance on loss prevention and claims procedures in a weather event (SGM) and (2) with airline passengers to inform them of expected delays at the airport in the lead-up to major public holidays. Implementation of this strategy in a particular context leads to additional cost that is incurred by outbound communication. For example, this includes advertising cost and the cost of modifying airport information displays.

- *Scope:* Modification of the state of entities in the system environment to mitigate risk
- *Reason:* Applies to both changes in the environmental and system context
- *Trade-off:* Incurs additional cost for outbound communication
- *Synonyms:* Escalation subprocess (van der Aalst et al., 2007); Insert process fragment, Add control dependency (Weber et al., 2008)
- *Evidence:* News reads and newspaper advertisements ahead of or after a severe weather event to provide instructions on how to prevent losses or how to make a claim (SGM); Advertising and information displays informing passengers of expected delays in the lead up to major public holidays (BTA)

Perform alternative functions (CAF-02). Second, context changes can prompt systems to perform alternative functions or sets of functions. This strategy increases system throughput by performing less functions or mitigates risk by performing different types of functions. Case respondents associate the strategy with (1) modifications made to claims processing to accelerate the processing of low-risk claims in a severe weather event (SGM) and the provisioning of food or temporary accommodation in a flight cancelation or severe weather event (BTA). Implementation of this strategy in a particular context can lead to lower quality and increased cost when removing process controls or performing different functions. This includes claims leakage and the cost of providing food and shelter.

- *Scope:* Modification of the state of the system to increase throughput or mitigate risk
- *Reason:* Applies to both changes in the environmental and system context
- *Trade-off:* May lead to changes in output quality and affects other views such as resources and data

- *Synonyms:* Alternative path (van der Aalst et al., 2007); Update condition (Weber et al., 2008)
- *Evidence:* Claims triage and cash settlement of low-risk claims in weather events (SGM); Provisioning of food and/or temporary accommodation in case of a flight cancellation or severe weather event (BTA)

Skip function (CAF-03). Third, systems may decide to skip a function in a particular context to speed up the processing of inputs. This strategy applies to one specific function and thereby differs from CAF-02. It leads to an increase in system throughput as a result of abandoning the performance of the function and is evidenced in the Sigma Group case in the skipping of the proof of ownership/proof of loss function in weather events. The strategy is absent in responses collected at Beta Corporation. Airport processes underly strict security and safety regulations which constrains the use of CAF-03. Implementation of the strategy can lead to a deterioration of output quality and data captured by the system. For example, skipping the proof of loss function can increase claims leakage.

- *Scope:* Modification of the state of the system to increase throughput
- *Reason:* Applies to both changes in the environmental and system context
- *Trade-off:* May lead to changes in output quality and affects other views such as the data view
- *Synonyms:* Delete process fragment, Remove control dependency (Weber et al., 2008)
- *Evidence:* Skipping of the requirement to provide proof of ownership and/or proof of loss in a weather event (SGM)

Wait before performing function (CAF-04). Fourth, systems may decide to wait before performing a specific function in a given context. The strategy has an adverse effect on system throughput but is required to mitigate risks arising in a specific context. It is evidenced in the postponement of boarding procedures following a thunderstorm event at the airport. Case respondents suggest the strategy is implemented to enforce occupational health and safety regulations and to mitigate the risk of safety incidents in a thunderstorm. Implementation of this strategy decreases the average throughput achieved by the system and leads to an increase in resource idle time. For example, postponement increases the idle time and cost incurred by blocking ground handling staff and parked aircraft.

- *Scope:* Modification of the state of the system to mitigate risk
- *Reason:* Applies to both changes in the environmental and system context

- *Trade-off*: Leads to a decrease in system throughput and increase in resource idle time
- *Synonyms*: Add control dependency (Weber et al., 2008)
- *Evidence*: Postponement of boarding procedures in a thunderstorm to enforce occupational health and safety regulations (BTA)

Wait before completing function (CAF-05). Finally, systems may decide to wait before completing a specific function in a given context. The implementation of this strategy mitigates risks that arise from completing the function too early (e.g., missing outputs). It is evidenced in (1) the mandatory coding of weather-related claims during claims lodgement (SGM) and (2) the temporary accommodation of passengers in the departure lounge facilities during a thunderstorm. The implementation of the strategy has adverse effects on system throughput and the utilisation of resources as a result of decreasing the number of events that are processed by a resource per unit of time. For example, lodgement consultants must spend additional time for coding and terminal infrastructure is blocked.

- *Scope*: Modification of the state of the system to mitigate risk.
- *Reason*: Applies to both changes in the environmental and system context
- *Trade-off*: Leads to a decrease in system throughput and increase in resource utilisation
- *Synonyms*: Add control dependency (Weber et al., 2008)
- *Example*: Catastrophe coding of weather-related claims to comply with government guidelines (SGM); Temporary accommodation of passengers during storm (BTA)

Table 6.5 summarises the five adaptation strategies for the function and control view of information systems. Each strategy is briefly summarised and mapped to empirical evidence from the exploratory case study presented in Chapter 4. Furthermore, an illustration of implementing a strategy is provided based on the syntactical constraints and semantics of a contemporary process modelling grammar (Object Management Group, 2011). Illustrations of adaptation strategies are non normative and are provided for additional clarity. Implementation of a strategy may differ based on the capabilities of a grammar.

Adaptation strategy	Description	Evidence		Illustration as flow-chart
		SGM	BTA	

Adaptation strategy	Description	Evidence		Illustration as flow-chart
Perform function fragment (CAF-01)	Perform a set of actions once or repeatedly in a given context in parallel to the normal course of action	<i>News reads and advertising to communicate claims process in a weather event</i>	<i>Advertising and information displays to communicate check-in time before major holiday</i>	
Perform alternative functions (CAF-02)	Select one among a number of alternative courses of action depending on context	<i>Separate call handling procedures for different contexts (storm, flooding, fire, normal loss)</i>	<i>Separate check-in and boarding procedures for different contexts (normal vs. groups)</i>	
Skip function (CAF-03)	Skip a particular function that is part of a normal course of action in a given context	<i>Skipping of proof of loss and proof of ownership requirement in weather context</i>	n/a	
Wait before performing function (CAF-04)	Wait before performing a function that is part of the normal course of action in a particular context	n/a	<i>Postponing boarding procedures in thunderstorm context</i>	
Wait before completing function (CAF-05)	Wait before completing a function that is part of the normal course of action in a particular context	<i>Claims lodgment cannot be completed without catastrophe coding</i>	<i>Passenger boarding cannot be completed in storm</i>	

Table 6.5. Adaptation strategies for function and control view

Adaptation strategies of the resource view. The resource view captures the organisational entities responsible for the transformation of inputs and other kinds of resources performing functions in information systems (Bernus, 2006). Adaptation strategies that apply to this view can modify two parameters: (1) the number of resources allocated to the processing of inputs in a system and (2) the types of resources (in terms of their skills or expertise) that are utilised by it. By modifying these parameters, a system may (1) achieve an increased throughput by increasing the number of resources that are of the same type or (2) modify the type of the resources and their allocation to functions in order to balance their workload.

Modify number of resources (CAR-01). First, context changes can prompt systems to allocate additional resources of the same type to a specific function. The strategy leads to an increase in the average throughput of systems by increasing their capacity to process inputs. It is evidenced by (1) recruiting and allocating temporary resources to call handling functions following weather events (SGM) and (2) allocating additional security screening resources and customs officers before public holidays or following changes in terminal peak hours (BTA). Implementation of this strategy leads to an increase in the cost of coordinating work between the existing and new resources. For example, the new resources must be trained, integrated into work units, and released after the context change.

- *Scope:* Modification of the state of the system to increase throughput
- *Reason:* Applies to both changes in the environmental and system context
- *Trade-off:* Increases coordination cost as a result of additional resources
- *Synonyms:* n/a
- *Evidence:* Allocation of temporary call handling resources following a weather event (SGM); Allocation of additional security screening resources and border protection resources before public holidays or following a change in terminal peak hours (BTA)

Modify type of resources (CAR-02). Second, context changes can prompt systems to allocate different types of resources to perform a specific function. The use of this strategy leads to an increase in throughput by either (1) balancing the workload between less or more qualified resources or (2) reallocating resources from other processes and functions. It is evidenced by (1) reallocating resources from sales or marketing processes to call handling functions following weather events (SGM) and (2) allocating management resources to perform routine security screening functions following a traffic event and drop in terminal throughput (BTA). Its imple-

mentation leads to opportunity cost as reallocated resources are unable to perform their routine tasks and thereby impact on other processes.

- *Scope*: Modification of the state of the system to increase throughput
- *Reason*: Applies to both changes in the environmental and system context.
- *Trade-off*: Incurs opportunity cost as a result of reallocating resources from other processes and functions
- *Synonyms*: Resource redeployment (van der Aalst et al., 2007)
- *Evidence*: Reallocation of sales and marketing resources to call handling functions following weather events (SGM); Reallocation of managers to security screening functions following traffic events and drop in terminal throughput (BTA)

Table 6.6 summarises the two adaptation strategies for the resource view of information systems. Each strategy is briefly summarised and mapped to empirical evidence from the exploratory case study presented in Chapter 4. Furthermore, an illustration of implementing a strategy is provided based on the syntactical constraints and semantics of a contemporary process modelling grammar (Object Management Group, 2011). Illustrations of adaptation strategies are non normative and are provided for additional clarity. Implementation of a strategy may differ based on the capabilities of a grammar.

Adaptation strategy	Description	Evidence		Illustration as flow-chart
		SGM	BTA	
Modify number of resources (CAR-01)	Increase or decrease the number of available resources in a particular context	<i>Allocation of temporary resources to call handling in a weather event</i>	<i>Allocation of additional security screening staff before major holiday</i>	

Adaptation strategy	Description	Evidence		Illustration as flow-chart
Modify type of resources (CAR-02)	Modify the type of resources allocated to perform a function in a particular context	<i>Allocation of sales or marketing staff to call handling in weather event</i>	<i>Allocation of management staff to security screening in traffic event</i>	<pre> graph TD subgraph Resource_1 [Resource 1] Start1(()) --> Task1[Task] Task1 --> End1((())) end subgraph Resource_2 [Resource 2] Start2(()) --> Task2[Task] Task2 --> End2((())) end subgraph Context [Context] Start1 Start2 end </pre>

Table 6.6. Adaptation strategies for resource view

Adaptation strategies of the data view. The data view specifies the data entities processed by information systems and the messages systems exchange with other systems in their environment. Adaptation strategies that apply to this view modify two parameters: (1) the quality of data captured, stored, and transmitted by the functions of an information system and (2) the point in time at which data is gathered in system processes. By modifying these parameters, information systems achieve an increase in system throughput as a result of (1) decreasing the quality of data that is being captured and thereby reducing the time required to complete a function or (2) postponing time-consuming data capture to later points in a process.

Gather different data (CAD-01). First, context changes can prompt systems to capture different data in a specific function. The use of this strategy leads to an increase in throughput as data capture and storage time is reduced for the function. It is evidenced by changes to call handling scripts and the types of questions asked by call centre consultants to policyholders in the Sigma Group case. Changing the types of questions that are being asked allows the insurer to reduce the time required to capture information by asking less questions. No evidence for this strategy could be found in the Beta Corporation case. This absence can be explained by security regulations that specify what data that must be captured. Use of the strategy leads to increased cost as a result of sub-optimal decisions.

- *Scope:* Modification of the state of the system to increase throughput
- *Reason:* Applies to both changes in the environmental and system context.
- *Trade-off:* Increases processing cost as a result of sub-optimal decisions
- *Synonyms:* Data degradation (van der Aalst et al., 2007)

- *Evidence:* Changes to call handling scripts and the types of questions asked by call centre consultants (SGM)

Gather data at a different time (CAD-02). First, a context change can prompt a system to perform a function or process fragment to modify the state of an entity in the system environment. The use of this strategy leads to a reduction in time of front-office functions but leads to an increase in workload in back-office functions. This is evidenced by the reduction in the time spent capturing data during a customer call to quickly lodge claims and the capturing of certain pieces of information such as evidence for loss and ownership at later points in the process. Again, no evidence for this strategy could be found in the Beta Corporation case, which can be explained by security regulations. The use of strategy increases the risk of detecting issues such as fraud only late in a work process.

- *Scope:* Modification of the state of the system to increase throughput
- *Reason:* Applies to both changes in the environmental and system context.
- *Trade-off:* Increases processing cost as a result of detecting issues late in the process
- *Synonyms:* Deferred data gathering (van der Aalst et al., 2007)
- *Evidence:* Reduction in the time spent capturing data during a customer call to quickly lodge claims (SGM)

Table 6.7 summarises the two adaptation strategies for the data view of information systems. Each strategy is briefly summarised and mapped to empirical evidence from the exploratory case study presented in Chapter 4. Furthermore, an illustration of implementing a strategy is provided based on the syntactical constraints and semantics of a contemporary process modelling grammar (Object Management Group, 2011). Illustrations of adaptation strategies are non normative and are provided for additional clarity. Implementation of a strategy may differ based on the capabilities of a grammar.

Adaptation strategy	Description	Evidence		Illustration as flow-chart
		SGM	BTA	

Adaptation strategy	Description	Evidence		Illustration as flow-chart
Gather different data (CAD-01)	Decrease amount of data captured by functions in a particular context	<i>Changes to call handling scripts and data requirements in weather context</i>	n/a	
Gather data at a different time (CAD-02)	Postpone the point in time at which data is captured by functions in a particular context	<i>Capture missing data in back-office functions by outbound communication</i>	n/a	

Table 6.7. Adaptation strategies for data view

Adaptation strategies of the output view. Finally, the output view captures all flows of tangible and intangible inputs and outputs between information systems and entities in the environment (Bernus, 2006). Adaptation strategies can modify two parameters: (1) the quality of inputs received or outputs produced by a system such as increasing the number of outputs that are processed in a standardised and automated manner and (2) the type of inputs received and outputs produced by a system. By modifying these parameters, an information system can (1) achieve an increased throughput by delivering outputs with less human intervention or (2) replace outputs it cannot deliver in a particular context by equivalent outputs it is able to deliver in that context.

Modify output quality (CAO-01). First, a context change can prompt a system to produce outputs in a different way such that the quality of outputs is reduced to essential properties. Use of the strategy leads to an increase in system throughput as functions that are not considered essential are abandoned in a particular context and outputs are produced in a streamlined fashion. The strategy is evidenced by (1) increasing the number of claims that are processed in an automated manner as opposed to by a claims handler (SGM) and (2) reducing the number of

special arrangements such as seating that can be made by passengers (BTA). The implementation of this strategy has a negative effect on customer experience. For example, individual requirements of policyholders cannot be considered.

- *Scope*: Modification of the state of entities in the system environment.
- *Reason*: Applies to both changes in the environmental and system context.
- *Trade-off*: Decrease in customer experience and long-term impact on advocacy
- *Synonyms*: n/a
- *Evidence*: Streamlining claims handling by increasing the amount of automatically handled claims (SGM); Reducing the number of special arrangements such as seating passenger can make (BTA)

Modify output type (CAO-02). Second, a context change can prompt a system to produce different outputs in a particular context as opposed to those that it produces in other contexts. Use of the strategy allows the system to continue delivering outputs in situations in which it would otherwise would have to cope with a decrease in the outputs it delivers. This is evidenced by (1) cash settlement of claims as opposed to the repair or replacement of lost items in a weather event (SGM) and (2) the provisioning of temporary accommodation to passengers in case of a flight cancelation that can be the result of, e.g., equipment failure or weather (BTA). Implementing the strategy has negative effects on customer experience. For example, cash settlement requires policyholders to organise the repair or replacement of items themselves.

- *Scope*: Modification of the state of entities in the system environment.
- *Reason*: Applies to both changes in the environmental and system context.
- *Trade-off*: Decrease in customer experience and long-term impact on advocacy
- *Synonyms*: n/a
- *Evidence*: Cash settlement of damaged items as opposed to their repair or replacement (SGM); Temporary accommodation provided to passengers that are affected by a flight cancelation following equipment failure or weather (BTA)

Table 6.8 summarises the two adaptation strategies for the output view of information systems. Each strategy is briefly summarised and mapped to empirical evidence from the exploratory case study presented in Chapter 4. Furthermore, an illustration of implementing a strategy is provided based on the syntactical constraints and semantics of a contemporary process modelling grammar (Object Management Group, 2011). Illustrations of adaptation strategies are

non normative and are provided for additional clarity. Implementation of a strategy may differ based on the capabilities of a grammar.

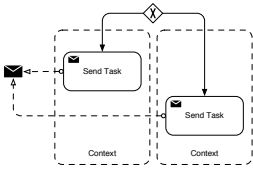
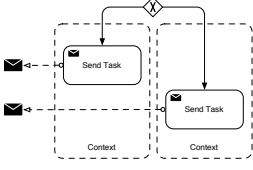
Adaptation strategy	Description	Evidence		Illustration as flow-chart
		SGM	BTA	
Modify output quality (CAO-01)	Increase or decrease the quality of outputs produced by functions in a particular context	<i>Streamline claim handling procedure in weather context</i>	<i>Streamline check-in procedure in weather or traffic context</i>	
Modify output type (CAO-02)	Modify the type of outputs produced by functions in a particular context	<i>Cash settlement of claims in weather context</i>	<i>Accommodation of passengers in flight cancellation context</i>	

Table 6.8. Adaptation strategies for output view

Finally, Table 6.9 summarises the mapping of meta-requirements concerning the planned adaptation of work systems to changes in the system context and maps each to the features presented in the discussion above. Each meta-requirement identified in the previous section is mapped to adaptation strategies of the respective information system viewpoint. For example, the meta-requirement to increase preparedness by planning the adaptation of the task variable of work systems is mapped to strategies for adapting the function, control, and output views. Based on the distinction between the cause, focus, and scope of context-driven adaptation (cf. Figure 6.8), system analysts can identify and select an appropriate adaptation option in response to a change in the relevant work system context. This significantly reduces the effort to design, identify, and implement context-driven adaptation.

Meta-requirement	Meta-design	Problem solution
Increase preparedness by defining how tasks are modified in response to a change in context	Output adaptation strategies	Capture the modification of outputs in response to changes in intrinsic and extrinsic variables
	Function adaptation strategies	Capture the modification of functions in response to changes in intrinsic and extrinsic variables
	Control adaptation strategies	Capture the modification of control in response to changes in intrinsic and extrinsic variables
Increase preparedness by defining how technology is modified in response to a change in context	Resource adaptation strategies	Capture the modification of technology in response to changes in intrinsic and extrinsic variables
	Data adaptation strategies	Capture the modification of data output in response to changes in intrinsic and extrinsic variables
Increase preparedness by defining how structure is modified in response to a change in context	Resource adaptation strategies	Capture the modification of responsibility in response to intrinsic and extrinsic work variables
Increase preparedness by defining how people are modified in response to a change in context	Resource adaptation strategies	Capture the modification of assignments in response to intrinsic and extrinsic work variables

Table 6.9. Mapping adaptation meta-requirements to meta-design features

6.5.4 ACTIVATION RULES COMPONENT

Modelling activation rules. Activation rules specify the contextual circumstances under which system adaptation strategies are activated and thereby modify the state of an information system. Activation is the process by which the representation of a real-world system affects the state of that system (Krogstie et al., 2006). In CAIS, the process of activation involves detecting changes in the state of context entities and selecting appropriate adaptation strategies to modify work system behaviour. Most approaches that deal with context-driven adaptation of system behaviour apply rules modelling to define the contextual circumstances in which behaviour is adapted (Hallerbach et al., 2009; Ramesh, Jain, Nissen, & Xu, 2005). Rules modelling captures the heuristics underlying human decision-making in "business rules" that guide or control organisational behaviour (Steinke & Nickolette, 2003). The IS literature distinguishes between

different types of rules and rule modelling approaches based on the rule structure. Known rule types include integrity rules, derivation rules, reaction rules, production rules, and transformation rules (Muehlen & Indulska, 2010).

I turn to and extend existing work on *rules modelling* in the information systems literature to capture the meta-requirement of context-driven activation in CAIS. Different types of rules logic can be identified in the IS literature (Muehlen & Indulska, 2010). The type of "reaction rule" or "event-condition-action" rule satisfies the meta-requirement of activating adaptation strategies on contextual situations as defined in Section 6.3. Event-Condition-Action (ECA) rules are statements whereby (1) the event part specifies changes in the state of internal or external entities of a system, (2) the condition part specifies conditions that must be met, and (3) the action part specifies modifications that need to be carried out or events that must be raised if the rule evaluates to true (Kovacic, 2004). In CAIS activation rules, the event part relates to the state of entities such as customers or environmental context. The condition part evaluates features of the entities at a given moment in time. Finally, the action part activates adaptation strategies or raises context events.

Figure 6.9 clarifies propositions in Ackoff's (1971) original text as ER model and extends the work system model to capture context activation rules. Work systems specify one or more activation rules (1) to react, respond, or act on events that affect their behaviour. Each activation rule defines an event part which specifies the internal or external event to which it applies (2). For example, an activation rule can "listen" to a change in the state of the environmental context (such as a change in weather). Furthermore, each activation rule defines a condition part which evaluates a set of expressions on the state of entities in the work system or the environment (3). Each condition comprises at least one expression and multiple expressions in a condition are joined by means of Boolean operators. Finally, each activation rule defines an action part which "activates" an adaptation strategy and "raises" an adaptation event to which other rules can register (4). Each entity introduced in the ER model represents a theory-driven extension of WSM on the basis of ECA and system theory.

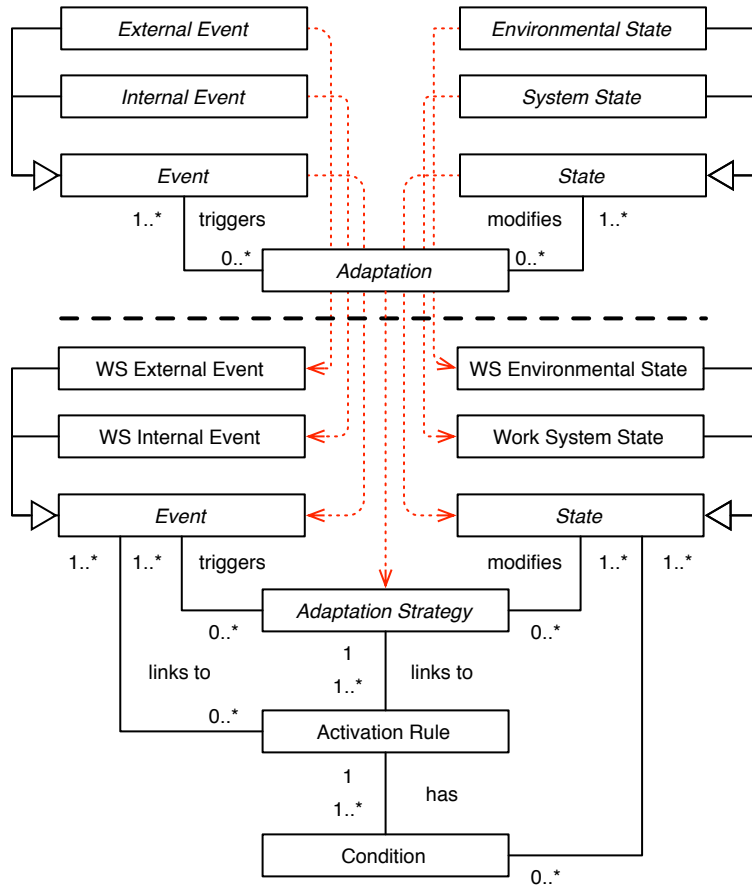


Figure 6.9. ER model of context activation rules

Specifying activation rules. Furthermore, the specification of activation rules in CAIS requires an interface by which such rules are defined and rendered. Several approaches exist, ranging from markup languages (Hebeler, Fisher, Blace, Perez-Lopez, & Dean, 2009) to natural language interfaces (Object Management Group, 2008) for rule definition. In the following, I turn to textual representations of ECA rules (Knolmayer, Endl, & Pfahrer, 2000) to define the interface for specifying context activation rules in CAIS. Each definition of an ECA rule contains three components: the event, condition, and action part. The event part of an ECA rule is denoted by the keyword "on" followed by the definition of the event activating the rule (in angle brackets). I introduce the keyword "on-ctx" as an extension of "on" to differentiate context events from other types of events occurring in information systems. Furthermore, the keyword "if" denotes the condition part of an ECA rule and contains expressions joined by Boolean operators (i.e., "and", "or"). Finally, the action part is specified by the keyword "then" and an optional "else" part can be used for alternative actions.

The following example demonstrates the specification of a context activation rule in extended ECA notation based on interpretations of context-driven activation in the Sigma Group case. A

rule header defines the unique name of the rule such that it can be identified in the set of activation rules specified in a system. In this example, the context rule "severe weather response rule" is evaluated upon occurrence of a state change in the weather context. If the probability of a storm event is high and the risk insured in the affected area is high or medium, then the adaptation strategy "modify call handling script" is activated to change the types of questions asked to inbound callers. The keywords "activate" and "raise" define the strategy to be activated and the type of event to be raised:

CONTEXT ACTIVATION RULE:

"Severe Weather Response Rule"

ON-CTX ⟨storm event⟩

IF ⟨storm probability = 'high'⟩ AND ⟨risk in force = 'medium'⟩ OR ⟨risk in force = 'high'⟩

THEN ACTIVATE (modify call handling script) AND

 RAISE (call handling script modified)

A second example demonstrates the specification of context activation rules in CAIS based on interpretations in the Beta Corporation case. The activation rule is evaluated in case of a change in the traffic context. In case the probability of a congestion on the road network leading to the airport's terminals is high (e.g., on the basis of historical traffic patterns on a workday) and the event occurs during the "peak hour" timeframe, then additional screening points are opened and the procedures for check-in are abbreviated by, e.g., not allowing last minute seating arrangements to be made by passengers. The example also demonstrates the use of an "else" part to activate alternative actions (such as deploying mobile traffic patrols for ongoing monitoring) in case the rule does not evaluate to true:

CONTEXT ACTIVATION RULE:

"Ground Traffic Response Rule"

ON-CTX ⟨traffic event⟩

IF ⟨probability of congestion = 'high'⟩ AND ⟨peak hour = 'true'⟩

THEN ACTIVATE (open additional screening points) AND

 ACTIVATE (streamline check-in procedure)

ELSE /* alternative actions */

Visualising activation rules. Finally, visual mechanisms facilitate the process of capturing, storing, and retrieving large sets of activation rules associated with an information system. The in-

clusion of a broad range of entities in the specification of system context and definition of behavioural adaptation can lead to a large number of activation rules that must be created and maintained. I turn to and extend approaches to store multi-dimensional context information (Adomavicius & Tuzhilin, 2010) and context-driven workflow adaptation (Hallerbach et al., 2009) using techniques known from On-Line Analytical Processing (OLAP). In OLAP, information cubes allow for the efficient storage and visualisation of multi-dimensional information. An information cube is defined as a set of dimensions $D_1, \dots, D_n$ each of which are characterised by a set of attributes $A_{ij} = \{j = 1, \dots, k\}$. For example, the set of dimensions required to capture and store activation rules in CAIS are the entities a change in which triggers rule activation. The Cartesian product $S = D_1 \times \dots \times D_n$ thus defines the space of possible activations.

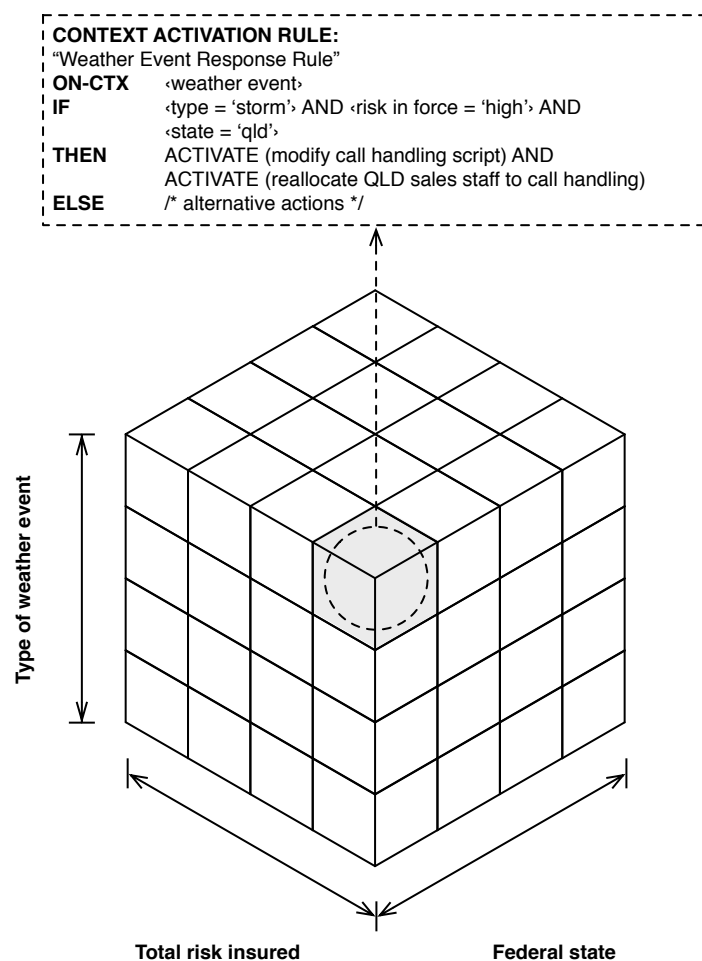


Figure 6.10. Visualisation of activation rules as context cube (Sigma Group case)

The following example illustrates the visualisation of multiple activation rules in a "context cube" based on interpretations in the Sigma Group case. Each dimension of the cube represents one entity in the claims handling system context to which system behaviour must be adapted. For

example, claims handling procedures respond to (1) the type of weather event that has or is predicted to occur at a particular point in time, (2) the insured risk in the area that is exposed to the weather event, and (3) the federal state within whose responsibility the affected area falls. The possible combinations of the attributes of these entities are visualised in Figure 6.10. Each point in the space defined by the three dimensions can store zero to many context activations.

The example presented in Figure 6.11 illustrates multiple activation rules for passenger facilitation based on data in the Beta Corporation case. In this example, passenger facilitation is adjusted based on (1) the type of traffic event that has or is predicted to occur on the airport road network at a particular moment in time, (2) the weekday at which the event occurs, and (3) the time of day at which the event occurs. By adding possible combinations of the attributes of these entities, a system analyst can specify adaptation strategies which are activated in different contextual situations. For example, an incident on the road network during peak hour traffic on a weekday prompts the airport to open screening points to speed up passenger processing.

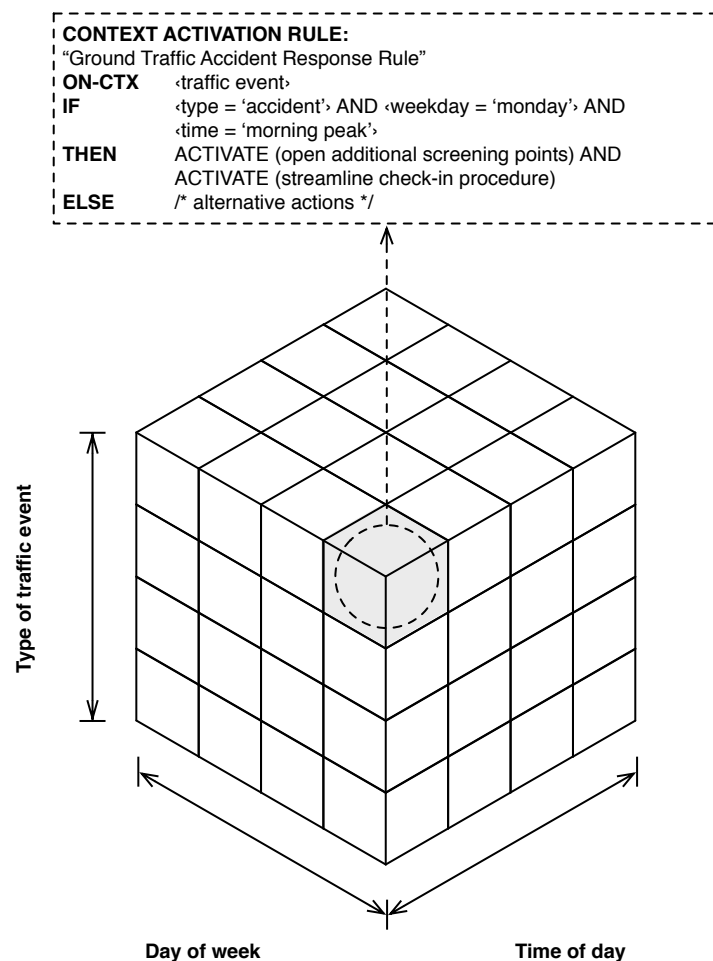


Figure 6.11. Visualisation of activation rules as context cube (Beta Corporation case)

Finally, Table 6.10 maps the meta-requirement concerning the context-driven activation of adaptation strategies to the meta-design features discussed above. Using context activation rules, system analysts can specify the triggers, conditions, and actions to be taken in response to a context change. The implementation of context activation rules in software systems can rely on existing contributions in the IS literature (cf. Kovacic, 2004).

Meta-requirement	Meta-design	Problem solution
Increase responsiveness by defining rules and thresholds for activating strategies in situation	Context activation rules	Capture the activation of adaptation strategies and thresholds on intrinsic and extrinsic variables

Table 6.10. Mapping context activation meta-requirements to meta-design features

6.5.5 LEARNING STRATEGY COMPONENT

Modelling context learning strategies. Context learning strategies capture modifications applied to the representation of system context, system behaviour, and context activation rules over time. Organisational learning is the process by which organisations articulate and amplify new knowledge developed by individuals that are engaged in it (Nonaka, 1994). Generally, two processes of organisational learning can be identified that are of relevance for the design of context-aware information systems (Argyris & Schon, 1978). First-order or single-loop learning describes the process by which individuals, groups of people, and organisations modify their actions in accordance with the difference in expected and obtained outcomes. For example, participants in work systems may learn to modify actions in a given work context by replacing them with different actions in the same context. Second-order or double-loop learning, on the other hand, describes the process by which these entities question the assumptions underlying their actions such as changing selections of context factors and context adaptation strategies.

With respect to the context-aware adaptation of information systems, the objective of learning is (1) to iteratively improve the representation of context in a system and (2) to adjust adaptation and activation strategies accordingly. Following the distinction between two types of learning, this maps to four learning strategies for improving and transforming context-aware information systems. First, single-loop learning can improve context representation by adjusting the information sources that lead to the activation of context-driven adaptation. Second, single-loop learning can also adjust the adaptation strategies defined in a system under constant conditions by modifying the rules of how they are instantiated. Third, double-loop learning can question the assumptions underlying the representation of context in a system, introduce new context factors, or remove deprecated ones. And fourth, double-loop learning can deprecate adaptation

strategies in a system that no longer lead to preferred outcomes and introduce new strategies of adaptation that lead to outcomes of relevance to the system's stakeholders.

Figure 6.12 visualises the *selection of context learning strategies* based on (1) the type of learning outcome that is desired and (2) the artefact to which learning in a context-aware information system applies. Nonaka (1994) emphasises the importance of creating and articulating new knowledge by crafting schemas or "mental models" of the world. Single-loop learning can thus be understood as modifying the rules according to which these models are "activated" and influence the reality of a system (Krogstie et al., 2006). Double-loop learning, on the other hand, does not modify activation of models but reevaluates the assumptions underlying a model and thus modifies its structure. Accordingly, a distinction can be made between strategies that lead to changes (1) to the rules governing instances of a model and (2) to the structure of the model itself. This maps to the requirement to create "models" of system context and adaptation strategies and to define rules according to which they can be activated. The relationship between learning strategies is expressed below.

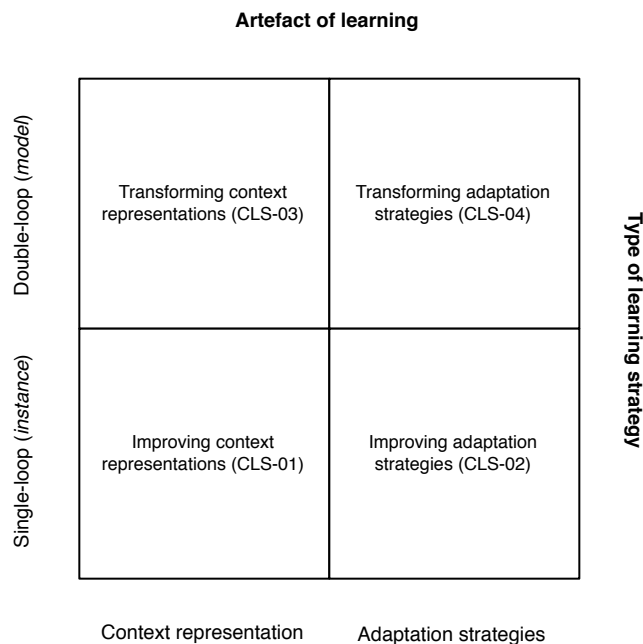


Figure 6.12. Selection of CAIS learning strategies

In the following, I introduce each of the four context learning strategies derived from theories of organisational learning and the types of artefacts in context-aware information systems. For each strategy, I define possible reasons that trigger learning processes and specify the type of learning and artefact to which learning applies. Furthermore, I provide empirical evidence for strategy implementation and point to the relevant literature on tools and techniques where this is applicable:

Improving context representations (CLS-01). Improving context representations is a context learning strategy that modifies context information sources and activation thresholds to increase the fidelity of context information. This is evidenced by (1) the modification of return call metrics as a trigger for context-driven adaptation of claims handling (SGM) and the integration of mobile traffic patrol information to increase the visibility of traffic flow in passenger facilitation (BTA). Most contributions in the IS literature focus on the data mining techniques to improve context adaptation strategies. However, contributions in the mobile and pervasive computing literature (Dey, 2001) define mechanisms by which some of the sensory load of detecting context can be "offloaded" onto devices.

- *Learning type:* Modifies context information sources and thresholds in accordance with differences in expected and obtained context information
- *Learning artefact:* Applies to rules governing activation of context models in CAIS
- *Implementation:* Context toolkit (Dey, 2001)
- *Evidence:* Modification of return call metrics to trigger adaptation of claims handling procedures (SGM); integration of traffic information from mobile traffic patrols to increase visibility of traffic flow in passenger facilitation (BTA)

Improving adaptation strategies (CLS-02). Improving context representations is a context learning strategy that modifies the parameters of context adaptation strategies in accordance with the difference between expected and obtained outcomes. This is evidenced by (1) the modification of cash settlement and builder allocation rules in accordance with differences in claims handling cost (SGM) and the increase or decrease of security screening resources allocated to passenger facilitation in different contexts (BTA). The use of data mining techniques to detect context-driven variations in system outcomes has received some attention in the IS literature (Ghattas et al., 2010; Günther, Rinderle-Ma, Reichert, van der Aalst, & Recker, 2008; Ramos & Santoro, 2010) and supports the optimisation and implementation of improved adaptation strategies.

- *Learning type:* Modifies parameters of context adaptation strategies in accordance with differences in expected and obtained outcomes
- *Learning artefact:* Applies to rules governing activation of adaptation models in CAIS
- *Implementation:* Context mining (Ghattas et al., 2010; Ramos & Santoro, 2010)

- *Evidence:* Modification of cash settlement and builder allocation rules in accordance with differences in expected and obtained claims handling cost (SGM); increase or decrease in security screening resources to match variations in terminal busy hours (BTA)

Transforming context representations (CLS-03). Transforming context representations is a context learning strategy that questions assumptions underlying the representation of context by identifying new context factors or deprecating old ones. This is evidenced by the context simulation meetings and post-incident review meetings held by system analysts and managers engaged in the claims handling work system at Sigma Group. No formal mechanism for double-loop context learning could be identified in the Beta Corporation case. Although double-loop learning strategies for context representation have received little attention in the IS literature (Rosemann et al., 2008), future work may look at adapting and extending environmental scanning techniques to context (Aguilar, 1967).

- *Learning type:* Questions the assumptions underlying the representation of work system context in CAIS
- *Learning artefact:* Applies to the representation of work system context in CAIS
- *Implementation:* adaptation of environmental scanning techniques (Aguilar, 1967) to context learning
- *Evidence:* Event simulation meetings and post-incident reviews of (SGM)

Transforming adaptation strategies (CLS-04). Finally, transforming adaptation strategies is a context learning strategy that questions assumptions underlying the design of system adaptation by continuously innovating how a system adapts to context. This is evidenced by (1) the implementation of an event leadership team that coordinates the design of event responses (SGM) and (2) the ongoing documentation of standard operating procedures for passenger facilitation and the introduction of new technology such as automated check-in kiosks (BTA). More research is required on how the innovation of adaptation strategies in work systems can be conducted. Some related research has been done on organisational level adaptation in high velocity environments (Brown & Eisenhardt, 1997).

- *Learning type:* Questions the assumptions underlying the representation of work system adaptation strategies in CAIS
- *Learning artefact:* Applies to the representation of work system adaptation strategies
- *Implementation:* adaptation of research on organisational level adaptation in high velocity environments (Brown & Eisenhardt, 1997) to context-driven work system adaptation

- *Evidence:* Event leadership team meetings (SGM); Documentation of standard operating procedures and technology-driven innovation (BTA)

Finally, Table 6.11 maps each meta-requirement concerning the context-driven activation of adaptation strategies to the meta-design features discussed above. System analysts select learning strategies based on the learning outcome that is to be achieved. If the quality of context information or the outcomes obtained by adapting work system variables fall short of expectations, then single-loop strategies can be implemented based on the techniques discussed above. If context models or adaptation plans must be adjusted to match permanent changes in the environment of work systems, then double-loop learning strategies can be applied. More research is required on applying knowledge in the IS reference fields to the problem of double-loop context learning.

Meta-requirement	Meta-design	Problem solution
Increase effectiveness of adaptation plans by measuring discrepancy between plan and outcome	Improving context representations	Improve context representation by comparing expected quality of information with obtained quality
	Improving adaptation strategies	Improve adaptation plans by comparing expected with obtained outcomes of adaptation
Increase relevance of adaptation plans by questioning the parameters that guide selection of plan	Transforming context representations	Transform context representations by questioning the selection of context variables in a model
	Transforming adaptation strategies	Transform adaptation plans by questioning the selection of actions and viewpoints in plan

Table 6.11. Mapping context learning meta-requirements to meta-design features

6.6 TESTABLE DESIGN PROPOSITIONS FOR EVALUATING CAIS

Finally, testable design propositions define assertions which evaluate whether or not the meta-design of ISDT satisfies the design goals specified by the meta-requirements in practice (Walls et al., 1992). For example, testable design propositions for the meta-design feature of process modelling assert that modelling the processes and tasks of work systems leads to an increase in transparency and thus improves communication between system participants such as managers, system analysts, and staff. Similarly, testable design propositions for the meta-design feature of context modelling assert that modelling context variables and interdependencies be-

tween variables in the context lead to increased awareness and thereby reduce the time from when a context change occurs to when information about it becomes available to responsible system participants. The notion of testable design propositions is viewed as a mandatory component in all contemporary descriptions of ISDT (Gregor & Jones, 2007; Kuechler & Vaishnavi, 2012). Their empirical evaluation leads to evidence confirming or rejecting the design theory.

First, CAIS asserts that *process modelling leads to transparency* by creating a shared understanding of important work system variables between members of the system (managers, system analysts, staff). Process modelling has been associated with an increase in awareness and knowledge about organisational processes and the reduction of organisation complexity by depicting the activities performed by work systems and their interdependencies (Bandara et al., 2005). Research in process modelling has led to an increased understanding of the representational capabilities of process modelling grammars (Recker et al., 2010) and guidelines for increasing the quality of process modelling outcomes (Mendling et al., 2010). Based on advances in our understanding of process modelling, system participants are increasingly in a position to select and evaluate appropriate modelling grammars and follow guidelines to improve the syntactical and semantic correctness of process models. This increases the probability of process models leading to increased transparency by allowing for consistent representations.

Second, CAIS asserts that *context modelling leads to increased awareness* by focussing the attention of system members and creating a shared understanding of context variable impact. A shared understanding of context has long been recognised as leading to more effective system design (Nardi, 1996). However, the degree to which context can be presented and encoded has been subject to some debate (Dourish, 2004). Advances in conceptual modelling have led to the creation of context modelling grammars that abstract from the underlying infrastructure used to acquire context information (Bettini et al., 2010) and thereby allow for the shared representation of relevant variables in the context of work systems. While more research is required on the representational capabilities of such context modelling grammars, their application to process modelling has the ability to increase the representational capabilities of such grammars and allow system participants to create, share, and communicate representations of relevant variables in the system context for increased awareness and understanding.

Third, CAIS asserts that *creating a repertoire of adaptation strategies* leads to better preparedness and a reduction in the time from when a context event occurs to when appropriate strategies are identified. The need for more adaptive system design has been recognised for various stages of the system lifecycle from when representations of processes in work systems are created (van der Aalst et al., 2007) to when they are realised in software systems and enacted

by system participants in different contexts (Hallerbach et al., 2009). CAIS recognises the importance of all four variables of socio-technical systems and thereby increases the preparedness of not only the technical but also the social subsystem of work. While many contemporary approaches to defining process flexibility in PAIS only address the technical system of work (Weber et al., 2008), socio-technical design calls for the joint optimisation of both the technical and social systems (Bostrom & Heinen, 1977). The classification scheme defined by CAIS allows system analysts to determine the degree of coverage achieved by adaptation strategies.

Furthermore, CAIS asserts that *defining activation rules reduces the time* from when an adaptation strategy is identified to when it is activated in the system by automating recurring decisions. The activation of models in systems to take action has been the subject of some research in the IS field (Krogstie et al., 2006). Three types of activations can be identified by which system participants enact their reality. Manual activation relies on the cognitive abilities of system participants to interpret and enact models prescribing actions. Partially or fully automated activation, on the other hand, externalises some portion of model activation by means of software systems. This leads to an increased effectiveness in the routing of tasks and transparency of progress in work systems (Dumas et al., 2005) and opens the potential to automate the heuristics applied by humans in decision-making situations in the form of business rules (Muehlen & Indulska, 2010). The application of rules modelling to the context-driven activation of work system adaptation allows for the partial automation of routine decisions made by system participants.

Finally, CAIS asserts that *supporting both single-loop and double-loop strategies* of learning increases effectiveness and relevance of context-aware information systems over time. The importance of capturing information about the performance of work systems in the two cycles of organisational learning have long been recognised in the process management and IS literature (Davenport & Beers, 1995). Still, many tools focus on single-loop learning by applying data mining tools to the problem of finding variances in system performance in different contexts (Ghataas et al., 2010; Ramos & Santoro, 2010). CAIS recognises the importance of both single and double-loop learning and thereby increases not only system effectiveness but also the renewal and adaptation of context and system representations over time.

Table 6.12 summarises each of the testable design propositions discussed above in a table and relates them to features in the meta-design of CAIS as described in the previous section. Each testable design proposition maps to exactly one meta-design feature in the CAIS design theory. Each testable design proposition, in turn, is an assertion or truth statement (Gregor & Jones, 2007) about the validity of CAIS that can be empirically evaluated.

CAIS meta-design feature	Testable design proposition	Mapping rationale
SYSTEM		
<i>Output view</i>	Modelling output variables leads to a measurable increase in system transparency and learning	Modelling creates shared understanding of output variables and improves communication
<i>Control view</i>	Modelling task control leads to a measurable increase in system transparency and learning	Modelling creates shared understanding of task and task control and improves communication
<i>Resource view</i>	Modelling resource variables leads to a measurable increase in system transparency and learning	Modelling creates shared understanding of system resources and improves communication
<i>Data view</i>	Modelling information variables leads to a measurable increase in system transparency and learning	Modelling creates shared understanding of information entities and improves communication
SYSTEM CONTEXT		
<i>Immediate context view</i>	Modelling immediate context variables leads to a measurable increase in context awareness	Modelling creates shared understanding of immediate context and its impact on work system
<i>Internal context view</i>	Modelling internal context variables leads to a measurable increase in context awareness	Modelling creates shared understanding of internal context and its impact on work system
<i>External context view</i>	Modelling external context variables leads to a measurable increase in context awareness	Modelling creates shared understanding of external context and its impact on work system
<i>Environmental context view</i>	Modelling environmental context variables leads to a measurable increase in context awareness	Modelling creates shared understanding of environment and its impact on work system
ADAPTATION STRATEGIES		
<i>Output adaptation plan</i>	Defining output adaptation plans leads to a measurable increase in preparedness for context change	Specifying boundaries of output adaptation improves preparedness and analysis latency
<i>Control adaptation plan</i>	Defining control adaptation plans leads to a measurable increase in preparedness for context change	Specifying boundaries of task adaptation improves preparedness and analysis latency

CAIS meta-design feature	Testable design proposition	Mapping rationale
<i>Resource adaptation plan</i>	Defining resource adaptation plan leads to a measurable increase in preparedness for context change	Specifying boundaries of resource adaptation improves preparedness and analysis latency
<i>Data adaptation plan</i>	Defining data adaptation plans leads to a measurable increase in preparedness for context change	Specifying boundaries of data adaptation improves preparedness and analysis latency
ACTIVATION RULES		
<i>Context activation rule</i>	Defining activation rules leads to a measurable increase in responsiveness to context change	Automating routine decisions improves work system responsiveness to context change
LEARNING		
<i>Improving context representations</i>	Defining strategies for improving context representations improves system effectiveness over time	Single-loop learning improves effectiveness by comparing outcomes and modifying actions
<i>Improving adaptation strategies</i>	Defining strategies for improving adaptation plans improves system effectiveness over time	Single-loop learning improves effectiveness by comparing outcomes and modifying actions
<i>Transforming context representations</i>	Defining strategies for transforming context representations improves system relevance	Double-loop learning improves relevance by questioning assumptions underlying actions
<i>Transforming adaptation strategies</i>	Defining strategies for transforming adaptation plans improves system relevance	Double-loop learning improves relevance by questioning assumptions underlying actions

Table 6.12. CAIS testable design propositions

6.7 DISCUSSION

The theory-driven design of CAIS has led me to suggest that they are a class of systems that facilitate the adaptation of work systems to intrinsic and extrinsic variables in the system context. Adaptation is achieved by representing and encoding information about such variables and activating adaptation strategies by modifying the variables classes task, technology, structure, and people. Furthermore, CAIS supports organisational learning by providing the basis on which the adaptation of system variables can be improved and by increasing the speed with which new context variables or system adaptation strategies can be defined.

In the following, I discuss how the CAIS construct enhances our understanding of context-aware system design and improves the theoretical completeness and clarity of artefact instantiation. Context has long been recognised as an important factor in system design (Nardi, 1996) but much of the contemporary literature on context-awareness in IS lacks a clear theoretical foundation (Rosemann et al., 2008). By viewing existing approaches through the lens of CAIS, deficiencies in the theoretical completeness and clarity can be identified and removed. The application of CAIS guidelines will thereby lead to more reliable system instantiations.

First, most contemporary contributions to context-aware design in the IS and CS literatures *focus on people and technology variables* such as user profiles and device properties. Following the classification of context presented in the CAIS meta-design, this results in only a partial coverage of the potential variables that affect work systems. For example, research in pervasive computing focusses on the properties of users, the devices they use to perform actions, and the time and location of device usage (Dey, 2001; Henriksen & Indulska, 2006; Schilit, Adams, & Want, 1994). Research in context-aware recommender systems, on the other hand, focusses on customer profiles, the properties of products or services consumed by these customers, and the time and location of consumption (Adomavicius et al., 2005; Palmisano, Tuzhilin, & Gorgoglione, 2008). Finally, research in process-aware information systems focusses on the properties of a case (Hallerbach et al., 2009) or the profile of resources (Saidani & Nurcan, 2007). The CAIS construct, on the other hand, provides a unifying view of context and the ability to define interdependencies between context layers.

Second, discussions of system adaptation in the PAIS literature often *focus on the task and technology components* of work systems without considering the implications of changes on the people and structure components. Following the mapping of socio-technical theory to CAIS, this leads to a partial adaptation of the technical system and ignores the requirement of joint optimisation in socio-technical systems (Bostrom & Heinen, 1977). For example, Weber et al. (2008) discuss change patterns that modify the task and technology variables of work systems by, e.g., changing the sequence of tasks in a process model or allowing for user-defined tasks to be inserted at selected points in the model. Hallerbach et al. (2009) discuss the context-driven configuration of process models by activating or deactivating certain options in process models based on the current context. However, they do not consider the implications of changes on people (e.g., fatigue or overwork) and structure variables (e.g., role definitions and training). In addition, more research must be conducted on the issue of adaptation trade-off.

Finally, much of the research done on context learning *focusses on single-loop learning strategies* by applying tools such as data mining but does not consider the relevance of double-loop

learning strategies. Research on the application of data mining to context learning (Ghattas et al., 2010; Ramos & Santoro, 2010) assumes that a fundamental issue of context-aware design is the lack of knowledge about relevant context variables. Accordingly, data mining techniques are suggested as tool to measure context-based deviations of processes and thus deduce the influence of context variables on process execution. However, empirical evidence in the multiple case study in Chapter 4 suggests that context-aware design is not (only) challenged by a lack of knowledge or information but by the failure to correctly interpret this information in the current context. For example, a surge in call volume may be unrelated to weather events and using call volume to trigger context-driven changes to claims handling can lead to false alarms (SGM). More research is needed to better support double-loop context learning strategies.

6.8 CONCLUSION

In this chapter, I developed a set of prescriptions consisting of five meta-requirements for CAIS, their corresponding meta-design features, and testable design propositions for their evaluation in practice. Each meta-requirement derives from design-relevant theory developed in Chapter 05 and refines these propositions to a set of actionable prescriptions based on which context-aware information systems can be constructed. These prescriptions are as follows. First, adaptation to context must consider both the technical and social component parts of information systems. Second, rapid system adaptation relies on predictable context and the modelling of context information. Third, adaptation strategies must be planned and evaluated in terms of their trade-off to increase the preparedness of systems with respect to context change. Fourth, adaptation and context models must be combined in the form of activation rules to define the thresholds and conditions under which system change is activated. Finally, learning strategies ensure context-driven adaptation is continuously improved and reevaluated.

Chapter 7: Theory Evaluation

7.1 INTRODUCTION

In this chapter, I evaluate the *utility of the design theory* of context-aware information systems (CAIS) by testing its prescriptive guidelines on the basis of data collected and presented in Chapter 04. In Chapter 06, I used theory-driven design to derive prescriptive guidelines for context-aware information systems from the behavioural classification of systems in Ackoff's (1971) system of systems concept. This lead me to conclude that CAIS are a class of systems that can be characterised by five distinct properties or "meta-design features" (Walls, Widmeyer, & Sawy, 1992). Applying these meta-design features to the empirical data that I collected and presented in Chapter 04 allows me to test whether the CAIS design theory generates net new insights into context-awareness and the adaptive behaviour of information systems. These insights have the potential to provide utility to system practitioners as they design, construct, and implement context-aware information systems. I define utility and place it among other important variables of DS in IS: the relevance and novelty of research studies.

In particular, I turn to the utility variables applicability and actionability to address the challenge of evaluating the properties of *classes of systems* as opposed to the properties of *system instances*. Much of the literature on DS evaluation has thus far focussed on expository instantiations (Gregor & Jones, 2007) as a vehicle for design evaluation. This is, in particular, evidenced in the field's most prominent treatise of the topic by Hevner *et al.* (2004) and Hevner (2007). Expository instantiation creates system instances from design theories which can then be tested and evaluated using empirical method such as focus groups (Tremblay, Hevner, & Berndt, 2010) or experiments (Arazy, Kumar, & Shapira, 2010). However, such evaluation generates statements about the quality of system instances but not the quality of the design theory itself (Baskerville & Pries-Heje, 2010). By reviewing recommended utility variables in the academic literature, I turn to applicability (Rosemann & Vessey, 2007) and actionability (Kuechler & Vaishnavi, 2011) as measures that can be applied to both the design theory level (system class) and artefact level (system instance).

The measurement of design theory applicability and actionability leads me to conclude that CAIS serves as a conceptual foundation for the *comparative analysis* of context awareness in work systems. The five meta-design features identified in Chapter 06 and their occurrence in empirical data collected across two case sites presented in Chapter 04 allows me to generate

concrete recommendations regarding the effectiveness of system design strategies followed by the two organisations. Furthermore, by comparing the strategies on the basis of CAIS, statements can be generated regarding the relative effectiveness and efficiency of a particular strategy. For example, investments into event monitoring and response systems as evidenced by data collected at one case site prove to be more effective than ad hoc and manual processes of information acquisition as evidenced by data in the second case. I conclude by providing guidelines for empirical validation of these statements and the expository instantiation of CAIS. This closes the cycle of problem identification, design theorising, and utility evaluation.

7.2 RESEARCH APPROACH

Evaluation of design outcomes is a fundamental aspect of the DS cycle of research in the IS discipline. Evaluating design outcomes for their utility in the domain of application is what sets DS apart from the behavioural cycle of research in IS (Hevner et al., 2004). The outcomes of theory-driven design in Chapter 06 represent a design theory, i.e., prescriptive statements about a class of systems that provide a generalisable solution to a class of IS design problems (Kuechler & Vaishnavi, 2012; Walls et al., 1992). Most established methods of design evaluation, however, focus on the expository instantiation (Gregor & Jones, 2007) of design theories in artefacts or the evaluation of relative artefact performance (Hevner et al., 2004). Before proceeding to a discussion of design theory evaluation, I therefore turn to the literature to identify methods of design evaluation and evaluate whether they apply to the level of design theory (Baskerville & Pries-Heje, 2010). Furthermore, I review the literature for utility variables of design theories and then proceed to a description of the procedural approach by which I test the applicability and actionability of CAIS.

7.2.1 EVALUATION CRITERIA

There is a common agreement among IS scholars that DSRIS outputs (i.e., design theories and artefacts) must be *relevant*, make *novel* contributions, and create *utility* in their domain of application. In their seminal contribution, Walls *et al.* (1992) argue that design theories deal with purposeful behaviour or goals, an aspect that is missing from behavioural research and natural or social science theories. According to the authors, design theories in the IS discipline have the goal of prescribing ways in which more effective information systems can be constructed. The requirement of goal-orientation is missing in classic descriptions of theory in the natural and social sciences (Kuhn, 1996). Similarly, Hevner *et al.* (2004) argue that utility is what sets the design science cycle of research apart from behavioural research. The authors continue to define the relevance of the problem addressed by DSRIS and novelty of the contribution given

the current state of knowledge in the literature as relevant measures for accepting DSRIS outputs. Relevance, novelty, and utility are thus key evaluation criteria.

Table 7.1 relates key evaluation criteria of DSRIS as discussed in the literature to issues in my research enquiry. Relevance measures the degree to which DSRIS research and outputs are likely to generate interest among the intended audience. Two such audiences can be identified: the scholarly and practitioner discussion of context in work systems and work system design. In my enquiry, this issue has been addressed by the collection and analysis of empirical evidence covering context in work systems as presented in Chapter 04. Second, novelty measures the contribution made by DSRIS research over and above the current body of knowledge in a discipline and research stream. This issue has been addressed by the mapping of system theory to our understanding of work systems in the IS literature in Chapter 05. This mapping revealed a conceptual gap that I addressed through theory-driven design in Chapter 06. In a last step, utility of the CAIS construct in practice remains as an outstanding issue not sufficiently addressed by research outcomes in any of the previous chapters.

Evaluation criterion	Description	Addressed by
Relevance	Relevance of the problem addressed by DS to IS practitioners	• Exploratory case study and data presented in <i>Chapter 04</i> • Empirical evidence of context and context-driven adaptation
Novelty	Novelty of the DS solution to both practitioners and academia	• Theory development and theory driven design of CAIS theory in <i>Chapters 05</i> and <i>06</i> • Mapping and design to address construct deficit in WST
Utility	Value generated by the DS solution by its application to problem	• Theory evaluation and utility measurement in <i>Chapter 07</i> • Applicability and recommendations to address issues

Table 7.1. DSRIS evaluation criteria

In the remainder, I focus on utility and the definition and selection of *utility variables* through which utility of DSRIS outputs can be measured. Although utility is widely accepted as an important attribute of DSRIS outputs, much less clarity exists on the kind of variables that should be selected in order to measure utility. Two kinds of variables classes can be distinguished in the DSRIS literature. First, some measurements operate on the level of artefacts. They measure the utility of instantiations and thus require concrete artefacts to be constructed before

evaluation can proceed. This is evidenced by comments on expository instantiation (Gregor & Jones, 2007) and examples of utility variables in the works of Hevner *et al.* (2004) or Davis (1989) that evaluate technical performance and the ease of use of design artefacts. Second, other measurements operate on the level of design theory and, by consequence, are more abstract. Examples include the applicability of IS and DSRIS outputs to practice (Rosemann & Vessey, 2007) and the actionability of DSRIS output in an application domain (Kuechler & Vaishnavi, 2011).

Table 7.2 summarises examples for utility variables extracted from seminal IS and DSRIS contributions and the variable class to which the variables belong. First, performance relates to the technical performance of design artefacts (Hevner *et al.*, 2004) and includes the measurement of increases in the speed of recording, processing, storing, or displaying information. Second, effectiveness measures the degree to which artefact outputs match user expectation (Hevner *et al.*, 2004). Together with efficiency, which measures the efficient use of technical resources by an artefact, these measures evaluate artefact utility in the technical component of work systems. Usefulness and ease of use, on the other hand, measure aspects of the social system such as the degree to which users accept an artefact and find it useful in the context of task performance (Davis, 1989). Finally, applicability and actionability measure the degree to which DSRIS outputs match practitioner needs, are feasible, and promise improvements (Kuechler & Vaishnavi, 2011). They can be measured on the theory and artefact level.

Utility variable	Variable class	Description	Illustration
Performance	<i>Artefact</i>	Degree to which an artefact delivers superior technical performance	Hevner <i>et al.</i> , 2004
Effectiveness	<i>Artefact</i>	Degree to which an artefact delivers outputs as expected by users	Hevner <i>et al.</i> , 2004
Efficiency	<i>Artefact</i>	Degree to which artefact makes efficient use of technical resources	Hevner <i>et al.</i> , 2004
Perceived usefulness	<i>Artefact</i>	Degree to which user thinks artefact will increase her performance	Davis, 1989

Utility variable	Variable class	Description	Illustration
Perceived ease of use	<i>Artefact</i>	Degree to which user thinks artefact will decrease her task effort	Davis, 1989
Applicability	<i>Design theory, artefact</i>	Degree to which a theory of artefact meets needs of practitioners	Rosemann & Vessey, 2007; Kuechler & Vaishnavi, 2011
Actionability	<i>Design theory, artefact</i>	Degree to which a theory of artefact meets needs of practitioners	Kuechler & Vaishnavi, 2011

Table 7.2. Utility variables in DSRIS literature

As the stated objective of my research is the theory-driven development of prescriptive statements about a *class of systems*, I cannot apply measurements that operate on the level of *artefact*. Performance, effectiveness, efficiency, perceived usefulness, and perceived ease of use operate on the level of design artefact. Measurements of these variables rely on the materialisation of design assumptions in a physical artefact, a process referred to as expository instantiation in the DSRIS literature (Gregor & Jones, 2007). For example, Müller-Wienbergen *et al.* (2011) provide an expository instantiation of their design artefact for creative group support in the form of a database for storing, organising, and retrieving a rich data set containing location-based information. Applicability and actionability, on the other hand, can operate both on the level of design theory and artefact. For example, Germonprez *et al.* (2007; Germonprez, Hovorka, & Gal, 2011) evaluate the applicability of their design theory for tailorable technology in a university setting and provide recommendations how to enact the theory.

7.2.2 EVALUATION METHOD

Further from the discussion of utility and utility variables, I turn to scientific methods of evaluating the utility of DSRIS outputs. Two prevalent views concerning the evaluation of DSRIS outputs can be identified in the literature. First, some scholars argue DSRIS outputs must be instantiated as material artefacts in order to evaluate utility. For example, Gregor and Jones (2007) argue expository instantiation is necessary to understand the utility of design theories. This view is reflected in Hevner *et al.*'s (2004) emphasis on artefact evaluation. Evaluation methods that subscribe to this view include experiments and simulation (Hevner *et al.*, 2004), focus groups (Tremblay *et al.*, 2010), survey instruments (Arazy *et al.*, 2010), and action research (Markus, Majchrzak, & Gasser, 2002). Second, others argue instantiation is an optional part of design theorising (Baskerville & Pries-Heje, 2010) and evaluation should focus on ana-

lytic generalisation (Walls et al., 1992). Examples for this view is Walls *et al.*'s evaluation of the design theory presented in their paper and Germonprez et al's. (2007) work on tailorability.

Table 7.3 summarises examples for evaluation methods extracted from the DSRIS literature and the class of DSRIS output they apply to. First, case studies can be used to confirm design assumptions by testing artefacts in the natural setting in which they are used (Müller-Wienbergen et al., 2011). This includes selecting appropriate sites and defining procedures for field work but generally assumes the existences of some material artefact. Experimentation is another approach of testing artefacts in a controlled clinical environment (Arazy et al., 2010). Other methods include the use of focus groups (Tremblay et al., 2010) or action research (Markus et al., 2002). Sein *et al.* (2011) apply action research (Susman & Evered, 1978) and the use of action research in IS (Baskerville & Wood-Harper, 1998) to a design context. Finally, Walls et al. (1992) and Baskerville and Pries-Heje (2010) propose a process of analytical generalisation by which design theory is evaluated based on prescriptive statements. Examples for this approach can be found in the work of Germonprez *et al.* (2007).

Evaluation method	Output class	Description	Illustration
Case study	<i>Artefact</i>	Confirm proposition by testing artefact in natural setting	Müller-Wienbergen <i>et al.</i> 2011
Experimentation	<i>Artefact</i>	Confirm proposition by testing artefact in controlled clinical setting	Arazy <i>et al.</i> 2010
Focus group	<i>Artefact</i>	Use focus groups for incremental improvement or confirmation	Tremblay <i>et al.</i> 2010
Action research	<i>Artefact</i>	Introduce artefact into setting and iteratively develop proposition	Markus <i>et al.</i> 2002
Analytic generalisation	<i>Design theory, artefact</i>	Evaluate truth value of design theory on the basis of prescriptions	Germonprez <i>et al.</i> 2007, 2011; Walls <i>et al.</i> 1992

Table 7.3. Common DSRIS evaluation methods

As the objective of my research enquiry is the theory-driven development of statements about a *class of systems*, I cannot apply evaluation methods that work on the level of *artefact*. I therefore turn to analytic generalisation (Baskerville & Pries-Heje, 2010; Walls et al., 1992) as method of CAIS evaluation. First, I map prescriptive CAIS statements to evidence collected as part of the case study in Chapter 04. The objective of this step is to demonstrate their "applicability" to

design problems in the case data, not to reevaluate case findings. Second, I measure the "actionability" of CAIS statements by mapping them to ongoing organisational challenges reported to me by respondents. The objective of this step is to generate a set of concrete, actionable recommendations for work system improvement based on CAIS and the observation of challenges defined in the previous step. For example, I map the notion of adaptation strategies to challenges in the coordination of changes to technical and social system components in both work systems. I then define how they can be resolved following statements in CAIS.

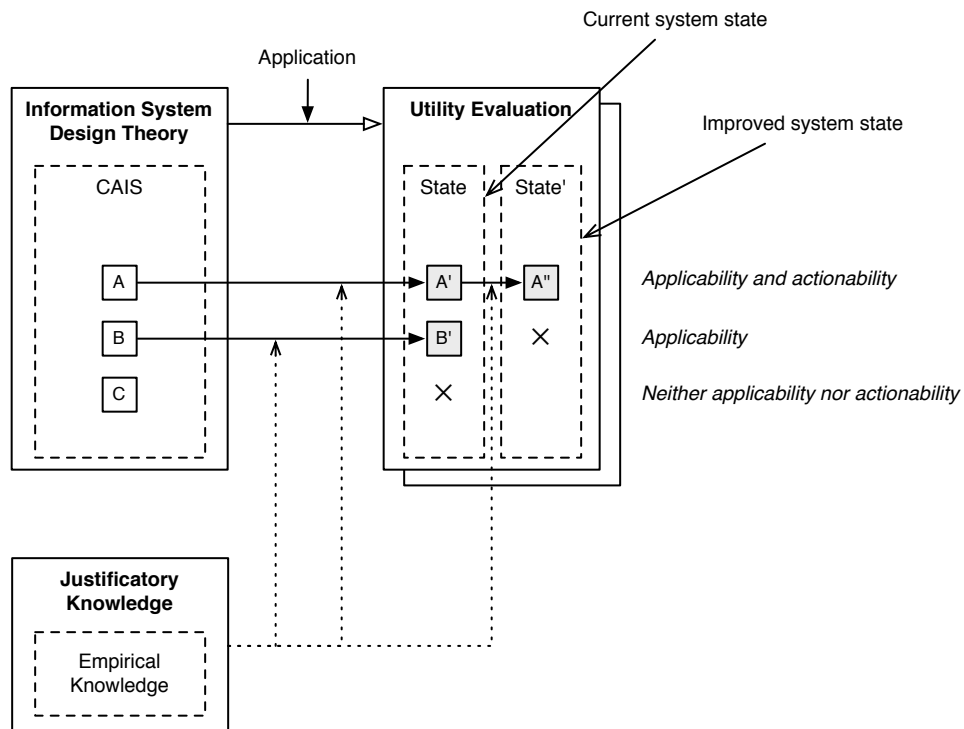


Figure 7.1. Illustration of evaluation process

Figure 7.1 illustrate the process of analytical evaluation based on evidence collected in the multiple-case study presented in Chapter 04. First, I map each meta-requirement defined in Chapter 06 to interpretations of relevant design problems in the case data. For example, I map interpretations of weather or traffic events to the meta-requirement of awareness. In a next step, I then map each meta-design feature defined in Chapter 06 to interpretations of the decisions taken by system members in order to address a particular problem. For example, I map the meta-design feature of context modelling to examples of tracking weather or traffic flow in the case data. Second, I map interpretations of recurring issues with system decisions in the case data to prescriptions of the CAIS design theory. I then define an improve system state by providing recommendations how system decisions can be improved on the basis of CAIS. I

then measure the applicability and actionability of CAIS by evaluating whether the prescriptions provide suitable solutions and recommendations in each case.

7.2.3 EVALUATION VALIDITY

The selection of data to evaluate the applicability and actionability of the CAIS design theory raises the question of *circular reasoning*. A circular argument is one in which the premises on which the argument rests are not separated from its conclusion. For example, a statement that holds that A is true because A is true is a circular one. In the context of the research study, this raises the question whether the evaluation of design propositions using the same data originally used to motivate the proposition is an instance of circular reasoning. However, three reasons speak against this concern and thus the evaluation can be considered valid.

First, the practice of theory-driven design as described in Chapter 06 guarantees that design propositions are derived from sound *academic kernel theory* and are generalisable across the concrete cases observed. The IS design theorising literature defines information system design theories (ISDT) as a set of prescriptions concerning a class of user requirements and a class of system solutions (Markus et al., 2002; Walls et al., 1992). By deriving meta-requirements from academic kernel theories, ISDT ensures problems for which system solutions are proposed rest on a sound theoretical foundation (Kuechler & Vaishnavi, 2012). In the context of the study, this foundation is defined by the concepts and general laws governing the class of open and adaptive systems (Ackoff, 1971). While data were collected and used to motivate the design theory, the purpose of data collection was to find instances of specific phenomena described by the system of systems concept as defined by Ackoff, not to introduce new concepts. Insofar, the propositions made by CAIS are generalisable across all instances of a particular phenomenon.

Second, where insufficient clarity existed in the original text describing the system of systems concept (Ackoff, 1971), the data were used to formulate meta-requirements but *not meta-design features*. The degree of abstraction in academic kernel theories can generally be considered high and insufficient to provide prescriptive guidance to system practitioners (Gregor, 2006). Kernel theory propositions must therefore be translated into the domain of application, i.e., the domain of system design and development (Kuechler & Vaishnavi, 2012). For example, Ackoff's notion of environment, state, and event explains environmental change and its impact on a system but does not provide clarity regarding the selection of context variables. I thus used data collected in the exploratory case study to motivate the meta-requirement of modelling context layers and variables. However, the data does not give indications regarding how

such variables are best selected and defined. On the basis of Ackoff's work, I turned to contextualisation (Rosemann, Recker, & Flender, 2008) and context modelling (Bettini et al., 2010).

Third, the specific approach taken during theory-driven design and evaluation leads to a high likelihood that applications of meta-design features to generalisable problems observed in the data produces the desired effect. First, the data collected in Chapter 04 were used to evaluate the applicability of the design theory to relevant problems reported by case respondents. Using CAIS, I tested whether the actions taken by respondents to resolve a relevant problem can be explained on the basis CAIS. For example, actions taken by respondents with respect to the selection and tracking of weather or traffic variables map to those prescribed by CAIS regarding the selection and tracking of context variables. Second, I used the prescriptions made by CAIS to formulate recommendations regarding how the current actions taken by respondents can be improved. For example, I argued based on CAIS that modelling context variables reveals interdependencies of weather and work system variables and help to identify information requirements. These prescriptions are not apparent on the basis of the collected data alone.

Figure 7.2 illustrates the approach taken in the research study to prevent circular statements during evaluation. First, constructs in kernel theory are measured in empirical data to frame system design problems observed in the exploratory case study presented in Chapter 04 (*arrows 1a, 1b*). The insights gained are then used to develop design hypotheses (Chapter 05) and prescriptions (Chapter 06) regarding user requirements (*arrow 2*) and normative system actions (*arrow 3*). Following this approach, a set of prescriptive statements such as "action A is hypothesised to lead to system effect B" is generated that collectively form the design theory for CAIS. The utility of the CAIS design theory is then evaluated on the basis of its applicability and actionability. This is achieved by finding instances of system effects (*arrows 4a, 4b*) and actions (*arrows 5a, 5b*) that can be explained on the basis of CAIS. In a second step, CAIS is then used to reason about which system actions A would have lead to more effective system outcomes B on the basis of interpretations made by respondents.

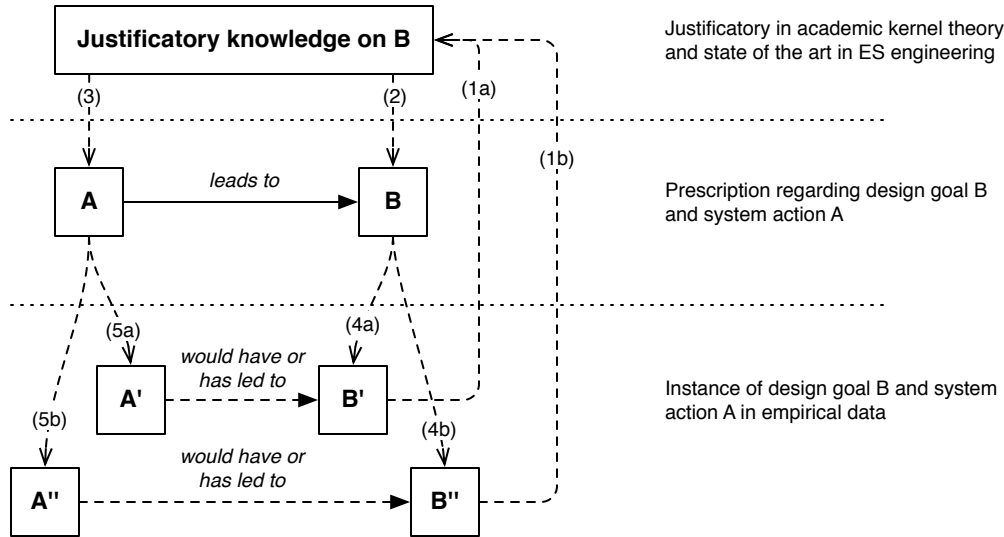


Figure 7.2. Illustration of dependencies between statements

7.3 APPLICABILITY

Applicability is a measure of utility that evaluates whether outcomes of DS address *relevant problems* and *meet the needs* of system practitioners (Rosemann & Vessey, 2007). In the following, I measure the applicability of CAIS by applying CAIS meta-design features to the case data presented in Chapter 04. My objective at this stage is twofold. First, I evaluate the applicability of CAIS meta-requirements to reason about relevant design problems experienced by system practitioners as per the case data. For example, I apply the CAIS meta-requirement of awareness to the problem of monitoring internal and external work system variables as reported by respondents in the case data. Second, based on the description of relevant design problems I then evaluate the ability of CAIS meta-design features to provide suitable solutions to a problem by matching them to design decisions taken by system members. For example, I match the CAIS meta-design feature of context modelling to interpretations of information needs and providing access context information as described by respondents in the case study.

7.3.1 SIGMA GROUP LTD

SYSTEM BEHAVIOUR COMPONENT

Contextual variance in system behaviour is a *relevant problem* for the claims department as evidenced by interpretations of a decrease in service levels and an increase in claims handling cost and leakage in different contextual circumstances. Three key issues emerge. First, weather events (storms, hail, flooding, and bushfire) impact on the number of calls received by the department from customers, the average complexity of claims, and their impact on the

claims backlog. This is evidenced by the following comment of respondent B: "You're handling 2 or 3 times the volume that you normally handle and so you get slower [...] so customers wait longer". Second, community attitudes toward insurance affect customer behaviour, as evidenced by the comment by respondent H: "claims is only one portion of that experience but it tends to be [...] the one that is talked about in the market". Third, the economic climate impacts on the organisation. This is evidenced by respondent B's comment: "because of the global financial crisis we've actually needed more resources than we've really budgeted".

Problem relevance is further evidenced by the following comment of respondent B, the executive manager for claims, on the impact of context variance on the satisfaction and advocacy of customers. She describes claims handling as a "test" of the ability of the organisation to respond to a customer based on context:

"I've bought the product, I was happy with the price and I believe I was adequately covered. The test comes when you put in a claim and we look to get between 1 and 2% differential between people who renew at a set rate, because it's different after a claim", **Respondent B, Sigma Group**

Furthermore, context variance analysis is a *suitable solution* to the problem of variance in behaviour as evidenced by interpretations of the event reporting and system improvement efforts undertaken by the department. First, the claims department measures call handling performance to understand variance in wait times and customer service levels. This is evidenced by the comment of respondent D, one of the managers responsible for the lodgement area: "the way that we measure that if you like, is through how quickly we answer the phone [...] So we have a target around how the percentage of calls that we answer within 60 seconds and then we have some quality type measures". Another concern is the correct routing of claims to claims handlers: "there's a business improvement framework [...] that measures how well or how accurately we path claims". Second, variance analysis depends on context. This is evidenced by respondent F's comment: "We have measurements for the business as usual pool, we've got the same measurements, some of them might be different, for the event pool".

Suitability of analysing system performance in different contexts is further evidenced by the following comment of respondent F, one member of the system design team. She points out the efforts undertaken by the team to extract information from transactional systems to analyse and improve performance:

"We also have a [...] reporting framework, so all the drivers linked to the process are in there [...] we write the report that we base business on, monthly, to actually pull out specific data that pertain to the process and out of that we can pick up a lot of things", **Respondent F, Sigma Group**

Table 7.4 illustrates instances of context-based variance in the department's claims handling system based on evidence collected during the case interviews. Six issues can be identified. First, the call volume experienced by the system increases dramatically in the wake of severe weather events (from 6000 to 20,000 weekly calls). This impacts on the service levels guaranteed by the organisation to its customers and prompts the modification of technical and social system variables. Second, repeat call volume increase over time as a result of a net increase in the department's workload, which further puts stress on the ability to fulfil service levels. Third, claims complexity increases following severe weather events such as storms, hail, flooding, and bushfire, impacting on the availability of qualified claims handlers, claims handling cost, and the potential for leakage. Fourth, information requests from local government and media increase during events. Finally, community attitudes and the economic climate lead to changes in claims behaviour (5) and instances of opportunistic fraud (6).

System variance	Context	Impact of variance on:		
		Goal attainment	Technical system	Social system
1. Volume of phone calls	"If a storm has hit, people generally <u>ring us straight away</u> ", Respondent E	Decrease in service levels	Truncate call handling procedures	Increase staffing levels of call centre
2. Volume of repeat calls	"Customers <u>start calling back again</u> , you know, I have not heard anything", Respondent D	Decrease in service levels	n/a	Increase staffing levels of event team
3. Complexity of claims	"We put them into our case managed team who deal with <u>big, difficult, complex claims</u> ", Respondent H	- Increase in handling cost - Increase in claims leakage	Claims triage matrix	Routing of claims based on expertise
4. Number of information requests from government	"You get <u>pressure from local MPs</u> to say, my constituents are affected, what are you doing to help them", Respondent B	Impact on public relations	- Activate catastrophe coding - Event reporting	n/a

System variance	Context	Impact of variance on:		
5. Number of fraud attempts	"Customers <u>pad</u> their claims when they're in financial hard times", Respondent B	• Increase in handling cost • Increase in claims leakage	• Route claims to forensic claims experts	• Increased vigilance of staff
6. Frequency of claims	" <u>Customers' behaviours have changed</u> so I guess it all relates back to the global financial crisis", Respondent B	• Increase in handling cost • Increase in claims leakage	• Increase policy excess	• n/a

Table 7.4. Context variance analysis

SYSTEM CONTEXT COMPONENT

Context awareness is a *relevant problem* for the claims department as evidenced by interpretations of weather patterns and the efforts undertaken by members of the department to track weather systems. Sudden changes in weather such as a thunderstorm, hailstorm, or flash flooding lead to a surge in call volume and an overall increase in claims on top of the department's current backlog. This is evidenced by the comment of respondent E, the manager that is responsible for the department's specialised event claims handling team: "people generally ring us straight away after a storm's finished, which would be 24 hours later, but when it's flood sometimes they're evacuated and they can't get to their phone line for a few days". The team thus continuously tracks weather information and holds regular meetings to assess the potential impact of events given its current staffing levels and backlog. Three issues emerge. First, service levels cannot be held as a result of call surges. Second, poor service levels lead to an increase in return calls and, third, unanswered calls ultimately affect customer satisfaction.

Problem relevance is further evidenced by the following comment of respondent E, the event team manager, on the handling of inbound weather-related calls and the issue of repeat calls from disgruntled customers. The repeat call matrix relates to a measure of inbound calls that relate to an existing claim:

"You've just got to look at the repeat call matrix, again one of my bugbears is every one uses the repeat call matrix as an indication of the health of claims, are we meeting our SLAs but no one takes the time to actually look and understand why are the calls coming in", Respondent E, Sigma Group

Furthermore, applying context modelling to the problem of context-awareness is a *suitable solution* as evidenced by interpretations of predictable context and the semi-automated response

to weather alerts. The claims department has started investing in an infrastructure which allows team members to subscribe to and receive weather alerts. The alerts are used, in combination with other information sources such as news media and the insurer's branch network, to assess the daily weather situation and the potential impact of weather events on the organisation at the current moment. This is evidenced by respondent A's note on the importance on tracking the current weather context: "every day I log onto the Bureau and I look at what's happening around the company, that's one, if I get an alert message for South East Queensland I'll be on there every hour". At the time of writing, information about weather and system throughput is being tracked manually and correlating events such as storms with information about the system state (current service levels, backlog, exposure) is a manual process.

Suitability of context modelling is further evidenced by the positive response of respondent A, the department's event coordinator, to tracking information about relevant contextual events such as weather. The Bureau relates to information and weather alerts released by the Australian Bureau of Meteorology:

"So, those mechanisms there are really useful and I never had a mechanism before and I found it very difficult to rely on self routine to check the Bureau daily, now, I've got a mechanism that can do it and send me a weather warning to say this is likely", **Respondent A, Sigma Group**

Table 7.5 illustrates the mapping of context modelling to interpretations of call answer time, service levels, net renewal rates, repeat call rates, risk in force, and storm category measures. Call handling performance is an instance of task context, which is an instance of immediate system context. The feature is measured by the caller wait time and cycle time measures which can be extracted from the call centre information system. Customer satisfaction is an instance of customer context, which in turn is an instance of external context. Satisfaction is measured by the metrics return call rate and net renewal rate which can be extracted from the claims management and policy systems respectively. Furthermore, thunderstorm is an instance of weather context, which is an instance of environmental context. The relevance of storm events is measured by the category of a storm event (numbering scheme) assigned by the Bureau of Meteorology and the exposure of the insurer to the event as measured by the risk in force. Context categories are interdependent and impact on each other.

Context feature	Illustration of feature	Representation as context model
TASK CONTEXT		
Call answer time	"We have a target around how the <u>percentage</u> of calls we <u>answer</u> within 60 seconds and then we have some quality type measures", Respondent D	<pre> graph BT CH[Call handling] --> TC[Task context] TC --> IC[Immediate context] CH --- CWT((Caller wait time)) CH --- CT((Cycle times)) </pre>
Service level	"Blows out the <u>SLAs</u> that are in <u>place</u> , so we probably aren't able to contact every customer within the timeframe we told them upfront", Respondent C	
CUSTOMER CONTEXT		
Net renewal rate	"Customers have a <u>high renewal rate</u> , but we need to test with external marketing agencies", Respondent B	<pre> graph BT CS[Customer satisfaction] --> CC[Customer context] CC --> EC[External context] CS --- NRR((Net renewal rate)) CS --- RCR((Return call rate)) </pre>
Repeat call rate	"Every one uses the <u>repeat call matrix</u> as an indication of health of claims", Respondent E	
WEATHER CONTEXT		

Context feature	Illustration of feature	Representation as context model
Risk in force	"I pull out our <u>risk in force data</u> , so that's going into our data-bases and finding out what our risk is in that area", Respondent A	<pre> graph BT EC[Environmental context] WC[Weather context] TS[Thunderstorm] RF([Risk in force]) SC([Storm category]) WC --> EC TS --> WC TS --- L L --- RF L --- SC </pre>
Storm category	"Everybody is watching this <u>category 4 to 5</u> to a 4 to a 3 and it's moving down the coast or it's across the sea", Respondent A	

Table 7.5. Vignette: context model of claims handling system

ADAPTATION STRATEGY COMPONENT

Preparedness is a *relevant problem* for the claims department as evidenced by interpretations of impacts on customer satisfaction, employee engagement, and leakage in the wake of a context change. Sudden changes in context following, e.g., a weather event, impact on the organisation's ability to fulfil service levels it promises to its customers. This is evidenced by respondent C's comment on the impact of surges in call volume following weather events: "that blows out the SLAs that are in place, so there probably aren't able to contact every customer within the timeframe we told them upfront". Sticking to procedures designed for a business as usual context negatively impacts on work system performance, as pointed out by respondent A in his comment: " it was quite in depth, spoke of work, what happened, policy obligations, it was quite lengthy". Context changes also impact on employee engagement and financial performance as a result of workload, as pointed out by respondent B: "when you're less busy and you have got time to think, you look at a claim in detail with a little bit more patience".

Problem relevance is further evidenced by respondent E's comment on the external view of the department by customers and the community. The respondent manages the department's event claims team and points to the reputational risk of poor performance when responding to a severe weather event:

"These guys are getting calls every week to tell them how their claim was going, and that's our biggest risk from a brand point of view, because these guys would ring up ABC and Channel 7 and say, look my roof is still not fixed, they have done nothing about.", **Respondent E, Sigma Group**

Furthermore, applying adaptation strategies to the problem of preparedness is a *suitable solution* as evidenced by interpretations of planned modifications to call handling procedures, staffing levels, and information systems in response to context changes. The claims department routinely monitors the preparedness of its systems with respect to context change. This is evidenced by respondent A's comment: "It's really looking at it as a health check for the business. What's our holdings, what are we doing, that kind of thing, let's mitigate the risks that we have, so that happens monthly". In the event of an impending context change, a range of options can be activated. These include process changes, as evidenced by respondent A's comment: "a lot of process attached to that, a lot of change process sometimes, depending on the type of event". It can also entail changes to staffing levels as evidenced by the following comment made by respondent E: "I would liaise with the recruiting company that provide us with temporary resources so we would be calling them with what they need to provide to us".

Planned adaptations are most apparent in the case of the "event module", an extension designed and developed by the organisation based on its information system. Respondent A, the department's event coordinator and initiator of the event module, notes its ability to centralise and classify claims in an event:

"One central queue where all the claims come into, one central manager for that, rather than spread them out to 4 or 5 different managers and team leaders, so that's another thing it gave us. It also gave us catastrophe coding, which is needed for reinsurance purposes", **Respondent A, Sigma Group**

Table 7.6 illustrates the mapping of adaptation strategies to interpretations of call handling, catastrophe coding, team allocation, and cash settlement changes in the system. First, the call handling strategy enables the rapid lodgement of claims by truncating process steps and reducing information requirements when handling customer interactions in a weather context. Second, the catastrophe coding strategy activates a coding scheme in accordance with guidelines for catastrophe reporting issued by government. This enables the organisation to code losses incurred and reported according to loss categories and thereby to provide timely updates to government. Third, the team allocation strategy modifies the assignment of claims captured and coded by the call centre to responsible teams depending on their experience or capacity. Finally, the cash settlement strategy activates or deactivates information requirements such as the proof of ownership and adjusts the limits for cash settlement to accelerate the handling of minor losses. More than one strategy is typically applied to a context.

Adaptation focus	Description	Representation as adaptation strategy
Call handling	"It gave us the ability to have upfront information to <u>change our process quickly</u> ", Respondent A	• Strategy: CAF-02 (<i>alternative function</i>) • Scope: <i>External variance</i> • Reason: <i>Change in weather context</i> • Trade-off: <i>Less information, more leakage</i>
Catastrophe coding	"It gave us <u>catastrophe coding</u> , which is needed for reinsurance purposes", Respondent E	• Strategy: CAF-05 (<i>wait before completing</i>) • Scope: <i>External variance</i> • Reason: <i>Government reporting requirements</i> • Trade-off: <i>n/a</i>
Team allocation	"Utilise these <u>guys in lower value</u> , less complex claims, food spoilage", Respondent H	• Strategy: CAR-02 (<i>modify resource type</i>) • Scope: <i>External variance</i> • Reason: <i>Increase in complexity and workload</i> • Trade-off: <i>Claims handling cost and leakage</i>
Cash settlement	"Let's <u>cash settle</u> [...] rather than asking people to send stuff in and wait", Respondent F	• Strategy: CAF-02 (<i>alternative function</i>) • Scope: <i>External variance</i> • Reason: <i>Change in weather context</i> • Trade-off: <i>More leakage</i>

Table 7.6. Vignette: adaptation strategies of claims handling system

ACTIVATION RULE COMPONENT

The problem of responsiveness is a *relevant problem* for the claims department as evidenced by interpretations of communication and consistency of changes in the work system. Communication relates to the effective communication of changes to organisational units such that they are implemented with urgency to respond to or anticipate an impending context change. This is evidenced by the following comment made by respondent F, a member of the system design team: "those guys at the front are so overloaded, the volume that keeps coming in. Calls that they have to field, plus they have to make sure that they follow the right rules, you expect those rules to change". Consistency relates to the consistent interpretation and implementation of changes across all organisational units in events. This is evidenced by respondent A's comment on inconsistent communication of changes: "one was telling the customer to throw the contents out and make a list and the other was telling our customers to retain their contents, which really was quite a failing of ours from a customer point of view".

Problem relevance is further evidenced by the following comment of respondent H, the executive manager of claims experience. She points to the "power" of staff in customer interactions and the impact of inconsistent communication of process or product changes on the system:

"However, our staff have amazing power because, if you ring up to say, hypothetically would I be covered for such and such, and they say yes, well regardless of what's in the policy booklet, you are", **Respondent H, Sigma Group**

Furthermore, applying activation rules to the problem of responsiveness is a *suitable solution* as evidenced by interpretations of efforts undertaken by the design team to streamline the communication and implementation of changes by organisational units. Respondent G, a member of the design team, describes her role: "our involvement in events is around the business rules that get changed when events come and how those business rules get applied throughout the business when an event happens and we get involved with that". The team then assesses the impact of events by "zoning" an event, i.e., by pulling policy data from the organisation's policy database for the affected postcodes. This is evidenced by respondent G's following comment: "we do risk in force data [...] this is the postcode zone that is affected and how many risks as in how many potential insured people do we have in that area". Together with the event coordinator, the team then activates the required changes. Respondent A elaborates: "there's about 25 business units that need to respond to an event when I declare it".

The suitability of activation rules as a concept is further evidenced by respondent F's following comment, which describes efforts undertaken by the department to improve the communication of changes. The system she refers to ensures changes, once activated, are consistently implemented by the respective units:

"Hopefully that's one of the things with communication we can solve, the communications response systems, when it goes out, it goes out to all and if it's only pertaining to this business area, we need to understand what's the impact of that in other areas.", **Respondent F, Sigma Group**

Table 7.7 illustrates the mapping of activation rules to interpretations of risk in force and repeat call rate rules as used by the department to respond to context changes. First, the risk in force rule defines the total exposure of the organisation in the postcodes affected by an event and activates the event module strategy for truncating claims lodgement processes to cut call handling time. The rule is activated once a weather alert is issued by the Bureau of Meteorology and the alert stage has been declared. At this stage, risk exposure and the current backlog of the department is evaluated. If the event is found to have a large impact on system performance, then the event module is activated. Second, the repeat call rate rule measures the rate of repeat calls received by call centres that relate to an existing event. The rule alerts work system participants to potential issues with service levels and customer satisfaction. If repeat calls relating to one event are found to exceed a defined threshold, then the recruitment of temporary staff to fill staff shortages in the event team is initiated.

respect to problems in responding to customer needs and exceptions in different contexts and making sure they are resolved:

"I think it's pretty good. You know, with all, I mean the points I've raised so far are more internal operational things that cause us problems, but where the problems are caused within our calls, either by us or we have to deal with it, and we always typically deal with it in the favour of the customer",

Respondent E, Sigma Group

Furthermore, applying context learning strategies to the problem of learning is a *suitable solution* as evidenced by the interpretation of recent changes made to context representation and adaptation strategies. First, changes to the department's event tracking system improved the accuracy and timeliness of context information. This is evidenced by the following comment of respondent A, the event coordinator, about an example for a recent change: "I found it very difficult to rely on self routine to check the Bureau daily, now, I've got a mechanism that can do it and send me a weather warning". Second, improvement of adaptation strategies ensures the effectiveness of changes to system variables. This is evidenced by comments of respondent C on changes to team structure: "we have a plan in place where we have a thing called a hybrid team that work across commitment and fulfilment". Third, the redesign of context models and strategies for adaptation ensures the ongoing relevance of the response system. Respondent E, the event team manager, provides an example: "my bugbears is every one uses the repeat call matrix as an indication of the health of claims [...] but no one takes the time to actually look and understand why are the calls coming in"

Suitability is further evidenced by the following comment of respondent H, the department's executive manager for claims experience, on the requirement of continuous learning and the redesign of claims processes, process controls, and strategies for adapting process controls in certain contextual situations:

"Once the market gets to know and understand how to break the process, you've got to change to put your controls, really strict controls, I'm serious, you put your strict controls back in there and you let something else go through it, so that's my view on how you manage it",

Respondent H, Sigma Group

Table 7.8 illustrates the mapping of learning strategies to interpretations of improvements to event tracking systems, the identification of call centre bottlenecks, and proposals to redesign activation rules. First, timely and accurate information on weather alerts is relevant to improve awareness of an impending event. Learning in this case corresponds to learning strategy CLS-01 (improving context) and is evidenced by respondent A's comment. Second, the timeliness of changes made to call handling processes and staffing levels in anticipation of an event im-

proves preparedness. This corresponds to learning strategy CLS-02 (improving adaptation) and is evidenced by the comment of respondent A. Third, false positive relate to issues in the activation of adaptation strategies in anticipation of or in response to context changes. Learning in this case can either seek to improve the existing context representation (CLS-01) or may define a new context (CLS-03). The latter maps to interpretations provide by respondent E on using the repeat call rate as a measure of system workload.

Learning issue	Illustration of issue	Representation as learning strategy
Weather alerts	"I found it very difficult to <u>rely on self routine</u> to check the Bureau daily", Respondent A	- Strategy: <i>CLS-01 (improving context)</i> - Learning type: <i>Single-loop context learning</i> - Learning artefact: <i>Context information sources</i> - Implementation: <i>Weather subscription feeds</i>
Call centre bottleneck	"If we <u>changed our process at the first re-sponse stage</u> we would not be in such a bad sit-uation", Re-spondent A	- Strategy: <i>CLS-02 (improving adaptation)</i> - Learning type: <i>Single-loop context learning</i> - Learning artefact: <i>Staff planning and rostering</i> - Implementation: <i>Changes to staffing model</i>
False positives	" <u>One of my bugbears</u> is every one uses the repeat call matrix as an indication of health of claims", Respondent E	- Strategy: <i>CLS-03 (transforming context)</i> - Learning type: <i>double-loop context learning</i> - Learning artefact: <i>Repeat call indicator</i> - Implementation: <i>Redesign of indicator</i>

Table 7.8. Vignette: context learning strategies

7.3.2 BETA CORPORATION

SYSTEM BEHAVIOUR

Contextual variance in system behaviour is a *relevant problem* for the terminal management unit as evidenced by interpretations of a decrease in service levels, the possibility of delays, and the impact of delays on airport reputation. First, variance in external variables such as weather and traffic flow impacts on terminal operations. This is evidenced by the following two comments. Respondent A notes that: "the aircraft can come in onto a gate but you cannot put a bridge onto it because of lightning strikes. So the passenger flow stops at front of the terminal in storms". Similarly, respondent B notes: "traffic can have a hell of an impact on what we do here at the airport. And that can affect facilitation, people running late, missing flights, causing delays". Second, such impacts can lead to build ups in passenger facilitation. Respondent B notes: "they would've come across people who arrive late, who got here early, some people

who couldn't find their way to the departure lounge". Ultimately, variance and delays impact on passenger satisfaction and the reputation of the airport with airlines and the public.

Problem relevance is further evidenced by the following comment of respondent E, the airport terminals manager, on the problem of delays and congestions at the airport. He points out that variance in traffic flow, passenger flow, and delays affect the reputation of the airport with passengers and the general public:

"We want it to be a good experience for the public and for the passengers so it's reputation which we take very seriously. We take our reputation very seriously [...] Because we don't want it to get out in the media that we're causing delays and congestion", **Respondent E, Beta Corporation**

Furthermore, context variance analysis is a *suitable solution* to variance as evidenced by interpretations of continuous improvement, performance monitoring, and documentation. First, tasks performed by participants and their ordering impact on terminal outputs. Respondent B notes "We want to get the person [...] into their seat by the appropriate time. We also want them here in sufficient time so that if there are problems, they can be sorted out at check-in or whatever it may be". Second, facilitation relies on critical technology assets such as check-in, security, and baggage systems. Respondent A points out: "your most critical systems for the terminal are your baggage system and your check-in system". Third, people are a critical but also costly element, as respondent D notes: "you invest in your people and provide better training it has to be cost efficient as well as business viable". Finally, respondent B describes the structural complexity of facilitation and the airport in general: "from a security perspective we deal with 240 odd entities just on aviation security alone [...] it's pretty complex".

Suitability is further evidenced by the following comment of respondent B, the security manager responsible for both terminals. He points out efforts undertaken by the airport to extract information from a variety of transactional systems in order to better analyse and improve terminal performance in different contexts:

"We're quite data hungry at the moment and we recognise the fact that there's a lot of work that has to be done in acquiring the data [...] I think understanding the patterns helps you also understand what data you need", **Respondent B, Beta Corporation**

Table 7.9 illustrates the conceptualisation of passenger facilitation as a work system and the impact of contextual variance on each of the system components. First, variance in passenger volume depends on the terminal capacity and flight schedule at a given day. The airport receives both data on expected and "reconciled" passengers from airlines that support it in planning terminal handling and capacity. Second, upstream delays such as traffic incidents impact

on terminal operations and require staff to handle disgruntled passengers and ensure delays do not impact on aircraft departures. Third, outage of critical equipment such as check-in or baggage systems lead to situations in which passengers must be manually processed. Fourth, security threats such as terrorism require the airport to increase the level of security screening applied to passengers. Finally, both downstream process delays (weather, aircraft breakdown) or security breaches bring terminal operations to a complete halt. In this case, the processing of passengers stops because of exposure to safety or security risks.

System variance	Context	Impact of variance on:		
		Goal attainment	Technical system	Social system
1. Volume of passengers	"We have approximately <u>four peaks</u> throughout the day [...] that's part of <u>patterns and scalability</u> ", Respondent B	• Increase in handling time • Decrease in service level	• Screening point capacity	• Staff rosters and overtime
2. Upstream process delays	"Traffic can have a hell of an <u>impact on what we do</u> [...] at the airport", Respondent B	• Increase in handling time • Potential impact on delays	• n/a	• Dealing with customer rage; staffing levels
3. Passenger throughput	"Critical <u>infrastructure loss</u> is a bit like weather, it just <u>slows the process down</u> ", Respondent A	• Increase in handling time • Decrease in service level	• Equipment outage; backup	• Manual check-in and boarding
4. Level of frisk searching	"If I turn [...] <u>security</u> to high everybody is [...] <u>slow going through facilitation</u> ", Respondent D	• Increase in queue times • Increase in handling times	• Technical configuration	• Training in frisk searching
4. Downstream process delays	"If it's a <u>storm</u> [...] everything will possibly be <u>delayed for hours</u> ", Respondent A	• Mitigation of safety risks	• Temporary shutdown	• Providing help to passengers
5. Security breach	"If you have a <u>security breach</u> [...] then <u>everything stops</u> ", Respondent D	• Mitigation of security risks	• Switch to surveillance mode	• Switch to surveillance mode

Table 7.9. Vignette: work system model of passenger facilitation

SYSTEM CONTEXT

Context awareness is a *relevant problem* for the terminal management unit as evidenced by interpretations of busy hours, traffic flow outside the terminal, and weather patterns. First, flight schedules determine terminal load on a given day. This is evidenced by the comment of the airport security manager, respondent B: "the airport operates on two schedules [...] and that's what the whole pattern of airport facilitation is based on". The pattern determines passenger volume to be expected at any given day of the year. Second, traffic flow outside the terminals impact on passenger arrival times and creates bottlenecks in facilitation at process points such as check-in and security. This is evidenced by the comment of respondent B: "traffic can have a hell of an impact on what we do here at the airport. And that can affect facilitation, people running late, missing flights, causing delays, those delays have a knock-on effects within the airline network". Finally, weather patterns such as storms affect facilitation as respondent A notes: "whenever there's a storm around with lightning, the apron closes down".

Most importantly, however, day to day operations at the terminal are affected by a wide variety of issues that the unit manages, not all of which can be tracked and monitored as contextual information. This is reflected in the following comment by respondent B, the manager responsible for overall airport security:

"Every day there are people who don't turn up for their flights or turn up late. Every day there's a flat tyre on a plane or every day there's a sick crew member. You know, something happens to put pressure on the ability of that airline and the airport to get that plane out on time", **Respondent B, Beta Corporation**

Furthermore, applying context modelling to the problem of context-awareness is a *suitable solution* as evidenced by interpretations of predictable context and the semi-automated response to changes in passenger flow and traffic flow. First, the unit tracks passenger flow to understand variations in passenger volume, busy hours at the terminal and potential bottlenecks in facilitation. This is evidenced by the comment made by respondent B: "I think the passengers are the key data. There's no doubt about that or it's probably the central part of our data. But the various aspects of the data they use what they're aiming to do is to be able to capture passenger data from the moment they enter the airport to the moment they go onto the airplane". Second, the unit monitors traffic flow outside the terminal infrastructure to understand the impact of traffic events on passenger facilitation. This is evidenced by respondent E's comment:

"we do a lot of survey so we've got a lot of data that we analyse so we've got a reasonably good idea when our peaks are [...] we're also looking at identifying actual [traffic] points".

Suitability is further evidenced by the following comment of respondent E, the airport terminals manager. He points to future investments made by the airport in improving kerbside traffic management and managing traffic information by integrating updates from mobile traffic patrols into decision-making:

"We're looking at improving, increasing the mobile patrols for example. And part of that will be also seeing what sort of information they can provide us with [...] say for example I've already started developing a traffic report and I get an update on a daily basis so I'm aware of what the issues are", **Respondent E, Beta Corporation**

Table 7.10 illustrates the mapping of context modelling to interpretations of passenger throughput, wait time, busy hour, traffic flow rate, visibility, and storm warning variables. First, passenger throughput and wait time are instances of immediate context and more specifically are a feature of task context. Both variables measure the performance of tasks that are performed either by passenger facilitation or other stakeholders such as airlines and security contractors. Second, busy hour and traffic flow rates are instances of external context and more specifically customer context. Both variables measure upstream and downstream factors that affect when passengers (i.e., customers in the sense of the facilitation system) arrive at one of the terminals. Finally, visibility and storm warnings are instances of environmental context and more specifically weather context. Both factors relate to natural phenomena such as the occurrence of fog in winter and thunderstorms in summer which do not directly impact on passenger facilitation but prevent aircraft from departing at the scheduled time.

Context feature	Illustration of feature	Representation as context model
TASK CONTEXT		
Throughput	"We have a people counting system that records throughput of passengers at certain points", Respondent C	<pre> graph BT IC[Immediate context] --> TC[Task context] TC --> CH[Call handling] CH --- T([Throughput]) CH --- W([Wait time]) </pre>
Wait time	"We don't want people waiting more than 10 minutes in queue, we want people processed through customs and immigration within at the very latest 10 minutes", Respondent B	
CUSTOMER CON-TEXT		
Busy hours	"We look at the <u>peak busy hour for the day</u> and typically that's morning peak and then at the domestic, there's a couple of those", Respondent C	<pre> graph BT EC[External context] --> CC[Customer context] CC --> FS[Flight schedule] CC --> FS FS --- BH([Busy hours]) FS --- TI1([Traffic incident]) TF[Traffic flow] --- TI2([Traffic incident]) </pre>
Traffic incident	"I get a traffic report but <u>at the moment I get it weekly</u> and if there's any incidents I get them immediately by text or even a phone call", Respondent E	
WEATHER CONTEXT		

Context feature	Illustration of feature	Representation as context model
Visibility	"That's something that air services make the call on and it's based on the visibility or a <u>measurement of visibility</u> as to whether they should close a runway", Respondent C	<pre> graph TD EC[Environmental context] --> WC[Weather context] WC --> F[Fog] F --> TS[Thunderstorm] F --> V([Visibility]) TS --> SW([Storm warning]) </pre>
Storm warning	"We've got <u>storm warnings</u> of course, like fog [...] and that happens on not a regular basis but you sort of expected it to happen at this time of year", Respondent E	

Table 7.10. Vignette: context model of passenger facilitation system

ADAPTATION STRATEGIES

Preparedness is a *relevant problem* for the passenger facilitation unit as evidenced by interpretations of terminal throughput, delays, passenger satisfaction, and security breaches. First, changes in external variables such as passenger volume and traffic flow impact on terminal throughput as evidenced by the comment of respondent B: "traffic can have a hell of an impact on what we do here at the airport. And that can affect facilitation, people running late, missing flights, causing delays, those delays have a knock-on effects within the airline network". The airport prepares for these changes and seeks to mitigate their impact on terminal performance and passenger satisfaction. This is evidenced by the comment of respondent E: "we take our reputation very seriously, that's right [...] because we don't want it to get out in the media that we're causing delays and congestion". Finally, the airport must manage security risks such as terrorism and respond to breaches: "if someone unintentionally does something wrong and then makes it through the primary line in security and disappears, then everything stops".

Problem relevance is further evidenced by the following comment of respondent B, the airport security manager. He points to the problem of responding to a wide range of operational issues that challenge the airport's ability to meet service levels with airlines operating flights:

"Even though it's business as usual, every day there are delays. Every day there are diversion. Every day there are people who don't turn up for their flights or turn up late. [...] something happens to put

pressure on the ability of that airline and the airport to get that plane out on time", **Respondent B, Beta Corporation**

Furthermore, applying adaptation strategies to the problem of preparedness is a suitable solution as evidenced by interpretations of planned modifications to the parts of the passenger facilitation system such as rostering, check-in, security, and boarding. First, context changes can require planned or ad hoc modifications to staffing levels as evidenced by the comment of respondent B: "screening is rostered and resourced to meet the projected schedule. As I said, there are other events that sometimes cause other issues. So those rosters can be altered. Extra people can be brought on board". Second, a change in context caused by the outage of equipment can lead to changes in check-in and boarding procedures. Respondent A notes: "if you loose your check-in system upstairs, then you got major delays. If you loose your baggage sortation system, you got major delays. if you loose gate boarding they just go manual". Third, more screening points can be opened to respond to context changes such as traffic as noted by respondent B: "we might have to keep a larger number of screening points open".

Suitability of adaptation strategies is further evidenced by the following comment of respondent B, the airport security manager. He points to the pressure on the airport to be prepared when responding to the wide range of issues that can affect passenger facilitation every day:

"Operationally the airport is on its front foot. Every day and every minute of the day, there are decisions being made because things are out of the ordinary. Operationally, it's very, very hard to define what's the norm and what's normal", **Respondent B, Beta Corporation**

Table 7.11 illustrates the mapping of adaptation strategies to interpretations of rostering, airline check-in, security screening, and boarding changes in the system. First, the airport collects weekly passenger data from a variety of information systems and provides rolling forecasts to immigration and security stakeholders for rostering. This enables stakeholders to include forecasts into their staff planning cycle. Second, system outages in airline check-in or the airport's baggage sortation system trigger the manual processing of passengers. The airport has created redundancies in these systems to mitigate minor glitches but can switch to manual processing should this be required. Third, changes to security screening and the number of screening points can be made to either respond to a security incident or to respond to changes in passenger volume following an external event such as an aircraft delay or deviation. This allows the airport to manager passenger throughput in a terminal as well as responding to external risks such as the exposure to terrorist attacks and criminal activity.

Adaptation focus	Illustration of focus	Representation as adaptation strategy
Rostering	“That’s our peak period for arrivals and for departures, this is what would’ve been processed. So that goes to <u>security to assist them in their resourcing</u> ”, Respondent C	• Strategy: CAR-01 (modify resource number) • Scope: External variance • Reason: Increase in passenger volume • Trade-off: Higher staff and training costs
Check-in processing	“So whereas you check in an aircraft in two hours now, if you were to <u>loose your systems and have to do it manually</u> that would probably take you three hours.”, Respondent A	• Strategy: CAF-02 (alternative function) • Scope: Internal variance • Reason: System outages or software failure • Trade-off: Decrease in passenger throughput
Security screening	“If I turn the switch <u>up on security to high</u> then everybody is going to be <u>slow going through the facilitation</u> . And high means everything is checked, every single thing will be checked to the highest level”, Respondent B	• Strategy: CAF-02 (alternative function) • Scope: External variance • Reason: Change in the security context • Trade-off: Decrease in passenger throughput
Security points	“We might to have to <u>contact more staff</u> to come on board. We might have to <u>keep a larger number of screening points open</u> ”, Respondent D	• Strategy: CAR-01 (modify resource number) • Scope: External variance • Reason: Traffic, aircraft delay or deviation • Trade-off: Higher staff and operating costs
Boarding	“Due to OH&S issues, the <u>work stops out there</u> . If it’s only raining, work keeps going on. It’s only the lightning that stops everything. That’s all to do with employees <u>but it impacts on passengers</u> ”, Respondent A	• Strategy: CAF-04 (wait before starting) • Scope: External variance • Reason: Occupational health and safety • Trade-off: Terminal utilisation

Table 7.11. Vignette: adaptation strategies of passenger facilitation system

ACTIVATION RULES

The problem of responsiveness is a *relevant problem* for the airport as evidenced by interpretations of rapid responses to safety or security incidents in the work system. The passenger facilitation system underlies both occupational safety and aviation security regulations maintained by authorities such as the Department of Infrastructure and Transport. This is evidenced by comment of the airport security manager, respondent B: "there's a little liaison and consultation, but within the federal compliance bodies, there's certainly an opportunity for them to require us to undertake additional compliance activity or regulatory activity". Terminal regulations broadly define responsibilities which must be interpreted and implemented in the local context. This is evidenced by respondent D's comment, the screening manager: "there's lots of government documents to say how it has to be and the minimum standard and that's it" and "if they ever standardise one approach then they have to standardise how it's done". Respondent A also points to workplace safety: "due to OH&S issues, the work stops [during lightning]".

Problem relevance is further evidenced by the following comment of respondent B, the airport security manager. He refers to the flexibility and responsiveness that according to him is a characteristic of working at the airport and the importance of having highly qualified and knowledgeable staff:

"From the seat of their pants what it really boils down to. It's really having people on the ground and dealing with people. So, the issue, the things you got to do, you got to have experienced, knowledgeable, skilled people in the jobs", **Respondent B, Beta Corporation**

Furthermore, applying activation rules to the problem of responsiveness is a *suitable solution* as evidenced by interpretations of defining triggers and rules for the activation of system changes. First, the airport closely monitors traffic flow to identify traffic build-ups and potential impacts on passenger facilitation. This is evidenced by the comment of respondent E: "the main issue with the traffic event is obviously I want to find out why it happened" and "the clean up is pretty important because there's a lot of environmental issues". Second, critical infrastructure failure requires an immediate response, as noted by respondent A: "if it's computer equipment that goes down, the hardware side of it, then we'll do everything we can to get it running". Third, in the event of lightning, labour must be withdrawn from airside facilities and passenger facilitation "stops" at the front of the terminal. Respondent A notes: "the aircraft can come in onto a gate but you cannot put a bridge onto it because of lightning". Respondent D points to security breaches: "everything stops [...] we then switch to our surveillance side".

Suitability of activation rules is further evidenced by the following comment of respondent D, the security screening manager. He demonstrates how changes in the security context can prompt a range of changes coordinated by the respondent such as updates to operational manuals and procurement processes:

"So then if it works out that they have to do it, they will then change the transport security program for Beta Corporation, I will then look at our screening manual and say erase the component of what we do at the oversize and change it to, this will have x-ray machine", **Respondent D, Beta Corporation**

Table 7.12 illustrates the mapping of activation rules to interpretations of traffic event, equipment failure, occupational safety, and security breach rules as used by the airport to respond to context changes. First, traffic events such as congestion or a vehicle accident trigger a response from the duty manager and activates changes to the passenger facilitation system such as opening additional screening points. Second, the failure of critical systems such as check-in and baggage sortation systems activate the manual processing of departing passengers and trigger a response from the airport to fix hardware issues. Third, thunderstorms create workplace safety issues for airport or airline ground handling staff working outside the terminal confines because of the possibility of lightning strikes. In case lightning strikes are likely, all labour is withdrawn from external facilities and passenger facilitation stops at the front of the terminal. Finally, security breaches trigger the evacuation of the terminal and activate surveillance measures to identify and track the person that caused the breach.

Activation trigger	Illustration of trigger	Representation as context activation rule
Traffic incident	"The main issue with the <u>traffic event</u> is obviously I want to find out why it happened", Respondent E	ON-CTX <traffic event> IF <incident risk = 'high'> OR <traffic volume = 'high'> THEN ACTIVATE (traffic management) AND RAISE (traffic management activated)

Activation trigger	Illustration of trigger	Representation as context activation rule
Equipment failure	"If you <u>lose gate boarding</u> , no brainer, they just go manual and they board and depart on time pretty well", Respondent A	ON-CTX <equipment event> IF <equipment type = 'gate boarding'> THEN ACTIVATE (manual departure) AND RAISE (manual processing activated)
OH&S risk	"Whenever there's a <u>storm around with lightning</u> , the apron closes down", Respondent A	ON-CTX <storm event> IF <lightning probability = 'high'> THEN ACTIVATE (apron shutdown) AND RAISE (apron temporarily closed)
Security breach	"If you have a <u>security breach</u> , so someone unintentionally does something wrong and then makes it through the primary line in security [...] then everything stops", Respondent D	ON-CTX <breach event> IF <location = 'terminal sterile area'> AND <passenger location = 'unknown'> THEN ACTIVATE (surveillance) AND RAISE (surveillance initiated)

Table 7.12. Vignette: context activation rules

LEARNING STRATEGIES

Context learning is a *relevant problem* to continuously improve and evaluate the relevance of system decisions as evidenced by variance in system performance and the impact of extraordinary events such as natural disasters or congestion. First, system performance can vary as a result of many factors, some of which depend on contextual situations such as downstream delays. This is evidenced by the following comment of respondent B: "well there's an overall delay where a plane might be late on departure. But there's also delays within or bottlenecks within the process". This requires the terminal unit to track information about system performance: "we're quite data hungry at the moment and we recognise the fact that there's a lot of work that has to be done". Second, extraordinary events such as natural disasters and congestion impact on terminal performance and lead to enquiries regarding certain system decisions.

Respondent E gives an example: "we deal with a lot of incidents. Sometimes maybe once in a lifetime, others on a fairly regular basis". Learning thus assumes an important role.

Problem relevance is further evidenced by the following comment of respondent A, the terminal facilitation manager. He describes his work experience over three decades at the airport and points out the problem of continuously learning and adapting to new types of issues and changes:

"There's never two days the same at the airport. I've been here 37 years or 38 year that I've been here and I never had two days the same. That's what makes it so good. And you might meet new changes and you might meet new things, but there's never two days the same", **Respondent A, Beta Corporation**

Furthermore, applying learning strategies to the problem of context learning is a *suitable solution* as evidenced by interpretations of reporting models, standard operating procedures, working groups, and lessons learnt. First, the systems unit designs information models provided to various stakeholders for tracking key system variables. Respondent C points out: "these are some of the dashboards that have been developed and this is fairly high level, but these are rolled out to people within the terminal but also to our executive team". If actual system performance falls short of the expectations of stakeholders, a working group is set up to gather information about the issue. Respondent B notes: "you might put a working group working with the airlines to address that particular problem". Decision that lead to system changes are documented: "when we do make changes [...] we do document it in form of a plan or a standard operating procedure". Finally, respondent E notes that every extraordinary event triggers a review by terminal stakeholders: "after every event [...] we always have a lesson learnt".

Suitability of learning strategies is further evidenced by the following comment of respondent B, the airport security manager. He refers to the airport continuous improvement program and the approach taken by system members to continuously evaluate and improve system performance:

"Part of our philosophy is having a continual improvement type approach to everything we do. So, even though it might be good, we always look at it again and say [...] how can we improve this, how can we make it cheaper, more efficient, reduce the costs, get greater outputs", **Respondent B, Beta Corporation**

Table 7.13 illustrates the mapping of learning strategies to interpretations of terminal information models, standard operating procedures, airline working groups, and emergency planning groups. First, terminal information models provide system participants with information about system variables and variables in the external context such as traffic flow. This maps to learning

strategy CLS-01 and respondents note that more work is required in developing such information models. Second, standard operating procedures document key system decisions for example in the facilitation or security area. This maps to learning strategy CLS-02 and respondents point out issues in the completeness of documentation. Third, airline working groups are set up to track specific issues that arise in terminal operations such as issues at the check-in counters or security screening points. This maps to learning strategy CLS-04. Finally, emergency planning groups plan terminal emergency response procedures and review issues that arise in the wake of emergencies. This maps to learning strategy CLS-04.

Learning issue	Illustration of issue	Representation as learning strategy
Terminal information models	"We're quite <u>data hungry</u> at the moment and we recognise the fact that there's <u>a lot of work that has to be done</u> ", Respondent B	- Strategy: <i>CLS-01 (improving context models)</i> - Learning type: <i>Single-loop context learning</i> - Learning artefact: <i>Terminal information models</i> - Implementation: <i>Data warehouse and reports</i>
Standard operating procedures	"When we do make changes [...] we do document it in form of a plan or a <u>standard operating procedure</u> ", Respondent B	- Strategy: <i>CLS-02 (improving adaptation)</i> - Learning type: <i>Single-loop context learning</i> - Learning artefact: <i>Operating procedures</i> - Implementation: <i>Documentation and implementation of system changes</i>
Airline working groups	"You might put a <u>working group</u> working with the airlines to address that particular problem", Respondent B	- Strategy: <i>CLS-04 (redesigning adaptation)</i> - Learning type: <i>Double-loop context learning</i> - Learning artefact: <i>Specific system issues</i> - Implementation: <i>Working group</i>
Emergency planning group	"Then you have your <u>emergency planning group</u> and that's already been activated three times this year", Respondent D	- Strategy: <i>CLS-04 (redesigning adaptation)</i> - Learning type: <i>Double-loop context learning</i> - Learning artefact: <i>Emergency response</i> - Implementation: <i>Post-event reviews</i>

Table 7.13. Vignette: context learning strategies

7.4 ACTIONABILITY

Applicability is a measure of utility that evaluates whether outcomes of DS address *recurring problems* and generate actionable recommendations for the design of more effective systems (Kuechler & Vaishnavi, 2011). In the following, I measure the actionability of CAIS by comparing system decisions taken by respondents in the case study presented in Chapter 04 and comparing them with the prescriptive statements made by CAIS. My objective at this stage is twofold.

First, I apply the lens of CAIS to view design problems as recurring problems as reported in the case data. For example, I identify the design problem of adaptation trade-off as a relevant and recurring problem. Second, I then map system solutions found by respondents to prescriptive statements made by the CAIS design theory about the design of effective, context-aware information systems. This mapping leads me to suggest actionable recommendations on the basis of CAIS concepts. If implemented, these recommendations promise to lead to an improvement in system performance as described by the issues identified in the previous step.

7.4.1 SIGMA GROUP

SYSTEM BEHAVIOUR COMPONENT

Respondents point to *recurring issues* in better managing system variance and context which result in unintended consequence during system adaptation. First, the adaptation of technical procedures for call handling in a weather context affects social system variables such as capacity for claims handling in the insurer's back office. This is evidenced by the following comment of respondent A: "one of the big failings was that the claims were being incorrectly categorised based on the questions the [event leadership team] had actually put it in". Incorrect classification leads to flow on effects in staffing and claims handling: "because our claims had been mis-categorised, it was eating our volume up so we were having to make decisions based on incorrect data that we were receiving". Second, changes in system behaviour following a context event also prompt to changes in customer behaviour. This is evidenced by the comment of respondent E: "we were telling customers to [...] take a picture of the damage" and then "what we were finding was happening was the neighbourhood were swapping photos".

Currency of the issue is further evidenced by the following comment of respondent H, the executive manager for claims experience. She points to the problem that changes in either system can lead to unintended consequences and that system decisions must be balanced and constantly revised:

"Once the market gets to know and understand how to break the process, you've got to change to put your controls, really strict controls, I'm serious, you put your strict controls back in there and you let something else go through it, so that's my view on how you manage it", **Respondent H, Sigma Group**

Following work system theory, these issues can be addressed by considering the joint optimisation of the technical and social components as basis of context adaptation. First, system decisions in response to context that affect the technical system must be aligned with corresponding changes to the social system. For example, frequent changes to system processes

and decision rules that are not adequately communicated to system members lead to a communication overhead and additional workload as they verify that actions correspond to prescribed rules in a particular context. Respondent F notes: "those guys at the front are so overloaded, the volume that keeps coming in [...] plus they have to make sure that they follow the right rules". Second, system decisions that affect the social system must also be reflected in system processes and rules. For example, call handling decisions impact on the quality of information in the information system. Respondent F notes: "[the decision] was a complete strain on the system and it was very manual for the guys [...] to pump out the correct number".

System component (cf. Section 7.3.1)	Recurring issue (Coded data)	Recommendation (CAIS concept)
Social system	"Those guys at the front are so <u>overloaded</u> , the volume that keeps coming in. Calls that they have to field, plus they have to <u>make sure that they follow the right rules</u> , you expect those rules to change", Respondent F	• Increase <u>transparency</u> of decisions of system members • Model <u>interaction</u> between organisational units • Define <u>responsibility</u> of each organisational unit • Define <u>resources</u> used and provided by each of the units
Technical system	"That was a complete strain on the system and it was <u>very manual for the guys</u> in the events processing team to be able to pump out the correct number [...] and it <u>compounded the issue</u> ", Respondent A	• Increase <u>transparency</u> of system processes in shared plan • Model system <u>functions</u> and system <u>processes</u> • Model flow of <u>information</u> between functions and technology <u>resources</u> used • Define system <u>outputs</u>

Table 7.14. Vignette: improving system behaviour based on CAIS

SYSTEM CONTEXT COMPONENT

Respondents point to *recurring issues* in the selection of context variables and the modelling of interdependencies between context variables as evidenced by interpretations of bushfire alerts and the repeat call problem. First, some context categories and variables are only partially covered by the department's event monitoring system. This is evidenced, for example, by respondent E's comments on missing support for bushfire alerts, i.e., one type of weather context that has an impact on the claims handling system: "bushfire is different because you can't predict that. It's not a [Bureau of Meteorology] issues, but certainly flood and storm and in particular we monitor that through the Bureau's website". Second, interdependencies between context categories are only partially defined in the department's response system. This is evi-

denced by respondent E's comment about the missing link between monitoring repeat calls and understanding the nature of calls: "no one takes the time to actually look and understand why are the calls coming in [...] and decisions that get made without consideration".

The need for addressing issues in the definition of context and interdependencies between several contextual variables is further evidenced by the comment of respondent E. He points to the use of the repeat call variables in activation rules that trigger the staffing of the departments event processing team:

"I went to my guys in operations saying, can you just do a quick and dirty and tell me at least what they're calling about and it turned out that 70% of the calls were from calls for BAU claims, not event. So we're about to resource an event team when it's the BAU team that's really hurting", Respondent E, Sigma Group

Furthermore, applying adaptation strategies to recurring issues in the case data generates *actionable recommendations* as evidenced by the mapping presented in Table 7.15. First, I select some of the context categories defined in Section 7.3.1. The weather context and task context categories are two of the categories defined to demonstrate the applicability of context modelling. Second, I map these categories to interpretations of recurring issues such as the bushfire and repeat call problems described above. For example, the bushfire problem maps to shortcomings in the specification of the claims handling context model. Similarly, the repeat call problem maps to shortcomings in the specification of interdependencies between context categories. Finally, I generate recommendations based on context modelling concepts introduced in Chapter 6. For example, the bushfire problem can be mitigated by adding new variables and information sources to the model. Similarly, the repeat call problem can be mitigated by creating a dependency between the weather and task categories.

Context category (cf. Section 7.3.1)	Recurring issue (Coded data)	Recommendation (CAIS concept)
Weather context	" <u>Bushfire is different because you can't predict that</u> . It's not a [Bureau of Meteorology] issues, but certainly flood and storm and in particular <u>we monitor that through the Bureau's website</u> ", Respondent E	- Model bushfire as a separate <u>context variable</u> under the <u>weather context category</u> - Define <u>context features</u> of the variable such as alerts issued by state emergency services

Context category (cf. Section 7.3.1)	Recurring issue (Coded data)	Recommendation (CAIS concept)
Task context	"One of my bugbears is every one uses the repeat call matrix as an indication of the health of claims, are we meeting our SLAs but <u>no one takes the time to actually look and understand why are the calls coming in</u> ", Respondent E	• Model repeat calls as one of the <u>context features</u> of the <u>task context category</u> • Model storm alerts as one of the <u>context features</u> of the <u>weather context category</u> • Define a <u>dependency</u> between the repeat call and the storm alert feature

Table 7.15. Vignette: improving context modelling based on CAIS

ADAPTATION STRATEGY COMPONENT

Respondents point to *recurring issues* in the implementation of adaptation strategies that affect system performance and prompt short-term decisions without proper weighing of adaptation trade-offs. This is evidenced by interpretations of call handling adaptation strategies such as the event module that is activated in a severe weather context. Respondent C, one of the department's call centre managers, notes: "one of the problems with that event module is that the event module compared to the business as usual discipline is very limited information". Respondents point to two trade-offs that are not considered when the module is activated. First, truncating information increases the potential for incorrect categorisation and miscommunication of service levels. Respondent A notes: "one of the big failings was that claims were being incorrectly categorised based on the questions [we] put it in". Second, truncation also strains backend resources, as noted by respondent E: "you then tell the customer we will get back to you in 7 days [...] and we don't have the people at the back end to manage that".

Currency of the issue is further evidenced by the following comment of respondent E, the event team manager of the claims department. He points to the need to balance customer satisfaction with back office capacity when activating the event module in a severe weather context to truncate claims intake:

"People should expect to hold for about 20 minutes, just as where we were answering the calls within 3 minutes, so great customer service at the front end, but then it suffers at the back end", Respondent E, Sigma Group

Furthermore, applying adaptation strategies to recurring issues in the case data generates *actionable recommendations* as evidenced by the mapping presented in Table 7.16. First, I select one of the adaptation strategies defined in Section 7.3.1. For example, the call handling strategy is maps to the CAF-02 (Perform Alternative Function) strategy. Second, I map interpreta-

tions of recurring issues such as the limited information and backend impact issues mentioned in the case data to the call handling adaptation strategy. Based on this mapping, two recommendations for immediate applicability can be identified. First, modelling adaptation trade-offs promises to provide system participants with the information that is required to balance adaptation decisions in a particular context. This is evidenced by the trade-off between immediate customer satisfaction and mid-term ability to satisfy service levels. Second, modelling trade-offs also opens the potential to explore combining strategies in a context. This is evidenced by the potential to combine call handling with team allocation strategies.

Adaptation strategy (cf. Section 7.3.1)	Recurring issue (Coded data)	Recommendation (CAIS concept)
Call handling	"People should expect to hold for about 20 minutes, just as where we were answering the calls within 3 minutes, so <u>great customer service at the front end</u> , but then it <u>suffers at the back end</u> ", Respondent E	• Balance <u>trade-off</u> between customer satisfaction and back end handling capacity • Decide to combine with back office <u>adaptation strategy</u> such as staffing event team

Table 7.16. Vignette: improving adaptation strategies based on CAIS

ACTIVATION RULE COMPONENT

Respondents point to *recurring issues* in the activation of adaptation strategies in a context that impact on customer satisfaction and claims handling cost as a result of inconsistent communication. This is evidenced by interpretations of claims intake and communication in a weather context. Respondent G, a member of the design team, points out: "so that's one improvement, those guys at the front are so overloaded, the volume that keeps coming in. Calls that they have to field, plus they have to make sure that they follow the right rules, you expect those rules to change". Respondent F concurs: "one minute you're getting told if you're doing this you've got to cover flood [...] there was so much stuff being heaped at the front end that you've got to wonder how much of it gets used". Communication of changes is mostly verbal and interpretation errors occur. Respondent A points out: "one was telling the customer to throw the contents out and make a list and the other was telling our customers to retain their contents, which really was quite a failing from a customer experience point of view".

The suitability of activation rules as a concept is further evidenced by respondent F's following comment, which describes efforts undertaken by the department to improve the communication of changes. The system she refers to ensures changes, once activated, are consistently implemented by the respective units:

"Hopefully that's one of the things with communication we can solve, the communications response systems, when it goes out, it goes out to all and if it's only pertaining to this business area, we need to understand what's the impact of that in other areas", **Respondent F, Sigma Group**

Furthermore, applying adaptation strategies to recurring issues in the case data generates *actionable recommendations* as evidenced by the mapping presented in Table 7.17. In the example, I select one of the activation rules defined in Section 7.3.1. For example, the repeat call rule activates staffing of the event team as a result of an increase in repeat calls from customers and a suspected decrease in customer satisfaction. Second, I map interpretations of recurring issues such as missing cross-checks concerning the nature of the calls and whether they relate to a weather event to the activation rule. Based on this mapping, three recommendations for immediate applicability can be identified. First, modelling a threshold for rule activation based on repeat call volume acts as the trigger event for the repeat call rule. Second, extracting information about the call reason, e.g., from call centre records, acts as a precondition that prevents system changes being activated in a non-event. Third, event team staffing strategies are only activated if both the rule's trigger event and precondition are met.

Activation rule (cf. Section 7.3.1)	Recurring issue (Coded data)	Recommendation (CAIS concept)
Repeat call	"One of my bugbears is every one uses the repeat call matrix as an <u>indication of the health</u> of claims, are we meeting our SLAs but <u>no one takes the time to actually look and understand</u> why are the calls coming in", Respondent E	• Model repeat call threshold as the trigger <u>event</u> for activation • Model call reason as required <u>condition</u> for rule activation • Activate event team staffing <u>action</u> only if the trigger event and rule condition are met

Table 7.17. Vignette: improving activation rules based on CAIS

LEARNING STRATEGY COMPONENT

Respondents point to *recurring issues* in the selection of learning strategies to specify learning based on feedback about system adaptation in a specific context or the redesign of context-driven adaptation. This is evidenced by interpretations of the effectiveness of post-incident reviews, one of the learning strategy used by the department to collect feedback information on the effectiveness of system adaptation in a weather context. Respondent E points out: "there is a debrief session that they have but it's too short, because it's an hour session, too many people involved. They spend half an hour talking about the weather and football and so it's not ef-

fective". In addition, respondents critique the lack of a systematic approach to specifying learning in a contextual situation. This is evidenced by respondent G's comment "I highlighted it at the end of this event season, I actually highlighted and said, I would love to go back over [...] go back over each of the [event meetings] and see the decisions that were made, the resourcing changes that were made and to try and map out how this impacted the business".

The need for a systematic approach to learning is further evidenced by the following comment of respondent E, the event team manager, regarding learning and analysis to improve decisions regarding context-driven adaptation. He stresses the prevalence of experience and the lack of evidence-based approaches:

"When a decision needs to be made or when an issue does come up and the solutions to get us out of them, there is no analysis, impact analysis that like empirical analysis or any data type analysis that takes place. It's gut feeling, it's experience with a gut feeling, that's how I see it", **Respondent E, Sigma Group**

Furthermore, applying learning strategies to recurring issues in the case data generates *actionable recommendations* as evidenced by the mapping presented in Table 7.18. First, I compare learning strategies defined in Chapter 06 with the learning strategies identified in Section 7.3.1. Second, I map interpretations of recurring issues such as the lack of systematic analysis and ineffective reviews with learning types and instances of learning in the data. Based on this mapping, four recommendations for immediate applicability can be identified. First, post-incident reviews can be used to improve context models by including new context sources. This is evidenced by the lack of information concerning bushfire alerts. Second, incident reviews can also be used to review and improve adaptation strategies. This is evidenced by the failure to include learning from previous events into the design of adaptation strategies. Third, event simulations and leadership team meetings can be used to include new variables, add adaptation strategies, or remove them on the basis of empirical analysis.

Learning strategy (cf. Section 7.3.1)	Recurring issue (<u>Coded data</u>)	Recommendation (<u>CAIS concept</u>)
Improving context representations (CLS-01)	"Bushfires for example [...] the first I heard about it was on <u>SGMday morning</u> when I turned on the TV and then you know, things will start flying [...] and then decisions get made from there", Respondent E	• Improve <u>context representations</u> by integrating new context information sources • For example, improve <u>claims context representation</u> by including bushfire alerts

Learning strategy (cf. Section 7.3.1)	Recurring issue (Coded data)	Recommendation (CAIS concept)
Improving adaptation strategies (CLS-02)	"I would love to go back over [...] go back over each of the [event meetings] and see the decisions that were made, the resourcing changes that were made and to try and map out how this impacted the business", Respondent G	• Improve <u>adaptation strategies</u> by evaluating the effectiveness of adaptation strategies • For example, improve <u>claims adaptation strategies</u> by reviewing event team decisions
Transforming context representations (CLS-03)	"One of my bugbears is every one <u>uses the repeat call matrix as an indication</u> of the health of claims [...] but no one takes the time to actually look and understand <u>why are the calls coming in</u> ", Respondent E	• Redesign <u>context representations</u> by adding context variables or interdependencies • For example, redesign <u>claims context</u> by adding an interdependency between storm and repeat call context variables
Transforming adaptation strategies (CLS-04)	"When a <u>decision needs to be made or</u> when an issue does come up and the solutions to get us out of them, there is no <u>analysis, impact analysis that like empirical analysis or any data</u> ", Respondent E	• Redesign <u>context adaptation strategies</u> by adding or removing adaptation strategies • For example, redesign <u>claims adaptation</u> by adding or removing strategies based on empirical analysis of decisions

Table 7.18. Vignette: improving learning strategies based on CAIS

7.4.2 BETA CORPORATION

SYSTEM BEHAVIOUR COMPONENT

Respondents point to *recurring issues* in better managing system variance and context which result from structural complexity and a lack of transparency in how the system operates. First, the airport environment is characterised by a large number of stakeholders such as airports, airlines, security, federal authorities, and contractors. Respondent B notes: "from a security perspective we deal with two hundred and forty odd entities just on aviation security alone". Second, while the terminal management unit seeks to make system processes "common users", some differences remain: "now check-in is always going to be a bit different between the airlines [...] and that's up to the airlines what they have on their software". Finally, information about system decisions and system variables such as tasks, technology, people, and structure is not readily available. Respondent B points out: "you certainly see uninformed decisions being

made because of poor data and poor information [...] people might struggle where to find this information and data or plans". Taken together, these issues affect system adaptation.

The currency of this issue is further evidenced in the following comment of respondent B, the airport security manager. He points to the lack of documentation about system decisions and the impact of transparency on the ability of the airport to implement rapid changes of the passenger facilitation system:

"So the more that sort of information is documented and tracked, practised, or trained, there's less likelihood of us having those issues. But, at the moment, we do have issues in that respect. Because the information is not really available", **Respondent B, Beta Corporation**

Following work system theory, these issues can be addressed by creating shared, usable maps of both the technical and social components as basis of context adaptation. First, increasing transparency of system decisions can be addressed by creating a shared understanding of the interactions and responsibility of each unit or organisation participating in the passenger facilitation system. For example, creating a shared understanding of system decisions taken by security and immigration stakeholders avoids situations in which process delays can occur. This is evidenced by the following comment of respondent B: "you certainly see uninformed decisions being made because of poor data and poor information". Second, increasing transparency of system processes can be addressed by the modelling of system functions, flow of information between functions, and the technology assets used in the process of transformation. This is evidenced by the comment of respondent B: "airlines are probably better than airports. Airlines actually document their systems quite well and their processes quite well".

System component (cf. Section 7.3.2)	Recurring issue (Coded data)	Recommendation (CAIS concept)
Social system	"But you certainly see <u>uninformed decisions</u> being made because of poor data and poor information [...] Now, people might <u>struggle where to find this information</u> and data or plans", Respondent B	• Increase <u>transparency</u> of decisions of system members • Model <u>interaction</u> between organisational units • Define <u>responsibility</u> of each organisational unit • Define <u>resources</u> used and provided by each of the units

System component (cf. Section 7.3.2)	Recurring issue (Coded data)	Recommendation (CAIS concept)
Technical system	"Airlines are probably better than airports. Airlines <u>actually document their systems quite well</u> and their processes quite well. From what I've seen at airports, <u>they're not as good as airlines</u> ", Respondent B	• Increase <u>transparency</u> of system processes in shared plan • Model system <u>functions</u> and system <u>processes</u> • Model flow of <u>information</u> between functions and technology <u>resources</u> used • Define system <u>outputs</u>

Table 7.19. Vignette: improving system behaviour based on CAIS

SYSTEM CONTEXT COMPONENT

Respondents point to *recurring issues* in the monitoring of predictable context as evidenced by interpretations of passenger flow, kerbside traffic, and departure times of aircraft. First, busy hours and passenger flow relies on indirect measures of how many passengers move through the terminal at a point in time. This is evidenced by following comment of respondent C: "[airlines] tap them into the computer, they won't give us that data as a feed. It'd be nice to know [passenger numbers] instead of having the bag data". Second, traffic information is not integrated into the department's decision-making tools: "the car parking data would be nice to pull in to get more of a holistic view of what's happening on the airport". Third, detailed security information is not shared by government agencies as is noted by respondent D: "ten years ago [...] we'd be fed straight from the BOM data centre and attorney generals. Now we're fed through a filtered process". Finally, runway data is incomplete and in arrears as lamented by respondent D: "we receive the runway data from air services virtually three months in arrears".

Currency of context information needs and the information access problem is most evident in the following comment by respondent B, the airport's security manager. He points to ongoing efforts by the airport to deliver technology to facilitate the acquisition of information and tracking of passengers in a terminal:

"That's what the airport of the future [project] is about. On the security-side, the airport of the future project is about making it more efficient and basically try to, well, through behavioural detection or you can do it through just knowledge of people who are coming through", **Respondent X, Beta Corporation**

While context modelling cannot mitigate information access problems, it provides *actionable recommendations* by identifying information needs and organising context variables into a consistent information model as shown in Table 7.20. First, passenger movement allows the de-

partment to have an accurate understanding of passenger flow through the terminal at different times. Improving data received from airline check-in systems, e.g., in an anonymised form, has the potential to improve awareness of variances. Second, passenger security profiles would enable the department to design more selective security screening procedures. While information challenges can be addressed through information technology, legal challenges remain. Third, kerbside traffic data at the moment is not integrated. Integration into decision-making tools will improve awareness of variance in traffic flow. Finally, runway data is needed to track actual departures of passengers from a terminal. Increasing the frequency of data extraction from air service systems will increase awareness of variance in departure.

Context feature (cf. Section 7.3.2)	Recurring issue (Coded data)	Recommendation (CAIS concept)
Passenger movement	"The airlines [...] won't give us that data as a feed. It'd be nice to know that instead of having the bag data [...] just data on queueing", Respondent C	• Increase awareness of passenger flow variance • Define <u>context feature</u> passenger movement • Extract <u>context information</u> from airline systems directly • Associate feature with customer <u>context category</u>
Passenger security profile	"Legally they're not allowed to [share passenger information] so they <u>could deny straight away</u> because it's personal information so by law they're not allowed to", Respondent D	• Increase <u>awareness</u> of security risks and exposure • Define <u>context feature</u> passenger security profile • Extract <u>context information</u> once standard security profiles become legally available • Associate feature with customer <u>context category</u>
Kerbside traffic flow	"The <u>car parking data</u> would be nice to pull in to get more of a holistic view of what's happening on the airport and the <u>road data</u> [...] We certainly have the system there but we <u>haven't brought that into the warehouse</u> ", Respondent C	• Increase awareness of kerbside traffic flow variance • Define <u>context feature</u> kerbside traffic flow • Extract and integrate <u>context information</u> from traffic and car parking systems directly • Associate feature with customer <u>context category</u>

Context feature (cf. Section 7.3.2)	Recurring issue (Coded data)	Recommendation (CAIS concept)
Runway movements	"This month here is a bad example because we receive the <u>runway data from air services virtually three months in arrears</u> . So, we've got some holes in that data", Respondent C	• Increase <u>awareness</u> of actual passenger departures • Define <u>context feature</u> runway movements • Extract and integrate <u>context information</u> from air service systems in higher frequency • Associate feature with customer <u>context category</u>

Table 7.20. Vignette: identifying context information needs based on CAIS

ADAPTATION STRATEGY COMPONENT

Respondents point to *recurring issues* in the implementation of adaptation strategies that affect system performance and which result from trade-offs between balancing security and terminal throughput. First, passenger facilitation adheres to strict occupational safety and aviation security regulations that are normative. This is evidenced by respondent B's comment: "there's certainly an opportunity for them to require us to undertake additional compliance activity or regulatory activity to meet those regulations and we have no choice". Second, meeting these regulations affects facilitation cost as a result of security training and slower terminal throughput. This is evidenced by respondent D's comment: "we are competing against Sydney and Melbourne and Perth to be an attractable port. If you're charging five dollars to land and Sydney is charging two dollars, where are they going to land". The objective is thus to enable selective processing, as noted by respondent B: "you meet certain requirements and that might mean you get the red, orange, or green light when you go through that process".

Currency of improving the responsiveness of passenger facilitation by defining selective screening processes is further evidenced by the following comment of respondent B, the airport security manager. He points to ongoing efforts in developing technology for information exchange and passenger tracking:

"So it's all in ... everybody is treated exactly the same. But that's why we look at technology and processes to potentially, to potentially, you know, provide assistance to identify people who are more of a threat", **Respondent B, Beta Corporation**

While adaptation strategies cannot mitigate legislative and regulatory issues, modelling adaptation strategies can define a pathway for enabling a more adaptive passenger facilitation system as show in Table 7.21. First, current adaptation strategies regarding security screening only work to increase the level of security screening applied to passengers. However, efforts in the

area of creating a passenger security profile and ongoing improvements such as electronic immigration cards already enable partially selective screening. This is evidenced by the comment of respondent B: "it doesn't change security at the moment, it gives you a quicker process in some areas like check-in". Future improvements in technology, as suggested by respondents, will provide the needed foundation for selective processing. Using adaptation strategies such as CAF-2 (perform alternative function), the department can model trade-offs between the level of security and improvements in throughput through selective processing. This depends on recommendations provided in the context modelling section.

Adaptation strategy (cf. Section 7.3.2)	Recurring issue (Coded data)	Recommendation (CAIS concept)
Security screening	"The current state is everybody is <u>treated exactly the same</u> and we don't even identify categories [...] But that's why we <u>look at technology and processes</u> to potentially [...] provide assistance to identify people who are more of a threat", Respondent B	• Increase <u>preparedness</u> by defining selective processes • Balance <u>trade-off</u> between security, passenger facilitation throughput, and cost • Model process as adaptation strategy <u>perform alternative functions</u> (CAF-02)

Table 7.21. Vignette: improving adaptation strategies based on CAIS

ACTIVATION RULE COMPONENT

While respondents do not point to immediate or recurring issues in the activation of system changes following a context change, activation is still *predominantly manual*. Most system changes are activated by the terminal duty manager, a role which coordinates the day to day activities at the airport including passenger facilitation. This is evidenced by the following comment of respondent E on the role of duty managers at the airport: "it could be a local standby, it could be a breach of sterile area, it could be a bomb threat [...] they're also the first point of contact for the airport so most of our airport stakeholders know that if there's some sort of issue or incident to call the duty managers". Where activation rules exist, they are provided with minimal automation and exist mainly to provide guidance to staff in decision-making situations. This is evidenced by following comment of respondent B, the manager responsible for airport security: "and most of the staff are equipped, where at least they can make decisions or seek some guidance or information to make those decisions to meet those changes".

However, continuous improvement and automation of manual decision-making processes rank high on the airport's agenda as work system participants seek to increase responsiveness and

decrease facilitation cost. This is evidenced by the following comment of respondent B, the airport's security manager:

"That's a strong business driver for us [...] You know, we might be able to use technology to take some of the people decisions out of the equation [...] you know, to hasten and speed it up so it's more efficient and more pleasing for the passengers", **Respondent B, Beta Corporation**

As part of increasing responsiveness and efficiency, modelling activation rules can define a pathway for reducing the amount of manual decisions that do not require the intervention of, e.g., the terminal duty manager. For example, aircraft deviations following weather events impact on terminal capacity and require system changes. This is evidenced by respondent B's comment: "we got to keep staff back. We might have to contact more staff to come on board. We might have to keep a larger number of screening points open". Aircraft delays are announced by aviation services and the activation of system changes can at least be partially automated. Similarly, thunderstorm warnings are issued by meteorological services and prompt the evacuation of staff from airside facilities. However, the responsibility to cater for passengers largely remains with airlines. Respondent A notes: "you got to start providing maybe pillows and blankets [...] but that's not the airport who has to find that". Defining a partially automated response by the airport has the potential to improve passenger satisfaction.

Activation rule (cf. Section 7.3.2)	Recurring issue (Coded data)	Recommendation (CAIS concept)
Aircraft deviation	"We got to keep staff back. We might have to <u>contact more staff to come on board</u> . We might have to <u>keep a larger number of screening points open</u> ", Respondent B	• Mitigate process build-ups by defining aircraft deviation as a trigger <u>event</u> for activation • Model capacity as required <u>condition</u> for rule activation • Activate shift changes, staffing, screening point <u>actions</u> if event and condition are met
Thunderstorm response	"You got to <u>start providing maybe pillows and blankets</u> , whatever you can find. But that's not the airport who has to find that, the pillows and blankets side of it, <u>it's the airlines' responsibility</u> ", Respondent A	• Increase satisfaction by defining thunderstorm as a trigger <u>event</u> for activation • Model passenger volume as <u>condition</u> for rule activation • Activate facilitation changes, and catering <u>actions</u> if event and condition are met

Table 7.22. Vignette: improving activation rules based on CAIS

LEARNING STRATEGY COMPONENT

Respondents point to *recurring issues* in the selection of learning strategies and the documentation of learning to improve and redesign passenger facilitation in accordance with changing needs and expectations. For example, the security side of passenger facilitation needs to be constantly updated to reflect changes to security policy and guidelines. This is evidenced by the following comment of respondent D, the screening manager and main contact for the screening contract managed by the airport: "I will then look at our screening manual and say erase the component of what we do at the oversize and change it to, this will have x-ray machine, that'll be done". However, not all respondents agree that issues, incidents, and the findings made are adequately documented. This is evidenced by the comment of respondent B, the overall security manager: "documentation isn't very good. We're very light on the ground. We do document more these days than we have in the past". This can create new issues: "you certainly see uninformed decisions being made because of poor data and poor information".

The currency of addressing learning and the continual improvement of passenger facilitation in light of context changes is further evidenced by the comment of respondent B, the airport's security manager. He points to the efforts undertaken by work system participants to continually improve system performance:

"Now part of our philosophy is having a continual improvement type approach to everything we do. So, even though it might be good, we always look at it again and say ... well how can we improve this, how can we make it cheaper, more efficient, reduce the costs, get greater outputs, etc.", **Respondent X, Beta Corporation**

Furthermore, applying learning strategies to recurring issues in the case data generates *actionable recommendations* as evidenced by the mapping presented in Table 7.23. First, the department can improve context models by integrating information from traffic management and car parking systems into its information infrastructure. This is evidenced by the comment of respondent C: "from the traffic lights, we're able to get throughputs [which would help] building a better model to understand traffic versus car parking versus the terminal". Second, the department has the ability to analyse and improve passenger facilitation changes by shifting from a largely undocumented to an at least partially documented repository of system adaptation strategies. This is evidenced by respondent B's comment: "documentation isn't very good. We're very light on the ground". Third, the redesign of how it receives security-relevant passenger information and how it manages traffic flow at the terminal has the potential to increase awareness of security context and better preparedness for traffic flow.

Learning strategy (cf. Section 7.3.2)	Recurring issue (Coded data)	Recommendation (CAIS concept)
Improving context representations (CLS-01)	"Building a better model to understand traffic versus car parking versus the terminal. But as far as analysing events like that <u>we don't do it at this stage</u> ", Respondent C	• Poor traffic information impacts on <u>context awareness</u> • Improve <u>context models</u> by integrating information from traffic and car park systems
Improving adaptation strategies (CLS-02)	" <u>Documentation isn't very good</u> . We're very <u>light on the ground</u> . We do document more these days than we have in the past", Respondent B	• Poor documentation impacts on terminal <u>preparedness</u> • Improve <u>adaptation strategies</u> by documenting standard operating procedures for different contextual situations
Transforming context representations (CLS-03)	"The <u>passenger information</u> and this is the one I have a <u>big problem with</u> , we are a national asset. We are a critical infrastructure asset", Respondent D	• Address lack of <u>awareness</u> by removing barriers to passenger information access • Redesign <u>context models</u> by collaborating with national and international authorities
Transforming adaptation strategies (CLS-04)	"One of the biggest issue we have up at the domestic is the fact that <u>we have a lot of pedestrian traffic</u> crossing the road which has a <u>huge impact on traffic flow</u> ", Respondent E	• Address lack of <u>preparedness</u> by removing traffic flow impediments at terminals • For example, <u>mitigate traffic evens</u> by building separate pedestrian access to terminal

Table 7.23. Vignette: improving learning strategies based on CAIS

7.5 DISCUSSION

In this chapter, I evaluated the applicability and actionability of the CAIS design theory by generating *novel insights* derived from the case data presented and analysed in Chapter 04. In Section 7.3, I evaluated the applicability of CAIS to design problems in the data and demonstrated its utility as a conceptual vocabulary to reason about context and context-driven system adaptation in the cases. For example, I demonstrated how context modelling provides a vocabulary to reason about extrinsic variables in the context of work systems (weather events, traffic flow, changes in customer needs and expectations) and the efforts undertaken by the organisations to collect, classify, and monitor context-related information (weather alert, risk in force, repeat call, busy hour, and traffic flow measures). I also evaluated the actionability of CAIS, i.e., its ability to generate immediate utility by proposing concrete and actionable recommendations for

ongoing system improvement. Examples include recommendations regarding the improvement of activation rules (Sigma Group) and adaptation strategies (Beta Corporation).

First, CAIS provides concrete and actionable recommendations regarding the analysis of system performance and the impact of context on the work objectives, technical component parts, and social component parts of information systems. In Chapter 04, I identified several recurring issues that arise as a result of the impact of context on information systems across both case sites. The application of socio-technical perspectives on information system design in Section 7.3 led me to conclude that such perspectives are a suitable solution to the issues reported by respondents in the case data. Furthermore, I realised that the application of socio-technical perspectives on information systems also lead to concrete, actionable recommendations regarding the creation of transparency and a shared understanding of system behaviour. For example, a socio-technical perspective on system design will enable Sigma Group to analyse interdependencies and potential impacts of change. Similarly, Beta Corporation will benefit from a shared understanding of system behaviour as a prerequisite to improved adaptation.

A socio-technical approach to viewing system behaviour highlights the interdependency of the technical and social system and emphasises the need for a joint adaptation of both systems. This addresses a range of issues reported by respondents in both case sites as evidenced by the following comments:

"One was telling the customer to throw the contents out and make a list and the other was telling our customers to retain their contents, which really was quite a failing of ours from a customer experience point of view", **Respondent A, Sigma Group**

"You certainly see uninformed decisions being made because of poor data and poor information [...] people tend to what we call reinvent the wheel or occasionally go over old ground before they get to a point where actually making a difference", **Respondent B, Beta Corporation**

Second, CAIS provides concrete and actionable recommendations regarding *context modelling* and the optimisation of context awareness in work systems. This is evidenced by interpretations of context awareness and context modelling in the data. In Chapter 04, I identified similarities in the exposure of the systems to context change in the form of weather patterns, traffic flow, and other variables. The application of context modelling in Section 7.3 lead me to conclude that context modelling is a suitable solution to increasing the context awareness of work system participants in both sites. However, I also realised that actionability is constrained by the ability of an organisation to gain access to context-related information. For example, Sigma Group has invested in technology by which it is supplied with regular information on the weather situation. It is thus aware of most weather events as they unfold. Beta Corporation, on

the other hand, has limited access to passenger information as a result of privacy. This affects its awareness of the security profiles of passengers that pass through its terminals.

Context modelling creates representations of context and relationships between external and internal work system variables to analyse the impact of context change on information system performance. This can help address a range of issues as evidenced by the following comments of respondents in both cases:

"So, those mechanisms there are really useful and I never had a mechanism before and I found it very difficult to rely on self routine to check the Bureau daily, now, I've got a mechanism that can do it and send me a weather warning to say this is likely", **Respondent A, Sigma Group**

"The passenger information and this is the one I have a big problem with, we are a national asset. [...] we're the biggest carrier of public transport out there apart from rail and bus. Except we're a private enterprise. So the information that comes from government is vetted", **Respondent D, Beta Corporation**

Third, CAIS provides concrete and actionable recommendations regarding *adaptation strategies* and the improvement of work systems preparedness. This is evidenced by interpretations of how work system participants describe preparedness and the management of system adaptation strategies. In Chapter 04, I identified similarities in the rapid adaptation of work system variables to context change such as process change, changes in staffing levels, and technology. The application of adaptation strategies in Section 7.3 lead me to conclude that such strategies are a suitable solution to the requirement of preparedness. However, there exist differences in the degree to which work system participants document, communicate, and activate such strategies. For example, Sigma Group documents adaptation strategies in its "event manual". This manual defines context-driven changes applied to the claims handling system and the responsibility for each change. Respondents in the Beta Corporation case, on the other hand, describe their approach to adaptation as less systematic and more pragmatic.

Adaptation strategies increase preparedness by modelling changes applied to different variables classes of the technical and social components of information systems. They furthermore emphasise the need to consider and define adaptation trade-off. This can help address a range of issues as evidenced below:

"Within an hour of the event actually striking [...] you won't have all the information to make correct determinations or decisions on, they'll be stock standard what we do every time [...] the event unfolds and the information is a bit more free flowing then we can actually talk about process", **Respondent A, Sigma Group**

"From the seat of their pants what it really boils down to. It's really having people on the ground and dealing with people. So, the issue ... the things you got to do, you got to have experienced, knowledgeable, skilled people in the jobs", **Respondent B, Beta Corporation**

Fourth, CAIS provides concrete and actionable recommendations regarding *activation rules* and fostering improvements in work system responsiveness. This is evidenced by interpretations of activation rules and the communication of system changes in the data. In Chapter 04, I identified similarities in the requirement to activate system changes in specific contextual circumstances such as a surge in call volume or aircraft delays. Furthermore, the application of activation rules in Section 7.3 lead me to conclude that they represent a suitable means to improve work system responsiveness. This is further evidenced by interpretations of the inconsistent implementation of system changes and the need to communicate system changes in a consistent fashion. First, Sigma Group operates two call centres and claims handling changes in a context must be implemented in speedy and consistent fashion. This is evidenced by instances of communication failure in the data. Second, Beta Corporation relies on stakeholder communication to realise system changes and to identify potential issues of such changes.

Activation rules create a shared understanding of the events, conditions, and actions associated with system changes in a particular context and offer potential for partial or full rule automation. This addresses a range of issues reported by respondents in both case sites as evidenced by the following comments:

"There's a lot of ad hoc communication happening, which is not communicated at the time necessarily, which causes some issues so, hopefully that's one of the things with communication we can solve, the communications response system, when it goes out, it goes out to all", **Respondent F, Sigma Group**

"But there's got to be very good stakeholder management. We actually have to understand what the other person's is about so we can anticipate the problems they might have and we actually have to be able to respond to their needs pretty quickly, which obviously means having the right people", **Respondent B, Beta Corporation**

Lastly, CAIS provides concrete and actionable recommendations regarding *learning strategies* and the efforts undertaken by organisations to continually improve and redesign context-driven adaptation. This is evidenced by interpretations of learning and learning approaches in the data. In Chapter 04, I identified similarities in the cycle of learning from context and how both case sites use tools such as event simulation and post-event reviews to continually improve context models and adaptation strategies. Furthermore, I demonstrated in Section 7.3 that context learning strategies are a suitable means to reason about different approaches to learning from

context. Structured approaches to learning from context are still very much in their infancy across both sites. Respondents at Sigma Group voice concerns regarding the effectiveness of post-event debrief sessions. Similarly, respondents at Beta Corporation suggest difficulties in identifying situations that are repetitive and for which learning can be specified. Learning strategies promise to provide more clarity about learning types and the scope of learning.

Learning strategies focus attention on two types of learning and provide guidance regarding the effectiveness and completeness of learning strategies applied to context learning in an information system. This addresses a range of issues reported by respondents in both case sites as evidenced in the comments below:

"There is a debrief session that they have but it's too short, because it's an hour session, too many people involved [...] It's not an effective debriefing session, in my view and in any event there's very little change that takes place with the decisions that are being made", **Respondent E, Sigma Group**

"Operationally the airport is on its front foot. Every day and every minute of the day, there are decisions being made because things are out of the ordinary [...] it's very, very hard to define what's the norm and what's normal", **Respondent B, Beta Corporation**

Table 7.24 summarises the recommendations and insights generated by the application of CAIS to case data collected at Sigma Group and Beta Corporation. Overall, Sigma Group has a more consistent approach to monitoring context variables and managing adaptation strategies by means of its event response system. This is evidenced by interpretations of the weather alert mechanism, event module, event manual, and the preplanning of system changes by the system design team. Beta Corporation, on the other hand, still relies very much on its staff to monitor context and take decisions regarding the adaptation of passenger facilitation to context changes. This is evidenced by interpretations of traffic flow, passenger flow, the terminal duty manager role, and the lack of documentation regarding context-driven system changes. The application of CAIS, however, provides applicability as it creates a standard vocabulary to reason about system decisions. Furthermore, it provide actionability as reasoning about system decisions leads to concrete and actionable recommendations.

System issue	Illustration of issue (Sigma Group)	Illustration of issue (Beta Corporation)	Recommendation (based on CAIS)
SYSTEM BEHAVIOUR			

System issue	Illustration of issue (Sigma Group)	Illustration of issue (Beta Corporation)	Recommendation (based on CAIS)
Lack of transparency of system behaviour	"This is the <u>first test of a brand new system</u> and brand new process that goes with that system", Respondent A	"Airlines are <u>probably better than airports</u>. Airlines actually document their systems quite well", Respondent B	<i>Use four system viewpoints to model system interdependencies</i>
SYSTEM CONTEXT			
Lack of awareness regarding relevant context variables	"The <u>first I heard about</u> it was on SGMday morning when I turned on the TV", Respondent E	"There's going to be times when <u>there might be some holes</u> in the data", Respondent C	<i>Use decomposition to decompose context into categories and features</i>
Lack of awareness regarding impact of context on system	"The neighbourhood were swapping photos [as proof] <u>and we'd pay it</u>", Respondent E	"We didn't become <u>aware of [the floods]</u> till probably [...] in the afternoon", Respondent E	<i>Based on decomposition, perform analysis of system impact areas</i>
ADAPTATION STGY.			
Lack of preparedness regarding relevant context change	"I think <u>we should clarify the roles of everybody</u> that's in the ELT team", Respondent D	"The company is <u>only just undertaking it's first step</u> into business continuity", Respondent B	<i>Using adaptation matrix, identify adaptation scenario and preparedness</i>
Lack of preparedness regarding adaptation trade-off	"<u>No one typically looks at leakage</u>, because the claims are so volatile", Respondent E	n/a	<i>Weigh benefit of adaptation strategy with cost of implementation</i>
ACTIVATION RULE			
Lack of responsiveness as a result of inconsistent trigger	"Every one uses the repeat call matrix [...] <u>but no one takes the time to look</u>", Respondent E	n/a	<i>Use heuristics to specify event and condition part of activation rule</i>
Lack of responsiveness from inconsistent implementation	"We need to improve moving forward, as an example around <u>what our staff say to customers</u>", Respondent A	"You certainly see <u>uninformed decisions</u> being made because of poor data and poor information", Respondent B	<i>Map activation rule to one or more adaptation strategies and events</i>
LEARNING STGY.			

System issue	Illustration of issue (Sigma Group)	Illustration of issue (Beta Corporation)	Recommendation (based on CAIS)
Lack of learning as a result of inconsistent learning strategies	"It's not an effective debriefing session, [...] in any event there's <u>very little change</u> that takes place", Respondent E	"Documentation isn't very good. We're <u>very light on the ground</u>. We do document more", Respondent B	<i>Select appropriate learning strategy and implementation using matrix</i>
	<i>Increased focus on understanding the impact of adaptation on cost</i>	<i>Increased focus on documentation and joint optimisation of systems</i>	
	Case recommendation (based on CAIS)		

Table 7.24. Summary of recommendations and insights generated by CAIS

7.6 CONCLUSION

In this chapter, I evaluated the applicability of CAIS to the needs of system practitioners and its ability to generate actionable recommendations for the design of more effective information systems. In particular, I found evidence for the ability of CAIS to create recommendations regarding the analysis of system variance, understanding context information needs, designing system adaptation strategies, defining activation rules for their activation in particular contextual situations, and the ability to improve and enhance context awareness through four types of context learning. In addition, I demonstrated the ability of CAIS to generate new insights and design knowledge through its application to the empirical evidenced collected in Chapter 04. Two issues remain that must be addressed by future research. First, the recommendations generated by the application of CAIS must be empirically validated on the basis of, e.g., focus groups (Tremblay et al., 2010) or experimentation (Arazy et al., 2010). Second, the expository instantiation (Gregor & Jones, 2007) of the CAIS meta-design may stimulate future research.

Chapter 8: Conclusion

8.1 REVIEW OF RESEARCH FINDINGS

In the following, I review the research findings generated by the study and map them to the research questions asked in Chapter 01 to assess the progress that was made in advancing our understanding. In Chapter 01, I asked three questions pertaining to context, the impact of context change on information systems, and how information systems should be changed to adapt to context change. I map each research question to findings generated in the research study.

Relevant context is *information about the situation of entities* such as customers, suppliers, workforce, places, products, or services that support an organisation in meeting its work objectives. In the beginning of the research study, I raised questions about the nature of context and its role in context-aware system design. As a result of a literature review, I identified two divergent views of context as information and occasioned property. Collecting empirical evidence about context then led me to suggest that context information is an important factor in system design. Based on the example of work systems in insurance and aviation passenger facilitation, I concluded that access to context information enables the rapid adaptation of work systems to context change. Furthermore, I analysed the support provided by contemporary methods for work system analysis and design (Alter, 2006) and found they provide little prescriptive guidance on considering context. Finally, I used theory-driven design to develop a set of prescriptive statements about systems and system design that are hypothesised to address this need.

I summarise the mapping of research question to research findings generated in the course of study as follows:

- **Research Question 1.** *What is relevant information system context?* Two divergent views of context can be identified of which the representational view and more specifically predictable context are relevant for achieving the rapid adaptation of information systems to a variety of contextual situations.

Context change impacts on both the *technical and social component of information systems* used by an organisation and reduces its ability to consistently meet work objectives. In the beginning of the research study, I raised questions about the nature of information systems and assumptions about the component parts which must be considered in system design. A review of the literature led me to conclude that two views of information systems are prevalent: as an

artefact and as a system. Collecting empirical evidence about context impact on the component parts then led me to suggest that both technical and social component parts must be considered. Examples include changes to information processing and awareness of system members of changes in processing rules. Comparing methods for work system analysis and design, I concluded that situations and interdependencies between component parts that affect adaptation are poorly understood. By applying theory-driven design, I developed prescriptive statements about systems and system design that are hypothesised to address this need.

I summarise the mapping of research question to research findings generated in the course of study as follows:

- **Research Question 2.** *Where does context change affect information systems?* Information systems are socio-technical systems comprising technical and social component parts. Context change equally impacts on both and their interdependency must be considered when defining system change.

Furthermore, adapting information systems to context change must *place equal weight* on adapting the technical and social components of information systems to rapidly recover from a loss in performance. In the beginning of the research study, I raised questions about the nature of adaptation and the assumptions that underly the adaptation of systems. Based on a review of the foundational literature of IS, I identified the concepts of adaptive systems (Ackoff, 1971) and the joint adaptation of technical and social system (Bostrom & Heinen, 1977) as justificatory knowledge for the prescription of context-aware information systems. Mapping these concepts to design-relevant theory for work system analysis and design, I found that work system adaptation to context is not defined. However, defining adaptation is a relevant problem as evidenced by the rapid and planned adaptation of work systems observed in the insurance and passenger facilitation cases. I turned to theory-driven design in Chapter 06 to develop prescriptive guidelines regarding system adaptation, change activation, and context learning strategies.

I summarise the mapping of research question to research findings generated in the course of study as follows:

- **Research Question 3.** *How should information systems be changed in order to adapt to relevant context change?* System change must place equal weight on the technical and social component parts of information systems to enable the rapid adaptation of information systems to context change.

Finally, I tested the utility claims made by the CAIS design theory and found it applicable (by providing a *vocabulary* for viewing system decisions) and actionable (by generating concrete

recommendations regarding system improvement). First, applicability is a measure of utility which applies both to design theory and design artefacts. It represents the ability of DS outcomes to meet the needs and expectations of system practitioners (Rosemann & Vessey, 2007). I evaluated the applicability of CAIS by mapping its concept to empirical data collected in Chapter 04 and found that it is suitable to view and reason about system decisions as evidenced by the data. Second, actionability is a measure of utility which applies both to design theory and design artefacts. It represents the ability of DS outcomes to define concrete and actionable recommendations that promise to provide immediate utility to system members (Kuechler & Vaishnavi, 2011). I evaluated CAIS actionability by applying it to the same data I had used to assess applicability. I found opportunities to improve the decisions of system members through the lens of CAIS.

8.2 CONTRIBUTIONS AND IMPLICATIONS

In the following, I summarise the contributions and implications generated by the research study for IS research, practice, and method. Research in IS must lead to the generation of novel knowledge that has the potential to provide relevance by answering relevant questions of system practitioners but which must also adhere to the strict rules of scientific enquiry (Benbasat & Zmud, 1999; Rosemann & Vessey, 2007). Furthermore, design theorising in IS research is still an emerging field and methods for conducting DS research studies in IS such that they can be communicated and evaluated by the academic community are sparse (Gregor & Jones, 2007; Kuechler & Vaishnavi, 2012). I argue that the research study makes contributions in three relevant areas as evidenced in the following: research, practice, and method.

8.2.1 IMPLICATIONS FOR RESEARCH

First, the research study integrates previously disparate research streams in the CS, HCI, and IS fields into *one classification* of context and information systems. In Chapter 03, I reviewed the academic literature of IS and its reference fields (Benbasat & Weber, 1996) and summarised divergent views about context and information systems. First, context can be viewed as information that can be encoded and represented or as an occasioned property that cannot be represented (Dourish, 2004). These two views explain the different assumptions of scholars about the nature of context and decisions regarding its consideration in system design. Second, information systems can be viewed as artefacts or as systems (Alter, 2008). Depending on the assumptions about the nature of information systems held by scholars, the emphasis of design shifts from the design of technology to the socio-technical design of work systems (McKay, Marshall, & Hirschheim, 2012). The combination of these views led me to define a descriptive

framework based on which research contributions in the investigated research fields can be uniquely organised.

Second, the research study identifies *deficiencies* in contemporary IS theory, namely the Work System Method (WSM), with respect to context and its role in system design. Based on the application of representation analysis (Weber, 2012) to the domain of design theorising (Kuechler & Vaishnavi, 2012), I identified five measures of the quality of design theory. I then applied these measures to an instance of IS design-relevant theory, i.e., WSM, to measure its completeness and clarity with respect to the concept of adaptive systems (Ackoff, 1971). This led me to identify 8 deficiencies that reduce the theoretical completeness and clarity of WSM with respect to reasoning about context-aware information systems. Furthermore, I measured system concepts in empirical evidence about context in system design that I had collected in Chapter 04 and identified three extensions to reason about context-aware information systems: predictable context, unpredictable context, and context change.

Finally, the research study combines adaptive systems and socio-technical views to make *prescriptive statements* about context-aware information systems that promise to lead to more effective system design. In Chapter 06, I argued that the representation of predictable context and the joint consideration of the technical and social information system components enable their rapid adaptation. I then proceeded to map design concepts generated in the previous chapter, Chapter 05, to one or more meta-requirements (Walls, Widmeyer, & Sawy, 1992) of the CAIS class. In a next step, I mapped each meta-requirement to a set of meta-design features and discussed the current state of research for each of these features in the academic literature. Where applicable, I provided illustrations based on the case data I collected in Chapter 04 and described how each feature can be realised. Finally, I mapped each meta-design feature to exactly one testable design proposition that relates the feature to a meta-requirement. I thus defined assertions that evaluate whether the meta-design of CAIS satisfies the goals specified by its meta-requirements.

8.2.2 IMPLICATIONS FOR PRACTICE

First, the research study generates prescriptive statements about the design of information systems that are hypothesised to lead to more effective outcomes in terms of their context-awareness. Based on the concepts of adaptive systems (Ackoff, 1971) and socio-technical systems (Bostrom & Heinen, 1977), I identified five meta-requirements and five corresponding meta-design features of the CAIS class of systems. The meta-requirements addressed by the CAIS design theory are the interdependency of technical and social system variables, the

awareness of context and context changes, the preparedness of systems with respect to such changes, the responsiveness of systems to context change, and their ability to continually learn from context-driven adaptation. Furthermore, the design theory prescribes five meta-design features including the socio-technical analysis of system variables, context modelling, defining adaptation strategies, activation rules, and context learning strategies. If followed, these prescriptions promise to lead to more effective decisions regarding work system design and implementation.

Second, the research study provides evidence regarding the applicability and actionability of its prescriptions on the basis of case sites selected from different industries. In Chapter 07, I provided empirical evidence regarding the ability of CAIS to generate suitable solutions to relevant problems and to provide actionable recommendations regarding recurring system issues. First, I demonstrated that CAIS provides a conceptual vocabulary to reason about relevant problems such as the consideration of predictable context in system design. This is evidenced by the applicability of CAIS to system decisions made by system members in two independent case studies. Second, I demonstrated that the prescriptions made by CAIS generate concrete, actionable recommendations regarding further improvement of system performance such as the inclusion and modelling of additional context variables. This is evidenced by shortcomings in both case sites regarding the selection of context variables and activation of context-driven changes. CAIS thereby provides prescriptive guidance on context in system design.

8.2.3 IMPLICATIONS FOR METHOD

First, the research study is characterised by the *explicit consideration* of empirical evidence in defining and reasoning about a design problem as evidenced by its use of the case method. Insights into design problems in design science often remain implicit and are generally not formalised (Kuechler & Vaishnavi, 2012). By turning to empirical methods of enquiry such as the case method (Yin, 2009), I provided complete traceability from research questions, to empirical insights, to decisions regarding the development of prescriptive statements. At each stage of the enquiry, I motivated decisions both on the basis of justificatory knowledge in kernel theory as well as their instantiation in the case data to provide a vivid account of the design challenges and issues faced by system practitioners in considering and utilising context information in system design. Furthermore, I demonstrated the utility of design theory by measuring its ability to generate new insights from the same data in Chapter 07. This provides a complete chain of evidence from coded data, emerging concepts, design prescriptions, and utility measurement.

Second, the research study is characterised by the *application* of methods derived from representation analysis (Weber, 2012) to design theorising in IS design science. In Chapter 05, I introduced and defined the notion of completeness and clarity of design theory on the basis of representation analysis (Weber, 2012). I identified five patterns of theory-driven design based on which the completeness and clarity of design theory with respect to its underlying kernel theory can be measured. First, construct mapping maps a single kernel-theory construct to exactly one design-theory construct. Second, construct extension introduces a new design-theory construct based on kernel-theory if missing from the design theory. Third, construct specialisation splits one design-theory construct into two or more constructs if it maps to two or more of the kernel-theory constructs. Fourth, construct reduction removes one or more redundant design-theory constructs from a design theory. Finally, construct elimination removes a design-theory construct if it maps to none of the constructs of the underlying kernel theory.

Finally, the combination of empirical and representational methods of analysis is the *first documented* instance of a prescriptive method for theory-driven design. Previous work on design theorising (Gregor & Jones, 2007; Walls et al., 1992) and its application to design science (Markus, Majchrzak, & Gasser, 2002; Müller-Wienbergen, Müller, Seidel, & Becker, 2011) recognises the need for the explicit representation of design knowledge in the form of prescriptive statements. Furthermore, recent contributions introduce and define a typology of design theories (Kuechler & Vaishnavi, 2012) and the relationships between different levels of reasoning. However, none of the described methods propose a prescriptive approach by which the mapping of justificatory knowledge to design-relevant explanatory concepts, to prescriptive statements, to the evaluation of utility of the prescriptions can be made explicit. By combining contributions on theory development in design science research, representation analysis and the measurement of theory goodness (Weber, 2012), and theory building from case data (Eisenhardt, 1989), I defined a new method for theory-driven design.

8.3 LIMITATIONS

Furthermore, limitations of the research study can be identified in the area of literature review, design and conduct of case study research, and the evaluation of design theory utility. First, the literature review as presented in Chapter 03 is concept-driven and selective rather than comprehensive. Second, the limitations of qualitative research and the case method in particular are well known (Benbasat, Goldstein, & Mead, 1987). In the following, I point out specific limitations of applying case study research in my study. Third, the evaluation of design theory utility is analytic rather than empirical and questions remain whether concrete instantiations of the

theory in practice will reliably lead to improvements if its recommendations are followed. Finally, I show how in each case I mitigated the issue of limitations.

8.3.1 LIMITATIONS IN LITERATURE REVIEW

The identification of sources and conduct of the literature review presented in Chapter 03 is selective but not comprehensive (Webster & Watson, 2002). Contributions in the review were selected based on their relevance to the research problem of developing prescriptive guidelines for the design and construction of CAIS. In this respect, the literature review was selective. Arguably, context has been the subject of research in other fields not sampled in the review. For example, research on context can also be found in the fields of natural language processing, databases and ontologies, computer vision, and machine learning (for an overview of context in these areas, cf. Brézillon, 1999). Similarly, context has also been recognised as a problem in philosophy and linguistics (cf. Akman & Surav, 1996). Finally, an understanding of context in its various shapes and definitions has been attempted by Bazire *et al.* (2005). Insofar as these fields of research have not been included in the review, it cannot be considered comprehensive. A complete review of context in the academic literature would need to cover these fields.

However, care was taken to include contributions from peer reviewed outlets that are *representative* for the respective fields of study and relate to the research problem of developing prescriptive guidelines. Context in system design is not a new phenomenon (Nardi, 1996). In fact, designing context-aware systems has been discussed in the fields of CS (Weiser, 1993) and HCI (Suchman, 1987) over an extended timeframe. Where these contributions relate to the study of problems that arise at the intersection of organisation, people, and technology as the core phenomenon of interest in the IS field (March & Storey, 2008), they were included in the review. Furthermore, I used tools such as ERA rankings and Web of Science in order to assess the quality of the contributions I selected. In particular, I consulted rankings for outlets in which a particular article appeared and traced referencing a) within an article and b) to the article from other articles following established guidelines (Webster & Watson, 2002). I argue that these measures ensured that the findings are an adequate reflection of the body of knowledge.

8.3.2 LIMITATIONS IN CASE STUDY DESIGN AND CONDUCT

Case sites, while following the case study protocol, were selected based on *access* through key contacts such that selection bias cannot be fully excluded. In the design of the exploratory case study in Chapter 04, I defined selection criteria following guidelines for case study research (Eisenhardt, 1989; Yin, 2009) and conducted a review of secondary evidence (Ploesser, Recker, & Rosemann, 2008) to compile a list of potential case candidates (presented in the

manuscript appendix). The final selection of case sites, however, was also made on the basis of convenience and access. Both case sites are Australian organisations headquartered in Brisbane and key contacts existed between Queensland University of Technology and the organisations on which I relied. While this approach allowed me to limit the impact of case selection on study resources and time, a selection bias cannot be fully excluded. However, I ensured that within these constraints, the selection of case sites is an adequate reflection of the selection criteria and that findings are generalisable by selecting cases from different industries.

Furthermore, the conduct of case interviews, coding of interview data, and analysis of coded text were conducted by a single researcher such that researcher bias cannot be fully excluded. In preparation of the exploratory case study presented in Chapter 04, I attended coursework offered by Queensland University of Technology in order to familiarise myself with methods of empirical enquiry (INN700, INN701). Furthermore, I consulted the academic literature on case study design (Yin, 2009), qualitative interviewing (H. J. Rubin & Rubin, 2011), qualitative data analysis (Miles & Huberman, 1994), and guidance on developing theory from data (Eisenhardt, 1989). I also consulted with my supervisors and other academics in the faculty who have experience with qualitative research and using the case method in IS research (Bandara, 2009). Finally, case study findings were published and presented at two major IS conferences (Ploesser, Recker, & Rosemann, 2010; 2011) to incorporate feedback from the academic community. Within the constraints, I cannot fully exclude a potential bias introduced by a single researcher.

8.3.3 LIMITATIONS IN UTILITY EVALUATION

Further empirical validation is necessary to test whether the recommendations made by the application of CAIS to case data will lead to lasting improvements in context-awareness for the case sites. The evaluation of utility presented in Chapter 07 works on the assumptions of theory applicability and actionability and the generation of new insights into case data and recommendations on the basis of CAIS concepts. Whether those recommendations, if implemented, will lead to a measurable impact on context-awareness of the studied work systems will, in practice, depend on a number of factors (Delone & McLean, 2003). It is thus necessary to conduct further empirical validation of the design theory by means of expository instantiation (Gregor & Jones, 2007) and the testing of artefact utility within the domain of application. Examples for such instantiations and their measurement can be found for creative group support (Müller-Wienbergen et al., 2011) and recommender systems (Arazy, Kumar, & Shapira, 2010). While this type of evaluation is part of the wider research program, it was not included given scope and time constraints of the study.

However, care was taken that the evaluation of the CAIS design theory corresponds to established principles in the DS literature and that the selection of utility variables are adequate for testing design theory. Kuechler and Vaishnavi (2012) suggest that we cannot evaluate the utility of design theory directly. Instead, we must evaluate the relevance of design theory in meeting the needs of system practitioners (Rosemann & Vessey, 2007) and by providing immediate utility through the generation of actionable recommendations (Kuechler & Vaishnavi, 2011). Within these parameters, I evaluated the CAIS design theory on the basis of empirical evidence and found it to be applicable and actionable. First, CAIS provides a conceptual vocabulary to consistently reason about a range of system decisions in the case sites. Second, mapping these decisions to CAIS prescriptions, one can generate concrete and actionable recommendations that promise to lead to an increase in system performance. As suggested earlier, whether or not they lead to factual improvements depends on a range of factors and must be evaluated.

8.4 FUTURE RESEARCH

The objective of the research study presented in this manuscript was to provide prescriptive guidelines to a relevant and current design problem through theory-driven design. Context and context in system design have long been the subject of research in the CS and HCI fields of research. However, the research has not translated into prescriptions regarding the design of effective context-aware *information systems*. From this point of view, the findings presented in the research study are the first documented attempt at defining such guidelines. Scientific research in the IS discipline as much as in other fields of research is characterised by a cumulative tradition (Webster & Watson, 2002). In this sense, the research study draws and builds on the work of others in system thinking (Ackoff, 1971), socio-technical design (Bostrom & Heinen, 1977; Mumford, 2000), conceptualising context (Dourish, 2004; Suchman, 1987), and design theorising (Kuechler & Vaishnavi, 2012). By the same token, I am confident my findings can provide early insight and stimulate future studies on a relevant and poorly understood phenomenon: *context*.

Going forward, I identify a number of pathways along which future research can build on and improve on the findings made by the research study. No research project can be considered complete and new insights are likely to emerge in future considerations of the topic. In the following, I present and briefly discuss five such pathways along which future research can generate additional insights that were not in scope of this study. These pathways are as follows:

- Examining in more detail the role theory-driven design, as defined in Chapter 02, can play in increasing the quality of design theorising and the utility of design outcomes by

tracing the link between justificatory knowledge and its manifestation in design artefacts.

- Expanding the collection and assessment of DS outputs in the areas of context and context-awareness using the classification scheme introduced and defined in Chapter 03 beyond the scope selected in the research study.
- Corroborating the findings generated during the empirical stage of the study in Chapter 04 with findings from other industries regarding the adaptiveness and context-awareness of work systems.
- Integrating the conceptual modelling outcomes in Chapter 05 and 06 with published work on the conceptualisation of work systems and the extension of WSM (Alter, 2009) toward context-awareness.
- Finally, creating expository instantiations of CAIS on the basis of the prescriptions presented in Chapter 06 and testing them in the domain of application by using methods for DS evaluation such as experimentation (Arazy et al., 2010) or focus group research (Tremblay, Hevner, & Berndt, 2010).

In summary, the manuscript narrates the course of my research study from initial problem formulation, to investigation of the academic literature, to exploration of the problem through empirical methods of research, to design theory development, theory-driven design, and the evaluation of design theory utility. While the study in itself is complete, many questions remain (as briefly discussed above) to which I hope future research will be able to provide satisfactory answers. The program of research into context in information systems and context-driven adaptation will therefore continue. Despite many years of research, context-awareness and context-aware computing are still in their infancy. However, the topic is increasingly attracting attention as evidenced by several publications by industry analysts such as the Gartner Group (cf. Clark, 2011; Lapkin, 2011). I am therefore confident that future research will be able to build on and extend the work presented in this thesis.

References

- Abowd, G. D., Mynatt, E. D., & Rodden, T. (2002). The Human Experience. *Pervasive Computing*, 1(1), 48–57.
- Ackoff, R. L. (1968). Management Misinformation Systems. *Management Decision*, 2(1), 4–8.
- Ackoff, R. L. (1971). Towards a System of Systems Concepts. *Management Science*, 17(11), 661–671.
- Ackoff, R. L. (1973). Science in the Systems Age. *Operations Research*, 21(3), 661–671.
- Adomavicius, G., & Tuzhilin, A. (2010). Context-Aware Recommender Systems. In F. Ricci, L. Rokach, B. Shapira, & P. B. Kantor (Eds.), *Recommender Systems Handbook*. Berlin, Germany: Springer.
- Adomavicius, G., Sankaranarayanan, R., Sen, S., & Tuzhilin, A. (2005). Incorporating Contextual Information in Recommender Systems Using a Multidimensional Approach. *ACM Transactions on Information Systems*, 23(1), 103–145.
- Aguilar, F. J. (1967). *Scanning the Business Environment*. New York, NY: Macmillan.
- Akman, V., & Surav, M. (1996). Steps Toward Formalizing Context. *AI Magazine*, 17, 1–18.
- Alter, S. (2001). Are the Fundamental Concepts of Information Systems Mostly about Work Systems? *Communications of the Association for Information Systems*, 5(11).
- Alter, S. (2003). 18 Reasons Why IT-Reliant Work Systems Should Replace “The IT Artifact” as the Core Subject Matter of the IS Field. *Communications of the Association for Information Systems*, 12(23).
- Alter, S. (2006). *The Work System Method*. Larkspur, CA: Work System Press
- Alter, S. (2006b). Work Systems and IT Artifacts-Does the Definition Matter. *Communications of the Association for Information Systems*, 17(14), 299–313.
- Alter, S. (2008). Defining Information Systems as Work Systems. *European Journal of Information Systems*, (17), 448–469.
- Alter, S. (2009). Metamodel for Understanding, Analyzing, and Designing Sociotechnical Systems. *Sprouts: Working Papers on Information Systems*, 9(59), 1–47.
- Alter, S. (2010). Bridging the Chasm Between Sociotechnical and Technical Views of Systems in Organizations (pp. 1–24). Presented at the International Conference on Information Systems 2010.
- Alter, S. (2013). Work System Theory: Overview of Core Concepts, Extensions, and Challenges for the Future. *Journal of the Association for Information Systems*, 14(2), Article 1.
- Analyti, A., Theodorakis, M., & Spyrtos, N. (2007). Contextualization as an Independent Abstraction Mechanism for Conceptual Modeling. *Information Systems*, 32(10), 24–60.
- Anderson, P. (1999). Complexity Theory and Organization Science. *Organization Science*, 10(3), 216–232.
- Arazy, O., Kumar, N., & Shapira, B. (2010). A Theory-Driven Design Framework for Social Recommender Systems. *Journal of the Association for Information Systems*, 11(9), 455–490.

- Argyris, C., & Schon, D. A. (1978). *Organizational Learning: A Theory of Action Perspective* (Addison-Wesley Series on Organization Development). Reading, MA: Addison-Wesley.
- Ashby, W. R. (1956). *An Introduction to Cybernetics*. New York, NY: J. Wiley.
- Ashmos, D. P., & Huber, G. P. (1987). The Systems Paradigm in Organization Theory. *The Academy of Management Review*, 12(4), 607-621.
- Bandara, W. (2009). *Process Modelling Success Factors and Measures*. (Doctoral dissertation). Queensland University of Technology.
- Bandara, W., Gable, G. G., & Rosemann, M. (2005). Factors and Measures of Business Process Modelling: Model Building through a Multiple Case Study. *European Journal of Information Systems*, 14(4), 347-360.
- Barney, J. (1991). Firm Resources and Sustained Competitive Advantage. *Journal of Management*, 17(1), 99-120.
- Baskerville, R., & Pries-Heje, J. (2010). Explanatory Design Theory. *Business & Information Systems Engineering*, 2(5), 271-282.
- Baskerville, R., & Wood-Harper, A. T. (1998). Diversity in Information Systems Action Research Methods. *European Journal of Information Systems*, 7(2), 90-107.
- Bazire, M., & Brézillon, P. (2005). Understanding Context Before Using It. In A. K. Dey, B. Kokinov, D. Leake, & R. Turner (Eds.), *Proceedings of The 5th International and Interdisciplinary Conference on Modeling and Using Context*, 05-08 Jul 2005, Paris, France.
- Benbasat, I., & Weber, R. (1996). Research Commentary: Rethinking Diversity in Information Systems Research. *Information Systems Research*, 7(4), 389-399.
- Benbasat, I., & Zmud, R. W. (1999). Empirical Research in Information Systems. *MIS Quarterly*, 23(1), 3-16.
- Benbasat, I., & Zmud, R. W. (2003). The Identity Crisis within the IS Discipline. *MIS Quarterly*, 27(2), 183-194.
- Benbasat, I., Goldstein, D. K., & Mead, M. (1987). The Case Research Strategy in Studies of Information Systems. *MIS Quarterly*, 11(3).
- Bernus, P. (2006). *Handbook on Architectures of Information Systems*. New York, NY: Springer.
- Bessai, K., Claudepierre, B., & Saidani, O. (2008). Context-aware Business Process Evaluation and Redesign. In: *Proceedings of The 9th Workshop on Business Process Modeling, Development, and Support*, 16-17 Jun 2008, Montpellier, France.
- Bettini, C., Brdiczka, O., Henriksen, K., Indulska, J., Nicklas, D., Ranganathan, A., & Riboni, D. (2010). A Survey of Context Modelling and Reasoning Techniques. *Pervasive and Mobile Computing*, 6(2), 161-180.
- Blomberg, J., Suchman, L., & Trigg, R. H. (1996). Reflections on a Work-Oriented Design Project. *Human-Computer Interaction*, 11(3), 237-265.
- Bostrom, R. P., & Heinen, J. S. (1977a). MIS Problems and Failures: A Socio-Technical Perspective. *MIS Quarterly*, 1(3), 17-32.
- Bostrom, R. P., & Heinen, J. S. (1977b). MIS Problems and Failures: A Socio-Technical Perspective, Part II. *MIS Quarterly*, 1(4), 11-28.
- Boulding, K. E. (1956). General Systems Theory-The Skeleton of Science. *Management Science*, 2(3), 197-208.

- Brdiczka, O., Crowley, J., & Reignier, P. (2007). Learning Situation Models for Providing Context-Aware Services. In: *Proceedings of the 4th International Conference on Universal Access in Human-Computer Interaction*, 22-27 Jul 2007, Beijing, China.
- Brézillon, P. (1999). Context in Problem Solving: A Survey. *The Knowledge Engineering Review*, 14(1), 47–80.
- Brown, S. L., & Eisenhardt, K. M. (1997). The Art of Continuous Change: Linking Complexity Theory and Time-Paced Evolution in Relentlessly Shifting Organizations. *Administrative Science Quarterly*, 1–34.
- Chaudhuri, S., Dayal, U., & Narasayya, V. (2011). An Overview of Business Intelligence Technology. *Communications of the ACM*, 54(8), 88–98.
- Chen, H., Perich, F., Finin, T., & Joshi, A. (2004). SOUPA: Standard Ontology for Ubiquitous and Pervasive Applications (pp. 258–267). In: *Proceedings of The 1st Annual International Conference on Mobile and Ubiquitous Systems: Networking and Services*, 22-26 Aug 2004, Boston, MA.
- Chen, P. P.-S. (1976). The Entity-Relationship Model. *ACM Transactions on Database Systems*, 1(1), 9–36.
- Cherns, A. (1976). The Principles of Sociotechnical Design. *Human Relations*, 29(8), 783–792.
- Cherns, A. (1987). Principles of Sociotechnical Design Revisited. *Human Relations*, 40(3), 153–162.
- Clark, W. (2011). *Key Issues for Context-Aware Computing, 2011* (No. G00211802), Stamford, CT: Gartner Group.
- Clegg, C. W. (2000). Sociotechnical Principles for System Design. *Applied Ergonomics*, 31, 463–477.
- Codd, E. F. (1970). A Relational Model of Data for Large Shared Data Banks. *Communications of the ACM*, 13(6), 377–387.
- Davenport, T. H. H. (1993). *Process Innovation*. Boston, MA: Harvard Business School Press.
- Davenport, T. H. H., & Beers, M. C. (1995). Managing Information about Processes. *Journal of Management Information Systems*, 12(1), 57–80.
- Davenport, T. H. H., & Stoddard, D. B. (1994). Reengineering: Business Change of Mythic Proportions? *MIS Quarterly*, 18(2).
- Davis, F. D. (1989). Perceived Usefulness, Perceived Ease of Use, and User Acceptance of Information Technology. *MIS Quarterly* 13(3): 319–340.
- de Bruin, T. (2007). Insights into the Evolution of BPM in Organisations. In: *Proceedings of the 18th Australasian Conference on Information Systems*, 5-7 Dec 2007, Toowoomba, Australia.
- Delone, W. H., & McLean, E. R. (2003). The DeLone and McLean Model of Information Systems Success: A Ten-Year Update. *Journal of Management Information Systems*, 19(4), 9–30.
- Dey, A. K. (2001). Understanding and Using Context. *Personal and Ubiquitous Computing*, 5(1), 4–7.
- Dey, A. K., Abowd, G. D., & Salber, D. (2001). A Conceptual Framework and a Toolkit for Supporting the Rapid Prototyping of Context-Aware Applications. *Human-Computer Interaction*, 16(2-4), 97–166.
- Dourish, P. (2004). What We Talk About When We Talk About Context. *Personal and Ubiquitous Computing*, 8(1), 19–30.
- Dreiling, A., Rosemann, M., van der Aalst, W. M. P., Heuser, L., & Schulz, K. (2006). Model-Based Software Configuration: Patterns and Languages. *European Journal of Information Systems*, 15(6), 583–600.

- Dumas, M., van der Aalst, W. M. P., & Hofstede, ter, A. H. M. (2005). *Process-Aware Information Systems*. Hoboken, NJ: Wiley-Interscience.
- Eisenhardt, K. M. (1989). Building Theories from Case Study Research. *The Academy of Management Review*, 14(4), 532–550.
- Eisenhardt, K. M. (1991). Better Stories and Better Constructs: The Case for Rigor and Comparative Logic. *The Academy of Management Review*, 16(3), 620–627.
- Eisenhardt, K. M., & Martin, J. A. (2000). Dynamic Capabilities: What Are They?. *Strategic Management Journal*, 21(10/11), 1105–1121.
- Galliers, R. D. (2003). Change as Crisis or Growth? Toward a Trans-Disciplinary View of Information System, *Journal of the Association for Information Systems*, 4(1), 360–376.
- Germonprez, M., Hovorka, D., & Collopy, F. (2007). A Theory of Tailorable Technology Design. *Journal of the Association for Information Systems*, 8(6), 351–367.
- Germonprez, M., Hovorka, D., & Gal, U. (2011). Secondary Design: A Case of Behavioral Design Science Research. *Journal of the Association for Information Systems*, 12(10), 662–683.
- Ghattas, J., Soffer, P., & Peleg, M. (2010). A formal model for process context learning (pp. 140–157). In: *Proceedings of the Workshops of The 8th International Conference on Business Process Management*, 13-16 Sep 2010, Hoboken, NJ.
- Gibbert, M., & Ruigrok, W. (2010). The “What” and “How” of Case Study Rigor: Three Strategies Based on Published Work. *Organizational Research Methods*, 13(4), 710–737.
- Glaser, B., & Strauss, A. (1967). *The Discovery of Grounded Theory: Strategies for Qualitative Research*. Chicago, IL: Aldine Pub. Co.
- Gregor, S. (2006). The Nature of Theory in Information Systems. *MIS Quarterly*, 30(3), 611–642.
- Gregor, S., & Jones, D. (2007). The Anatomy of a Design Theory. *Journal of the Association for Information Systems*, 8(5), 312–335.
- Gu, T., Pung, H. K., & Da Qing Zhang. (2005). A Service-Oriented Middleware for Building Context-Aware Services. *Journal of Network and Computer Applications*, 28(1), 1–18.
- Günther, C. W., Rinderle-Ma, S., Reichert, M., van der Aalst, W. M. P., & Recker, J. C. (2008). Using Process Mining to Learn from Process Changes in Evolutionary Systems. *International Journal of Business Process Integration and Management*, 3(1), 61–78.
- Hackman, R. J., & Oldham, G. R. (1975). Development of the Job Diagnostic Survey. *Journal of Applied Psychology*, 60(2), 159–170.
- Hallerbach, A., Bauer, T., & Reichert, M. (2009). Guaranteeing Soundness of Configurable Process Variants in Provop. In: *Proceedings of The 11th IEEE Conference on Commerce and Enterprise Computing*, 20-23 Jul 2009, Vienna, Austria.
- Hallerbach, A., Bauer, T., & Reichert, M. (2010). Capturing Variability in Business Process Models: The Provop Approach. *Journal of Software Maintenance and Evolution: Research and Practice*, 22(6-7), 519–546.
- Hammer, M. (1990). Reengineering Work. *Harvard Business Review*, 68(7-8), 104–112.
- Harmon, P. (2007). *Business Process Change*. Amsterdam, The Netherlands; Boston, MA: Elsevier/Morgan Kaufmann Publishers.
- Hebeler, J., Fisher, M., Blace, R., Perez-Lopez, A., & Dean, M. (2009). *Semantic Web Programming*. Indianapolis, IN: Wiley.
- Henricksen, K., & Indulska, J. (2006). Developing Context-Aware Pervasive Computing Applications: Models and Approach. *Pervasive and Mobile Computing*, 2(1), 37–64.

- Hesse-Biber, S., Dupuis, P., & Kinder, T. S. (1991). HyperRESEARCH: A Computer Program for the Analysis of Qualitative Data with an Emphasis on Hypothesis Testing and Multimedia Analysis. *Qualitative Sociology*, 14(4), 289–306.
- Hevner, A. R. (2007). The Three Cycle View of Design Science Research. *Scandinavian Journal of Information Systems*, 19(2), 87–92.
- Hevner, A. R., & Chatterjee, S. (2010). *Design Research in Information Systems: Theory and Practice*. New York, NY ; London, UK: Springer.
- Hevner, A. R., March, S. T., Park, J., & Ram, S. (2004). Design Science in Information Systems Research. *MIS Quarterly*, 28(1), 75–105.
- Holland, J. H. (1992). *Adaptation in Natural and Artificial Systems*. Cambridge, MA: MIT Press.
- Huber, G. (1982). Organizational Information Systems. *Management Science*, 28(2), 138–155.
- Iivari, J. (2007). A Paradigmatic Analysis of Information Systems as a Design Science. *Scandinavian Journal of Information Systems*, 19(2), 39–64.
- Kasper, G. M. (1996). A Theory of Decision Support System Design for User Calibration. *Information Systems Research*, 7(2), 215–232.
- Kast, F. E., & Rosenzweig, J. E. (1972). General Systems Theory. *Academy of Management Journal*, 15(4), 447–465.
- Katz, D., & Kahn, R. L. (1978). *The Social Psychology of Organizations*. New York, NY: John Wiley & Sons Inc.
- Kendall, K. E., & Kendall, J. E. (2010). *Systems Analysis and Design*. Upper Saddle River, NJ: Prentice Hall.
- Kling, R. (1980). Social Analyses of Computing: Theoretical Perspectives in Recent Empirical Research. *Computing Surveys (CSUR)*, 12(1), 61–110.
- Knolmayer, G., Endl, R., & Pfahrer, M. (2000). Modeling Processes and Workflows by Business Rules. In: *Business Process Management, Models, Techniques, and Empirical Studies. Lecture Notes in Computer Science 1806*. New York, NY ; London, UK: Springer.
- Kovacic, A. (2004). Business Renovation: Business Rules (Still) the Missing Link. *Business Process Management Journal*, 10(2), 158–170.
- Krogstie, J., Sindre, G., & Jørgensen, H. (2006). Process Models Representing Knowledge for Action. *European Journal of Information Systems*, 15(1), 91–102.
- Kuechler, W., & Vaishnavi, V. (2008). The Emergence of Design Research in Information Systems in North America. *Journal of Design Research*, 7(1), 1–16.
- Kuechler, W., & Vaishnavi, V. (2011). Promoting Relevance in IS Research: An Informing System for Design Science Research. *Informing Science: the International Journal of an Emerging Trans-discipline*, 14, 125–138.
- Kuechler, W., & Vaishnavi, V. (2012). A Framework for Theory Development in Design Science Research. *Journal of the Association for Information Systems*, 13(6), 395–423.
- Kuhn, T. S. (1996). *The Structure of Scientific Revolutions*. Chicago, IL: University of Chicago Press.
- La Rosa, M., van der Aalst, W. M. P., Dumas, M., & Hofstede, ter, A. H. M. (2008). Questionnaire-Based Variability Modeling for System Configuration. *Software and Systems Modeling*, 8(2), 251–274.
- Lapkin, A. (2011). *Hype Cycle for Context-Aware Computing (No. G00213793)*. Stamford, CT: Gartner Group.

- Lieberman, H., & Selker, T. (2000). Out of Context: Computer Systems That Adapt To, And Learn From, Context. *IBM Systems Journal*, 39(3.4), 617–632.
- London, K. (1976). *People Side of Systems*. London, UK; New York, NY: McGraw Hill.
- Mackay, W. E. (1990). Patterns of Sharing Customizable Software. In: *Proceedings of The 1990 ACM Conference on Computer-Supported Cooperative Work*, 07-10 Oct 1990, Los Angeles, CA.
- MacLean, A., Carter, K., Lövsstrand, L., & Moran, T. (1990). User-Tailorable Systems: Pressing the Issues with Buttons. *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems: Empowering People*, 01-05 Apr, 1990, Seattle, WA.
- March, J. G., & Shapira, Z. (1987). Managerial Perspectives on Risk and Risk Taking. *Management Science*, 33(11), 1404–1418.
- March, J. G., & Simon, H. A. (1993). *Organizations*. New York, NY: Wiley-Blackwell.
- March, S. T., & Smith, G. F. (1995). Design and Natural Science Research on Information Technology. *Decision Support Systems*, 15, 251–266.
- March, S. T., & Storey, V. C. (2008). Design Science in the Information Systems Discipline. *MIS Quarterly*, 32(4), 725–730.
- Markus, M. L. (1983). Power, Politics, and MIS Implementation. *Communications of the ACM*, 26(6), 430–444.
- Markus, M. L., Majchrzak, A., & Gasser, L. (2002). A Design Theory for Systems That Support Emergent Knowledge Processes. *MIS Quarterly*, 26(3), 179–212.
- McGregor, D. (1960). *The Human Side of Enterprise*. New York, NY: McGraw-Hill.
- McKay, J., Marshall, P., & Hirschheim, R. (2012). The Design Construct in Information Systems Design Science. *Journal of Information Technology*, 27(2), 125–139.
- Mendling, J., Reijers, H., & van der Aalst, W. M. P. (2010). Seven Process Modeling Guidelines (7PMG). *Information and Software Technology*, 52(1), 127–136.
- Merali, Y. (2006). Complexity and Information Systems: The Emergent Domain. *Journal of Information Technology*, 21(4), 216–228.
- Miles, M. B., & Huberman, A. M. (1994). *Qualitative Data Analysis*. Thousand Oaks, CA: Sage Publications, Inc.
- Miller, E. J. (1959). Technology, Territory, and Time. *Human Relations*, 12(3), 243–272.
- Miller, J. G. (1995). *Living Systems*. New York, NY: McGraw-Hill.
- Mills, C. W., & Gitlin, T. (2000). *The Sociological Imagination*. New York, NY: Oxford University Press, USA.
- Muehlen, zur, M., & Indulska, M. (2010). Modeling Languages for Business Processes and Business Rules: A Representational Analysis. *Information Systems*, 35(4), 379–390.
- Mullins, G., & Kiley, M. (2002). 'It's a PhD, not a Nobel Prize': How Experienced Examiners Assess Research Theses. *Studies in Higher Education*, 27(4), 369–386.
- Mumford, E. (1971). *Systems Design for People*. Manchester, UK: National Computing Centre.
- Mumford, E. (2000). A Socio-Technical Approach to Systems Design. *Requirements Engineering*, 5(2), 125–133.
- Mumford, E. (2003). *Redesigning Human Systems*. Hershey, PA: Information Science Pub.

- Müller-Wienbergen, F., Müller, O., Seidel, S., & Becker, J. (2011). Leaving the Beaten Tracks in Creative Work—A Design Theory for Systems that Support Convergent and Divergent Thinking. *Journal of the Association for Information Systems*, 12(11), 714-740.
- Mørch, A. I., & Mehandjiev, N. D. (2000). Tailoring as Collaboration: The Mediating Role of Multiple Representations and Application Units. *Computer Supported Cooperative Work (CSCW)*, 9(1), 75-100.
- Nan, N. (2011). Capturing Bottom-Up Information Technology Use Processes. *MIS Quarterly*, 35(2), 505-532.
- Nardi, B. A. (1995). *Context and Consciousness: Activity Theory and Human-Computer Interaction*. Cambridge, MA: MIT Press.
- Nardi, B. A. (1996). Studying Context: A Comparison of Activity Theory, Situated Action Models, and Distributed Cognition. In: B. A. Nardi (Ed.), *Context and Consciousness: Activity Theory and Human-Computer Interaction*. Cambridge, MA: MIT Press.
- Neiger, D., Churilov, L., & Muehlen, M. (2006). Integrating Risks in Business Process Models With Value Focused Process Engineering. In: *Proceedings of the 14th European Conference on Information Systems*, 12-14 Jun 2006, Göteborg, Sweden.
- Nonaka, I. (1994). A Dynamic Theory of Organizational Knowledge Creation. *Organization Science*, 5(1), 14-37.
- Object Management Group (2008). *Semantics of Business Vocabulary and Business Rules (SBVR)*, v1.0. Needham, MA: Object Management Group.
- Object Management Group (2011). *Business Process Model and Notation, 2.0*. Needham, MA: Object Management Group.
- Orlikowski, W. J. (2007). Sociomaterial Practices: Exploring Technology at Work. *Organization Studies*, 28(9), 1435-1448.
- Orlikowski, W. J., & Iacono, C. S. (2001). Research Commentary: Desperately Seeking the "IT" in IT research—A Call to Theorizing the IT Artifact. *Information Systems Research*, 12(2), 121-134.
- Palmisano, C., Tuzhilin, A., & Gorgoglione, M. (2008). Using Context to Improve Predictive Modeling of Customers in Personalization Applications. *IEEE Transactions on Knowledge and Data Engineering*, 20(11), 1535-1549.
- Ploesser, K., Recker, J. C., & Rosemann, M. (2008). Towards a Classification and Lifecycle of Business Process Change. In S. Nurcan, R. Schmidt and P. Soffer (Eds.): *Proceedings of the 9th Workshop on Business Process Modeling, Development and Support*. 16-17 Jun 2008, Montpellier, France
- Ploesser, K., Recker, J. C., & Rosemann, M. (2010). Building a Methodology for Context-Aware Business Processes: Insights from an Exploratory Case Study. In: *Proceedings of the 18th European Conference on Information Systems*. 7-9 Jun 2010, Pretoria, South Africa.
- Ploesser, K., Recker, J. C., & Rosemann, M. (2011). Challenges in the Context-Aware Management of Business Processes: A Multiple Case Study. In: *Proceedings of the 19th European Conference on Information Systems*. 09-11 Jun 2011, Helsinki, Finland.
- Ramesh, B., Jain, R., Nissen, M., & Xu, P. (2005). Managing Context in Business Process Management Systems. *Requirements Engineering*, 10(3), 223-237.
- Ramos, E., & Santoro, F. (2010). Process Improvement Based on External Knowledge Context. In: *Proceedings of the 21th Australasian Conference on Information Systems*. 01-03 Dec 2010, Brisbane, Australia.

- Recker, J. C., Indulska, M., Rosemann, M., & Green, P. (2010). The Ontological Deficiencies of Process Modeling in Practice. *European Journal of Information Systems*, 19(5), 501–525.
- Rosemann, M., & Muehlen, zur, M. (2005). Integrating Risks in Business Process Models. In: *Proceedings of The 16th Australasian Conference on Information Systems*. 30 Nov-02 Dec 2005, Sydney, Australia.
- Rosemann, M., & Recker, J. C. (2006). Context-aware Process Design: Exploring the Extrinsic Drivers for Process Flexibility. Faculty of Science and Technology. In: T. Latour and M. Petit (Eds.), *Proceedings of The 18th International Conference on Advanced Information Systems Engineering Workshops*. 05-09 Jun 2006, Luxembourg, Luxembourg.
- Rosemann, M., & van der Aalst, W. M. P. (2007). A Configurable Reference Modelling Language. *Information Systems*, (32), 1–23.
- Rosemann, M., & Vessey, I. (2007). Toward Improving the Relevance of Information Systems Research to Practice. *MIS Quarterly*, 32(1), 1–22.
- Rosemann, M., Recker, J. C., & Flender, C. (2008). Contextualisation of Business Processes. *International Journal of Business Process Integration and Management*, 3(1), 47–60.
- Rubin, H. J., & Rubin, I. S. (2011). *Qualitative Interviewing: The Art of Hearing Data*. Thousand Oaks, CA: Sage Publications, Inc.
- Russell, N., van der Aalst, W. M. P., & Hofstede, ter, A. (2006). Workflow Exception Patterns. In: E. Dubois & K. Pohl (Eds.), *Proceedings of The 18th Conference on Advanced Information Systems Engineering*. 5-9 June 2006, Luxembourg, Luxembourg.
- Russell, N., van der Aalst, W. M. P., Hofstede, ter, A. H. M., & Edmond, D. (2005). Workflow Resource Patterns. In: *Proceedings of The 17th Conference on Advanced Information Systems Engineering*. 13-17 Jun 2005, Porto, Portugal.
- Rüegg-Stürm, J. (2005). *The New St. Gallen Management Model*. New York, NY: Palgrave Macmillan.
- Saidani, O., & Nurcan, S. (2007). Towards Context-Aware Business Process Modelling. In: *Proceedings of The 8th Workshop on Business Process Modeling, Development, and Support*. 13-17 Jun 2005, Porto, Portugal.
- Saldana, J. (2009). *The Coding Manual for Qualitative Researchers*. Thousand Oaks, CA: Sage Publications Ltd.
- Scheer, A.-W., Thomas, O., & Adam, O. (2005). Process Modeling Using Event-Driven Process Chains. In M. Dumas, W. M. P. van der Aalst, & A. H. M. ter Hofstede (Eds.), *Process-Aware Information Systems*. Hoboken, NJ: Wiley-Interscience.
- Schilit, B. N., & Theimer, M. M. (1994). Disseminating Active Map Information to Mobile Hosts. *IEEE Network*, 8(5), 22–32.
- Schilit, B. N., Adams, N., & Want, R. (1994). Context-Aware Computing Applications. In: *Proceedings of The 1st Workshop on Mobile Computing Systems and Applications*. Dec 1994, Santa Cruz, CA.
- Seidel, S., Recker, J. C., & Brocke, vom, J. (2013). Sensemaking and Sustainable Practicing. *MIS Quarterly*, Forthcoming.
- Sein, M., Henfridsson, O., Purao, S., & Rossi, M. (2011). Action Design Research. *MIS Quarterly*, 35(1), 37–56.
- Smith, H., & Fingar, P. (2006). *Business Process Management: The Third Wave*. Tampa, FL: Meghan Kiffer Press.
- Soffer, P., & Wand, Y. (2005). On the Notion of Soft-Goals in Business Process Modeling. *Business Process Management Journal*, 11(6), 663–679.

- Steinke, G., & Nickolette, C. (2003). Business Rules as the Basis of an Organization's Information Systems. *Industrial Management & Data Systems*, 103(1), 52–63.
- Strang, T., & Linnhoff-Popien, C. (2004). A Context Modeling Survey. In: *Proceedings of The 6th International Conference on Ubiquitous Computing*. 07-10 Sep 2004, Nottingham, UK.
- Suchman, L. A. (1987). *Plans and Situated Actions*. Cambridge, MA: Cambridge Univ Press.
- Suchman, L., Blomberg, J., Orr, J. E., & Trigg, R. (1999). Reconstructing Technologies as Social Practice. *American Behavioral Scientist*, 43(3), 392–408.
- Susman, G. I., & Evered, R. D. (1978). An Assessment of the Scientific Merits of Action Research. *Administrative Science Quarterly*, 23(4), 582–603.
- Taylor, H., Dillon, S., & Van Wingen, M. (2010). Focus and Diversity in Information Systems Research: Meeting the Dual Demands of a Healthy Applied Discipline. *MIS Quarterly*, 34(4), 647–667.
- Treacy, M., & Wiersema, F. (1997). *The Discipline of Market Leaders*. Cambridge, MA: Perseus Books.
- Tremblay, M. C., Hevner, A. R., & Berndt, D. J. (2010). Focus Groups for Artifact Refinement and Evaluation in Design Research. *Communications of the Association for Information Systems*, 26(1).
- Trist, E. L., & Bamforth, K. W. (1951). Some Social and Psychological Consequences of the Longwall Method. *Human Relations*, 4(1951), 3–38.
- Vaishnavi, V., & Kuechler, W. (2004). *Design Research in Information Systems*. 20 Jan 2004, last updated September 30, 2011. URL: <http://desrist.org/desrist>
- van der Aalst, W. M. P., & Hofstede, ter, A. H. M. (2003). Workflow Patterns. *Distributed and Parallel Databases*, 14, 5–51.
- van der Aalst, W. M. P., Pesic, M., & Schonenberg, H. (2009). Declarative Workflows: Balancing Between Flexibility and Support. *Computer Science - Research and Development*, 23(2), 99–113.
- van der Aalst, W. M. P., Rosemann, M., & Dumas, M. (2007). Deadline-based Escalation in Process-Aware Information Systems. *Decision Support Systems*, 43, 492–511.
- Wade, M., & Hulland, J. (2004). Review: The Resource-Based View and Information Systems Research. *MIS Quarterly*, 28(1), 107–142.
- Walls, J. G., Widmeyer, G. R., & Sawy, El, O. A. (1992). Building an Information System Design Theory for Vigilant EIS. *Information Systems Research*, 3(1), 36–59.
- Walls, J. G., Widmeyer, G. R., & Sawy, El, O. A. (2004). Assessing Information System Design Theory in Perspective. *Journal of Information Technology Theory and Application*, 6(2), 43–58.
- Walsham, G. (1995). Interpretive case studies in IS research: nature and method. *European Journal of Information Systems*, 4(2), 74–81.
- Wand, Y., & Weber, R. (1995). On the deep structure of information systems. *Information Systems Journal*, 5(3), 203–223.
- Weber, B., Reichert, M., & Rinderle-Ma, S. (2008). Change Patterns and Change Support Features—Enhancing Flexibility in Process-Aware Information Systems. *Data & Knowledge Engineering*, 66(3), 438–466.
- Weber, B., Reichert, M., & Wild, W. (2009). Providing Integrated Life Cycle Support in Process-Aware Information Systems. *International Journal of Cooperative Information Systems*, 18(1), 115–165.

- Weber, R. (1997). *Ontological Foundations of Information Systems*. Melbourne, Vic.: Coopers & Lybrand and the Accounting Association of Australia and New Zealand.
- Weber, R. (2012). Evaluating and Developing Theories in the Information Systems Discipline. *Journal of the Association for Information Systems*, 13(1), 2.
- Webster, J., & Watson, R. T. (2002). Analyzing the Past to Prepare for the Future: Writing a Literature Review. *MIS Quarterly*, 26(2), xiii-xxiii.
- Weill, P., & Olson, M. H. (1989). An Assessment of the Contingency Theory of Management Information Systems. *Journal of Management Information Systems*, 6(1), 59-79.
- Weiser, M. (1993). Some Computer Science Issues in Ubiquitous Computing. *Communications of the ACM*, 36(7), 75-84.
- Yin, R. K. (2009). *Case Study Research*. Thousand Oaks, CA: Sage Publications, Inc.

Appendix A: Case Shortlist

#	Prio	Organisation			Contact		Engagemt	Rationale	Comments
		Name	Sector	Location	Channel	Name	Role	Contact Q	
1	1	Brisbane Airport	Aviation	ANZ	QUT			High dependency on crude, demand elasticity for leisure travel, weather patterns	
2	1	Qantas	Aviation	ANZ	?			"	
3	1	Lufthansa	Aviation	EMEA	SAP			"	
4	1	Emirates	Aviation	ANZ	QUT			"	
5	1	Virgin Blue	Aviation	ANZ	QUT			"	
6	1	Suncorp	Insurance & Bankin	ANZ	QUT/SAP			Mitigating financial risk, increased risk from natural disasters	
7	1	Commonwealth	Insurance & Bankin	ANZ	SAP			"	
8	1	NAB	Insurance & Bankin	ANZ	SAP			"	
9	1	IAG	Insurance & Bankin	ANZ	?			"	
10	1	Australia Post	Logistics	ANZ	SAP/QUT			High dependency on crude, weather patterns	
11	1	Toll	Logistics	ANZ	?			"	
12	1	Rio Tinto	Mining	ANZ	SAP			Environmental, health & safety a key priority in a high-risk environment	
13	1	Woolworths	Retail	ANZ	SAP/QUT			Seasonal sourcing, drought, campaign management, reputation	
14	1	Coles	Retail	ANZ	SAP/QUT			"	
15	2	BHP Billiton	Mining	ANZ	?			Environmental, health & safety a key priority in a high-risk environment	
16	2	IGA	Retail	ANZ	?			Seasonal sourcing, drought, campaign management, reputation	
17	2	Origin	Utilities	ANZ	SAP			Managing the grid, energy mix	
18	3	Australian Defer	Public Security & D	ANZ	SAP			Natural disasters, emergency response, terrorist threat	
19	3	Melbourne Fire	Public Security & D	ANZ	SAP			"	
20	3	Myer	Retail	ANZ	?			Trends, fashion, taste, campaign management, demand elasticity for luxury goods	
21	3	David Jones	Retail	ANZ	?			"	

Appendix B: Ethics Clearance Forms



COVERSHEET FOR RESEARCH INVOLVING HUMAN PARTICIPANTS

This coversheet must be completed and attached to your application

OFFICE USE ONLY

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VERSION: 1.1 07-08-2007

PLEASE NOTE

If you do not see the red "hidden text" (which provides guidance to the questions) please go to:

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PLEASE ENSURE HIDDEN TEXT IS NOT PRINTED WITH YOUR FINAL SUBMISSION

PROJECT TITLE Context-awareness in Business Processes

CHIEF INVESTIGATOR

Surname	First name
Ploesser	Karsten
STAFF / STUDENT #	CONTACT NUMBER: 0424169255
AOU # (if known)	EMAIL ADDRESS: k.ploesser@student.qut.edu.au

SUPERVISOR / ALTERNATE

Surname	First name
Rosemann	Michael
STAFF / STUDENT #	CONTACT NUMBER: (07) 3138 9473
AOU # (if known)	EMAIL ADDRESS: m.rosemann@qut.edu.au

OTHER MEMBERS OF THE RESEARCH TEAM

Surname	First name
Recker	Jan
STAFF / STUDENT # OR EXTERNAL ORGANISATION	
STAFF / STUDENT # OR EXTERNAL ORGANISATION	
STAFF / STUDENT # OR EXTERNAL ORGANISATION	

LEVEL OF APPLICATION ☒ LOW RISK

☐ COMMITTEE

PROJECT TYPE Please indicate which applies to your application and provide details as appropriate.

☐ UNDERGRADUATE RESEARCH (Honours) ☒ POSTGRADUATE RESEARCH (PhD, MSc) ☐ OTHER (Please specify below)

PROJECT FUNDING

If you have a grant / contract application with Office of Research or Commercialisation Office, please provide the application reference number here so that we can link with your ethics.

N/A

RESEARCH SAFETY

- ☒ Does not require consideration. (Please provide brief reason below for not requiring consideration.)
- ☐ Has been considered and approved by a Faculty / School approved process.
- ☐ Is yet to be considered by a Faculty / School approved process.
- ☐ I believe that this project requires an expert / external research safety review.

The research is concerned with research topics in Business Process Management. The main data collection techniques is qualitative interviews. There are no anticipated risks involved in participating in this project.

CONFLICT OF INTEREST

Please indicate if there are any perceived or actual conflicts of interest for the *applicant / research team / supervisor / Head of School* in regard to this project. For definitions refer to Code of Conduct for Research.

- ☒ NO ☐ YES With any funding body.
- ☒ NO ☐ YES With the supervision of the research student.
- ☒ NO ☐ YES With any commercial activity.
- ☒ NO ☐ YES Other conflict

(If you answer "YES" to any of the above, please provide details and indicate how they will be managed.)

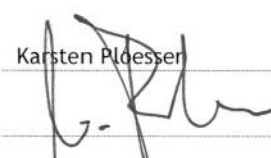
DECLARATIONS SECTION

APPLICANT — In submitting this application I declare that:

- All ethical implications of the proposed research have been considered and all relevant local, state and national guidelines, regulations and legislation have been considered and the measures taken are appropriate and in accordance with University policy and the National Statement on Ethical Conduct in Research Involving Humans.
- The research / scientific merit of this project has been considered and validated (Question 3.4 of application).
- The Research Safety of the project has been considered and the measures taken are appropriate.
- There are no perceived or actual conflicts of interest (or these have been declared above).
- The qualifications and experience of all investigators are appropriate to the study to be undertaken.
- All support staff involved in the project (eg administrative staff who undertake data entry) will be required to understand and observe the assurances provided to subjects (eg to safeguard their anonymity).
- Approval from University Human Research Ethics Committee will be requested for any divergence from the protocol stated in this proposal.
- University Human Research Ethics Committee will be advised immediately of any adverse effects arising from this study (eg unexpected adverse outcomes, unexpected community / subject risk factors or complaints, etc).
- A Progress Report will be provided at least annually.

Name: Karsten Ploesser

School: BPM Research Group, Faculty of Science and Technology

Signature: 

Date: 01/06/2009

SUPERVISOR (if applicable) —

I have considered this application and believe:

- All ethical implications of the proposed research have been considered and all relevant local, state and national guidelines, regulations and legislation have been considered and the measures taken are appropriate and in accordance with University policy and the National Statement on Ethical Conduct in Research Involving Humans.
- The research / scientific merit of this project has been considered and validated
- The Research Safety of the project has been considered and the measures taken are appropriate
- There are no perceived or actual conflicts of interest (or these have been declared above)
- The qualifications and experience of all investigators are appropriate to the study to be undertaken

Name: Michael Rosemann

School: BPM Research Group, Faculty of Science and Technology

Signature: _____

Date: / /

HEAD OF SCHOOL / CENTRE DIRECTOR / HEAD OF DISCIPLINE —

I have considered this application and believe:

When the Head of School, Centre Director or Head of Discipline is also a listed applicant (or is a supervisor for the project) the Dean or Dean of Research should sign below.

- All ethical implications of the proposed research have been considered and all relevant local, state and national guidelines, regulations and legislation have been considered and the measures taken are appropriate and in accordance with University policy and the National Statement on Ethical Conduct in Research Involving Humans.
- The research / scientific merit of this project has been considered and validated.
- The Research Safety of the project has been considered and the measures taken are appropriate.
- There are no perceived or actual conflicts of interest (or these have been declared above).
- The qualifications and experience of all investigators are appropriate to the study to be undertaken.

Name: _____

School: _____

Signature: _____

Date: ____ / ____ / ____

PLEASE FORWARD COMPLETED AND SIGNED FORM WITH RELEVANT ATTACHMENTS TO:

Research Ethics Officer Office of Research Level 3 O Block Podium Gardens Point Campus



APPLICATION FOR REVIEW OF LOW RISK RESEARCH INVOLVING HUMAN PARTICIPANTS

PLEASE NOTE

If you do not see the red **"hidden text"** (which provides guidance to the questions) please go to:

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If you wish to view hidden text when you print the document please go to:

| Tools | Options | Select "Print" tab | Check "Hidden Text" box |

PLEASE ENSURE HIDDEN TEXT IS NOT PRINTED WITH YOUR FINAL SUBMISSION

APPLICATION SECTIONS: A. Research Proposal Overview | B. Participant Overview | C. Data Management

SECTION A: RESEARCH PROPOSAL OVERVIEW

1. Summary Information

1 Project Title

Context-awareness in Business Processes

1.1 Research summary

The emergence of Business Process Management (BPM) and Enterprise Resource Planning (ERP) systems has led to significant improvements in the design, implementation and overall management of business processes (Hammer & Champy, 2003). So far, the focus of BPM research and practice has been on internal business and internal operations, assuming a static, implicit context in which processes are designed and operated. However, corporations are increasingly facing challenges emerging from a dynamic, changing context. A lack of context-awareness often leads to processes, and supporting information systems, that are unable to react appropriately and quickly enough to external disturbances and other change drivers (Knoll & Jarvenpaa, 1994). This study is being conducted as part of Mr Karsten Ploesser's doctoral work, at the Faculty of Science and Technology, Queensland University of Technology. Following a systemic approach, this study understands business processes as complex systems embedded in and interdependent on their context. The candidate suggests that an increased awareness of this context can lead to improved designs of business processes. Research in related fields such as human factors and system thinking has led to insights into the phenomenon of awareness (Endsley, 1995; Norman, 1976; Wickens & Hollands, 2000) and the behaviour of dynamic systems (Serman, 2001; Holland, 1992). However, to this date little empiric evidence and practical guidance exists on how to attain context-awareness in business processes in a systematic manner (Rosemann, Recker, & Flender, 2008). Through this study, the candidate seeks to achieve two objectives: (1) a comprehensive understanding of context-awareness in a real world, business setting; and (2) the collaborative development of a systematic framework for attaining context-awareness in business processes.

1.2 Participant summary

Participants will be involved in two stages of research. The first phase, stage (1), involves the application and empirical testing of the conceptual framework. Upon successful completion of stage (1), participants will be given the opportunity to extend the collaboration with the candidate into the next phase of research, stage (2). During this stage, the candidate seeks to collaboratively develop a systematic framework for attaining context-awareness in business processes. This involves a project agreement between researcher and receiving organisation with the intent of implementing the framework in several cycles of analysis, design, and testing. Participation is at any time voluntary. The research will make use of qualitative interviews with key participants and stakeholders of a business process such as the business owner, business analyst, enterprise architect, and line of business expert. Interview data may triangulated with other sources of evidence such as handbooks, process documentation, and interviews). The findings will serve to describe current practice, elicit requirements and potential for improvement. The interviews will be led in an open-ended, conversational manner to avoid distress on the informant's part (Yin, 2003).

1.3 Please provide a brief justification for considering this a Low Risk application.

This study proposes to develop an empirical understanding of context-awareness and a systematic approach to attaining it, which is of immediate relevance and benefit to its participants. The applicant consulted the literature on ethical issues related to the research methods. No direct harm can be expected from the use of case study research. The research does not manipulate the participants, but seeks to provide a faithful and non-biased account of their experiences. The questions asked and general approach of the case study will be documented in a case study protocol, which will be made available to its participants upfront (Yin, 2003). As for the action research cycle, each action undertaken in the action research cycle will be carefully planned with the participants, and risks and benefits will be balanced before the action is implemented in the client-system (Susman & Evered, 1978).

2. Potential Risks and Benefits

2 Potential Risks — please indicate if there are any potential risks associated with the project?

There are no anticipated risks involved in participating in this project. In particular, no substances whatsoever are going to be administered to the participants. The researcher will mainly use interviews as data collection technique. There is a very low probability of these interviews causing distress, e.g. in case participants are unwilling to share information.

2.1 Managing the risk

Participants will be asked to review the informant information sheet and sign the letter of consent before the commencement of interviews and will be given the opportunity to opt out of interviews at any time and at their own discretion should they cause them distress.

2.2 Potential Benefits — please indicate if there are any potential benefits associated with the project and who benefits?

This research benefits two audiences: the community of academic research in BPM and the community of BPM practice. On the one hand, the collection of empiric evidence in stage (1) on how organisations attain context-awareness in their business processes will support the advancement of our understanding of this phenomenon. It also provides an indication on how the techniques currently employed by firms to attain context-awareness may be improved. On the other hand, organisations may directly benefit from the output of stage (2), i.e. the joint development of a systematic framework for attaining context-awareness. Through their participation, these firms receive a framework through which they may attain context-awareness in a systematic, repeatable manner.

2.3 Balancing against the risks

The investigation has the potential to considerably support the participants in doing their work and may improve existing means to attain context-awareness. The candidate suggests that these benefits outweigh the investments made by the participants in terms of interview time.

3. Other General Information

3 Location of research — Location where the research will be conducted

The research will be conducted at a location assigned by the case study partner. The candidate will ask each case study partner to send a letter of support for conducting the study in that place before the start of the case study. This letter will be forwarded to QUT once it is available.

3.1 Role of QUT HREC — Please state if QUT is the primary / only HREC reviewing this proposal

QUT HREC will be the only HREC reviewing this proposal. At least one and at most two more case studies will be conducted in the context of this research that are of similar nature. There is a potential to streamline the application process for these in the future as the study set up will be the same. Note that the candidate has agreed to sign a confidentiality statement with the first case study partner.

3.2 Estimated timeframes for the project

START OF PROJECT: 03 / 03 / 2008

END OF PROJECT: 02 / 03 / 2011

START OF DATA COLLECTION: 08 / 06 / 2009

**COMPLETION OF DATA
COLLECTION:** 31 / 12 / 2010

SECTION B: PARTICIPANT OVERVIEW

1 Who will be approached to participate in the project?

The applicant will approach a minimum of 4 participants (business manager, business analyst, events specialist, subject matter experts) in each case site. Additional participants may be recruited within the unit should this be required and given the consent by the case site.

1.1 How will the participants be approached and provide their consent to participate?

The participants will be nominated by the business manager of the respective unit of the case site. Each participant will be asked to sign a letter of consent based on the template provided by QUT.

1.2 Will the study involve participants who are unable to give informed consent?

☒ NO ☐ YES

1.3 Will the potential participants be screened?

The applicant will rely on the nomination provided by the business manager.

1.4 Will participants be offered reimbursements, payments or incentives?

The individual participants will not be offered reimbursements.

1.5 Is there an existing relationship with participants?

No, there is no such relationship

1.6 Is it proposed to conduct a debriefing session at the end of the research (or at the end of each participant's involvement)?

The research does not address issues that could cause distress amongst participants

1.7 Will feedback, the outcome / results of this research be reported to participants?

The participants will be provided with a case study report outlining the findings of the case study. This will comprise a documentation of the current situation, as well as a register of key issues and improvement areas observed during the study. In addition, a presentation to the participants will be given at the end of the case study.

SECTION C: DATA MANAGEMENT

1. Procedures & Protection

1.1 What data collection procedures will be utilised?

QUESTIONNAIRE	☒ NO	☐ YES	SURVEY	☒ NO	☐ YES
INTERVIEWS	☐ NO	☒ YES	FOCUS GROUPS	☒ NO	☐ YES
ARCHIVAL RECORDS	☐ NO	☒ YES	OTHER INSTRUMENT	☒ NO	☐ YES

1.2 Have the data collection procedures been previously approved by QUT or are they and academic standard instrument? ☐ NO ☒ YES

1.3 Please provide brief details on prior approval / where instruments have been used previously eg under a similar context to this proposal.

Similar techniques have been applied in the BPM group and have received clearance in the past. Examples include the use of interviews and archival records in the work of Wasana Bandara, Jan Recker, and Tonia de Bruin.

1.4 How will the data be recorded?

Individually Identifiable	☒ NO	☐ YES
Re-Identifiable or Potentially Re-Identifiable	☒ NO	☐ YES
Non-Identifiable	☐ NO	☒ YES

1.5 Data Ownership & Access

The data ownership will remain with the research student. Only the research team (i.e. the student and his supervisors) will have access to the data.

1.6 Protecting Confidentiality — How will confidentiality of the records of the study be protected during the study and in the publication of results?

A confidentiality agreement will be signed with cases for information that is considered classified by the organisation.

2. Storage & Security

2.1 Records stored for required period ☐ NO ☒ YES

2.2 Location of storage

Computer of chief investigator.

Note that through the restructuring of the department of information technology, the candidate is currently enquiring who is responsible for approval of off-site data storage (i.e. on the laptop of the candidate). Thus the candidate could not make a final selection for questions 2.2. However, the candidate will send a variation of this form with 2.2 answered appropriately once approval was granted.

2.2 Approval from Faculty for storage if off site ☐ N/A ☐ NO ☐ YES

2.4 Who will have access? The research team

2.5 How will access be controlled? Password protected computer

3. Privacy of Information Held by Commonwealth Agencies

3.1 Is this a medical research proposal (including epidemiological research)? ☒ NO—Please go to Part 4 ☐ YES

3.2 Does the proposal require the use or disclosure of information from a Commonwealth agency? ☐ NO ☐ YES

3.3 Does the proposal require use or disclosure of personal information, ☐ NO ☐ YES

3.4 Does the proposal involve not obtaining consent from the individuals to whom the information related? ☐ NO ☐ YES

4. Privacy of Information Held by Private Sector

- 4.1 Does the proposal involve:
- Research relevant to public health or safety? **OR**
 - The compilation or analysis of statistics relevant to public health or safety? **OR**
 - The management, funding or monitoring of a health service?
- ☒ NO—Please go to Part 5 ☐ YES
- 4.2 Does the proposal involve collection, use or disclosure of information from a **private sector organisation**? ☐ NO ☐ YES
- 4.3 Was it **necessary** to collect, use or disclose **health information**? ☐ NO ☐ YES
- 4.4 Was it **impracticable** for **consent to be obtained** from the individuals to whom the health information related? ☐ NO ☐ YES

5. Specific details

Where you have answered “Yes” to any of the questions in Parts 3 and 4 above (access to Commonwealth or Private Sector Health Data), please provide the following information.

5.1 **Agency**

5.2 **Number of records**

5.3 **Information Privacy Principles** — Will this access constitute a breach of an Information Privacy Principle (eg access to this data without the prior approval of the participants)?

Appendix C: Information Consent Forms



PARTICIPANT INFORMATION for QUT RESEARCH PROJECT

Context-awareness in Business Processes

Research Team Contacts

Karsten Ploesser, Chief Investigator	Professor Michael Rosemann, Principal Supervisor	Dr Jan Recker, Associate Supervisor
Business Process Management Research Group	Business Process Management Research Group	Business Process Management Research Group
Faculty of Science and Technology	Faculty of Science and Technology	Faculty of Science and Technology
Queensland University of Technology	Queensland University of Technology	Queensland University of Technology
Level 5 / 126 Margaret Street	Level 5 / 126 Margaret Street	Level 5 / 126 Margaret Street
Brisbane QLD 4000, Australia	Brisbane QLD 4000, Australia	Brisbane QLD 4000, Australia
Mobile: 04 2416 9255	Phone: (07) 3138 9473	Phone: (07) 3138 9479
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Description

The emergence of Business Process Management (BPM) and Enterprise Resource Planning (ERP) systems has led to significant improvements in the design, implementation and overall management of business processes. So far, the focus of BPM research and practice has been on internal business and internal operations, assuming a static, implicit context in which processes are designed and operated. However, corporations are increasingly facing challenges emerging from a dynamic, changing context. A lack of context-awareness thus often leads to processes, and supporting information systems, that are unable to react appropriately and quickly enough to external disturbances and other change drivers.

This study is being conducted as part of Mr Karsten Ploesser's doctoral work, at the Faculty of Science and Technology, Queensland University of Technology. Following a systemic approach, this study understands business processes as complex systems embedded in and interdependent on their context. The candidate suggests that an increased awareness of this context can lead to improved designs of business processes. Research in related fields such as human factors and system thinking has led to insights into the phenomenon of awareness. However, to this date little empiric evidence and practical guidance exists on how to attain context-awareness in business processes in a systematic manner.

Through this study, the candidate seeks to achieve two objectives: (1) a comprehensive understanding of context-awareness in a real world, business setting; and (2) the collaborative development of a systematic framework for attaining context-awareness in business processes. For this purpose, a conceptual framework has been developed through a thorough analysis of the existing literature. The first phase, stage (1), involves the application and empirical testing of this framework. Your participation in the research plays a crucial part for this phase of the study.

Upon successful completion of stage (1), you will be given the opportunity to extend the collaboration with the candidate into the next phase of research, stage (2). During this stage, the candidate seeks to collaboratively develop a systematic framework for attaining context-awareness in business processes. This involves a project agreement between researcher and receiving organisation with the intent of implementing the framework in several cycles of analysis, design, and testing. Your participation in this stage is voluntary and will be considered separately from stage (1).

Participation

Your participation in this project is voluntary. If you do agree to participate, you can withdraw from participation at any time during the project without comment or penalty. Your decision to participate will in no way impact upon your current or future relationship with QUT.

Your participation will involve interviews in an open-ended, conversational manner. All interviews will typically be conducted at your location, preferably in a meeting room. You will be contacted at a maximum of three instances. The interviews will generally cover the topics information about yourself related to the study topic (such as your role, responsibilities and experience), general information about a selected business process, information about

procedures applied by you or your team to attain context-awareness. Depending on your availability, these interviews can be conducted in one session of 2h or two separate session of 1h each. A final contact will be made for information justification and follow up.

Expected benefits

Your involvement in this project may not directly benefit you. However, it is hoped that by obtaining insights into context-awareness in business processes, organisations and individuals concerned with the management and design of such processes will benefit significantly.

Risks

There are no anticipated risks involved in participating in this project. However, if you are concerned with any issues pertaining to your involvement in the project, you can contact any of the research team members of this project (see details above) or the QUT Research Ethics Officer on (07) 3138 2340 or ethicscontact@qut.edu.au.

AV Recordings of Proceedings

The research team would appreciate the permission of the respondents to audio record the interviews, for better data capture. The decision to grant permission to record will solely be made by the respondent and they may still participate in the project, without being audio recorded. The recordings of those contacts being audio recorded will be destroyed after the contents have been transcribed. All interview transcripts will be handled in a fully confidential manner.

Confidentiality

All comments and responses are anonymous and will be treated confidentially. The names of individual persons are not required in any of the responses. Only the research team will have access to the information you provide. Your anonymity and confidentiality will be safeguarded in any publication of the results of this research. No individual will be referred to (except through the use of pseudonyms), only aggregated results will be reported. All information gathered will reside securely with the research team and will be subject to the audits of the QUT's Research Ethics Review Committee.

Consent to Participate

We would like to ask you to sign a written consent form (enclosed) to confirm your agreement to participate.

Questions / further information about the project

Please contact the researcher team members named above to have any questions answered or if you require further information about the project.

Concerns / complaints regarding the conduct of the project

QUT is committed to researcher integrity and the ethical conduct of research projects. However, if you do have any concerns or complaints about the ethical conduct of the project you may contact the QUT Research Ethics Officer on 3138 2340 or ethicscontact@qut.edu.au. The Research Ethics Officer is not connected with the research project and can facilitate a resolution to your concern in an impartial manner.



CONSENT FORM for QUT RESEARCH PROJECT

Context-awareness in Business Processes

Statement of consent

By signing below, you are indicating that you:

- have read and understood the information document regarding this project
- have had any questions answered to your satisfaction
- understand that if you have any additional questions you can contact the research team
- understand that you are free to withdraw at any time, without comment or penalty
- understand that you can contact the Research Ethics Officer on 3138 2340 or ethicscontact@qut.edu.au if you have concerns about the ethical conduct of the project
- agree to participate in the project, and ... *(please choose only one of the options below)*
 - grant permission to audio record interviews ☐
 - do not grant permission to audio record any interviews ☐

Name

.....

Signature

.....

Date

..... / /

Interview Guide

Who – The Team

Question	Supporting Documents
1. Can you briefly describe yourself? - What is your role, responsibility, and expertise in the team? - How long have you been working in this team?	
2. How would you describe your team? - How many team members are involved? - What are their roles and responsibilities? - What is their respective expertise?	
3. Who are your main contacts in the team or other units - With what frequency do you interact with each of them? - What makes them significant to your work? - Is it possible for me to contact them?	

What – The Selected Business Process

Question	Supporting Documents
1. What are the process goals and objectives? - Can you enumerate these goals and objectives? - How are they managed? - How are these objectives measured? - What data is collected for the purpose of measurement? - Can you determine typical patterns in this data that correlate with events? - Following this scheme, how would you classify this pattern? [<i>presentation of exhibit 1</i>]	Balanced scorecards, samples of key performance indicators
2. What are relevant events for this process? - Can you enumerate these events? - Following the following context taxonomy, how would you classify these events? [<i>presentation of exhibit 2</i>] - What criteria do you apply to determine whether an event is relevant? - How do you document events and information about their relevancy?	Templates for documentation of events
3. What are the main components of the process? - Can you enumerate these components (e.g. documents, systems, activities, resources, ...)? - How are these components documented? - Do events affect these components? If yes, how? - How is the effect of events on components documented?	Process specifications, process models

How – Procedures For Attaining Context-awareness

Question	Supporting Documents
1. How would you describe the overall approach to events - What are the main activities (please describe in detail)? - Can you relate these to the following lifecycle model? [<i>presentation of exhibit 3</i>]	
2. Which out of these activities do you perform yourself? - How does this activity relate to events? - What information do you require (documents, data, ...)? - Where do you source them from (people, data sources, ...)? - What devices do you use to support you in the activity	Sample documents, training manuals for manual activities or software applications

-
- (visual aids, analytical tools, ...)?
- e. What outputs do you produce (documents, data, ...)?
 - f. Who receives these outputs?
 - g. What is the next activity that is performed after you complete your work?
3. Are these procedures documented? Procedural guidelines
- a. Is documentation provided in a systematical manner?
 - b. What variation points from the guidelines can you perceive?
4. What could be improved? What would support you in this work beyond what is available today?
-

Appendix

Exhibit 1: Measurement Patterns



Exhibit 2: Context Taxonomy

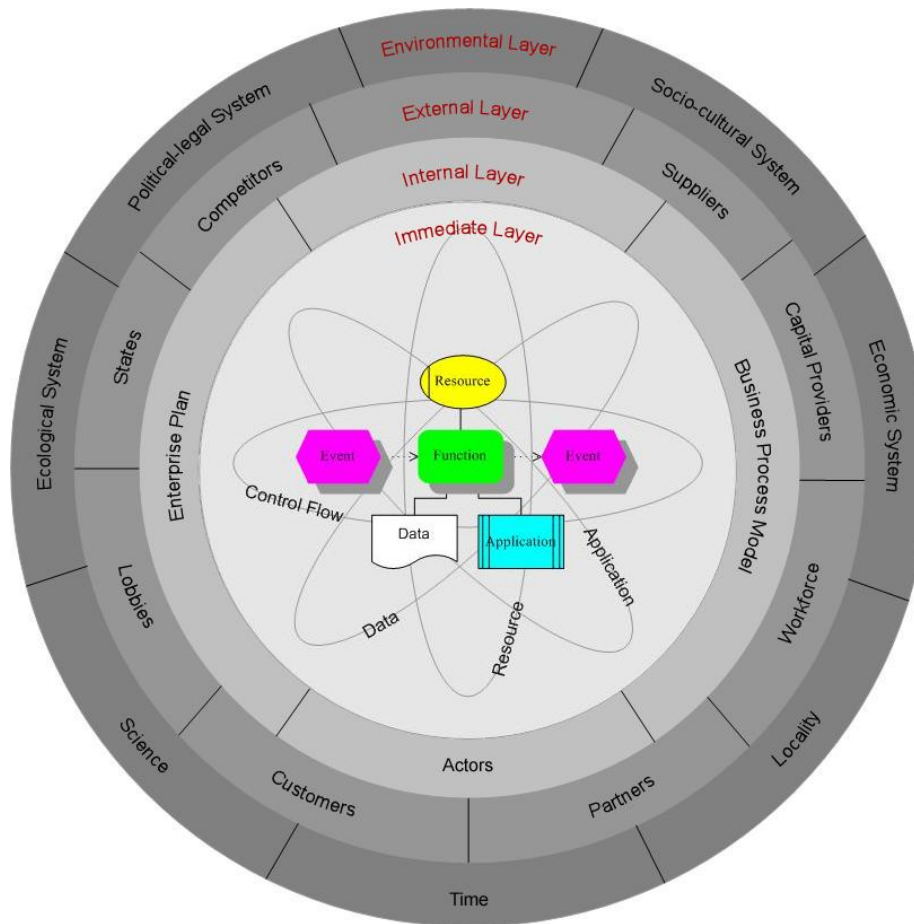


Exhibit 3: Lifecycle Model Context-awareness

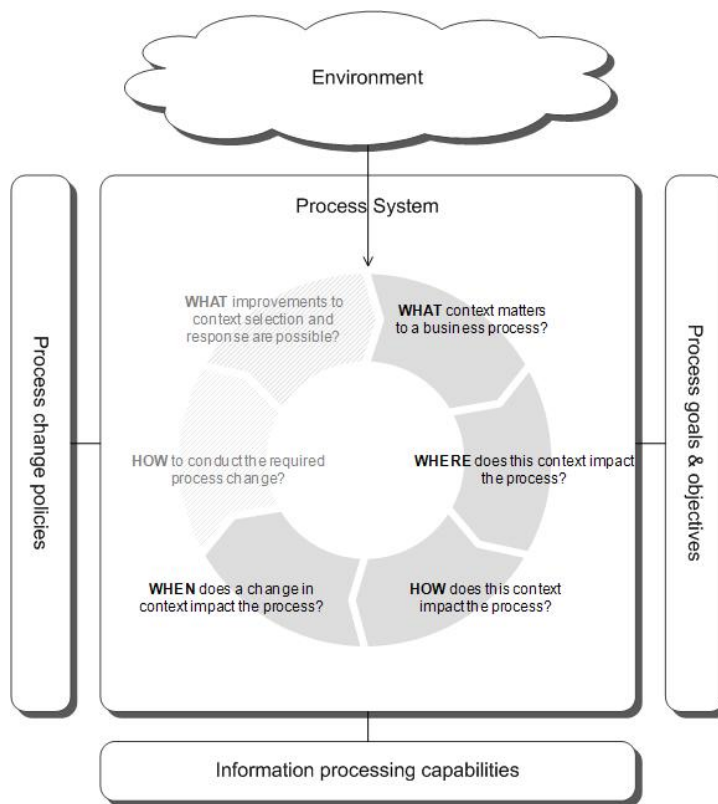


Exhibit 4: Brainstorming Diagram of Interview Questions

